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**UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., Defendant.	Case No. 3:26-cv-00705
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SUPPLEMENTAL TO THE ADDENDUM

SUPPLEMENTAL EXHIBIT APPENDIX

VOLUME I

SUPPLEMENTAL EXHIBITS SA THROUGH SJ

Organized as a two-volume supplemental exhibit appendix

**UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., Defendant.	Case No. 3:26-cv-00705
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PRELIMINARY STATEMENT

PURPOSE OF THIS SUPPLEMENTAL EXHIBIT APPENDIX

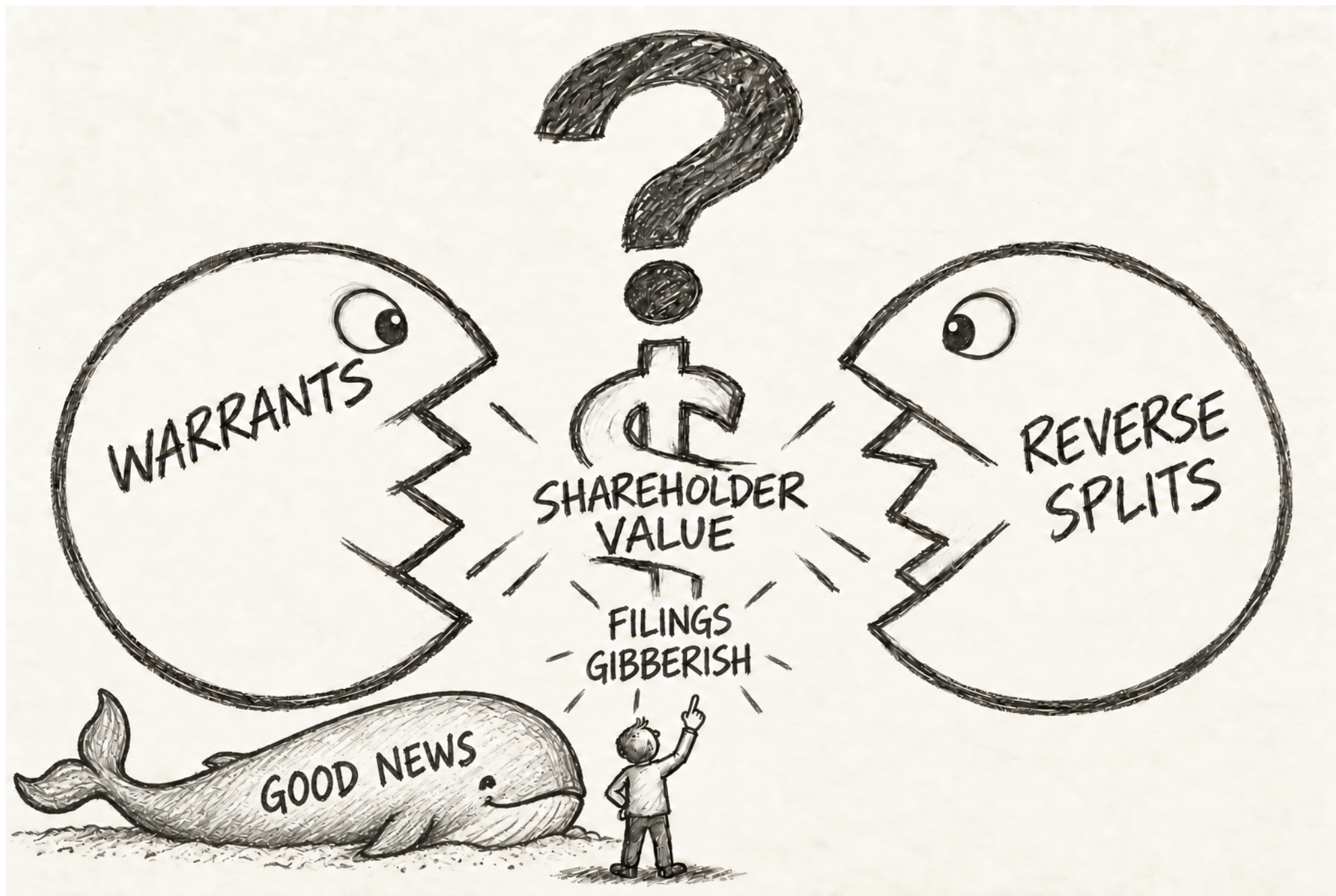
This supplemental appendix assembles public filings, company releases, patent records, financing records, market data, demonstratives, and Plaintiff's identified questions concerning Theriva's capitalization, warrants, reverse splits, cash use, debt, intellectual property, strategic review, and potential change-of-control mechanisms. It is submitted so the Court may evaluate those materials together rather than in isolation.

Plaintiff respectfully seeks leave to add this supplemental appendix to the previously submitted Addendum. In a separate emergency motion, Plaintiff requests temporary preservation of the status quo concerning the disputed warrants and any adopted or contemplated shareholder-rights plan or control-directed preferred-stock mechanism while the Court obtains verified disclosure and considers the requested relief.

Plaintiff does not ask the Court to accept his questions or inferences as established fact. He asks the Court to review the cited record, consider the materials in context, and decide for itself whether temporary preservation, further disclosure, or other relief is warranted.

Please review the following record and decide for yourself.

This statement identifies the requested procedure; it is not a finding of fact or liability.



Opening illustrative demonstrative moved from original Part A - submitted as a question, not as a finding of intent.

SUPPLEMENTAL EXHIBIT INDEX - VOLUME I

Nejmeh v. Theriva Biologics, Inc. - Case No. 3:26-cv-00705

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SB	Patent Disclosures Market Activity Warrants and Reverse Splits	SB-1 to SB-12	18-30
SC	Financial Program and Strategic Record	SC-1 to SC-11	31-42
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SE	Reverse Splits Warrant Financing Market Capitalization and Cash Use Analysis	SE-1 to SE-8	47-55
SF	Grifols Deferred Contingent Consideration and Strategic Transaction Liquidity Sequence	SF-1 to SF-4	56-60
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Every supplemental exhibit begins with a descriptive cover sheet. Content pages carry only a bottom-margin identifier.

**UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., Defendant.	Case No. 3:26-cv-00705
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SUPPLEMENTAL EXHIBIT SA

Reverse Splits Warrants Related Person Inquiry and Capital Use

Original Part A

The following pages are separately paginated under this supplemental exhibit identifier.

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**UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., a Nevada corporation, Defendant.	Case No. 3:26-cv-00705 PROPOSED SUPPLEMENTAL ADDENDUM CONCERNING REVERSE SPLITS, WARRANT TRANSACTIONS, FINANCING SEQUENCES, AND CAPITAL USE
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QUESTION PRESENTED

**Did the Company initiate reverse splits and warrant transactions in a repeated
sequence to facilitate later financings?**

LIMITED PURPOSE: This proposed addendum concerns only the existing warrant-related action. It identifies federal and Nevada legal guideposts only to frame limited preservation, discovery, and fact-finding questions. Plaintiff does not ask the Court to adjudicate any claim not pleaded in the verified complaint through this memorandum.

Submitted as proposed supplemental material, subject to the Court's permission.

I. PURPOSE AND LIMITED SCOPE

1. Plaintiff submits this proposed addendum to place the existing warrant dispute in a documented chronology of reverse splits, warrant-related transactions, capital raising, common-share expansion, and cash use.
2. The addendum does not ask the Court to infer fraud or unlawful purpose from chronology alone. It identifies a repeated sequence that may bear on motive, necessity, disclosure, irreparable-harm analysis, and the need for targeted records concerning the challenged warrants.
3. The figures below are derived from the Company's SEC filings and daily market prices. Market capitalization is a calculated proxy equal to closing price multiplied by reported common shares outstanding. It excludes preferred shares and unexercised warrants.

II. ANSWER TO THE QUESTION PRESENTED

4. The public record establishes a repeated chronology: three reverse splits were followed within 66, 3, and 32 days by financings that either issued new warrants or altered the economics of outstanding warrants.
5. The 2018 and 2024 post-split transactions issued new warrants. The July 2022 transaction was a preferred-stock placement and did not issue new warrants, but it repriced warrants issued in 2018 to \$1.22 per share and produced a reported \$340,000 noncash deemed dividend.
6. The filings state that each reverse split was undertaken to satisfy NYSE American per-share-price requirements. The filings reviewed do not expressly state that any reverse split was approved to facilitate a later financing. Accordingly, the repeated sequence supports a factual inquiry; it does not by itself establish subjective intent.

III. REVERSE-SPLIT AND FINANCING CHRONOLOGY

Split effective	Ratio	Next event	Days	Documented event
2018-08-10	1-for-35	2018-10-15	66	\$18.6M gross unit offering; common/preferred plus common warrants
2022-07-25	1-for-10	2022-07-28	3	\$3.0M gross Series C/D preferred placement; existing 2018 warrants repriced to \$1.22
2024-08-26	1-for-25	2024-09-27	32	\$2.5M gross common/pre-funded warrant offering plus common warrants

7. The three reverse splits compound to 1-for-8,750. One pre-2018 common share therefore represents 1/8,750 of one current-basis common share, before considering later issuances.
8. The chronology is consistent with reverse splits restoring a higher quoted share price before later capital-market transactions. Whether that sequencing was planned for financing, required by exchange compliance, or both requires contemporaneous board materials, banker communications, term sheets, and capitalization analyses.

IV. LATER WARRANT SEQUENCE LEADING TO THE CHALLENGED WARRANTS

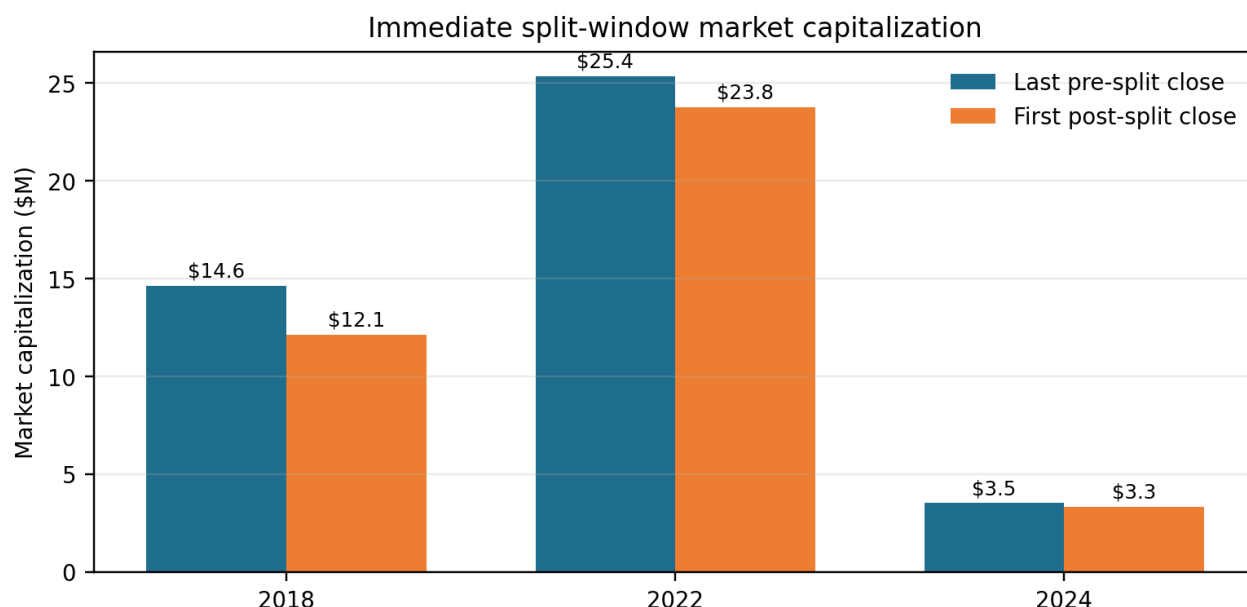
Transaction	Gross	Net	Instrument / result
September 27, 2024	\$2.500M	\$1.960M	Common/pre-funded warrants plus common warrants
May 2025	\$7.500M	\$6.690M	Common/pre-funded warrants plus common warrants
October 16, 2025	\$4.400M	\$3.942M	Exercise of existing warrants; 16,184,560 new warrants issued

9. The October 2025 inducement reported approximately \$4.400 million of gross exercise proceeds, \$356,000 of placement-agent fees, \$72,000 of other expenses, and approximately \$3.942 million of net cash proceeds.

10. The same transaction issued 16,184,560 new warrants and produced a reported \$5.880 million noncash deemed dividend. Cash fees, noncash accounting value, and potential future dilution are separate measurements.

V. IMMEDIATE MARKET-CAPITALIZATION MEASUREMENTS

Year	Pre date	Pre cap	Post date	Post cap	Change
2018	2018-08-10	\$14.63M	2018-08-13	\$12.12M	-17.1%
2022	2022-07-22	\$25.35M	2022-07-25	\$23.77M	-6.3%
2024	2024-08-23	\$3.52M	2024-08-26	\$3.32M	-5.7%



Demonstrative 1. Immediate pre/post split market-capitalization proxy.

11. A reverse split is mechanically value-neutral: it changes share count and per-share price in reciprocal proportions. The measured declines of approximately 17.1%, 6.3%, and 5.7% reflect market-price movements across the selected dates, not value destroyed by the arithmetic of the split itself.

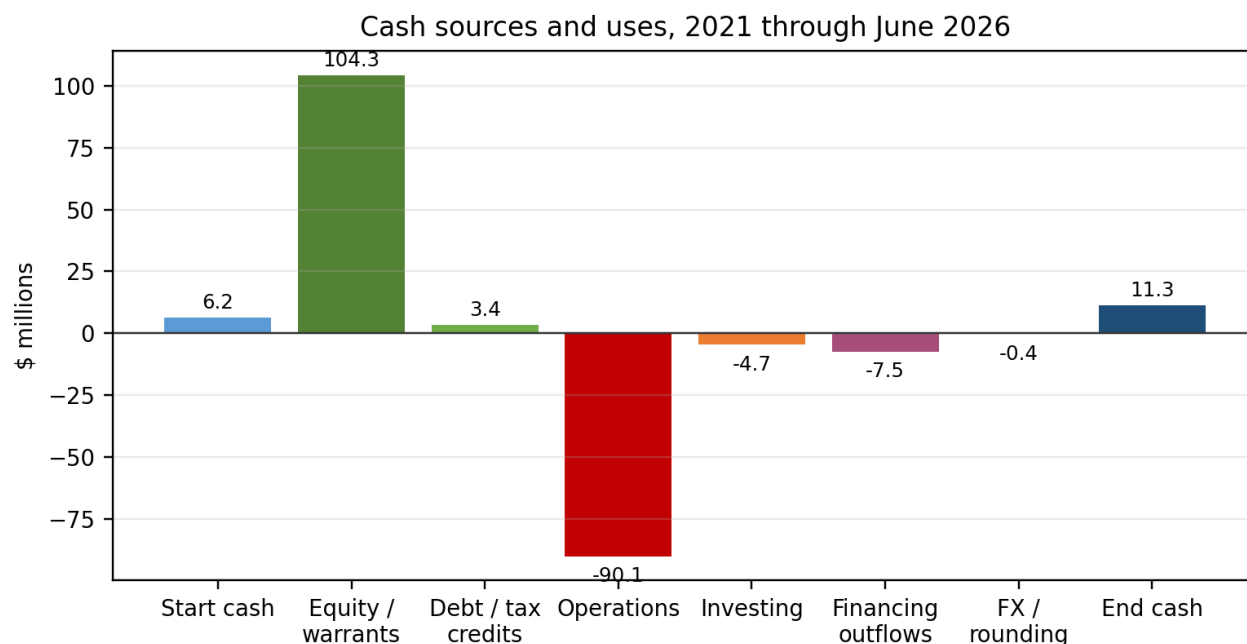
VI. CAPITAL RAISED AND REPORTED CASH USE, 2021 THROUGH JUNE 2026

12. The Company's statements of cash flows report \$104.290 million of equity- and warrant-related proceeds and \$3.403 million of debt or tax-credit financing proceeds during the period. After \$7.455 million of financing outflows, net financing cash flow was \$100.238 million.
13. During the same period, the statements report \$90.115 million of cash used in operations and \$4.651 million used in investing activities. Cash increased from \$6.227 million at December 31, 2020 to \$11.337 million at June 30, 2026, with approximately \$0.362 million attributable to foreign-exchange effects and rounding in the bridge.
14. The approximately \$70 million 2021 financing inflow was \$74.002 million net: \$65.960 million of net at-the-market offering proceeds and \$8.042 million of warrant-exercise proceeds. Money reported as cash raised came from investors and warrant holders; it was not operating revenue.
15. Reported research-and-development expense totaled \$56.092 million and reported general-and-administrative expense totaled \$50.394 million for 2021 through June 2026. These expense totals are accrual measures and therefore are not identical to cash used in operations.

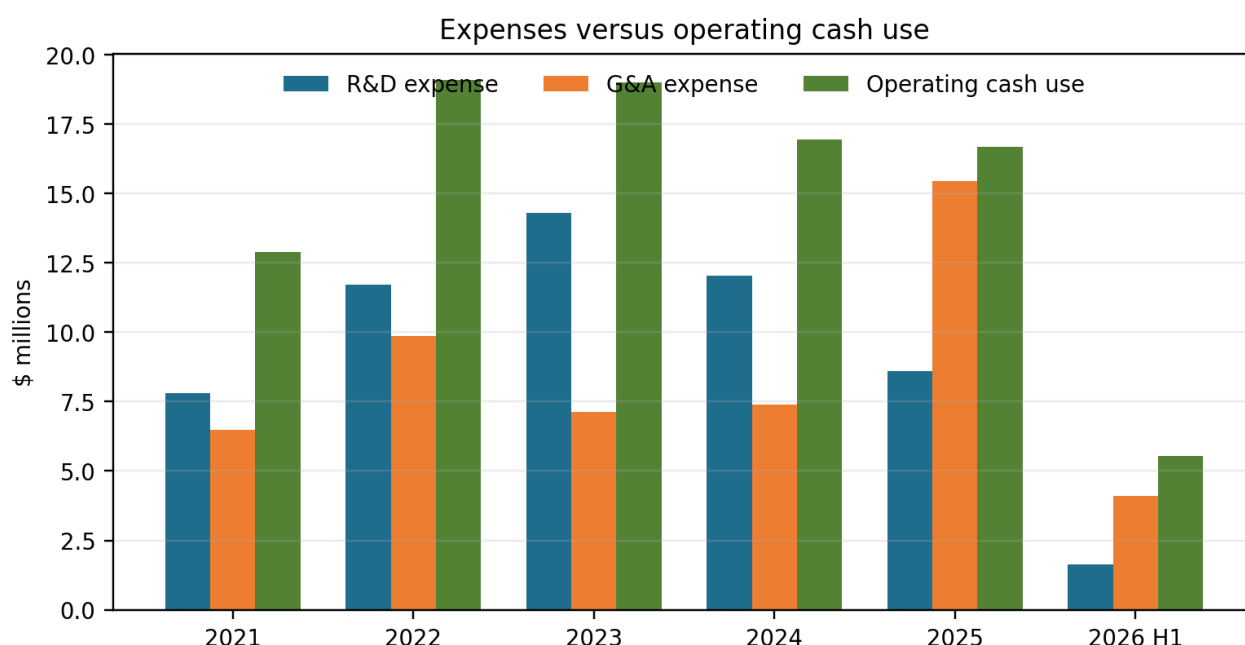
2021 bridge	Amount	What the filed statements establish
Beginning cash	\$6.227M	Cash and cash equivalents at December 31, 2020

2021 bridge	Amount	What the filed statements establish
ATM proceeds	\$65.960M	Net cash supplied through at-the-market equity sales
Warrant exercises	\$8.042M	Cash supplied by exercising warrant holders
Operating cash used	(\$12.890M)	Cash-flow measure; not identical to R&D plus G&A expense
Investing cash used	(\$0.014M)	Reported investing cash outflow
Ending cash	\$67.325M	Cash and cash equivalents at December 31, 2021
R&D expense	\$7.800M	Accrual expense shown for context; do not add again to operating cash use
G&A expense	\$6.474M	Accrual expense shown for context; do not add again to operating cash use
Deemed dividend	\$7.402M	Noncash conversion-inducement value; not a cash disbursement

- 16.** The 2021 cash bridge reconciles exactly: \$6.227 million beginning cash, plus \$74.002 million of financing inflow, less \$12.890 million of operating cash use and \$0.014 million of investing cash use, equals \$67.325 million ending cash. The public statements do not trace fungible financing dollars to individual laboratories, vendors, employees, programs, or fees.
- 17.** For scale only, the \$7.800 million of reported 2021 R&D expense equaled approximately 10.5% of the \$74.002 million financing inflow. That ratio is not a tracing conclusion: financing cash is fungible, R&D expense is accrual-based, and the Company also began 2021 with cash and used \$12.890 million of total operating cash during the year. The filed statements therefore do not establish that a particular financing dollar paid a particular research cost.
- 18.** The factual allocation question is therefore not whether \$74.002 million disappeared in 2021. It is what portions of investor-supplied capital were later devoted to scientific development, administration, acquisitions or other investing activity, cash financing costs, noncash security-holder inducements, and remaining liquidity. Transaction-level bank records, ledgers, invoices, payroll allocations, and board-approved budgets are required to answer that question without double counting.



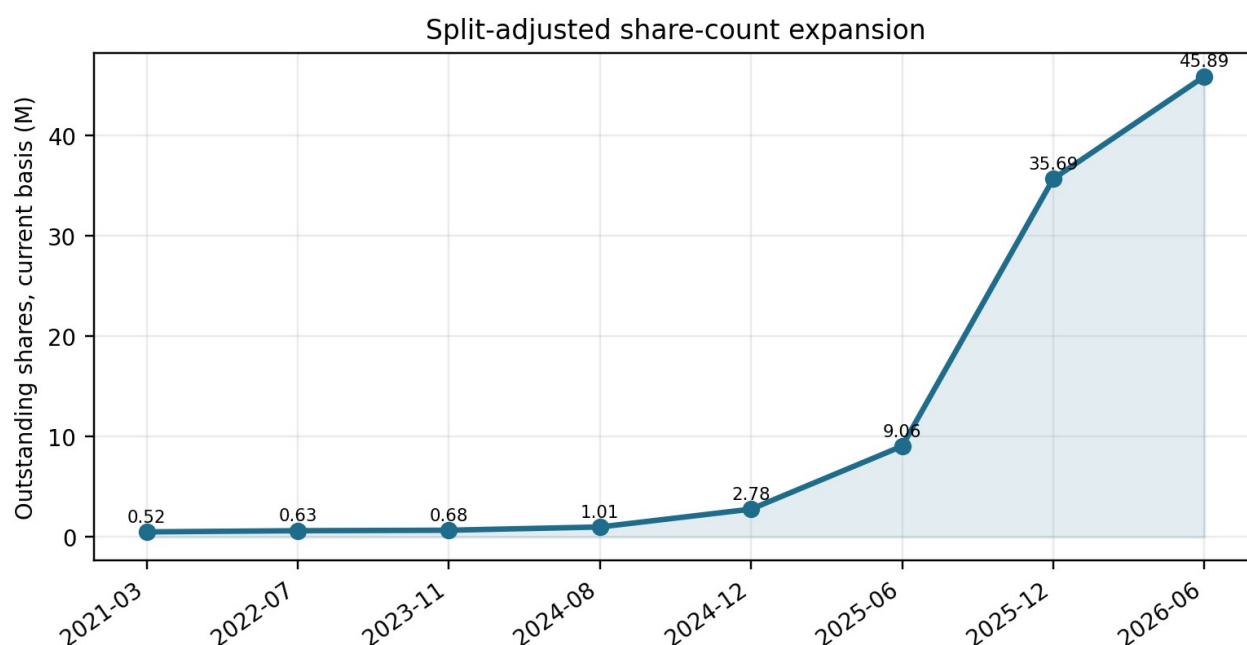
Demonstrative 2. Reconciliation of reported cash sources and uses, 2021 through June 2026.



Demonstrative 3. Reported R&D, G&A, and operating cash use by period.

VII. FINANCING COSTS, WARRANT-RELATED VALUE TRANSFERS, AND DILUTION

- 19.** For four comparable transactions for which both gross and net proceeds were identified, the known gross-to-net difference was approximately \$2.074 million. This is a lower bound and does not purport to include every placement-agent fee, legal fee, accounting fee, or other issuance cost during the full period.
- 20.** The filings separately report noncash deemed dividends of \$7.402 million in 2021 for a conversion inducement, \$0.340 million in 2022 for warrant repricing, and \$5.880 million in 2025 for a warrant inducement, totaling \$13.622 million. These are accounting measures of value transferred to security holders, not additional cash paid out in those amounts.
- 21.** On a current split-adjusted basis, reported common shares increased from approximately 518,618 at the March 2021 snapshot to 45,892,668 at June 30, 2026—approximately 88.5 times. Reverse splits reduce the displayed share count but do not reverse the economic dilution caused by later issuances.



Demonstrative 4. Split-adjusted expansion of reported common shares outstanding.

VIII. FEDERAL AND NEVADA LEGAL GUIDEPOSTS

22. Section 9(a)(2) of the Securities Exchange Act, 15 U.S.C. § 78i(a)(2), addresses a series of transactions creating actual or apparent active trading or raising or depressing a security’s price when performed for the purpose of inducing another person to buy or sell. Rule 10b-5, 17 C.F.R. § 240.10b-5, separately prohibits deceptive schemes, material misstatements or omissions, and fraudulent courses of business in connection with a securities transaction. A financing sequence, price decline, reverse split, or warrant exercise is not unlawful merely because it occurred; purpose, deception, materiality, scienter, transaction causation, and loss causation depend on the claim and proof presented.
23. *Santa Fe Industries, Inc. v. Green*, 430 U.S. 462, 476-77 (1977), treats “manipulation” as a term of art concerning practices intended to mislead investors by artificially affecting market activity and holds that unfair corporate conduct without deception or manipulation does not itself state a federal Rule 10b-5 claim.
24. *Dura Pharmaceuticals, Inc. v. Broudo*, 544 U.S. 336, 342-47 (2005), requires a private securities-fraud plaintiff to connect the challenged conduct to an actual economic loss; purchase at an allegedly inflated price alone is insufficient. *Desai v. Deutsche Bank Securities Ltd.*, 573 F.3d 931, 940-41 (9th Cir. 2009), further cautions that reliance is not presumed merely from an asserted loss of “market integrity.” These decisions are controlling guideposts in this federal action.
25. A stay of warrant exercises or share issuance is preliminary injunctive relief governed by Federal Rule of Civil Procedure 65. Under *Winter v. Natural Resources Defense Council, Inc.*, 555 U.S. 7, 20-24 (2008), the applicant must make the required showing concerning likely success, likely irreparable harm without relief, the balance of equities, and the public interest. Rule 65(c) permits the Court to require security, and Rule 65(d)(2) limits the persons bound by an injunction. The requested accounting examination does not eliminate those requirements.
26. *ATSI Communications, Inc. v. Shaar Fund, Ltd.*, 493 F.3d 87, 99-106 (2d Cir. 2007), is not controlling in Nevada, but is a factually relevant caution involving convertible financing, alleged short sales, and an alleged price-depressing sequence. The Second Circuit required particularized facts supporting manipulative acts and scienter and affirmed dismissal. Plaintiff cites *ATSI* for the proof boundary, not as a finding concerning *Theriva*.

27. Nevada incorporation does not change the elements of a federal Exchange Act claim. For state corporate-law issues, NRS 78.207 governs specified changes in authorized and outstanding shares; NRS 78.140 addresses transactions involving interested directors or officers; and NRS 78.138 supplies Nevada’s statutory business-judgment framework. *Chur v. Eighth Judicial District Court*, 136 Nev. 68, 458 P.3d 336 (2020), and *Guzman v. Johnson*, 137 Nev. 126, 483 P.3d 531 (2021), require more than a poor outcome or gross negligence to impose personal fiduciary liability. Plaintiff includes these authorities because they limit, rather than presume, any inference of wrongdoing.

IX. RASAYANA AND RELATED-PERSON DISCLOSURE INQUIRY

28. Theriva’s 2025 Form 10-K reports an exclusive worldwide, sublicensable SYN-020 license to Rasayana Therapeutics, Inc., a \$300,000 upfront payment, contingent development and sales milestones, royalties, and a percentage of sublicense revenue. The filing states that Rasayana assumed development responsibility commencing February 7, 2026 in one pipeline footnote, while the agreement and other narrative identify February 17, 2026.
29. Item 13 of that Form 10-K states that, except for the disclosed executive-compensation matter involving MaryAnn Shallcross, there were no related-party transactions during the stated period. It does not identify the Rasayana license as a related-party transaction. That is a fact about the filing’s presentation; it does not establish either that Rasayana was a related person or that disclosure was legally required.
30. Regulation S-K Item 404, 17 C.F.R. § 229.404, requires specified disclosure when the registrant participates in a transaction above the applicable threshold and a director, executive officer, covered security holder, or specified immediate family member has a direct or indirect material interest. Item 404 also requires disclosure of policies for review, approval, or ratification of covered transactions. A prior employment, scientific, educational, professional, or personal connection alone does not satisfy that test; beneficial ownership, control, compensation, economic interest, and the actual decision process must be established.
31. Plaintiff therefore asks, without alleging the answer, whether any Theriva director, officer, covered holder, immediate family member, adviser, or agent had a direct or indirect material interest in Rasayana or the license; whether the Audit Committee reviewed the transaction under the Company’s disclosed Item 404 policy; whether any conflict questionnaire identified a relationship; and whether additional information was necessary to make the filed description not misleading.
32. The requested records are Rasayana’s formation and beneficial-ownership records; capitalization and fundraising records; officer, director, employee, consultant, and adviser lists; the license-negotiation chronology; Theriva conflict questionnaires; Audit Committee materials; valuation and diligence materials; communications concerning counterparty selection; and records showing any reversion, sublicense, transfer, reimbursement, shared service, or continuing Theriva support.

X. SHARED PERSONNEL SERVICES AUDITORS AND INTERLOCKING ROLES

Person or firm	Theriva or Synthetic record	Heat NightHawk or Scorpius record	Earliest verified cross-company chronology
Jeffrey J. Kraws	Director since January 2006; part-time VP of business development during 2006-February 2007; chair since May 2012	No Heat/Scorpius role established in the reviewed record	Served with Wolf on the Synthetic/Theriva board from 2006
Jeffrey Wolf	Director since 2006	Founder in 2008; CEO and chair since 2010	Dual-company connection documented from 2008
John Monahan	R&D executive 2010-2015; adviser 2015-2020; director since 2020	Director since November 2009	Roles at both company groups documented from 2010
Leslie Marlow	Named in 2022 Synthetic/MSD agreement at Blank Rome	Named as issuer counsel in May 2018 Heat S-1/A at Gracin & Marlow	Professional roles documented in 2018 and 2022
BDO USA	Named auditor in 2022 reporting and March 2023 non-reliance filing	Named auditor in 2022 reporting and March 2023 non-reliance filing	Same firm name and reporting period; engagement staff not established

33. This chronology establishes a long-running network of overlapping roles. Theriva’s FY2025 Form 10-K states that Kraws and Wolf have both served on the Synthetic/Theriva board since 2006, and that Kraws also served part-time

as vice president of business development during 2006 through February 2007. Wolf then founded Heat in August 2008, creating the earliest verified bridge from that pre-existing board relationship to the Heat/NightHawk/Scorpius company group. Monahan joined the Scorpius board in November 2009 and held a Synthetic research-and-development role beginning in 2010. These dates do not establish that Marlow, BDO personnel, Kraws, or every identified person acted as one coordinated group beginning in 2008. Each person's conduct, knowledge, engagement, and dates must be proved separately.

34. Theriva's FY2025 director disclosures identify Jeffrey Wolf and John Monahan as Theriva directors and also describe their Scorpius roles. Wolf founded Heat Biologics, now Scorpius, in 2008 and has served as its chief executive officer and board chairman since 2010; he has served on Theriva's board since 2006. Monahan has served on Scorpius's board since November 2009 and Theriva's board since November 2020. These are two documented overlapping board relationships.
35. Heat's May 2, 2018 Form S-1/A identifies Leslie Marlow of Gracin & Marlow as issuer counsel. Synthetic Biologics' July 28, 2022 agreement with MSD separately directs company-side copies to Leslie Marlow at Blank Rome. The records establish professional roles for the same attorney in identified transactions of both issuers; they do not establish that she represented both companies at the same time, participated in every later transaction, or supplied nonprivileged advice common to both.
36. Theriva filed an Item 4.02 non-reliance Form 8-K on March 10, 2023; NightHawk filed one on March 23, 2023. Both concerned the June and September 2022 interim periods, identified BDO USA, LLP, addressed acquisition-related deferred-tax accounting and management-review controls over income taxes, and stated that amended quarterly reports would not be filed. The underlying acquisition adjustments and amounts differed. The matching dates, audit-firm name, periods, control category, and reporting method establish a pattern appropriate for document comparison; they do not establish shared engagement staff, coordination, or commingling.
37. Service by one person on more than one board, employment by more than one company, or use of the same independent audit firm by two issuers is not, standing alone, unlawful or proof that assets or expenses were commingled. Auditor independence is governed principally by Regulation S-X Rule 2-01 and related professional standards. Section 8 of the Clayton Act, 15 U.S.C. § 19, concerns certain interlocking directorates between competing corporations and applies only when its statutory competition and financial thresholds are satisfied. The present public record does not establish those predicates.
38. Sections 13(a) and 13(b)(2) of the Exchange Act and implementing rules address periodic reporting, books and records, and internal accounting controls. These provisions may guide preservation and discovery, but their existence does not by itself create a private damages claim for every accounting question. The factual inquiry is whether Theriva and Heat Biologics/NightHawk Biosciences/Scorpius Holdings shared personnel, laboratories, equipment, vendors, invoices, services, expenses, reimbursements, or allocations during overlapping service periods, and if so, whether each issuer's books accurately recorded its own assets, liabilities, expenses, and transactions.
39. The public HTBX/NightHawk/Scorpius balance sheets and SEC filings may be used as public comparators. Internal ledgers, bank records, auditor workpapers, invoices, and allocation schedules belonging to a nonparty may be obtained only through discovery authorized by the Federal Rules, including the relevance and proportionality limits of Rule 26 and, where appropriate, a Rule 45 subpoena. Plaintiff requests preservation and, if the Court finds the subject relevant to an existing claim or defense, narrowly tailored production rather than an unrestricted examination of another company's books.

XI. RECORDS NEEDED TO TEST PURPOSE CAUSATION AND ALLOCATION

40. Board and committee minutes, presentations, written consents, and capitalization tables concerning each reverse split and the financings that followed;
41. Communications with A.G.P./Alliance Global Partners and other placement agents, financial advisers, warrant holders, and prospective investors concerning minimum share price, reverse splits, offering structure, warrant repricing, or inducement terms;
42. Term sheets and drafts showing when each financing was first proposed, negotiated, approved, priced, and closed;
43. Schedules reconciling gross proceeds to net proceeds and identifying placement-agent fees, legal and accounting fees, warrant-agent fees, and other issuance costs;
44. Warrant ledgers identifying issuance, repricing, exercise, cancellation, beneficial-ownership blockers, replacement warrants, and shares delivered;
45. Monthly cash forecasts and use-of-proceeds reports comparing stated uses with actual disbursements; and
46. Documents analyzing whether the post-split authorized-share capacity or quoted share price was necessary to execute each later financing.
47. Records sufficient to identify any Theriva cost, employee time, laboratory, equipment, intellectual property, vendor invoice, or service shared with, paid by, reimbursed by, or allocated to Heat Biologics, NightHawk Biosciences, Scorpius Holdings, Rasayana, or an entity affiliated with a director or officer, limited to periods and subjects relevant to an existing claim or defense;
48. Audit-committee minutes, conflict questionnaires, independence communications, management-representation letters, related-party search procedures, and nonprivileged schedules concerning the Rasayana license or any identified shared-service or cost-allocation arrangement; and
49. Public HTBX/NightHawk/Scorpius annual and quarterly financial statements for relevant overlapping periods, followed only if justified by targeted nonparty discovery identifying the particular account, transaction, person, asset, or allocation sought.

XII. LIMITS OF THE ANALYSIS

50. Timing is evidence of sequence, not proof of causation. The 2022 event differs materially from the 2018 and 2024 events because it issued preferred stock rather than new warrants, although it repriced outstanding warrants.
51. Market capitalization is sensitive to the chosen date, share-count snapshot, liquidity, and price volatility. The calculations are transparent measurements, not an expert valuation opinion.
52. The cash-flow bridge explains reported sources and uses at the financial-statement level. It does not identify every vendor, employee, program, bank transfer, or restricted-use account. Transaction-level tracing requires the records listed above.
53. No reviewed public record presently establishes that Theriva and HTBX/NightHawk/Scorpius commingled funds, employees, expenses, laboratories, equipment, or auditor services; that the shared audit-firm name identifies the same engagement personnel; that Rasayana was controlled by an Item 404 related person; or that a director interlock violated the Clayton Act. Those are questions requiring admissible records, not conclusions supplied by overlapping names.

XIII. CONCLUSION AND LIMITED REQUEST

54. The record shows a repeated split-to-financing sequence and substantial subsequent common-share expansion. It also shows that the Company publicly described exchange-price compliance as the stated purpose of each split. The appropriate factual question is whether the later warrant-related financings were already contemplated when the splits were proposed or approved. That question should be resolved from contemporaneous records and testimony rather than allegation.
55. Plaintiff therefore asks the Court, only to the extent relevant to the pleaded warrant claims and requested emergency relief, to preserve the identified records and permit targeted discovery sufficient to determine: (a) whether the split and warrant sequence was planned to induce or facilitate securities transactions; (b) how investor-supplied capital was allocated among technology, administration, financing friction, security-holder inducements, acquisitions, and remaining cash; (c) whether any Rasayana relationship met the legal definition of a related-person transaction; and (d) whether any shared personnel, service, asset, cost, or auditor relationship was accurately recorded and disclosed. Plaintiff does not ask the Court through this memorandum to adjudicate a new manipulation, related-party, antitrust, or accounting claim.
56. As alternative or supplemental relief, if the Court finds the accounting allocation relevant and proportional to an existing claim or defense, Plaintiff requests an order requiring Theriva to retain, at its expense and subject to reasonable Court-approved independence criteria, a qualified forensic accounting firm that is not Theriva's current independent auditor and did not participate in the challenged financings. A firm with capabilities comparable to Citrin Cooperman is an example of the requested qualifications; Plaintiff does not request the exclusive appointment or endorsement of that particular firm.
57. The independent Theriva report should be filed with the Court, supported by a responsible professional's declaration, and should present a separate reconciliation for each of 2021, 2022, 2023, 2024, and 2025 and for the six months ended June 30, 2026. For each period it should: (a) reconcile cash and noncash capital raised; (b) trace material disbursements to research and development, acquired technology, general and administrative activity, financing fees, debt or contingent consideration, and ending cash; (c) identify and separately quantify noncash deemed dividends and warrant-related accounting charges; (d) compare actual uses with filed use-of-proceeds disclosures and board-approved budgets; (e) test material related-person, shared-service, reimbursement, employee-time, laboratory, equipment, vendor, and cost-allocation entries involving Rasayana, Heat Biologics, NightHawk Biosciences, Scorpius Holdings, or an entity affiliated with a Theriva director or officer; and (f) state the records reviewed, procedures performed, assumptions, exceptions, and limitations. The report should also provide a cumulative reconciliation for the full period without substituting that total for the annual schedules.
58. Pending completion and judicial review of that report, Plaintiff requests, only if the Court finds the Rule 65 requirements satisfied and subject to any security the Court sets, an order preserving the status quo as to any portion of the 16,184,560 New Warrants issued under the October 16, 2025 inducement agreement that remains unexercised. The requested order would restrain Theriva and the persons within Rule 65(d)(2) who receive actual notice from accepting, processing, settling, or completing further exercises and from issuing, registering, delivering, or transferring the corresponding underlying common shares until further order. It would not finally adjudicate or cancel an absent holder's rights.
59. If the Court finds a stay through completion of the forensic report broader or longer than warranted, Plaintiff alternatively requests an immediate sworn report stating the number of New Warrants exercised, shares issued or delivered, proceeds received, and warrants remaining; expedited production of the annual accounting and warrant records identified above; and the shortest temporary stay the Court finds necessary to permit meaningful review and an early noticed preliminary-injunction hearing.
60. Plaintiff requests a parallel forensic examination of Heat Biologics/NightHawk Biosciences/Scorpius Holdings only if that entity is properly joined, consents, or is otherwise subject to an order supported by the Court's jurisdiction and due process. Until then, Plaintiff requests leave to seek narrowly tailored existing records from that nonparty under

Rule 45, limited to identified overlapping directors, officers, services, assets, vendors, payments, reimbursements, accounting entries, audit personnel, and allocation schedules during relevant periods. Similar financing patterns or overlapping personnel alone do not authorize unrestricted discovery or an order compelling a nonparty to create a new report. If the Court finds a forensic report premature or unduly burdensome, Plaintiff requests the narrower production listed in Section XI.

SOURCE SCHEDULE

1. 2018 reverse-split Form 8-K: https://www.sec.gov/Archives/edgar/data/894158/000114420418043710/tv500716_8k.htm
2. 2018 warrant-unit offering filing: https://www.sec.gov/Archives/edgar/data/894158/000114420418049861/tv503005_s1.htm
3. 2022 reverse-split Form 8-K: https://www.sec.gov/Archives/edgar/data/894158/000110465922082251/tm2221681d1_8k.htm
4. 2022 MSD preferred placement and warrant repricing Form 8-K: https://www.sec.gov/Archives/edgar/data/894158/000110465922084301/tm2222078d1_8k.htm
5. 2024 reverse-split Form 8-K: https://www.sec.gov/Archives/edgar/data/894158/000110465924092644/tm2422579d1_8k.htm
6. 2024 warrant offering prospectus supplement: https://www.sec.gov/Archives/edgar/data/894158/000110465924103658/tm2425095d1_424b5.htm
7. May 2025 common/warrant offering prospectus supplement: https://www.sec.gov/Archives/edgar/data/894158/000110465925045934/tm2514445d1_424b5.htm
8. October 2025 warrant-inducement Form 8-K: https://www.sec.gov/Archives/edgar/data/894158/000110465925100468/tm2528131d2_8k.htm
9. Theriva 2025 Form 10-K, including Item 13 and Rasayana disclosures: <https://www.sec.gov/Archives/edgar/data/894158/000110465926026767/tmb-20251231x10k.htm>
10. Rasayana License Agreement, Exhibit 10.40: <https://www.sec.gov/Archives/edgar/data/894158/000110465926026767/tmb-20251231xex10d40.htm>
11. Theriva March 10, 2023 Item 4.02 non-reliance Form 8-K: https://www.sec.gov/Archives/edgar/data/894158/000110465923031255/tm239041d1_8k.htm
12. NightHawk March 23, 2023 Item 4.02 non-reliance Form 8-K: <https://www.sec.gov/Archives/edgar/data/1476963/000155837023004449/nhwk-20230317x8k.htm>
13. Heat May 2, 2018 Form S-1/A identifying issuer counsel: https://www.sec.gov/Archives/edgar/data/1476963/000155335018000461/htbx_s1a.htm
14. Scorpis FY2024 Form 10-K, directors and financial statements: <https://www.sec.gov/Archives/edgar/data/1476963/000155837025005986/scpx-20241231x10k.htm>
15. Regulation S-K Item 404, 17 C.F.R. § 229.404: <https://www.ecfr.gov/current/title-17/chapter-II/part-229/subpart-229.400/section-229.404>
16. Rule 10b-5, 17 C.F.R. § 240.10b-5: <https://www.ecfr.gov/current/title-17/chapter-II/part-240/subpart-A/subject-group-ECFRbda83517ce4377f/section-240.10b-5>
17. Nevada Revised Statutes Chapter 78: <https://www.leg.state.nv.us/nrs/nrs-078.html>
18. Federal Rule of Civil Procedure 26: https://www.law.cornell.edu/rules/frcp/rule_26
19. Federal Rule of Civil Procedure 45: https://www.law.cornell.edu/rules/frcp/rule_45
20. Federal Rule of Civil Procedure 65: https://www.law.cornell.edu/rules/frcp/rule_65
21. Winter v. Natural Resources Defense Council, Inc., 555 U.S. 7 (2008): <https://supreme.justia.com/cases/federal/us/555/7/>
22. Santa Fe Industries, Inc. v. Green, 430 U.S. 462 (1977): <https://supreme.justia.com/cases/federal/us/430/462/>
23. Dura Pharmaceuticals, Inc. v. Broudo, 544 U.S. 336 (2005): <https://tile.loc.gov/storage-services/service/lj/usrep/usrep544/usrep544336/usrep544336.pdf>
24. Desai v. Deutsche Bank Securities Ltd., 573 F.3d 931 (9th Cir. 2009): <https://law.justia.com/cases/federal/appellate-courts/ca9/08/55081/08-55081-2011-02-25.html>
25. ATSI Communications, Inc. v. Shaar Fund, Ltd., 493 F.3d 87 (2d Cir. 2007): <https://caselaw.findlaw.com/court/us-2nd-circuit/1232023.html>
26. Chur v. Eighth Judicial District Court, 458 P.3d 336 (Nev. 2020): <https://law.justia.com/cases/nevada/supreme-court/2020/78301.html>
27. Guzman v. Johnson, 483 P.3d 531 (Nev. 2021): <https://law.justia.com/cases/nevada/supreme-court/2021/79818.html>
28. FY2021 Form 10-K: <https://www.sec.gov/Archives/edgar/data/894158/000141057822000405/syn-20211231x10k.htm>
29. FY2022 Form 10-K: <https://www.sec.gov/Archives/edgar/data/894158/000141057823000428/tmb-20221231x10k.htm>
30. FY2023 Form 10-K: <https://www.sec.gov/Archives/edgar/data/894158/000141057824000293/tmb-20231231x10k.htm>
31. FY2024 Form 10-K: <https://www.sec.gov/Archives/edgar/data/894158/000141057825000285/tmb-20241231x10k.htm>
32. FY2025 Form 10-K: <https://www.sec.gov/Archives/edgar/data/894158/000110465926026767/tmb-20251231x10k.htm>
33. 2026 Q2 Form 10-Q: <https://www.sec.gov/Archives/edgar/data/894158/000110465926093786/tmb-20260630x10q.htm>

Dated: _____

Signature: _____

Mark Joseph Nejme, Plaintiff, self-represented

P.O. Box 589, Clifton, New Jersey 07012

Telephone: 732-995-3914 | Email: research@warrantwire.com

DEMONSTRATIVE SEQUENCE OF THE QUESTION PRESENTED

This sequence identifies a hypothesis to be tested against transaction records, market data, contemporaneous communications, and testimony. It does not itself establish causation or intent.



PERFECT TIMING, BUSINESS NECESSITY, OR COINCIDENCE

?

**UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., Defendant.	Case No. 3:26-cv-00705
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SUPPLEMENTAL EXHIBIT SB

Patent Disclosures Market Activity Warrants and Reverse Splits

Original Part B

The following pages are separately paginated under this supplemental exhibit identifier.

1 Mark Joseph NejmeH
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5 Email: research@warrantwire.com
6 Plaintiff, self-represented

7 **UNITED STATES DISTRICT COURT**
8 **DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., a Nevada corporation, Defendant.	Case No. 3:26-cv-00705 PROPOSED SUPPLEMENTAL INFORMATION — PART B PATENT ASSETS, PUBLIC DISCLOSURES, MARKET ACTIVITY, WARRANTS, FINANCINGS, AND REVERSE SPLITS
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9 **QUESTION PRESENTED**

10 **Do public records show recurring temporal sequences in which patent or scientific**
11 **developments appeared near registration statements, reverse splits, warrant**
12 **transactions, financings, and material changes in trading—and what**
13 **contemporaneous records are needed to explain those sequences?**

SEPARATE DOCUMENT AND LIMITED PURPOSE. This Part B does not amend, replace, withdraw, or revise the separately prepared supplemental-warrant document (Part A), including its opening. Part B presents a public-record chronology and proposed demonstratives. Timing, price movement, volume, a patent grant, a reverse split, or a warrant transaction does not by itself prove causation, intent, materiality, scienter, manipulation, or liability. This document is marked PROPOSED because District of Nevada LR 7-2(g) generally requires leave before supplemental briefing.

14 **VERSION 1 — FOR REVIEW BEFORE ANY FILING**

I. SCOPE AND GUARDRAILS

1. Part B addresses a different evidentiary direction from Part A. It compares selected patent and intellectual-property events with public trading data, SEC filings, reverse splits, warrant-related transactions, and financings. It is self-contained so that the Court can evaluate the chronology without altering Part A.
2. The charts are demonstratives generated from cited public records. They are not independent evidence. Before trial use, the underlying records and calculations must be authenticated, and any market-data summary should be supported by an admissible data source or a competent declaration.
3. The review does not assume that an issuer must issue a separate press release for every patent. Regulation S-K Item 101 addresses material business developments, development status, and the duration and effect of patents and licenses in Commission filings. Rule 10b-5 addresses materially misleading statements or omissions in context. Whether a particular matter required additional disclosure is a fact- and context-specific legal question.
4. The market windows are descriptive, not causal event studies. They do not adjust for a market index, biotechnology-sector performance, simultaneous news, bid-ask spread, offering terms, dilution, short interest, warrant exercises, or trading by any identified person.

II. PATENT AND PRODUCT-ASSET RECORD

5. Theriva's FY2025 Form 10-K states that the Company and its subsidiaries have more than 135 U.S. and foreign patents and more than 50 pending applications, and that VCN controls more than 50 patents and more than 15 pending applications. The filing describes patents and licenses as resources supporting the Company's product candidates.
6. SYN-004 (ribaxamase) remains investigational. The record identifies a 2017 Phase 2b result and later development work; the Company's August 2026 update states that additional enrollment in the HCT trial is paused and contingent on grant funding or a strategic partnership. It is not an approved cure for C. difficile infection.
7. SYN-020 is an investigational oral intestinal alkaline phosphatase program. U.S. Patent 12,318,434, issued June 3, 2025, concerns alkaline-phosphatase-based oncology treatment and identifies gemcitabine and cancer treatment in the patent record. It does not establish an approved early-detection test for pancreatic cancer. Theriva later granted Rasayana an exclusive worldwide, sublicensable license; the FY2025 Form 10-K reports a \$300,000 upfront payment, contingent milestones, royalties, and a sublicense-revenue share.
8. VCN-01 remains an active clinical program. The May 7, 2025 Form 8-K furnished positive topline VIRAGE Phase 2b results in metastatic pancreatic cancer. The Company's August 2026 update states that dosing began in a new VIRAGE2 Phase 2a trial intended to inform a future Phase 3 program.

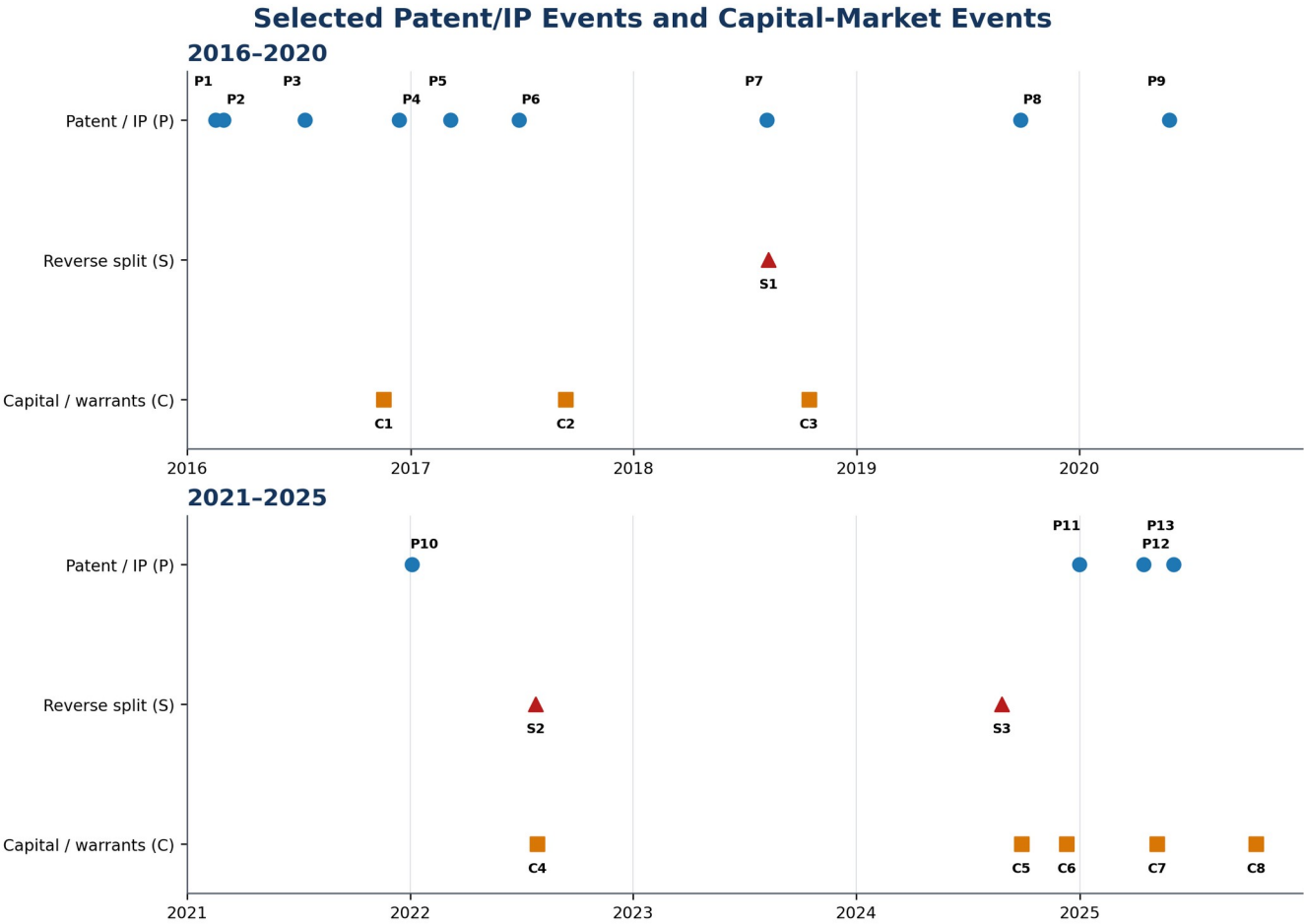
Dedicated patent/IP release-title review

A review of 361 entries in the Company's press-release archive identified seven titles from 2016 forward containing "patent" or "intellectual property." The last such title was May 27, 2020. This title-based result does not mean later patents were secret or undisclosed: U.S. patent grants are public, later annual reports discuss the patent estate, and other releases may mention patents in body text.

Date	Program	Dedicated company release
2016-02-17	SYN-010	Company press release: allowance of U.S. application 14/211,197
2016-03-01	SYN-004	Company press release: three additional U.S. patent allowances
2016-07-12	SYN-004 / ribaxamase	Company press release: EP 2,576,776 and U.S. 9,376,673; another U.S. allowance
2016-12-13	SYN-005	Company press release: U.S. 9,512,204 issued
2017-06-27	SYN-010	Company press release: allowance of U.S. application 14/776,465
2019-09-26	SYN-010	Company press release: U.S. 10,328,151 and an additional allowance
2020-05-27	SYN-020	Company press release: exclusive option to license intellectual property from MGH

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III. PUBLIC-RECORD CHRONOLOGY



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3

Demonstrative B-1. Selected patent/IP (P), reverse-split (S), and capital-market (C) events; dates show sequence only.

4

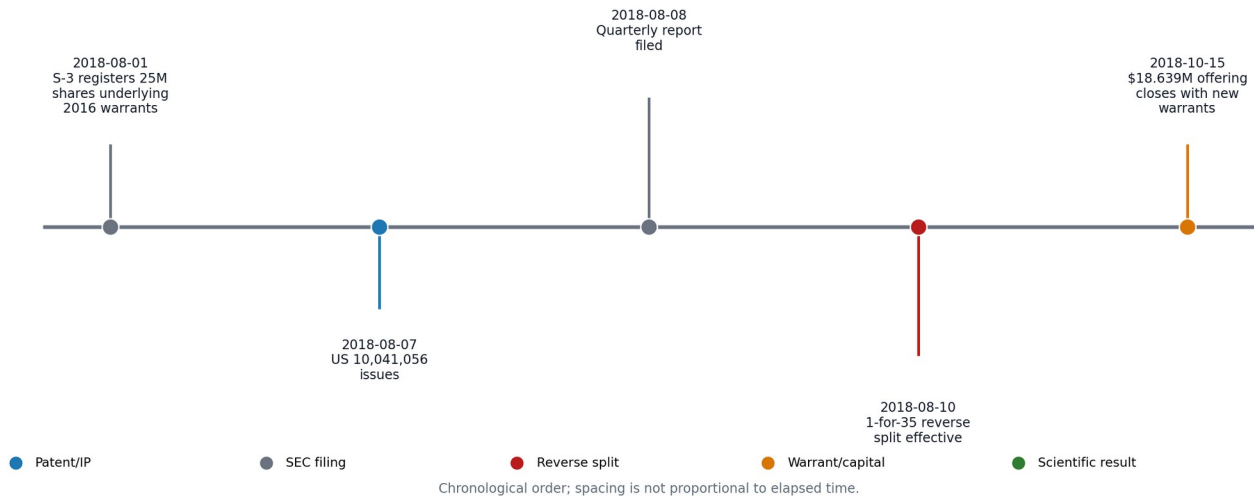
5

The overview is selective, not a claim that every patent, filing, financing, or market event is included. The detailed source table later in Part B identifies the selected patent/IP events and the closest SEC filings.

IV. THE 2018 SEQUENCE

9. On August 1, 2018, Synthetic filed a Form S-3 registering 25,000,000 common shares issuable upon exercise of outstanding warrants issued in November 2016. The filing stated a \$1.43 exercise price and potential aggregate cash proceeds of \$35.75 million if all covered warrants were exercised for cash.
10. U.S. Patent 10,041,056, a SYN-004/ribaxamase-family patent titled “Modified beta-lactamases and methods and uses related thereto,” issued on August 7, 2018. The press-release title archive does not show a dedicated patent-titled release for that grant.
11. The Company filed its Form 10-Q on August 8, 2018. It described a growing patent portfolio, listed representative SYN-004 patents, described SYN-020 work, and reported the May 1, 2018 SYN-010 patent. A text search did not locate U.S. 10,041,056 by number in that filing.
12. The 1-for-35 reverse split became effective at 11:00 p.m. Eastern on August 10, 2018, with post-split trading beginning August 13. The Form 8-K states that the purpose was compliance with NYSE American per-share-price requirements.
13. On October 15, 2018, the Company closed an approximately \$18.639 million gross unit offering that included new common warrants. The sequence spans 75 days from the August 1 warrant-share registration statement to the October 15 closing, and 69 days from the patent grant to the closing.

Documented 2018 Sequence

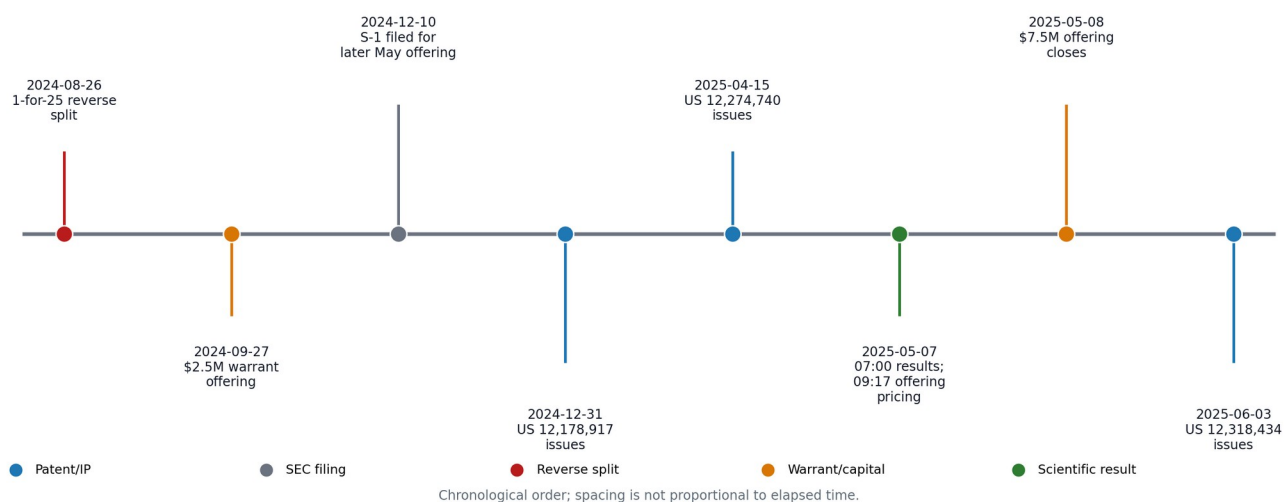


Demonstrative B-2. The documented 2018 sequence; sequence is not proof of a coordinated plan.

V. THE AUGUST 2024–JUNE 2025 SEQUENCE

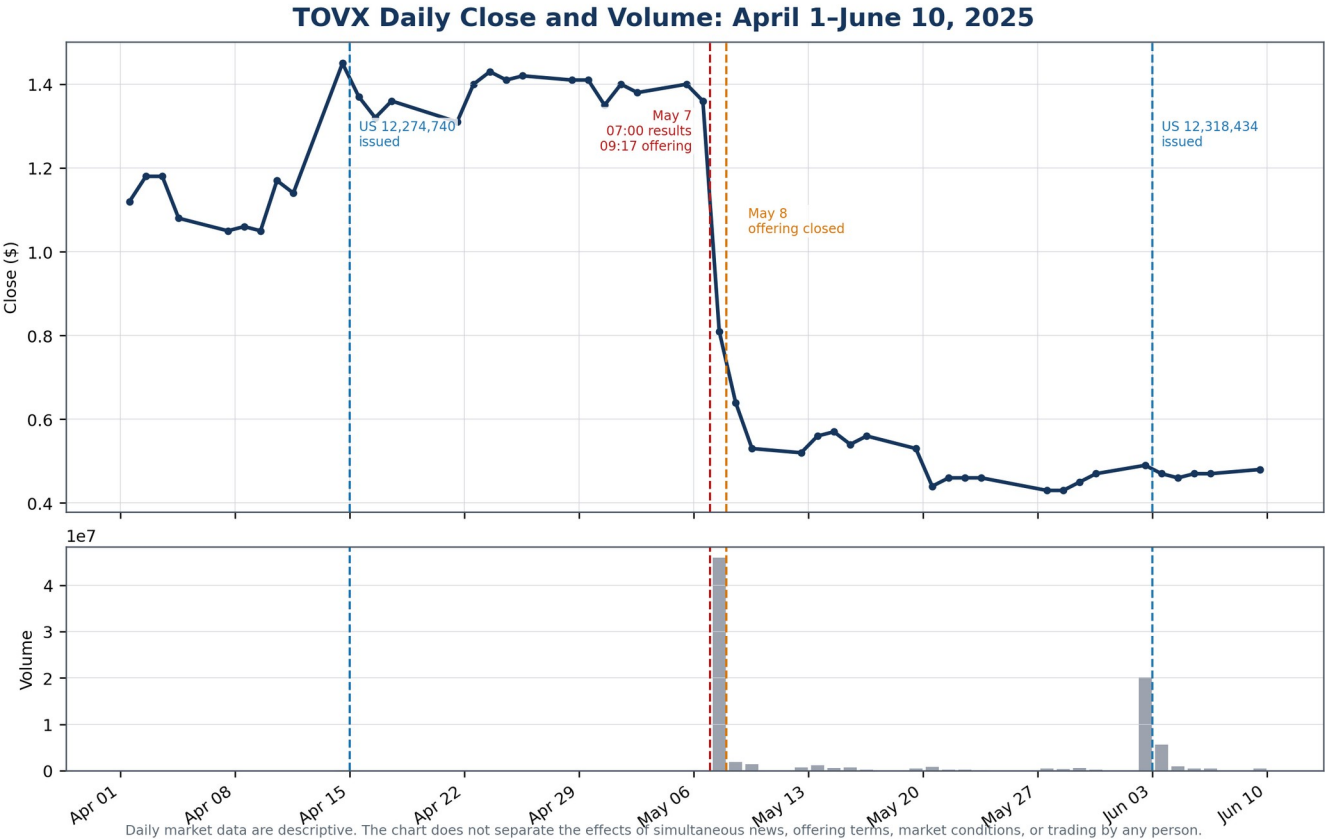
14. Theriva effected a 1-for-25 reverse split on August 26, 2024 and closed a \$2.5 million gross common/pre-funded-warrant and common-warrant offering on September 27, 2024.
15. Theriva filed the registration statement for a later offering on December 10, 2024. U.S. Patent 12,178,917 issued December 31, 2024, and U.S. Patent 12,274,740 issued April 15, 2025. Both concern SYN-004/ribaxamase-family technology. No dedicated patent-titled company release was located for either grant.
16. At 7:00 a.m. Eastern on May 7, 2025, Theriva announced positive VIRAGE Phase 2b results for VCN-01. At 9:17 a.m. Eastern the same day, Theriva announced pricing of a \$7.5 million offering of common stock or pre-funded warrants together with common warrants. The May 8 Form 8-K states that the offering included 1,990,900 common shares, pre-funded warrants for up to 4,827,280 shares, and common warrants for up to 6,818,180 shares.
17. On May 7, the stock traded 45,957,600 shares, approximately 2,038 times the prior 20-session average in the downloaded series; the close declined from \$1.36 on May 6 to \$0.81 on May 7. The high, low, and close that day were \$1.45, \$0.74, and \$0.81. These figures do not identify who traded or why.
18. U.S. Patent 12,318,434, concerning alkaline-phosphatase-based oncology treatment and associated with SYN-020, issued June 3, 2025. No dedicated patent-titled company release was located for that grant. The event-session close moved from \$0.49 on June 2 to \$0.47 on June 3, and volume was approximately 1.5 times the prior 20-session average.

Documented August 2024–June 2025 Sequence



Demonstrative B-3. The documented August 2024–June 2025 sequence.

VI. MAY 7, 2025 SAME-DAY SCIENTIFIC-DISCLOSURE AND OFFERING RECORD



Demonstrative B-4. Daily close and volume surrounding the April–June 2025 events.

Date	Public event	Price	Volume	Market-cap note
May 6, 2025	Prior session	\$1.36 close	10,700 shares	\$3.78M
May 7, 2025	07:00 results; 09:17 offering pricing	\$1.45 high / \$0.74 low / \$0.81 close	45,957,600 shares	\$2.25M
May 8, 2025	Offering closed; Form 8-K filed	\$0.64 close	1,881,700 shares	Old-share-count proxy not used as post-closing capitalization
May 14, 2025	Five sessions after May 7	\$0.57 close	606,500 shares	Share count changed through offering

The pre-closing market-cap proxy holds the March 4, 2025 reported common-share count constant. It is not used after the offering closed because the capitalization changed. The May 8 Form 8-K separately reported 4,773,358 common shares outstanding after closing, assuming no pre-funded- or common-warrant exercise.

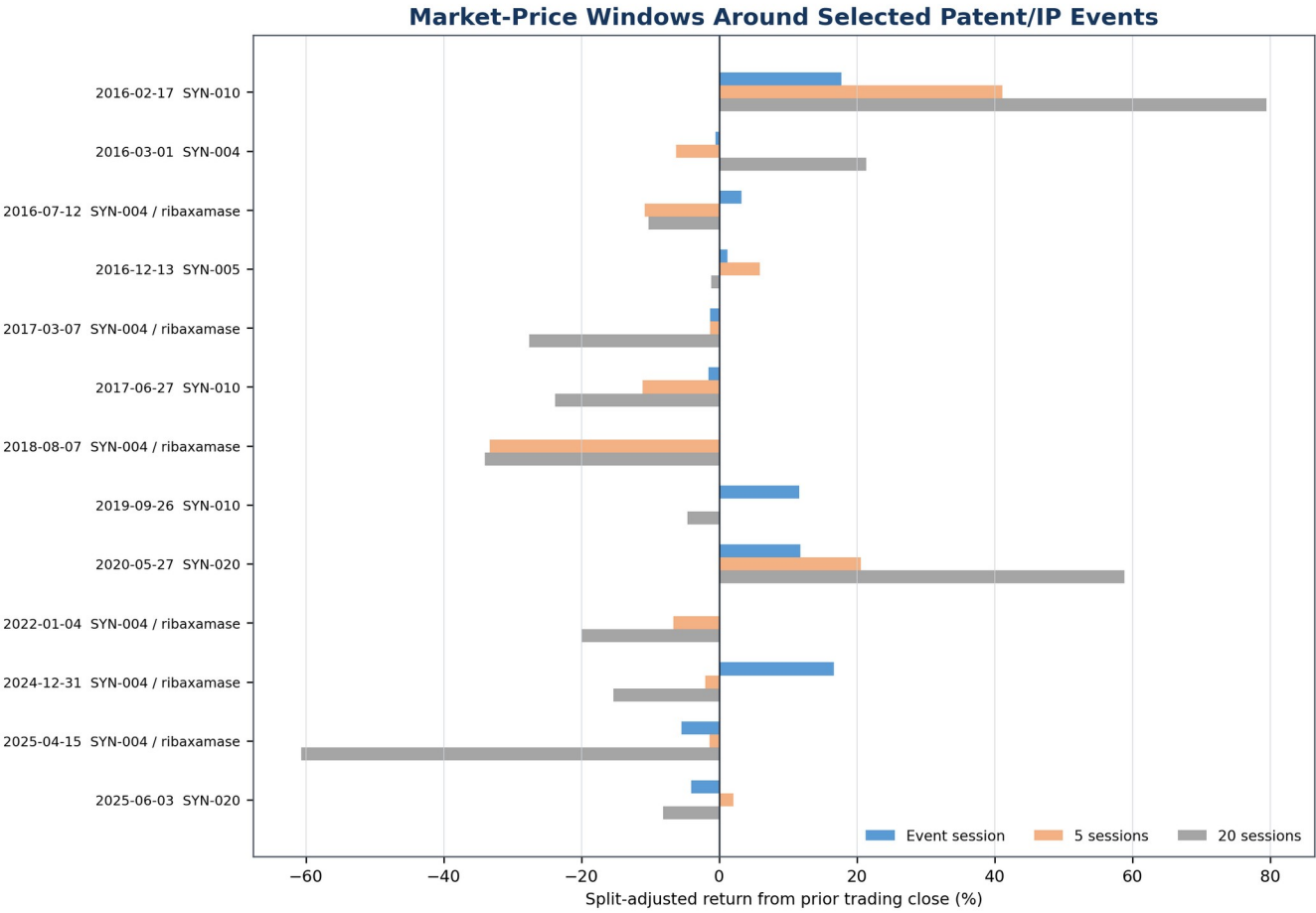
VII. PATENT/IP EVENT-WINDOW METHODOLOGY

19. Event set. Selected dedicated patent/IP company releases from 2016 forward and later verified U.S. patent grants associated with SYN-004 or SYN-020. The set is not a complete patent docket and was not selected by return size.
20. Prices and volume. Yahoo Finance daily data downloaded through September 18, 2026. Returns use the fully split-adjusted series. Contemporaneous quoted prices are shown by reversing later split adjustments. Volume comparisons use a consistent split-adjusted basis.
21. Trading-day convention. The event session is the stated date if it was a trading day; otherwise it is the next trading session. Returns are measured from the prior trading close to the event close and to closes 5 and 20 sessions later.
22. Market-cap proxy. Split-adjusted close multiplied by the latest reported common-share count dated on or before the event, converted to the same share basis. It is an estimate and may omit intraperiod issuance, preferred stock, warrants, conversions, and other securities.
23. No controls. The calculation does not control for market, sector, simultaneous company news, financing announcements, bid-ask spread, liquidity, short interest, or general conditions. A qualified financial expert should determine whether a litigation-grade event study is appropriate.

Date	Program	Mode	Prior → event close	+5	+20	Volume
2016-02-17	SYN-010	Company release	\$1.07 → \$1.26	+41.1%	+79.4%	0.7×
2016-03-01	SYN-004	Company release	\$1.92 → \$1.91	-6.2%	+21.4%	2.0×
2016-07-12	SYN-004 / ribaxamase	Company release	\$1.85 → \$1.91	-10.8%	-10.3%	0.5×
2016-12-13	SYN-005	Company release	\$0.85 → \$0.86	+5.9%	-1.2%	0.6×
2017-03-07	SYN-004 / ribaxamase	Patent record	\$0.76 → \$0.75	-1.3%	-27.6%	1.2×
2017-06-27	SYN-010	Company release	\$0.63 → \$0.62	-11.1%	-23.8%	0.5×
2018-08-07	SYN-004 / ribaxamase	Patent record	\$0.12 → \$0.12	-33.3%	-34.0%	2.6×
2019-09-26	SYN-010	Company release	\$0.43 → \$0.48	+0.0%	-4.7%	7.4×
2020-05-27	SYN-020	Company release	\$0.34 → \$0.38	+20.6%	+58.8%	7.4×
2022-01-04	SYN-004 / ribaxamase	Patent record	\$0.30 → \$0.30	-6.7%	-20.0%	0.4×
2024-12-31	SYN-004 / ribaxamase	Patent record	\$1.50 → \$1.75	-2.0%	-15.3%	0.5×
2025-04-15	SYN-004 / ribaxamase	Patent record	\$1.45 → \$1.37	-1.4%	-60.7%	1.6×
2025-06-03	SYN-020	Patent record	\$0.49 → \$0.47	+2.0%	-8.2%	1.5×

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VIII. MARKET-WINDOW DEMONSTRATIVES



Source: Yahoo Finance daily series; methodology and limitations stated in Part B. No market-index adjustment was applied.

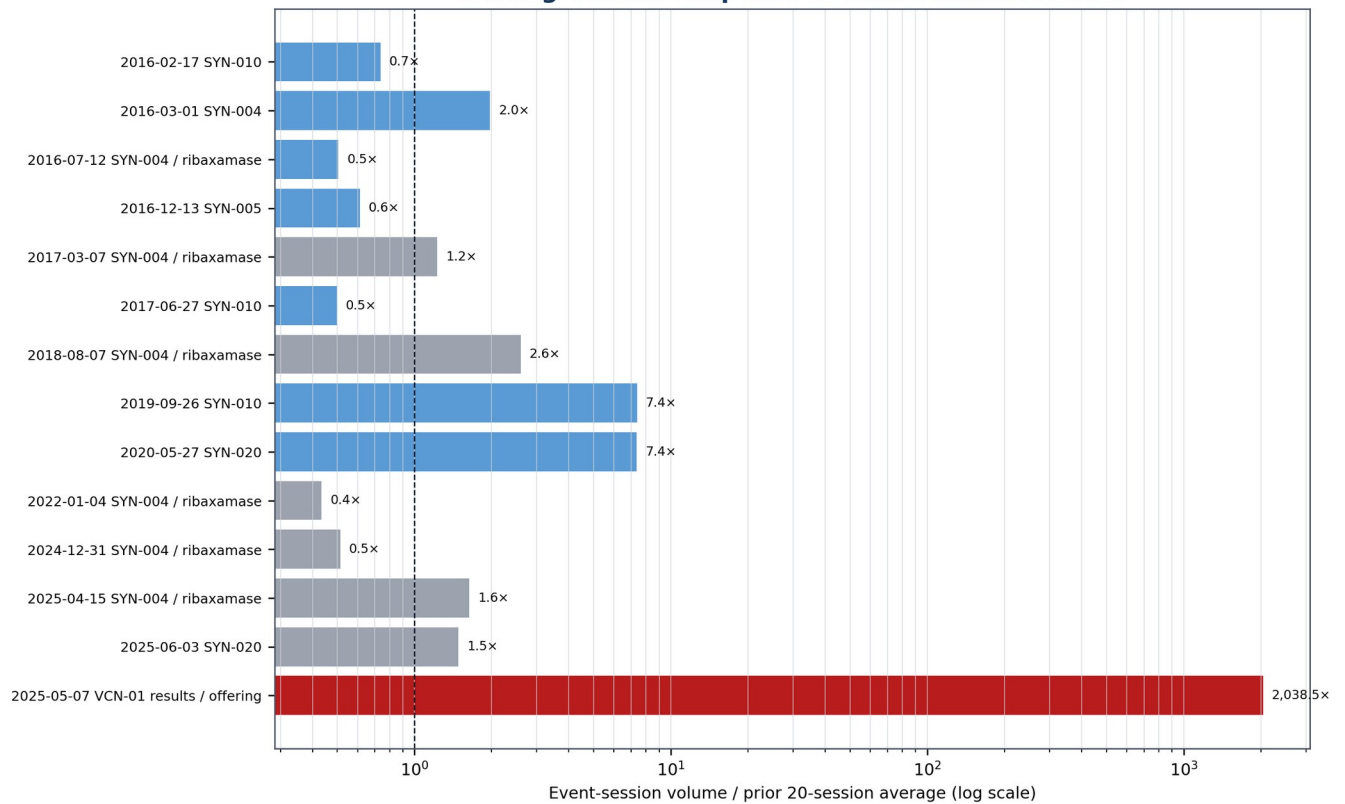
Demonstrative B-5. Split-adjusted return windows for selected patent/IP events.

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3

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Trading-Volume Response Around Selected Events



Demonstrative B-6. Event-session volume relative to the prior 20-session average.

IX. WHAT THE PUBLIC RECORD DOES AND DOES NOT ESTABLISH

24. The selected patent/IP events do not show one uniform market response. Some dedicated company releases coincided with positive short-term returns or elevated volume; others did not. Later patent-record events likewise show mixed responses.
25. The 2018 chronology is unusually compact: an S-3 covering shares underlying old warrants, a patent grant, a quarterly report, a reverse split, and a new warrant offering occurred within approximately eleven weeks. That sequence supports a focused request for contemporaneous planning records; it does not prove that the patent, split, and financings were one plan.
26. The May 7, 2025 chronology is the clearest same-day scientific-disclosure/financing example in this Part B. It establishes the two public timestamps, offering terms, and market response. It does not identify investor expectations, the offering-planning timeline, who bought or sold, or whether any trade was unlawful.
27. No inference should be drawn merely from the absence of a dedicated patent-titled release. Patent grants are public records; SEC reports may describe the patent estate; and management retains discretion over voluntary press releases, subject to applicable disclosure duties and the prohibition against materially misleading statements.
28. The public record cannot answer whether announcement timing, warrant terms, reverse-split timing, or financing calendars were coordinated. That question requires contemporaneous internal documents, communications, capitalization analyses, and testimony.

X. FOCUSED RECORDS REQUESTED FOR FACT DEVELOPMENT

- a. The complete patent docket and asset inventory for SYN-004, SYN-020, and VCN-01, including issue notices, maintenance decisions, assignments, licenses, valuations, and board presentations from January 1, 2016 forward;

- b. Board and committee minutes, written consents, presentations, calendars, and communications concerning whether and when to announce the selected patent and scientific events;
- c. Drafts, metadata, distribution lists, embargo instructions, and approval records for the May 7, 2025 VIRAGE results release and offering-pricing release, together with the timeline for negotiating and approving the offering;
- d. Communications with investor-relations firms, placement agents, financial advisers, warrant holders, prospective investors, and counsel concerning patent news, clinical news, reverse splits, trading liquidity, warrant exercise, warrant repricing, offering timing, or use of proceeds;
- e. Daily capitalization tables, warrant ledgers, exercise notices, settlement records, pre-funded-warrant activity, beneficial-ownership notices, and issuance records for the periods surrounding the identified events;
- f. Program-specific budgets and actual expenditures for SYN-004, SYN-020, and VCN-01, sufficient to compare financing proceeds with product-development spending and asset disposition; and
- g. Native market, transfer-agent, and financing records sufficient for a qualified expert to test price, volume, dilution, warrant exercise, market capitalization, and loss-causation questions without relying solely on a public website download.

XI. LEGAL GUIDEPOSTS AND PROCEDURAL LIMITATION

29. Regulation S-K Item 101 requires disclosure of material business developments and, when material to understanding the business, development status and the duration and effect of patents and licenses. It does not create a general rule requiring a separate press release for every patent.
30. Exchange Act Section 9(a)(2) addresses specified series of transactions undertaken for the purpose of inducing purchases or sales by creating actual or apparent activity or raising or depressing price. Rule 10b-5 addresses deceptive schemes and materially misleading statements or omissions in connection with securities transactions. The required elements cannot be supplied by chronology alone.
31. This Part B does not ask the Court to adjudicate a new unpleaded securities claim. It is proposed factual background for preservation and focused discovery related to the existing warrant dispute. If the Court concludes that any material exceeds the operative pleadings or authorized briefing, Plaintiff requests that it be treated only as a proposed offer of proof or denied without prejudice rather than treated as an amendment by implication.
32. Under LR 7-2(g), supplemental briefs or evidence ordinarily require leave of Court. This Version 1 should therefore be lodged as a proposed attachment to a motion for leave, or held for later use, rather than filed as a free-standing merits brief without procedural authorization.

Dated: _____

Signature: _____

Mark Joseph Nejme, Plaintiff, self-represented
P.O. Box 589, Clifton, New Jersey 07012
Telephone: 732-995-3914 | Email: research@warrantwire.com

1

APPENDIX A — SELECTED PATENT/IP EVENTS AND CLOSEST SEC FILINGS

Event date	Program	Patent/IP event	Closest filing on/before	Closest filing on/after
2016-02-17	SYN-010	Company press release: allowance of U.S. application 14/211,197	2016-02-02 8-K	2016-03-10 10-K
2016-03-01	SYN-004	Company press release: three additional U.S. patent allowances	2016-02-02 8-K	2016-03-10 10-K
2016-07-12	SYN-004 / ribaxamase	Company press release: EP 2,576,776 and U.S. 9,376,673; another U.S. allowance	2016-07-06 DEF 14A	2016-08-03 10-Q
2016-12-13	SYN-005	Company press release: U.S. 9,512,204 issued	2016-12-02 8-K	2017-01-05 8-K
2017-03-07	SYN-004 / ribaxamase	U.S. 9,587,234 issued: modified beta-lactamases	2017-03-02 8-K	2017-05-04 10-Q
2017-06-27	SYN-010	Company press release: allowance of U.S. application 14/776,465	2017-06-27 8-K	2017-06-27 8-K
2018-08-07	SYN-004 / ribaxamase	U.S. 10,041,056 issued: modified beta-lactamases	2018-08-01 S-3	2018-08-08 10-Q
2019-09-26	SYN-010	Company press release: U.S. 10,328,151 and an additional allowance	2019-09-06 8-K	2019-11-04 10-Q
2020-05-27	SYN-020	Company press release: exclusive option to license intellectual property from MGH	2020-05-05 8-K	2020-08-04 DEF 14A
2022-01-04	SYN-004 / ribaxamase	U.S. 11,214,787 issued: modified beta-lactamases	2022-01-04 8-K	2022-01-04 8-K
2024-12-31	SYN-004 / ribaxamase	U.S. 12,178,917 issued: beta-lactamase formulations and uses	2024-12-10 S-1	2025-01-21 S-1/A
2025-04-15	SYN-004 / ribaxamase	U.S. 12,274,740 issued: microbiome protection from oral antibiotics	2025-03-31 8-K	2025-05-07 8-K
2025-06-03	SYN-020	U.S. 12,318,434 issued: alkaline phosphatase-based oncology treatments	2025-06-02 8-K	2025-06-20 424B5

2 The closest-filing columns show temporal proximity only. They do not state that a listed filing concerned the patent event. Same-day filings may
3 concern another subject.

1

1. Theriva FY2025 Form 10-K (patent estate, programs, Rasayana license): <https://www.sec.gov/Archives/edgar/data/894158/00011046592060>
2. Theriva press-release archive: https://therivabio.com/investors/news/?post_types=press_releases
3. U.S. Patent 10,041,056: <https://patents.google.com/patent/US10041056B2/en>
4. U.S. Patent 12,178,917: <https://patents.google.com/patent/US12178917B2/en>
5. U.S. Patent 12,274,740: <https://patents.google.com/patent/US12274740B2/en>
6. U.S. Patent 12,318,434: <https://patents.google.com/patent/US12318434B2/en>
7. August 1, 2018 Form S-3 covering warrant shares: https://www.sec.gov/Archives/edgar/data/894158/000114420418041398/tv498567_s3.htm
8. August 8, 2018 Form 10-Q: https://www.sec.gov/Archives/edgar/data/894158/000114420418042887/tv499157_10q.htm
9. August 10, 2018 reverse-split Form 8-K: https://www.sec.gov/Archives/edgar/data/894158/000114420418043710/tv500716_8k.htm
10. October 2018 offering registration statement: https://www.sec.gov/Archives/edgar/data/894158/000114420418049861/tv503005_s1.htm
11. August 2024 reverse-split Form 8-K: https://www.sec.gov/Archives/edgar/data/894158/000110465924092644/tm2422579d1_8k.htm
12. September 2024 warrant-offering prospectus: https://www.sec.gov/Archives/edgar/data/894158/000110465924103658/tm2425095d1_424b5.htm
13. December 10, 2024 Form S-1: https://www.sec.gov/Archives/edgar/data/894158/000110465924127275/tm2430478d1_s1.htm
14. May 7, 2025 VIRAGE-results Form 8-K: https://www.sec.gov/Archives/edgar/data/894158/000110465925045309/tm2514050d1_8k.htm
15. May 8, 2025 offering Form 8-K: https://www.sec.gov/Archives/edgar/data/894158/000110465925046170/tm2514447d1_8k.htm
16. Theriva August 11, 2026 Q2 update: https://therivabio.com/press_releases/theriva-biologics-reports-second-quarter-2026-operational-highlights-and
17. Yahoo Finance TOVX historical data page: <https://finance.yahoo.com/quote/TOVX/history/>
18. 17 C.F.R. § 229.101 (Regulation S-K Item 101): <https://www.ecfr.gov/current/title-17/chapter-II/part-229/subpart-229.100/section-229.101>
19. 17 C.F.R. § 240.10b-5: <https://www.ecfr.gov/current/title-17/chapter-II/part-240/subpart-A/subject-group-ECFRbda83517c4e4377f/section-240.10b-5>
20. 15 U.S.C. § 78i: <https://uscode.house.gov/view.xhtml?req=granuleid:USC-prelim-title15-section78i&num=0&edition=prelim>
21. District of Nevada Local Rules: <https://www.nvudcourts.gov/court-information/rules-and-orders/>

23 Research file generated with Part B

24 The computational source file is part_b_event_study.json. It records each selected event, the market-window calculations, reported-share snapshot,
25 nearest SEC filings, and nearby capital events. It should be retained with the working papers and produced if a chart is offered as evidence.

**UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., Defendant.	Case No. 3:26-cv-00705
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SUPPLEMENTAL EXHIBIT SC

Financial Program and Strategic Record

Original Part C

The following pages are separately paginated under this supplemental exhibit identifier.

1 Mark Joseph Nejme
2 P.O. Box 589
3 Clifton, New Jersey 07012
4 Telephone: 732-995-3914
5 Email: research@warrantwire.com
6 Plaintiff, self-represented

7 UNITED STATES DISTRICT COURT
8 DISTRICT OF NEVADA

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., a Nevada corporation, Defendant.	Case No. 3:26-cv-00705 PROPOSED SUPPLEMENTAL INFORMATION — PART C TEN-YEAR FINANCIAL, PROGRAM-SPENDING, INSTITUTIONAL-TRANSACTION, AND STRATEGIC- RECORD SYNTHESIS
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9 **QUESTION PRESENTED**

10 What do the cited public records establish—and leave unanswered—when the ten-year expenditure record,
11 program-level spending, institutional financing, asset divestiture and licensing, non-dilutive funding opportunities,
12 capitalization changes, and the already-pleaded strategic-review disclosures are read together?

SEPARATE PROPOSED DOCUMENT; NO DUPLICATION OR NEW MERITS CLAIM. This Part C does not amend, replace, withdraw, or revise the Verified Complaint or proposed Parts A and B. It incorporates the Verified Complaint’s paragraphs 90–94 rather than repeating its strategic-review allegations. It synthesizes public records and identifies records that may be relevant if discovery is authorized. Chronology and association do not establish intent, manipulation, personal knowledge, control, or liability. District of Nevada LR 7-2(g) ordinarily requires leave before supplemental briefing.

13 VERSION 3 — FOR REVIEW BEFORE ANY FILING

I. SCOPE, CROSS-REFERENCES, AND NONDUPLICATION

1. The Verified Complaint already alleges that Theriva’s June 30, 2026 quarterly report disclosed ongoing discussions about strategic alternatives that may include a business combination, merger, or reverse merger; that July and August 2026 compensation documents addressed transaction-related vesting and change in control; and that those disclosures did not establish a signed merger, counterparty, or selected course. Verified Complaint paragraphs 90–94 are incorporated by reference and not repeated here.
2. Proposed Part A contains the detailed reverse-split, financing, warrant, capital-use, Rasayana, and professional-overlap record. Proposed Part B contains the detailed patent and intellectual-property chronology, event windows, and May 7, 2025 scientific-disclosure/offering sequence. Part C does not reproduce those tables. It adds a ten-year financial view, program-level R&D allocation, the August 2026 incentive-plan registration, a transaction-specific MSD witness trail, the Grifols divestiture and risk-allocation record, a limited non-dilutive-funding comparison, and a balanced synthesis.
3. All charts are demonstratives derived from the cited records. They are not independent evidence. Financial statements are accrual- and cash-flow records; they do not trace fungible financing proceeds to a particular vendor, scientist, employee, patent, or transaction without ledgers, invoices, bank records, payroll allocations, and testimony.

Record	Subject	Treatment in Part C
Verified Complaint ¶¶90–94	Strategic alternatives; July/August 2026 transaction-readiness disclosures	Incorporated once; not restated
Proposed Part A	Reverse splits; financings; warrants; 2021–H1 2026 cash bridge; Rasayana; overlaps	Detailed figures remain there
Proposed Part B	Patent/IP record; trading windows; May 7, 2025 sequence	Detailed event study remains there
Part C	Ten-year trends; 2024–2025 program allocation; Aug. 24 S-8; MSD witness trail; Grifols divestiture; non-dilutive funding comparison; synthesis	Incremental material

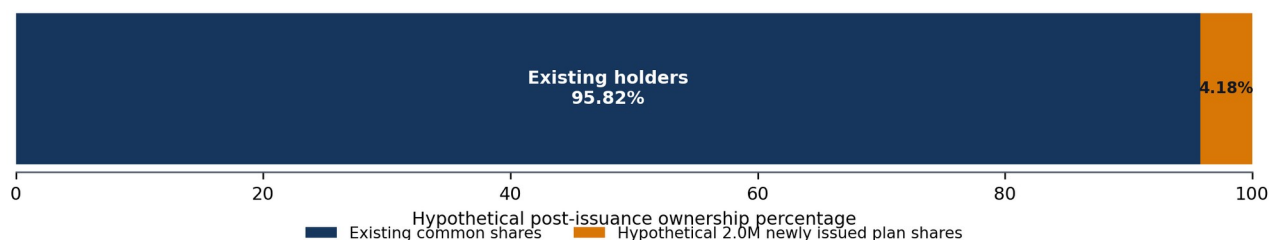
II. AUGUST 2026 INCENTIVE-PLAN REGISTRATION — WHAT IT DID AND DID NOT DO

4. On August 3, 2026, stockholders approved Amendment No. 4 to the 2020 Stock Incentive Plan, increasing the plan ceiling from 4,500,000 to 6,500,000 common shares. Theriva reported 3,110,068 votes for, 2,840,658 against, 105,575 abstentions, and 10,823,826 broker non-votes on that proposal.
5. On August 24, 2026 at 5:00:35 p.m. Eastern, Theriva filed a Form S-8 registering the additional 2,000,000 plan shares. The S-8 identifies Steven A. Shallcross as agent for service and Leslie Marlow of Blank Rome LLP as counsel receiving copies.
6. The S-8 is not a reverse-merger announcement and does not itself establish that any employee received, accepted, vested in, exercised, sold, or was required to accept any share. It registers securities for possible plan issuance. The separate reverse-merger language appears in the Form 10-Q filed August 11, 2026 and is already addressed in Verified Complaint paragraphs 90–94.

7. The 2,000,000-share increment equals approximately 4.36% of the 45,892,668 common shares reported outstanding on August 7, 2026. If all 2,000,000 were newly issued and no other capitalization changed, existing holders collectively would own approximately 95.82% of the resulting total—a percentage reduction of approximately 4.18%. Actual dilution depends on grants, vesting, exercises or settlements, forfeitures, withholding, treasury-share use, and other issuances.

Registration is not issuance. This illustration assumes all 2,000,000 additional shares are newly issued and ignores every other capitalization change.

Scale of the August 24, 2026 S-8 Registration



Demonstrative C-1. Hypothetical scale of the 2,000,000-share S-8 increment; registration is not issuance.

III. TEN-YEAR FINANCIAL RECORD

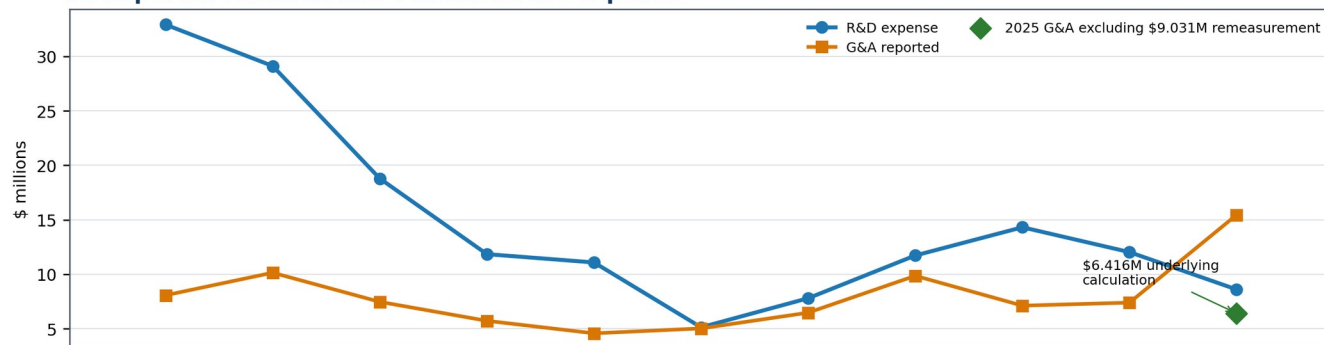
8. The annual statements for 2015 through 2025 report cumulative R&D expense of approximately \$163.326 million, cumulative G&A expense of approximately \$87.315 million, cumulative operating cash use of approximately \$214.889 million, and cumulative net financing cash inflow of approximately \$216.701 million. Expense totals and cash-flow totals are different accounting measures and must not be added together as separate uses of the same cash.
9. Reported R&D declined from \$32.906 million in 2015 to \$8.604 million in 2025, a decline of approximately 73.9%. The largest reported R&D years in this period were 2015, 2016, and 2017. From 2021 forward, the highest annual R&D expense was \$14.311 million in 2023.
10. Reported 2025 G&A was \$15.447 million, but the FY2025 filing includes a \$9.031 million change in the fair value of contingent consideration in that category. Subtracting that remeasurement produces \$6.416 million, compared with \$6.697 million after the analogous \$0.699 million 2024 remeasurement—an approximately 4.2% decrease. The remeasurement is an accounting amount, not proof of a \$9.031 million cash administrative payment.
11. Cash and cash equivalents were \$20.818 million at December 31, 2015, peaked at \$67.325 million at December 31, 2021 after the 2021 financing inflow, and were \$13.056 million at December 31, 2025. Part A contains the separate 2021–June 2026 cash bridge and should be consulted rather than duplicating it here.

FY	R&D	G&A	Operating cash used	Net financing cash flow	Year-end cash
2015	32.906	8.074	38.890	+42.684	20.818
2016	29.109	10.143	27.901	+26.707	19.055
2017	18.784	7.467	20.204	+18.477	17.116
2018	11.844	5.727	17.279	+29.088	28.918
2019	11.083	4.580	13.873	+0.000	15.045
2020	5.131	5.029	12.168	+3.359	6.227
2021	7.800	6.474	12.890	+74.002	67.325
2022	11.723	9.858	19.082	-1.930	41.786
2023	14.311	7.120	18.996	+0.625	23.177
2024	12.031	7.396	16.937	+5.496	11.609
2025	8.604	15.447 (6.416 excl. remeasurement)	16.669	+18.193	13.056

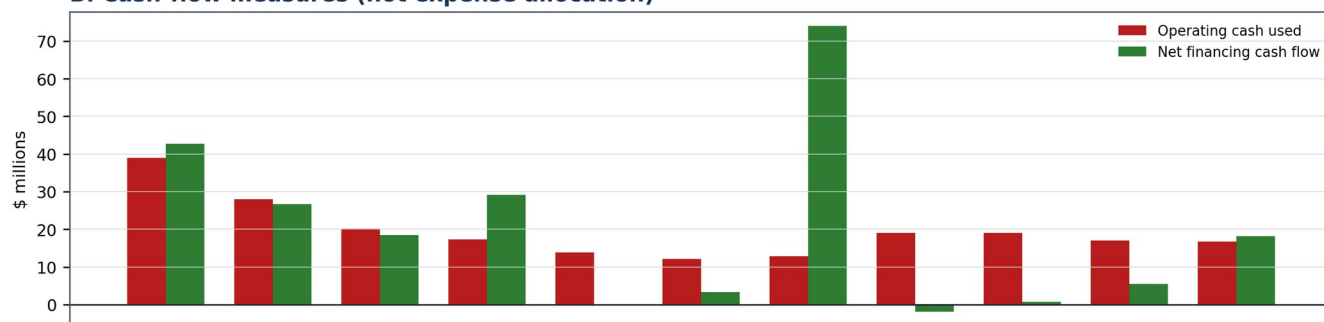
All amounts in \$ millions. Parentheses are not used in the cash-used column because the table presents the magnitude of reported operating outflow. The 2025 G&A parenthetical is a calculation, not a GAAP line item.

Theriva / Synthetic: Ten-Year Financial Record

A. Reported research and administrative expense

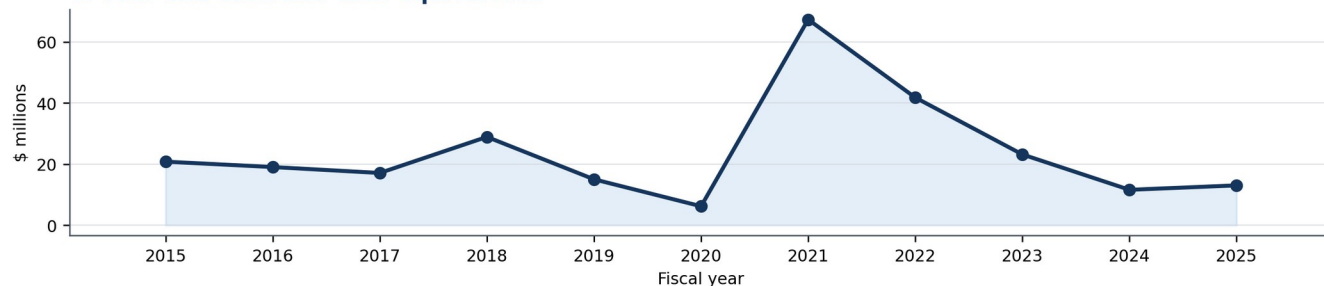


B. Cash-flow measures (not expense allocation)



2015–2025 totals: USD 214.889M operating cash used; USD 216.701M net financing cash inflow.

C. Year-end cash and cash equivalents

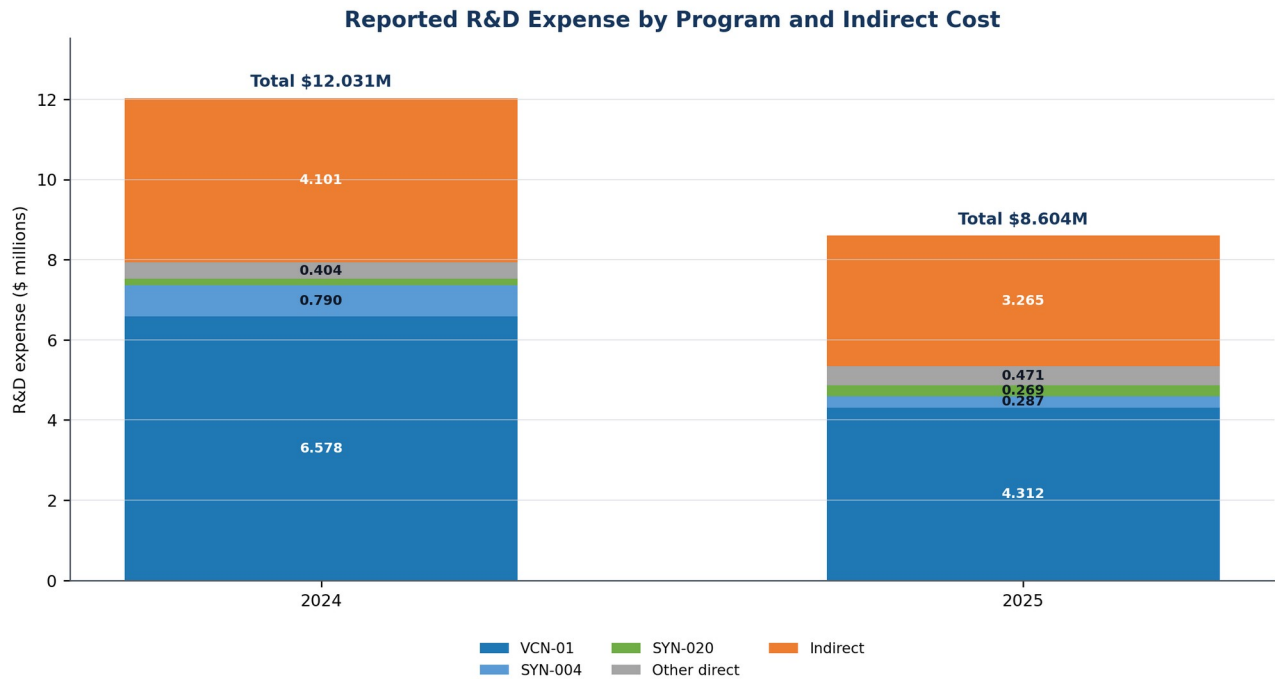


Demonstrative C-2. Ten-year expense, cash-flow, and ending-cash record from annual SEC filings.

IV. PROGRAM-LEVEL R&D ALLOCATION, 2024–2025

12. The FY2025 Form 10-K provides program-level R&D for 2024 and 2025. Total R&D declined from \$12.031 million to \$8.604 million. Direct VCN-01 expense declined from \$6.578 million to \$4.312 million; direct SYN-004 expense declined from \$0.790 million to \$0.287 million; direct SYN-020 expense increased from \$0.158 million to \$0.269 million; and indirect R&D declined from \$4.101 million to \$3.265 million.
13. The filing attributes the overall decline primarily to reduced VCN-01 and SYN-004 clinical-trial expense and lower indirect compensation, partly offset by higher SYN-020 patent expense. It states that operating cash was used primarily for VCN-01 development. Those statements identify reported categories but do not establish why each scientific or capital-allocation decision was made.

1 14. Theriva’s February 2026 SYN-020 license placed future development responsibility and cost with Rasayana. Public
2 records therefore support asking for the program budgets, patent-cost detail, valuation work, and board materials that
3 preceded the license. They do not establish that SYN-020 was transferred for use in a later reverse merger or that any
4 counterparty was selected.



Demonstrative C-3. Reported R&D allocation by program and indirect cost, 2024–2025.

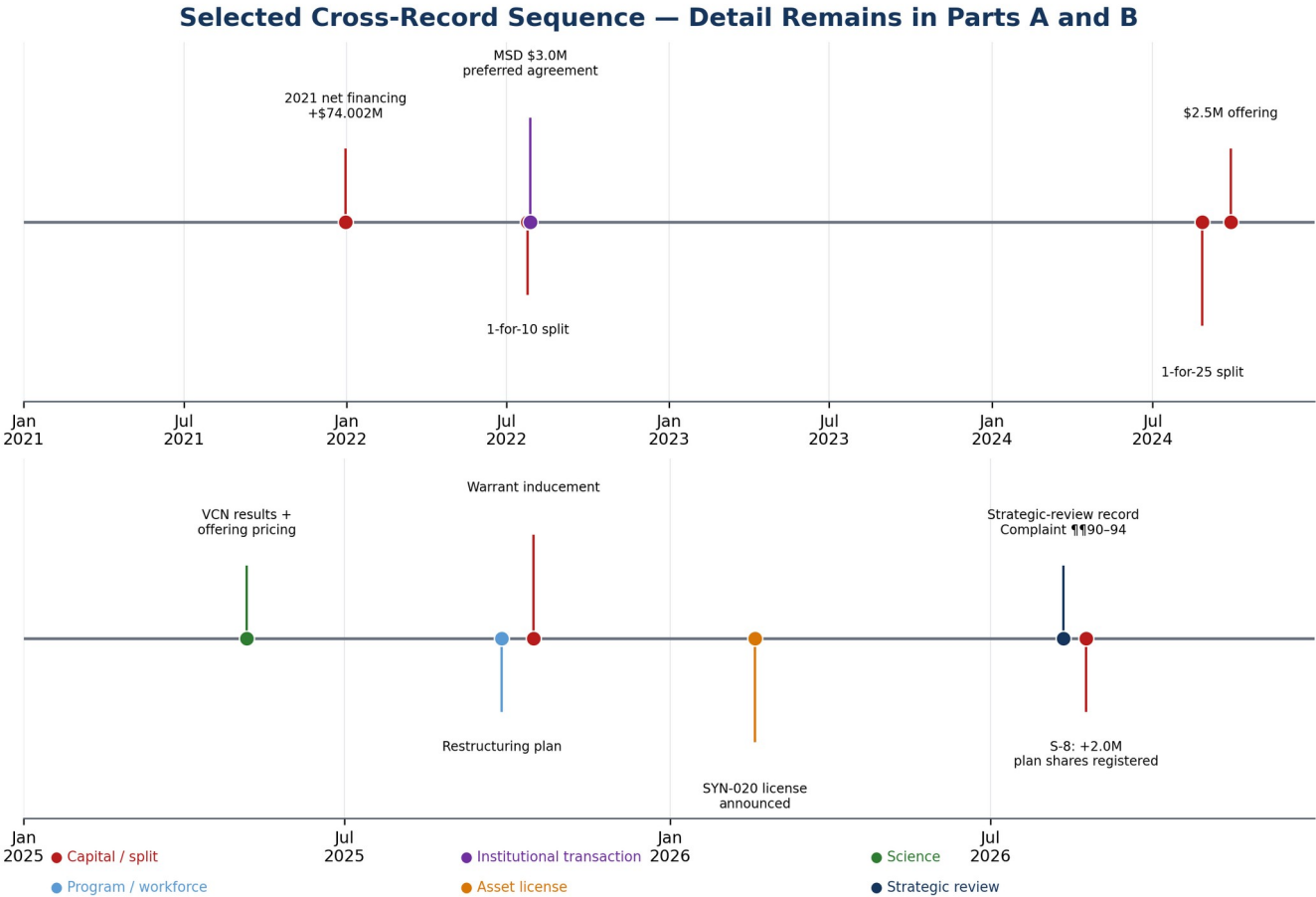
V. MSD TRANSACTION AND PROPER WITNESS TRAIL

15. On July 28, 2022, Synthetic Biologics and MSD Credit Opportunity Master Fund, L.P. executed a Securities Purchase Agreement. The fund agreed to purchase \$2.200 million of Series C preferred stock and \$0.800 million of Series D preferred stock, for \$3.000 million gross in the aggregate. The agreement identifies A.G.P./Alliance Global Partners as financial adviser.
16. Steven A. Shallcross signed for Synthetic. Marcello Liguori, identified as a Managing Director, signed for MSD Credit Opportunity Master Fund. Company-side notices copied Leslie Marlow of Blank Rome LLP. These are transaction-specific identities supported by the agreement; the document does not identify Michael Dell as a signatory or state that he personally negotiated, approved, monitored, or directed the investment.
17. The agreement includes company representations, purchaser representations, a pre-announcement confidentiality and no-trading covenant, resale-registration obligations, and stockholder-approval obligations tied to an authorized-share increase and name change. Those provisions make the fund's transaction files potentially relevant to the 2022 capitalization sequence; they do not establish unlawful trading or participation in the later challenged warrant transaction.
18. Theriva's FY2025 Form 10-K reports that all 275,000 Series C and 100,000 Series D preferred shares were converted during 2024 into 72,132 and 26,230 post-split common shares, respectively, and that no Series C or D preferred remained outstanding at December 31, 2025. The reviewed public record does not establish whether MSD retained or disposed of the resulting 98,362 common shares.
19. If transaction-specific discovery is permitted, the supported witnesses are the fund's records custodian or designated corporate representative and Mr. Liguori, subject to relevance, personal knowledge, jurisdiction, proportionality, and proper process. The present record does not supply a factual basis to characterize Michael Dell personally as a participant or a necessary witness.

Role / issue	Documented fact	Limit
Investor	MSD Credit Opportunity Master Fund, L.P.	Party to purchase agreement
Investor signatory	Marcello Liguori, Managing Director	Signed for purchaser
Issuer signatory	Steven A. Shallcross, CEO	Signed for Synthetic
Financial adviser	A.G.P./Alliance Global Partners	Defined in agreement
Company counsel copied	Leslie Marlow, Blank Rome LLP	Notice provision
2024 conversion	98,362 common shares in aggregate	72,132 Series C conversion + 26,230 Series D conversion
Current holding	Not established by reviewed public filings	Preferred no longer outstanding; common disposition unknown

1

VI. CROSS-RECORD SYNTHESIS



Demonstrative C-4. Selected cross-record sequence; the underlying detail and limitations remain in Parts A and B.

20. The combined record shows recurring reliance on external capital; decreasing long-run R&D expense; repeated reverse splits and securities transactions; a later restructuring and workforce reduction; licensing of SYN-020; continued but narrowed VCN-01 development; an active strategic review; and a later registration of additional incentive-plan shares.
21. Those facts are consistent with more than one explanation. They may reflect capital constraints, clinical priorities, ordinary efforts to preserve runway, transaction preparation, or a combination. The same chronology does not by itself establish that management accumulated patents for a reverse merger, intentionally depressed the stock price, coordinated trading with any holder, concealed assets, or abandoned product development for an improper purpose.
22. The public record also contains evidence inconsistent with a conclusion that all scientific work ceased: Theriva reported continued VCN-01 trials, the August 2026 first patient in VIRAGE2, planned manufacturing scale-up, regulatory interactions, and limited discovery work. The record simultaneously states that cash is insufficient for broader trials and that a \$5.0 million Grifols obligation would significantly deplete liquidity.

1 **Balanced hypothesis test**

Analytical category	Record-based assessment
Facts consistent with transaction preparation	Active strategic review; July/August 2026 compensation changes; program narrowing; workforce reduction; SYN-020 license; additional plan-share registration.
Facts consistent with ordinary capital constraints	Going-concern uncertainty; limited runway; stated inability to fund broader trials; continuing VCN-01 work; Grifols obligation; disclosed need for financing or partners.
Not established by public record	Counterparty; signed reverse-merger agreement; plan to depress price; coordinated warrant trading; hidden patent valuation; Michael Dell’s personal knowledge; employee-share issuance under the Aug. 24 S-8; whether Theriva applied for the PanCAN award; undisclosed conflict in Grifols’ selection of Synthetic.
Records that could distinguish explanations	Board/adviser materials, budgets and forecasts, patent valuations, transaction outreach, capitalization models, award ledger, warrant/transfer-agent records, bank/ledger/invoice detail, Grifols sale and milestone files, non-dilutive-funding applications, and witness testimony.

2 **VII. GRIFOLS DIVESTITURE, DEVELOPMENT RISK, AND NON-DILUTIVE FUNDING**

3 **QUESTIONS**

- 4 **23.** On December 14, 2021, Grifols announced that it had agreed to sell its entire 86.83% interest in VCN Biosciences to
5 Synthetic Biologics. Grifols described VCN as a non-strategic activity within an announced divestment plan, stated that it
6 had financed VCN research since 2012, and stated that Synthetic would continue to finance VCN’s research programs
7 and maintain its employee pool.
- 8 **24.** The disclosed structure did not consist only of a \$4.7 million cash sale. Grifols stated that its equity interest was valued at
9 up to \$75 million. The transaction included \$4.7 million in initial cash, assumption of \$2.4 million of VCN liabilities, and
10 up to \$70.3 million in contingent clinical-development and regulatory milestone payments. The remaining VCN holders
11 received Synthetic common stock representing 19.99% of Synthetic’s then-outstanding common shares.
- 12 **25.** At announcement, VCN-01 had been evaluated in four Phase 1 trials and a controlled Phase 2 pancreatic-cancer trial was
13 planned. Synthetic also disclosed a post-closing commitment to fund approximately \$27.8 million of VCN research,
14 development, and necessary general and administrative costs. The relatively modest cash payment and larger contingent
15 component are consistent with allocating early-stage clinical and regulatory risk between seller and buyer. Standing
16 alone, that structure establishes neither that VCN-01 lacked value nor that the transaction was fraudulent.
- 17 **26.** Theriva’s FY2025 Form 10-K reports that \$3.0 million and \$3.25 million milestone payments were made after specified
18 regulatory and first-patient events. It further reports that a \$6.0 million Phase 2 milestone was achieved in 2025, that \$1.0
19 million was paid, and that the remaining \$5.0 million was deferred pending discussions with Grifols. The same filing
20 reports a \$10.004 million non-current fair-value liability for other contingent consideration at December 31, 2025. These
21 are different accounting categories and should not be added or described as cash already paid without a complete
22 milestone ledger.
- 23 **27.** A.G.P./Alliance Global Partners served as Synthetic’s exclusive financial adviser and Tungsten Advisors served as
24 VCN’s exclusive financial adviser. The public announcements reviewed do not disclose the bidder universe, alternative
25 proposals, valuation analyses, fairness work, adviser compensation, the criteria used to select Synthetic, or the diligence
26 performed on Synthetic’s ability to satisfy the development budget and milestone obligations. Adviser participation does
27 not by itself establish a conflict, deceptive purpose, or wrongdoing.
- 28 **28.** On September 26, 2023, while Theriva’s VIRAGE Phase 2b trial of VCN-01 was underway, the Pancreatic Cancer
29 Action Network awarded Oncolytics Biotech, Inc. a \$5.0 million Therapeutic Accelerator Award for a different
30 pancreatic-cancer oncolytic-virus program. PanCAN stated that Oncolytics was selected through a rigorous competitive
31 scientific, business, and programmatic review. The reviewed public record does not establish whether Theriva applied,

was eligible, was evaluated, was rejected, or elected not to seek that award. The award therefore supplies a comparator for non-dilutive funding inquiry, not proof that Theriva was bypassed or that its science was inferior.

29. Plaintiff's concern, framed expressly as a question rather than an accusation, is whether the acquisition and later financing choices were selected solely to advance the clinical programs and preserve shareholder value, or whether transaction incentives also materially influenced the financing path. The present record does not establish fraud, a sham asset, fabricated science, coordinated trading, or an arrangement to benefit any financial intermediary. The question can be tested only through contemporaneous valuation, buyer-selection, adviser-compensation, financing, grant-application, milestone, and board records, together with competent testimony.
30. Before the VCN transaction, Synthetic's February 19, 2021 prospectus supplement summarized the SYN-004 record. It reported positive topline results from a Phase 2b proof-of-concept trial in the first quarter of 2017, including achievement of the primary endpoint of significantly reducing *Clostridioides difficile* infection. It also reported additional 2017 results concerning protection of the gut microbiome and antimicrobial-resistance genes, and preliminary agreement with the FDA in the first half of 2018 on a proposed Phase 3 trial synopsis. The December 14, 2021 VCN acquisition announcement came nearly five years after the reported Phase 2b result and continued to identify SYN-004 and SYN-020 as Synthetic's lead candidates. The reviewed record therefore does not establish that SYN-004 was sold or terminated when VCN was acquired.
31. The acquisition announcement described pancreatic ductal adenocarcinoma as an aggressive and lethal malignancy, emphasized a significant unmet medical need, described early VCN-01 clinical results as encouraging, and stated that the combined programs could drive shareholder value. Those public statements may have broadened investor interest in the company, including interest among retail investors affected by pancreatic cancer. The reviewed record does not establish how many investors purchased because of those statements or whether expected retail interest influenced the acquisition or later financing choices. That question requires contemporaneous investor-relations materials, capital-allocation analyses, event-level market data, and competent testimony; the seriousness of the disease and the appeal of the scientific program do not themselves prove efficacy, valuation, or intent.

VIII. FOCUSED RECORDS AND TESTIMONY REQUESTED

- a. Strategic-review engagement letters, counterparty lists, indications of interest, capitalization models, board presentations, minutes, written consents, and communications, limited to materials bearing on the challenged warrants, potential share issuance, asset disposition, or a transaction affecting the requested status quo;
- b. The 2020 Stock Incentive Plan reserve and award ledger through the present, including grants, recipients, vesting terms, exercises, settlements, cancellations, forfeitures, tax withholding, treasury-share use, and shares remaining available after the August 24, 2026 S-8;
- c. Program budgets versus actual expenditures for VCN-01, SYN-004, and SYN-020; general-ledger detail; vendor and laboratory invoices; payroll and time allocations; patent prosecution costs; and board-approved capital-allocation plans for 2021 through the present;
- d. Records supporting the SYN-020 valuation and Rasayana license, including diligence, alternative bids or counterparties, relationship questionnaires, approvals, milestone accounting, sublicense economics, and post-license development reporting;
- e. MSD transaction files held by the fund or its custodian concerning the July 2022 investment, due diligence, valuation, conversion, voting, resale registration, 2024 conversion, disposition of the resulting common shares, and later Theriva communications, to the extent relevant and obtainable through proper process;
- f. A.G.P. records concerning the 2022 financing and any later engagement connected to the challenged warrants or a strategic transaction, with privileged material protected and relevance limited to nonprivileged transaction facts;
- g. Transfer-agent and warrant-agent ledgers, exercise notices, settlement instructions, share-issuance records, beneficial-ownership notices, and daily capitalization tables needed to determine whether and when potential dilution became actual; and
- h. If the Court finds the accounting issues relevant and proportional, an independent forensic accounting examination or neutral expert report reconciling financing sources, issuance costs, operating uses, program costs, asset transactions, warrant-related noncash amounts, and remaining liquidity without double counting.

- i. The complete Grifols milestone ledger; payment notices; invoices; proof of payment; deferral, waiver, amendment, or security agreements; valuation workpapers; probability assumptions; and communications concerning the \$6.0 million milestone achieved in 2025, the \$1.0 million payment, and the \$5.0 million deferred balance;
- j. Nonprivileged records concerning the VCN sale process, including bidder and outreach lists, indications of interest, alternative proposals, valuation and fairness materials, buyer-capacity diligence, board approvals, and A.G.P. and Tungsten engagement letters, compensation, and transaction-specific communications; and
- k. Records showing whether Theriva or VCN sought PanCAN’s Therapeutic Accelerator Award, Precision Promise participation, governmental grants, foundation support, collaborative trial funding, or other non-dilutive capital for VCN-01, including applications, eligibility reviews, correspondence, decisions, and board consideration.
- l. If testimony is authorized, organizational testimony from a Grifols corporate designee under Federal Rule of Civil Procedure 30(b)(6), obtained through Rule 45 or other lawful process applicable to a foreign nonparty, concerning the stated divestiture rationale, the buyer-selection process, valuation and alternative proposals, diligence concerning Synthetic’s ability to fund its existing programs and VCN, the approximately \$27.8 million post-closing funding covenant, adviser roles and compensation, milestone administration, and the later \$5.0 million deferral. Any trial appearance or use of deposition testimony remains subject to the Court’s jurisdiction, scheduling, evidentiary rulings, and Federal Rule of Civil Procedure 32; and
- m. Nonprivileged board, finance, and investor-relations materials analyzing how the VCN acquisition would affect cash runway, SYN-004 and SYN-020 funding, public positioning, capital needs, and anticipated investor response, limited to the acquisition and later financing choices that bear on the pleaded warrant and proxy issues.

IX. WHAT PART C DOES NOT ASK THE COURT TO DECIDE

32. Part C does not ask the Court to find manipulation, fraud, breach of fiduciary duty, a related-party transaction, commingling, or personal liability from chronology, professional association, an investment, a plan registration, or a poor shareholder outcome alone.
33. Part C does not assert that every dollar raised was available for research, that operating cash use equals investor market loss, that R&D expense equals cash paid, or that a patent grant necessarily required a separate press release. It does not treat contingent milestone amounts as cash received.
34. Part C does not enlarge the operative complaint by implication. If the Court concludes that any subject exceeds the pleaded warrant and proxy issues or authorized briefing, Plaintiff requests that the material be treated only as a proposed demonstrative or offer of proof, or denied without prejudice, rather than as a basis to dismiss the existing claims.
35. Because LR 7-2(g) generally requires leave for supplemental briefing or evidence, this document should be lodged as a proposed attachment to a motion for leave or retained for later authorized use. It should not be filed as an unauthorized free-standing merits brief.

Dated: _____

Signature: _____

Mark Joseph Nejme, Plaintiff, self-represented
P.O. Box 589, Clifton, New Jersey 07012
Telephone: 732-995-3914 | Email: research@warrantwire.com

1 APPENDIX A — CALCULATION NOTES

Calculation	Result	Method
Ten-year cumulative R&D	\$163.326M	Sum of reported 2015–2025 R&D expense
Ten-year cumulative G&A	\$87.315M	Sum of reported 2015–2025 G&A expense; includes reported remeasurements
Ten-year operating cash used	\$214.889M	Absolute value of summed 2015–2025 cash from operations
Ten-year net financing cash flow	\$216.701M	Sum of 2015–2025 net financing cash flow
R&D decline, 2015 to 2025	73.9%	1 – \$8.604M ÷ \$32.906M
2025 underlying G&A calculation	\$6.416M	\$15.447M reported – \$9.031M contingent-consideration remeasurement
2024 underlying G&A calculation	\$6.697M	\$7.396M reported – \$0.699M contingent-consideration remeasurement
S-8 increment / Aug. 7 shares	4.36%	2,000,000 ÷ 45,892,668
Hypothetical post-issuance new-share percentage	4.18%	2,000,000 ÷ (45,892,668 + 2,000,000)
MSD conversion total	98,362	72,132 + 26,230 post-split common shares

2 Rounding may cause immaterial differences. These calculations should be checked by a qualified accountant or financial expert before evidentiary
3 use.

4 APPENDIX B — SOURCE SCHEDULE

- 5 **1. Verified Complaint, submitted September 16, 2026:** Retained filed copy; paragraphs 90–94; ECF number not inserted in the retained
6 copy.
- 7 **2. Theriva August 4, 2026 Form 8-K — annual-meeting vote and plan increase:**
8 https://www.sec.gov/Archives/edgar/data/894158/000110465926090064/tm2619846d1_8k.htm
- 9 **3. Theriva August 24, 2026 Form S-8 — additional 2,000,000 plan shares:** [https://www.sec.gov/Archives/edgar/data/894158/000110465926100340/
10 tm2623752d1_s8.htm](https://www.sec.gov/Archives/edgar/data/894158/000110465926100340/tm2623752d1_s8.htm)
- 11 **4. Theriva August 11, 2026 Form 10-Q — outstanding shares, program status, strategic alternatives, liquidity:**
12 <https://www.sec.gov/Archives/edgar/data/894158/000110465926093786/tmb-20260630x10q.htm>
- 13 **5. Synthetic / MSD July 28, 2022 Securities Purchase Agreement:** [https://www.sec.gov/Archives/edgar/data/894158/000110465922084301/
14 tm2222078d1_ex10-1.htm](https://www.sec.gov/Archives/edgar/data/894158/000110465922084301/tm2222078d1_ex10-1.htm)
- 15 **6. Theriva FY2025 Form 10-K — program R&D, contingent consideration, preferred conversion, Rasayana:**
16 <https://www.sec.gov/Archives/edgar/data/894158/000110465926026767/tmb-20251231x10k.htm>
- 17 **7. Grifols December 14, 2021 VCN divestiture announcement — stated rationale and transaction structure:**
18 <https://www.grifols.com/en/view-news/-/news/grifols-moves-forward-with-its-divestment-plan-by-agreeing-to-sell-vcn-biosciences>
- 19 **8. Synthetic Biologics December 14, 2021 planned-acquisition announcement — program status, consideration, advisers, and risks:**
20 [https://therivabio.com/press_releases/synthetic-biologics-announces-planned-transformative-acquisition-of-vcn-biosciences-developer-of-a-novel-
21 oncolytic-virus-platform-targeting-pancreatic-and-other-solid-tumors/](https://therivabio.com/press_releases/synthetic-biologics-announces-planned-transformative-acquisition-of-vcn-biosciences-developer-of-a-novel-oncolytic-virus-platform-targeting-pancreatic-and-other-solid-tumors/)
- 22 **9. Synthetic Biologics March 11, 2022 acquisition-closing announcement — milestones, \$27.8 million funding covenant, and
23 advisers:** https://therivabio.com/press_releases/synthetic-biologics-completes-acquisition-of-vcn-biosciences/
- 24 **10. PanCAN September 26, 2023 Therapeutic Accelerator Award announcement — \$5.0 million award to Oncolytics and
25 competitive review:** [https://pancan.org/press-releases/the-pancreatic-cancer-action-network-selects-oncolytics-biotech-inc-to-receive-5-million-
26 therapeutic-accelerator-award-to-develop-leading-edge-treatments/](https://pancan.org/press-releases/the-pancreatic-cancer-action-network-selects-oncolytics-biotech-inc-to-receive-5-million-therapeutic-accelerator-award-to-develop-leading-edge-treatments/)
- 27 **11. ClinicalTrials.gov NCT05673811 — VIRAGE Phase 2b study record:** <https://clinicaltrials.gov/study/NCT05673811>
- 28 **12. Synthetic Biologics February 19, 2021 prospectus supplement — SYN-004 Phase 2b results and proposed Phase 3 synopsis
29 chronology:** https://www.sec.gov/Archives/edgar/data/894158/000110465921025447/tm217135d1_424b5.htm

30 Annual filings used for the ten-year schedule

- 31 • **Theriva/Synthetic FY2016 Form 10-K:** https://www.sec.gov/Archives/edgar/data/894158/000114420417012545/v459847_10k.htm
- 32 • **Theriva/Synthetic FY2017 Form 10-K:** https://www.sec.gov/Archives/edgar/data/894158/000114420418010286/tv485536_10k.htm
- 33 • **Theriva/Synthetic FY2018 Form 10-K:** https://www.sec.gov/Archives/edgar/data/894158/000114420419010592/tv513906_10k.htm
- 34 • **Theriva/Synthetic FY2019 Form 10-K:** https://www.sec.gov/Archives/edgar/data/894158/000110465920023415/tm205312-1_10k.htm
- 35 • **Theriva/Synthetic FY2020 Form 10-K:** https://www.sec.gov/Archives/edgar/data/894158/000110465921032150/tm211106d1_10k.htm
- 36 • **Theriva/Synthetic FY2021 Form 10-K:** <https://www.sec.gov/Archives/edgar/data/894158/000141057822000405/syn-20211231x10k.htm>
- 37 • **Theriva/Synthetic FY2022 Form 10-K:** <https://www.sec.gov/Archives/edgar/data/894158/000141057823000428/tmb-20221231x10k.htm>
- 38 • **Theriva/Synthetic FY2023 Form 10-K:** <https://www.sec.gov/Archives/edgar/data/894158/000141057824000293/tmb-20231231x10k.htm>
- 39 • **Theriva/Synthetic FY2024 Form 10-K:** <https://www.sec.gov/Archives/edgar/data/894158/000141057825000285/tmb-20241231x10k.htm>
- 40 • **Theriva/Synthetic FY2025 Form 10-K:** <https://www.sec.gov/Archives/edgar/data/894158/000110465926026767/tmb-20251231x10k.htm>

41 Prepared-material cross-references

- 42 • Proposed Supplemental Addendum — Part A, Version 5 (reverse splits, warrants, related-person inquiry, and capital use).
- 43 • Proposed Supplemental Information — Part B, Version 1 (patent assets, public disclosures, market activity, warrants, financings, and
44 reverse splits).

**UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., Defendant.	Case No. 3:26-cv-00705
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SUPPLEMENTAL EXHIBIT SD

Funding Since 2021

Original Part D

The following pages are separately paginated under this supplemental exhibit identifier.

**UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., Defendant.	Case No. 3:26-cv-00705 PROPOSED SUPPLEMENTAL INFORMATION PART D — A LOOK AT ALL FUNDING SINCE 2021 Breakdown of ATM sales, public offerings, warrant proceeds, R&D reimbursements, and special financing
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Purpose and measurement period

This Part D identifies and categorizes cash that Theriva Biologics, Inc. reported receiving from January 1, 2021 through June 30, 2026. It separates at-the-market stock sales, other public offerings, warrant exercises, Spanish research-and-development reimbursements, and special financing transactions so that unlike sources are not combined.

Methodology. Amounts use cash actually received after disclosed issuance costs where the filings provide that information. Gross amounts are shown separately where useful. Categories are intended to be non-overlapping. This schedule identifies reported sources of cash; it does not by itself trace or determine the ultimate use of those funds.

1. At-the-market stock sales

Period	Net cash received
2021	\$65.960 million
2022	\$0
2023	\$2.219 million
2024	\$3.603 million
2025	\$6.833 million
First half 2026	\$2.303 million
Total ATM proceeds	\$80.918 million

2. Public offerings separate from the ATM

Transaction	Gross proceeds	Net cash received
September 2024 common/pre-funded shares with 1,428,600 accompanying warrants	\$2.500 million	\$1.960 million
May 2025 common/pre-funded shares with 6,818,180 accompanying warrants	\$7.500 million	\$6.690 million
Total	\$10.000 million	\$8.650 million

The initial proceeds from these units are classified as offering proceeds. Subsequent cash exercises are classified separately as warrant proceeds.

3. Cash from warrant exercises

Transaction	Cash received
January–February 2021 exercise of 11,655,747 warrants originally issued in 2018	\$8.042 million
October 2025 induced exercise of 8,092,280 warrants	\$3.942 million net
2025 exercise of pre-funded warrants at \$0.001 per share	\$0.004 million
Total warrant cash	\$11.988 million

The October 2025 inducement generated approximately \$4.370 million gross, less \$356,000 of placement-agent fees and \$72,000 of other expenses. The 16,184,560 replacement warrants generated no exercise cash through June 30, 2026.

4. Spanish research-and-development reimbursements

Cash receipt	Amount
First half 2025 direct R&D reimbursement	\$1.798 million
January 22, 2026 R&D rebate payment	\$1.609 million
Total Spanish R&D cash received	\$3.407 million

The program is sponsored by the Spanish government and administered through Theriva Biologics S.L. It reimburses qualifying R&D conducted in Spain or the European Union through tax credits or direct refunds. Applications covered qualifying expenses incurred in 2022, 2023, and 2024. At June 30, 2026, another \$1.681 million remained recorded as a tax-credit receivable and is not included as cash received.

5. Special financing transactions

Transaction	Gross/face amount	Net or reported proceeds
July 2022 MSD Credit Opportunity Master Fund Series C and Series D preferred private placement	\$3.000 million	\$2.734 million
January 2025 Spanish THERICEL government loan	approximately EUR 1.3 million	\$1.507 million in 2025 annual cash-flow statement
Total special financing received		\$4.241 million

The MSD net proceeds comprise \$2.006 million from Series C preferred stock and \$728,000 from Series D preferred stock. The THERICEL loan supports a Theriva/UAB manufacturing-platform collaboration, bears 4.015% interest, and is repayable over seven years after an initial three-year period. The June 2026 comparative interim statement reported \$1.458 million for the 2025 receipt; the annual statement reported \$1.507 million. This schedule uses the annual figure and discloses the interim difference.

6. Consolidated totals

Source category	Cash received
ATM stock sales	\$80.918 million
Other public offerings	\$8.650 million
Warrant exercises	\$11.988 million
Spanish R&D reimbursements	\$3.407 million
MSD preferred financing and THERICEL loan	\$4.241 million
Total identified capital and government-financing receipts	\$109.204 million

Securities subtotal

\$80.918 million ATM + \$8.650 million other public offerings + \$11.988 million warrant exercises + \$2.734 million MSD preferred financing = \$104.290 million of shareholder-securities proceeds.

Separate non-financing receipt

Theriva reported receiving \$300,000 in February 2026 as the up-front payment under the Rasayana Therapeutics license for SYN-020. This is license revenue, not financing. Including that separate receipt produces \$109.504 million of identified receipts from the listed sources.

What this schedule establishes—and what remains to be traced

The schedule establishes the amounts Theriva reported receiving from the identified sources during the measurement period. It does not establish where every dollar was spent. A separate use-of-funds reconciliation should compare these receipts with annual operating cash outflows, investing activity, financing costs, acquisitions and license obligations, compensation, research-and-development spending, general and administrative spending, debt service, and ending cash balances.

This standalone Part D preserves the source schedule for later incorporation into a supplemental filing or appendix. It does not replace or modify any previously submitted paper, and it states no conclusion about intent or wrongdoing.

Source filings

- [Theriva/Synthetic Biologics Form 10-K for 2021](#)
- [Theriva Biologics Form 10-K for 2023, covering 2022 and 2023](#)
- [Theriva Biologics Form 10-K for 2024](#)
- [Theriva Biologics Form 10-K for 2025](#)
- [Theriva Biologics Form 10-Q for the six months ended June 30, 2026](#)

**UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., Defendant.	Case No. 3:26-cv-00705
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SUPPLEMENTAL EXHIBIT SE

**Reverse Splits Warrant Financing Market Capitalization and Cash Use
Analysis**

Original Part E

The following pages are separately paginated under this supplemental exhibit identifier.

TOVX / Synthetic Biologics / Theriva Biologics

Reverse Splits, Warrant Financing, Market Capitalization and Cash Use

Forensic Accounting Analytical Report — Version 1

Market data through 2026-09-21; financial statements through June 30, 2026

Purpose: trace the relationship in time among reverse splits, warrant-related financings, market capitalization, capital raised, operating expenses and cash use. This report states documented facts and distinguishes calculations from conclusions about intent.

Executive findings

- Three reverse splits occurred from 2016 through the present: 1-for-35 effective August 10, 2018; 1-for-10 effective July 25, 2022; and 1-for-25 effective August 26, 2024. Their cumulative factor is 1-for-8,750.
- Each split filing stated that the split was effected to meet NYSE American per-share-price requirements.
- Each split was followed within 3 to 66 days by a financing that either issued new warrants or affected existing warrants. The 2018 and 2024 transactions issued new warrants. The July 2022 preferred placement did not issue new warrants, but it repriced the 2018 warrants to \$1.22 and produced a reported \$0.340 million noncash deemed dividend.
- This repeated chronology is a basis for further inquiry. It does not, standing alone, establish that the board adopted the splits to prepare for warrant financings.
- The immediate pre/post split market-capitalization comparisons fell by approximately 17.1%, 6.3%, and 5.7%. A split itself is mechanically value-neutral; these differences are market-price movements across the measurement dates.
- From January 1, 2021 through June 30, 2026, the statements of cash flows report \$100.238 million of net financing cash inflow, \$90.115 million of operating cash use, \$4.651 million of investing cash use, and \$7.455 million of financing outflows embedded in the financing section. Cash increased from \$6.227 million at December 31, 2020 to \$11.337 million at June 30, 2026.
- The approximately \$70 million 2021 event was \$74.002 million of net financing cash: \$65.960 million of net ATM proceeds plus \$8.042 million of warrant exercise proceeds.
- On a current split-adjusted basis, outstanding common shares increased from approximately 518,618 at the March 2021 snapshot to 45,892,668 at June 30, 2026, an 88.5-fold increase.

1. Scope and method

The analysis uses Theriva/Synthetic SEC filings (CIK 894158), filed share counts, filed statements of cash flows and operations, and daily TOVX/SYN closing prices. Market capitalization is calculated as closing price multiplied by common shares outstanding. Preferred shares and unexercised warrants are excluded from common-market-capitalization calculations.

Yahoo Finance back-adjusts older price history for all later reverse splits. To recover contemporaneous quoted prices, the historical series was divided by the remaining future split factor: 8,750 before August 13, 2018; 250 from August 13, 2018 through July 22, 2022; 25 from July 25, 2022 through August 23, 2024; and 1 thereafter.

For financing windows, the report uses the nearest disclosed point-in-time common-share count. Those windows are labeled proxies because issuers do not publish daily share counts. Raw price changes are not abnormal-return estimates and do not control for other news or market movements.

2. Reverse splits and the next warrant-related financing

Effective	Ratio	Pre shares	Post shares	Pre cap	Post cap	Change	Interval	Next financing / warrant effect
2018-08-10	1:35	132,969,743	3,799,136	\$14.6 million	\$12.1 million	-17.1%	66 days	\$18.6M gross unit offering; common/preferred plus common warrants
2022-07-25	1:10	158,437,840	15,843,784	\$25.4 million	\$23.8 million	-6.3%	3 days	\$3.0M gross Series C/D preferred placement; existing 2018 warrants repriced to \$1.22
2024-08-26	1:25	25,131,230	1,005,249	\$3.5 million	\$3.3 million	-5.7%	32 days	\$2.5M gross common/pre-funded warrant offering plus common warrants

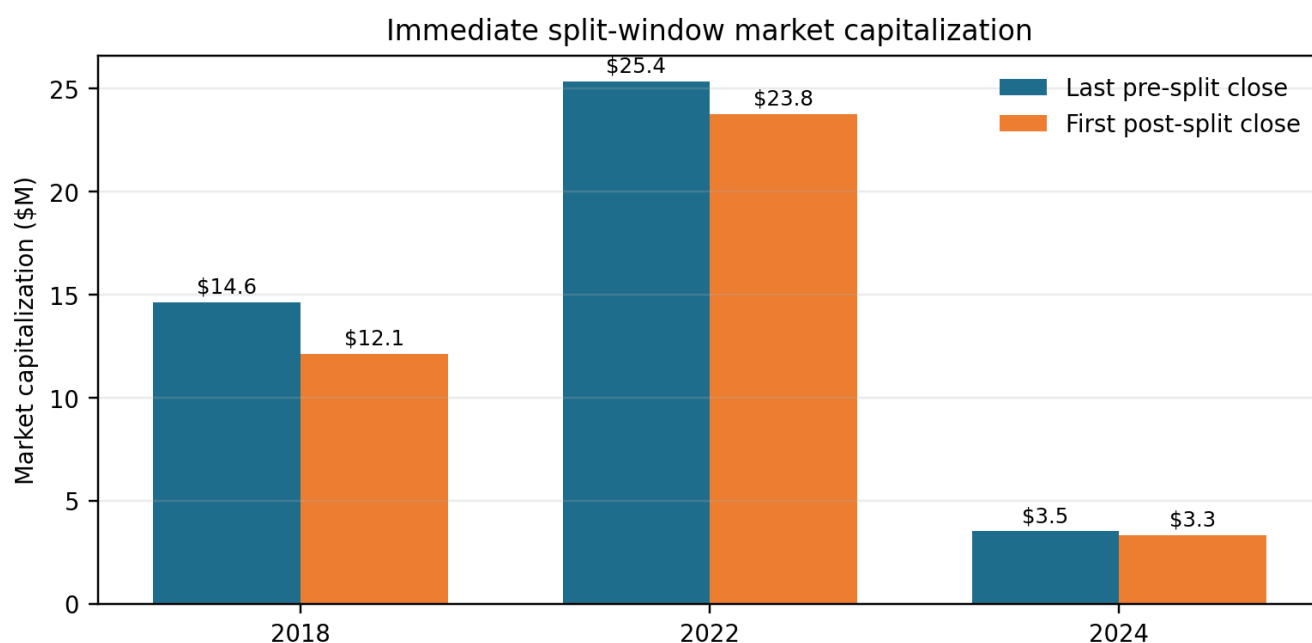


Figure 1. Immediate market-capitalization comparison using the exact pre/post share counts stated in each split filing and the closing prices on the last pre-split and first post-split trading days.

Documented sequence

- 2018: the 1-for-35 split became effective after the August 10 close. On October 15, 66 days later, the company closed an \$18.6 million gross offering of common/preferred units accompanied by common warrants.
- 2022: the 1-for-10 split became effective July 25. On July 28, three days later, the company signed a \$3.0 million gross preferred-stock placement with MSD Credit Opportunity Master Fund. The placement repriced the existing 2018 warrants to \$1.22; the company later reported a \$0.340 million deemed dividend for that change.
- 2024: the 1-for-25 split became effective August 26. On September 27, 32 days later, the company closed a \$2.5 million gross offering of common stock/pre-funded warrants and accompanying common warrants.

Assessment: the consistent public purpose of the splits was listing-price compliance. The repeated financing proximity is documented and warrants examination of contemporaneous board materials and financing negotiations. It does not by itself prove pretext or improper purpose.

3. Warrant and financing windows

Event	Pre date	Pre cap	Post date	Post cap	Cap change	Gross	Gross-net
2018 split then warrant offering	2018-08-13	\$12.1 million	2018-11-07	\$12.0 million	-1.0%	\$18.6 million	—
2022 split then MSD preferred placement	2022-07-25	\$23.8 million	2022-08-10	\$22.5 million	-5.3%	\$3.0 million	\$0.266 million
2024 split then warrant offering	2024-08-26	\$3.3 million	2024-09-30	\$3.6 million	8.9%	\$2.5 million	\$0.540 million
May 2025 common/warrant offering	2025-05-07	\$2.3 million	2025-05-12	\$4.3 million	88.9%	\$7.5 million	\$0.810 million
October 2025 warrant inducement	2025-09-30	\$4.2 million	2025-11-07	\$9.4 million	123.6%	\$4.4 million	\$0.458 million

The table demonstrates why price, market capitalization and dilution must be kept separate. In May and October 2025, the price per share declined across the selected windows while total market capitalization rose because the common-share count expanded. A higher market capitalization does not mean that a pre-existing holder avoided dilution or price loss.

The October 2025 inducement reported approximately \$4.4 million gross exercise proceeds, \$356,000 of placement-agent fees, \$72,000 of other expenses, and approximately \$3.942 million of net cash proceeds. It also issued 16,184,560 new warrants and reported a \$5.880 million noncash deemed dividend. Cash fees, noncash value transfer and potential future dilution are separate measurements.

4. Where the approximately \$70 million went

The filings provide a complete cash reconciliation, although they do not provide vendor-level tracing. The 2021 financing inflow was \$74.002 million, not exactly \$70 million: \$65.960 million net from the ATM program and \$8.042 million from warrant exercises. The company used \$12.890 million in operating cash during 2021 and ended that year with \$67.325 million of cash.

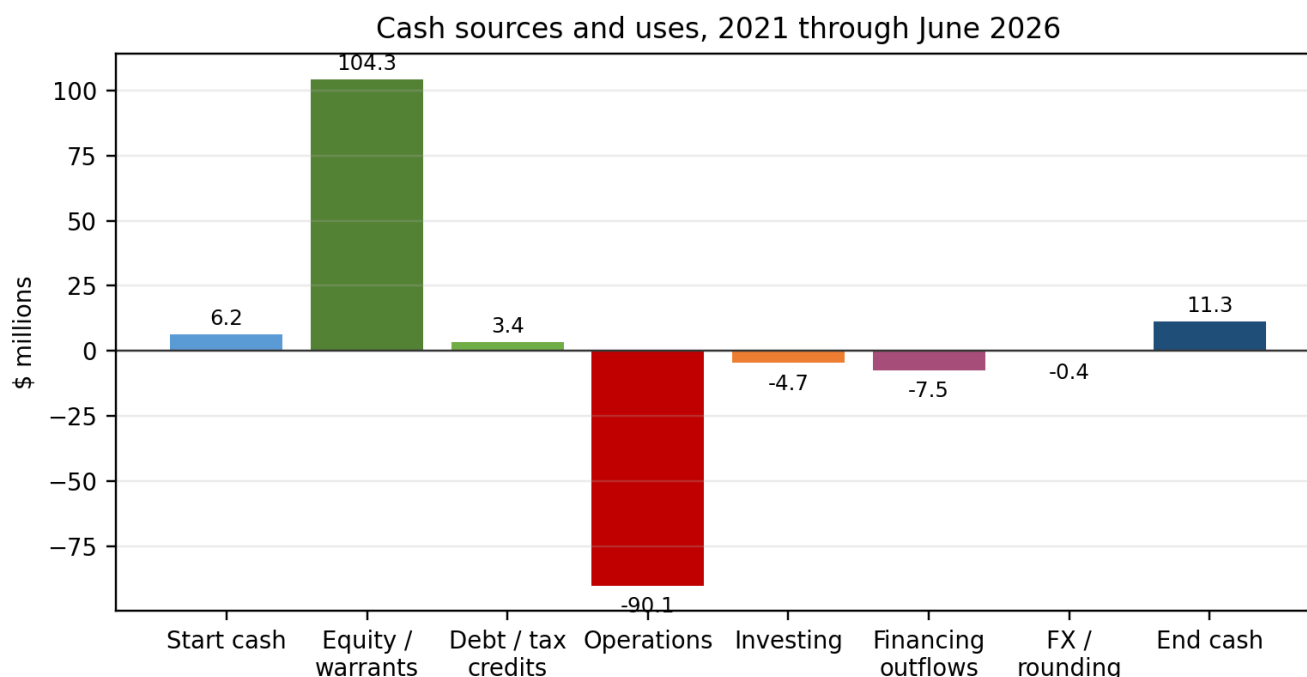


Figure 2. Cumulative cash reconciliation from the December 31, 2020 balance through June 30, 2026. Equity/warrant proceeds and debt/tax-credit financing are shown gross of the financing outflows listed separately.

Period	Net financing	Operating cash used	Investing cash used	R&D expense	G&A expense	Ending cash
2021	\$74.0 million	\$12.9 million	\$0.0 million	\$7.8 million	\$6.5 million	\$67.3 million
2022	\$-1.9 million	\$19.1 million	\$4.4 million	\$11.7 million	\$9.9 million	\$41.8 million
2023	\$0.6 million	\$19.0 million	\$0.2 million	\$14.3 million	\$7.1 million	\$23.2 million
2024	\$5.5 million	\$16.9 million	\$0.0 million	\$12.0 million	\$7.4 million	\$11.6 million

Period	Net financing	Operating cash used	Investing cash used	R&D expense	G&A expense	Ending cash
2025	\$18.2 million	\$16.7 million	\$0.0 million	\$8.6 million	\$15.4 million	\$13.1 million
2026 H1	\$3.9 million	\$5.5 million	\$0.0 million	\$1.6 million	\$4.1 million	\$11.3 million

Cash bridge, January 2021 through June 2026

- Beginning cash at December 31, 2020: \$6.227 million.
- Equity and warrant cash proceeds: \$104.290 million.
- Debt and tax-credit financing proceeds: \$3.403 million.
- Operating cash used: \$90.115 million.
- Investing cash used: \$4.651 million; the largest identified component was the 2022 VCN acquisition/pre-acquisition loan.
- Financing outflows: \$7.455 million, including loan, contingent-consideration and treasury-stock payments reported in the financing section.
- Foreign-exchange effect and rounding: approximately \$0.362 million reduction.
- Ending cash at June 30, 2026: \$11.337 million.

5. Expenses versus financing loss

The phrase “loss to financing” can refer to several different measurements. They should not be combined without labels:

- Direct cash issuance cost: gross proceeds minus net proceeds for transactions where both are reported on a comparable basis. Four reviewed events produce an approximately \$2.074 million gross-to-net difference on approximately \$17.4 million gross. This is a lower bound because it excludes ATM transactions and offerings without a comparable disclosed net figure.
- Financing-section cash outflows: \$7.455 million, principally payments associated with acquired loans, contingent consideration and treasury stock. These are not all issuance costs.
- Noncash deemed dividends/value transfers: \$7.402 million in 2021, \$0.340 million in 2022, and \$5.880 million in 2025, totaling \$13.622 million. These affect loss available to common holders or equity accounting but are not cash expenditures.
- Dilution: the reduction in a static holder's ownership percentage caused by share issuance, conversion and warrant exercise. Dilution is an ownership effect, not an operating expense.
- Market-price loss: change in the trading price or market capitalization. It is not equal to capital raised or cash spent and requires an event-study design before causation is attributed.

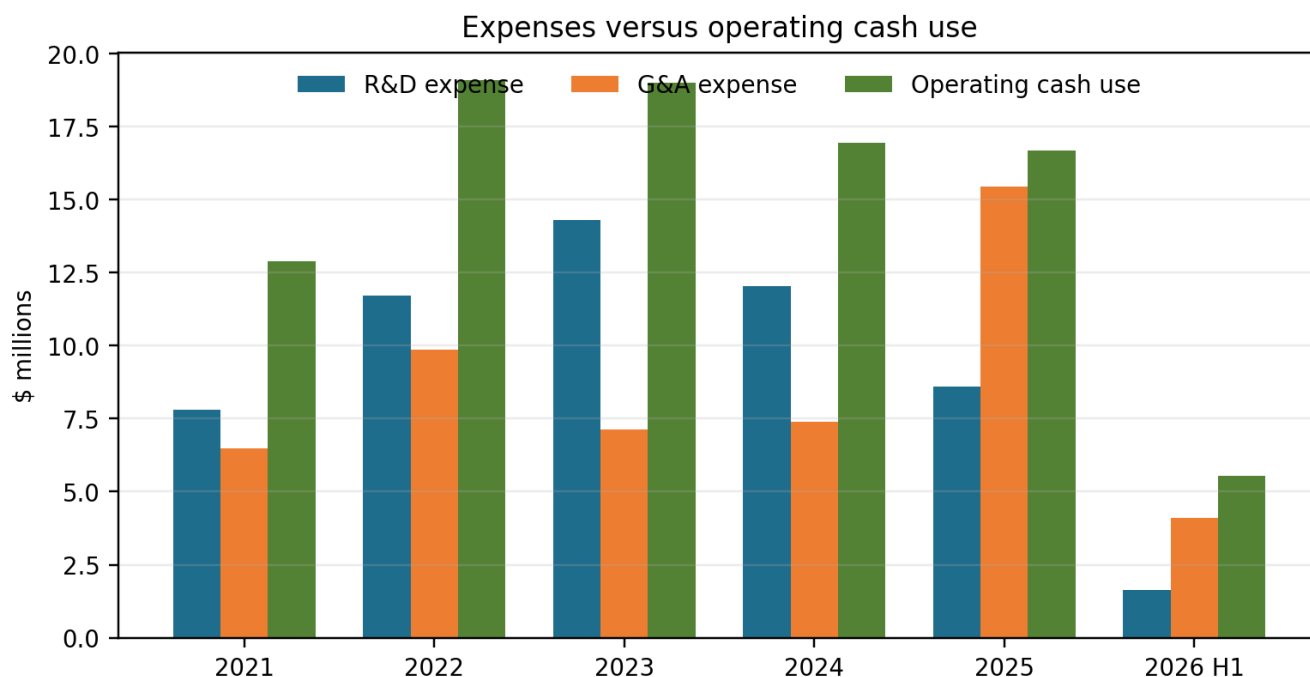


Figure 3. GAAP R&D and G&A expense compared with operating cash use. Expense categories contain accruals and noncash items, so they cannot be added to operating cash use as separate uses of funds.

Expense totals

For 2021 through June 2026, reported R&D totaled \$56.092 million, G&A totaled \$50.394 million, and operating expense totaled \$113.405 million. Net loss totaled \$106.955 million. Operating cash use totaled \$90.115 million. The differences arise from accruals, noncash valuation changes, stock compensation, working-capital movements and other reconciliation items.

The 2025 G&A increase should not be described entirely as cash overhead. The filed statements report \$15.447 million of G&A, including a \$9.031 million change in fair value of contingent consideration. Subtracting that remeasurement produces \$6.416 million, compared with \$6.697 million for 2024 after subtracting the corresponding \$0.699 million remeasurement.

6. Dilution after removing reverse-split optics

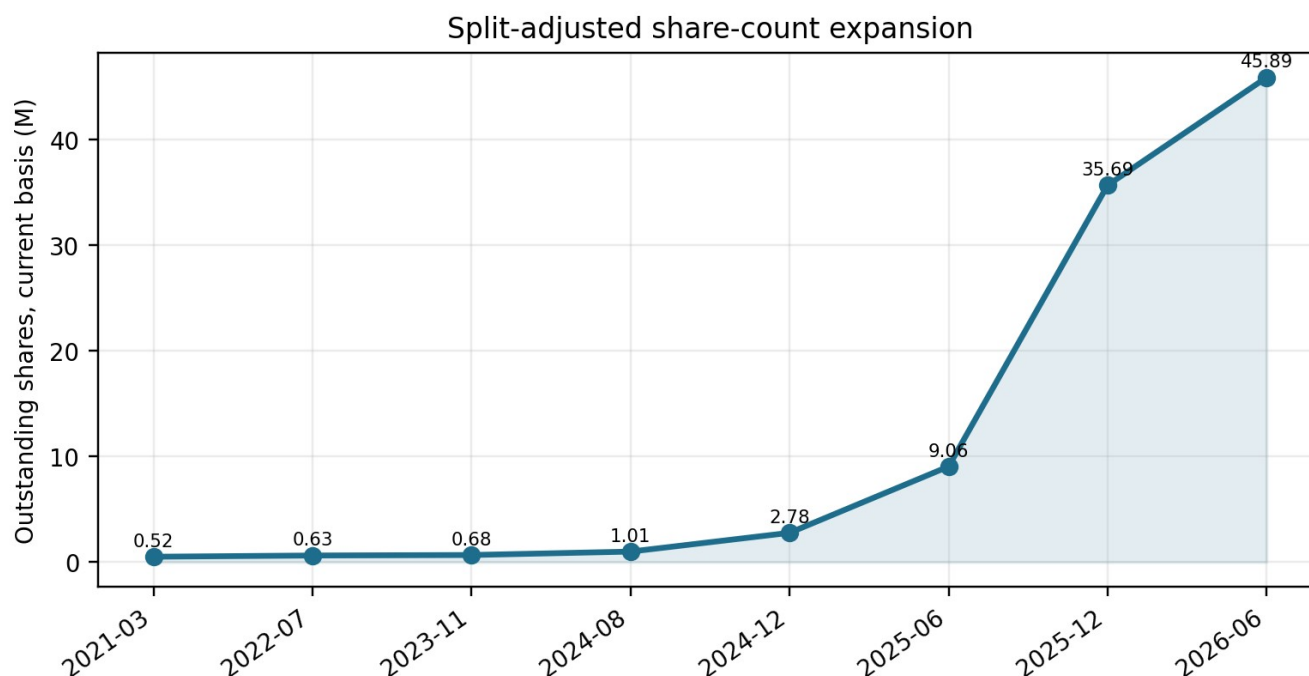


Figure 4. Outstanding common shares converted to the current post-August-2024 basis.

The March 3, 2021 share count of 129,654,575 converts to approximately 518,618 shares after the later 1-for-10 and 1-for-25 splits. The June 30, 2026 outstanding count was 45,892,668. The ratio is approximately 88.5 to 1. A static holder would retain approximately 1.13% of the ownership percentage held at the March 2021 snapshot, before considering additional future exercises or preferred-stock conversion rights.

From immediately after the August 2024 split—1,005,249 shares—to June 30, 2026—45,892,668 shares—the outstanding count increased approximately 45.65 times. That change, rather than the split itself, is the direct dilution measure.

7. Forensic accounting approach

The public-file analysis supports screening and reconciliation. A true forensic accounting conclusion requires transaction-level records. The procedures below are designed to test competing explanations rather than assume a result.

Procedure	Work performed / requested	Evidentiary objective
Cash proof	Tie every bank account to monthly reconciliations and the general ledger.	Confirms completeness and identifies transfers, timing differences and unrecorded accounts.
Use-of-proceeds	Trace financing receipts to subsequent disbursements by vendor, program, entity and approver.	Tests whether actual use matches stated R&D/general-corporate purposes.
Offering close	Reperform gross-to-net proceeds from closing statements, agent fees, legal bills and reimbursements.	Measures direct financing cost precisely.
Cap-table roll-forward	Reconcile transfer-agent records, warrants, preferred conversions and stock-plan issuances daily.	Tests the completeness of dilution disclosures.
Warrant valuation	Reperform Black-Scholes or other models using the original and modified terms.	Tests noncash deemed-dividend measurements and value transfer.
VCN / contingent consideration	Tie acquisition cash, assumed loans, milestones, liability remeasurement and payments to source documents.	Separates acquired-program costs from ordinary operations.
Cost allocation	Test payroll, contractors, lab/equipment invoices and intercompany allocations across Theriva and VCN.	Tests commingling or misallocation without presuming it.
Related-party screen	Match vendor/holder beneficial owners against officers, directors, advisers and conflict questionnaires.	Tests whether relationships were identified and reviewed.
Event study	Measure abnormal return against a biotech index and market index, with confounding news logged.	Tests whether observed returns are statistically associated with splits/financings.
Restatement/control review	Use the final restated numbers and inspect audit	Avoids relying on superseded figures and tests

Procedure	Work performed / requested	Evidentiary objective
	adjustments and remediation evidence.	recurring control failures.

Priority records

- Monthly bank statements and bank reconciliations for every Theriva and VCN account, 2021–present.
- General-ledger detail and trial balances with legal entity, cost center, program, vendor and journal-entry fields.
- Transfer-agent ledgers; warrant and preferred-stock registers; exercise, conversion and beneficial-ownership-blocker notices.
- Financing closing binders, settlement statements, engagement letters, invoices and expense reimbursements.
- VCN acquisition accounting workbook, contingent-consideration valuation models, milestone notices and payment support.
- Payroll registers, contractor lists, time allocations, laboratory/equipment invoices and fixed-asset registers.
- Board and Audit Committee minutes and conflict questionnaires concerning splits, financings, warrant modifications, VCN and material licenses.

8. Conclusions stated at the proper level

What the public record establishes:

- Three reverse splits with an aggregate 1-for-8,750 factor, each expressly tied to NYSE American price compliance.
- A financing or warrant-pricing event within 3–66 days after every split.
- \$100.238 million of net financing cash flow from 2021 through June 2026 and a complete aggregate cash reconciliation to \$11.337 million ending cash.
- \$90.115 million of operating cash use, \$4.651 million of investing cash use, and \$7.455 million of financing outflows over the same period.
- At least \$2.074 million of known gross-to-net cash issuance difference for four comparable events, plus \$13.622 million of separately identified noncash deemed dividends.
- An 88.5-fold split-adjusted increase in common shares from the March 2021 snapshot to June 2026.

What the public record does not establish by itself:

- That any board adopted a reverse split for the purpose of enabling a later warrant financing.
- That a particular financing caused a specified stock-price decline.
- The vendor-level destination of every dollar, or whether any expenditure was improper.
- Undisclosed relationships, commingling or misallocation without the underlying records.

Source index

1. SEC company filings and XBRL facts, CIK 894158: [Open source](#)
2. Yahoo Finance daily TOVX price history; prices cross-checked against split events: [Open source](#)
3. FY2021 Form 10-K: [Open source](#)
4. FY2022 Form 10-K: [Open source](#)
5. FY2023 Form 10-K: [Open source](#)
6. FY2024 Form 10-K: [Open source](#)
7. FY2025 Form 10-K: [Open source](#)
8. 2026 Q2 Form 10-Q: [Open source](#)
9. 2018 reverse split Form 8-K: [Open source](#)
10. 2018 post-split financing registration statement: [Open source](#)
11. 2022 reverse split Form 8-K: [Open source](#)
12. 2022 preferred financing Form 8-K: [Open source](#)
13. 2024 reverse split Form 8-K: [Open source](#)
14. 2024 warrant offering prospectus: [Open source](#)
15. May 2025 warrant offering prospectus: [Open source](#)
16. October 2025 warrant inducement Form 8-K: [Open source](#)

Prepared as an analytical research aid. Not an audit opinion, valuation opinion, or legal conclusion.

**UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., Defendant.	Case No. 3:26-cv-00705
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SUPPLEMENTAL EXHIBIT SF

**Grifols Deferred Contingent Consideration and Strategic Transaction
Liquidity Sequence**

Original Part F

The following pages are separately paginated under this supplemental exhibit identifier.

Nejmeh v. Theriva Biologics, Inc. | Part F

UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., Defendant.	Case No. 3:26-cv-00705 Judge initials: _____ PROPOSED SUPPLEMENTAL INFORMATION PART F GRIFOLS DEFERRAL CONTINGENT CONSIDERATION AND STRATEGIC TRANSACTION LIQUIDITY SEQUENCE
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Grifols Deferral Contingent Consideration and Strategic Transaction Liquidity Sequence

Purpose and principal finding

This Part F isolates a narrow sequence reported in Theriva Biologics filings. Achievement of the stated VCN-01 Phase 2b endpoints made a \$6.0 million milestone payable to Grifols. Theriva paid \$1.0 million and deferred \$5.0 million. Later filings reported approximately \$9.819 million of additional probability-weighted contingent consideration, \$9.6 million of cash in early August 2026, and a strategic review that could include a business combination, merger, or reverse merger.

The public record establishes the obligations, the changing wording used to describe the deferral, the Rasayana license, the cash position, and the announced strategic alternatives. It does not establish that management manufactured the liquidity condition, that Rasayana satisfied the deferral condition, or that any proposed transaction is unfair. Those questions require the underlying agreements, communications, forecasts, minutes, and capitalization models.

Scope. This Part F presents financial and transactional background only. It does not assert motive, characterize a contingent fair-value estimate as cash already paid or presently due, or assert a separate disclosure or readability claim.

1 Accounting distinctions and liquidity snapshot

Reported item	Amount and classification	What the filed record supports
Grifols milestone earned after VCN-01 Phase 2b endpoints	\$6.000 million accrued milestone	\$0.500 million was paid in August 2025 and \$0.500 million in December 2025. The remaining \$5.000 million was earned but deferred.
Deferred portion of earned milestone	\$5.000 million in accrued expenses	This is the actual deferred payment. At early-August 2026 cash of \$9.6 million, it equaled about 52.1 percent of reported cash before considering other cash flows.
Additional VCN acquisition contingent consideration	\$9.819 million Level 3 fair value	\$2.650 million was current and \$7.169 million was noncurrent at June 30, 2026. This is a probability-weighted accounting estimate of possible future milestone payments, not an additional \$9.819 million already earned or deferred.

Reported item	Amount and classification	What the filed record supports
Combined recorded VCN acquisition related obligations	\$14.819 million arithmetic total	This combines the separate \$5.000 million accrued milestone and \$9.819 million contingent-consideration fair value. The total is not a statement that all amounts were immediately payable.
Cash and working capital	\$11.3 million cash at June 30 2026; \$9.6 million cash in early August 2026; \$1.6 million working capital at June 30 2026	Theriva stated that paying the deferred \$5.0 million would significantly deplete cash and could limit operations or other obligations.

Source note. The June 30, 2026 Form 10-Q separately reports the \$5.000 million accrued milestone and the \$9.819 million contingent-consideration fair value. The distinction must be preserved in any argument or demonstrative.

2 Date ordered public record

Date or period	Filed fact	Significance of the record
By June 30 2025	Theriva reported that achievement of the primary survival and safety endpoints in the VCN-01 Phase 2b trial obligated it to pay Grifols \$6.0 million.	The liability arose from a disclosed acquisition milestone tied to trial results.
August 5 2025	Theriva and Grifols agreed to three payments: \$0.5 million by the end of August, \$0.5 million by the end of December, and \$5.0 million deferred until a licensing or business-development transaction was secured.	This is the first filed description of the deferral condition.
August 2025 filings	The Form 10-Q stated the deferral and its transaction condition. The earnings release furnished as Exhibit 99.1 did not identify Grifols or state the \$6.0 million payment schedule.	The deferral was publicly disclosed in the SEC report but not repeated in the same-day earnings release.
November 2025 Form 10-Q	The filing repeated that \$5.0 million would remain deferred until a licensing or business-development transaction was secured.	The original transaction-based condition remained in the filed description after the first \$0.5 million payment.
February 17 2026	Theriva granted Rasayana an exclusive worldwide license covering SYN-020 and related licensed intellectual property. Theriva received \$0.3 million upfront and retained contingent milestone, royalty, and sublicense-revenue rights.	The public record does not state whether this license was the licensing or business-development transaction contemplated by the Grifols deferral.
March 12 2026 Form 10-K	The filing stated that \$1.0 million had been paid and that the remaining \$5.0 million was deferred pending ongoing discussions with Grifols. It did not repeat the earlier phrase tying deferral to a licensing or business-development transaction.	The description changed. The public filings do not identify whether the written deferral terms were amended or only summarized differently.
June and August 2026	Theriva reported \$11.3 million of cash and \$1.6 million of working capital at June 30 and \$9.6 million of cash in early August. The company stated that paying \$5.0 million would significantly deplete cash.	The deferred payment had become material to the stated liquidity analysis.
August 2026 Form 10-Q	Theriva stated that strategic alternatives could include a business combination, merger, or reverse merger and that the board had not approved a definitive course of action.	The filing connects the strategic review to going-concern and financing conditions, but does not establish the cause, purpose, counterparty, or terms of any transaction.

3 What the sequence establishes and does not establish

- 1 **Earned payment** A clinical milestone became payable. The amount was not a discretionary estimate after the endpoint achievement: \$6.0 million was reclassified from contingent consideration to accrued expenses, \$1.0 million was later paid, and \$5.0 million remained accrued.
- 2 **Contractual deferral** The August 2025 filed description tied the remaining payment to a licensing or business-development transaction. Later filings used different language. The public record does not include the complete August 5 deferral agreement or any amendment that would show the controlling condition.
- 3 **Rasayana license** Theriva entered a licensing transaction in February 2026 and received \$0.3 million. The annual report filed after that agreement continued to report the \$5.0 million as deferred. The filings do not explain whether Grifols treated the Rasayana agreement as a qualifying transaction, rejected that characterization, waived payment, or agreed to a new condition.
- 4 **Strategic alternatives** The 2026 Form 10-Q identifies a merger or reverse merger as a possible response to liquidity and going-concern uncertainty. It also states that the board had not approved a definitive course. The filing does not prove that a reverse merger was inevitable or that the Grifols payment was created to force one.
- 5 **Accounting boundary** The separate \$9.819 million Level 3 amount represents probability-weighted future milestones. It bears on the acquisition's recorded obligations but cannot fairly be described as another cash payment already owed in full.

4 Factual questions requiring the underlying record

- 1 What are the complete written terms of the August 5, 2025 Grifols deferral, including any side letter, amendment, waiver, or payment trigger?
- 2 Did the February 17, 2026 Rasayana License Agreement qualify as the licensing or business-development transaction described in the August and November 2025 filings? If not, how was that phrase defined and applied?
- 3 Why did the filed description change from a transaction-based condition to pending ongoing discussions with Grifols, and was that change supported by a written amendment or only by revised disclosure language?
- 4 What cash forecasts did the board, management, auditors, and advisers review before and after the payment was deferred, and how did those forecasts treat the timing of the \$5.0 million payment?
- 5 What standalone financing, licensing, asset-sale, expense-reduction, merger, and reverse-merger alternatives were evaluated, and what assumptions were used for each?
- 6 How would each contemplated transaction treat existing common stockholders, outstanding warrants, employee equity, the Grifols obligations, and the remaining VCN assets?
- 7 Which underlying milestones and expected payment dates compose the \$2.650 million current and \$7.169 million noncurrent contingent-consideration balances?
- 8 When were possible strategic-transaction counterparties first contacted, and what instructions, valuation materials, or capitalization models were provided to financial and legal advisers?

5 Focused records needed for examination

If the Court permits targeted discovery, the following categories would allow the sequence to be tested without treating unanswered questions as established facts. Legitimate confidential material may be produced under an appropriate protective order.

- 1 The executed August 5, 2025 deferral agreement with Grifols and every amendment, side letter, waiver, demand, invoice, payment confirmation, and communication concerning the \$6.0 million milestone.

- 2 The accounting memoranda and journal-entry support for reclassifying \$6.0 million from contingent consideration to accrued expenses and for classifying the remaining contingent consideration between current and noncurrent liabilities.
- 3 Board, Audit Committee, and any strategic-committee minutes, presentations, consents, and supporting materials addressing the milestone, deferral, liquidity forecast, going-concern analysis, Rasayana license, and strategic review.
- 4 Weekly or monthly cash forecasts, including any 13-week cash-flow model, showing the effect of paying or deferring the \$5.0 million and the assumptions used for financing and clinical spending.
- 5 Communications among Theriva, Grifols, Rasayana, auditors, placement agents, investment banks, and counsel concerning whether a license or strategic transaction triggered payment.
- 6 Valuation, fairness, exchange-ratio, pro forma capitalization, and dilution analyses for each merger, reverse merger, asset sale, license, financing, or other strategic alternative considered.
- 7 A milestone-by-milestone schedule supporting the \$9.819 million Level 3 fair value, including probability assumptions, expected payment dates, discount rates, and changes reviewed by management and the Audit Committee.
- 8 Documents comparing a standalone plan with each contemplated transaction, including treatment of VCN-01, SYN-004, warrants, options, employee equity, and existing common stockholders.

6 Limited conclusion

Theriva's filings show a documented sequence: a \$6.0 million VCN-01 milestone became payable; \$1.0 million was paid; \$5.0 million was deferred; the filed description of the deferral condition later changed; a separate licensing transaction occurred; cash declined; and the company identified merger and reverse-merger alternatives while warning that payment of the \$5.0 million would significantly deplete cash. These facts support focused examination of the contemporaneous record. They do not, without that record, prove that any person intentionally created the condition, concealed the liability, or selected an unfair transaction.

Official source index

The source documents below are official SEC filings or exhibits. The cited amounts are stated in dollars unless a filed table identifies them as amounts in thousands.

- [Form 10-Q for the quarter ended June 30 2025](#)
- [Earnings release furnished August 11 2025](#)
- [Form 10-Q for the quarter ended September 30 2025](#)
- [Form 10-K for the year ended December 31 2025 filed March 12 2026](#)
- [Rasayana License Agreement dated February 17 2026](#)
- [Form 10-Q for the quarter ended June 30 2026](#)

Dated:

Signature:

Mark Joseph Nejme, Plaintiff, self-represented

**UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., Defendant.	Case No. 3:26-cv-00705
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SUPPLEMENTAL EXHIBIT SG

May 2025 Warrant Financing and Accounting

Original Part G

The following pages are separately paginated under this supplemental exhibit identifier.

**UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., Defendant.	Case No. 3:26-cv-00705 PROPOSED SUPPLEMENTAL INFORMATION PART G MAY 2025 WARRANT FINANCING AND ACCOUNTING
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Part G May 2025 Warrant Financing and Accounting

Purpose. This Part G answers a narrow factual question: what securities Theriva sold in May 2025, what cash the company reported receiving, how the public financial statements recorded the transaction, and what records are needed to verify the allocation among the shares and warrants. It preserves Parts A through F and does not repeat the market-price analysis presented elsewhere. It states no conclusion that the accounting was improper.

I. Verified transaction terms

1. On May 7, 2025, Theriva announced the pricing of a reasonable-best-efforts public offering. The company announced the closing on May 8, 2025. A.G.P./Alliance Global Partners acted as sole placement agent.
2. The June 30, 2025 Form 10-Q identifies three securities in the May offering: 1,990,900 common shares; pre-funded warrants for 4,827,280 common shares; and common warrants for 6,818,180 common shares.

Security delivered	Number	Up-front combined price	Later exercise terms
Common share plus one common warrant	1,990,900 units	\$1.10 per unit	Common warrant: \$1.10 per share; immediate; five years
Pre-funded warrant plus one common warrant	4,827,280 units	\$1.099 per unit	Pre-funded: \$0.001 per share; immediate; until exercised
Total common-warrant coverage	6,818,180 shares	No separate up-front price stated	\$1.10 per share at original issuance

3. The public terms use a combined unit price. They do not state a separate up-front cash price for the accompanying common warrant. That is different from saying that the common warrant had no economic or accounting value: the company states that it valued the common warrants at the grant date with a Black-Scholes model and classified both warrant types in equity.

II. Cash received and gross-to-net bridge

The contractual price arithmetic produces \$7,495,170.72, which the company reported as approximately \$7.5 million gross. The audited 2025 Form 10-K reports \$6.690 million of net proceeds after placement-agent fees and offering expenses.

Calculation	Arithmetic	Amount
Common-share units	1,990,900 x \$1.10	\$2,189,990.00
Pre-funded-warrant units	4,827,280 x \$1.099	\$5,305,180.72
Contractual gross cash	Sum of unit proceeds	\$7,495,170.72

Calculation	Arithmetic	Amount
Net proceeds in financial statements	After issuance costs	\$6,690,000.00
Gross-to-net difference	\$7,495,170.72 - \$6,690,000	\$805,170.72

4. The gross-to-net difference is approximately \$805,171. The public financial statements describe that difference collectively as placement-agent fees and other offering expenses. They do not provide a transaction-level closing statement that separately identifies each recipient and invoice.

III. The reported net-proceeds inconsistency

The June 2025 quarterly report contains two different net-proceeds descriptions. Its financial statements report \$6.690 million, while its liquidity narrative states \$6.9 million. The later 2025 Form 10-K reports \$6.7 million, which is consistent with rounding \$6.690 million.

Public statement	Reported amount	Reconciliation
June 30, 2025 stockholders' equity statement	\$6.690 million	Exact statement entry
June 30, 2025 cash-flow statement	\$6.690 million	Matches equity statement
June 30, 2025 liquidity narrative	\$6.9 million	Does not equal ordinary rounding of \$6.690 million
December 31, 2025 Form 10-K	\$6.7 million	Consistent with rounded \$6.690 million

5. For reconciliation purposes, this Part G uses \$6.690 million because that is the amount carried in the financial statements. The public record does not explain why the June 2025 narrative used \$6.9 million. The closing statement and general-ledger entry would resolve that difference without inference.

IV. Accounting shown in the public statements

The following entries reconstruct the public financial-statement presentation. They are not represented as copies of Theriva's internal journal entries.

Event	Debit shown or implied	Credits shown in equity statement	What remains aggregated
May 8 offering, net presentation	Cash \$6.690M	Common stock \$0.002M; additional paid-in capital \$6.688M	Allocation among common shares, pre-funded warrants, and common warrants
Pre-funded warrant exercises through June 30	Cash approximately \$0.004M	Common stock \$0.003M; additional paid-in capital \$0.001M	Cash versus cashless exercises and holder-level share calculation

6. The company classified the May common warrants and pre-funded warrants as equity. The company also states that the May common warrants were valued at the grant date using a Black-Scholes model. With equity classification, the warrant amount is recorded within stockholders' equity rather than as a recurring fair-value liability in the income statement, subject to later modification accounting.

7. The public statements aggregate the transaction in common stock and additional paid-in capital. They do not disclose the separate dollar value assigned to the common shares, the pre-funded warrants, or the common warrants; nor do they disclose the complete May warrant valuation workpaper and its inputs. Therefore, the precise debit-and-credit subledger allocation cannot be independently reconstructed from the public filings alone.

V. Share issuance and warrant coverage

Measure	Filed amount	Meaning
Common shares issued at closing	1,990,900	Issued immediately
Pre-funded warrant shares	4,827,280	Issuable upon exercise; \$0.001 exercise price
Common-warrant shares	6,818,180	Additional potential shares; original \$1.10 exercise price
Total shares or potential shares in May package	13,636,360	6,818,180 primary/pre-funded plus 6,818,180 common-warrant coverage; not all outstanding on May 8
Common shares outstanding at Dec. 31, 2024	2,782,449	Before the May offering
Common shares outstanding at June 30, 2025	9,059,232	After closing and substantial pre-funded-warrant exercise

8. The June 2025 warrant note states that 4,287,374 May pre-funded warrants had been exercised, while the June equity statement reports 4,285,883 shares from conversion of warrants, a difference of 1,491 shares. For the full year, the warrant note states that all 4,827,280 May pre-funded warrants were exercised, while the equity statement reports 4,824,603 shares from conversion of pre-funded warrants, a difference of 2,677 shares.

9. The pre-funded warrants permitted cashless exercise. A net-share calculation is therefore a possible explanation for the differences, but the public filings do not show the exercise notices or the holder-level cashless-exercise arithmetic. Those records are needed before characterizing the differences as error.

VI. Later use of the May common warrants

10. The 2025 Form 10-K states that 6,747,280 of the May common warrants were later exercised through the October 2025 warrant-inducement transaction. The October terms and replacement warrants are addressed in Part A. This Part G includes the fact only to establish why the May grant-date valuation, warrant register, and later modification accounting are connected records.

VII. Records needed for complete verification

The following records would permit a direct reconciliation without attributing motive or assuming wrongdoing:

1. the May 2025 placement-agent agreement, final closing statement, bank settlement record, and every offering-expense invoice;
2. the May closing journal entry, general-ledger detail, trial balance, and additional-paid-in-capital and warrant subledgers;
3. the grant-date Black-Scholes valuation workpaper for the 6,818,180 common warrants, including stock price, exercise price, term, volatility, risk-free rate, dividend assumption, and any other inputs;
4. the purchaser schedule and warrant register, including the number of common shares, pre-funded warrants, and common warrants issued to each purchaser;
5. all pre-funded-warrant exercise notices, cash or cashless-election forms, and calculations supporting the shares actually issued;
6. the transfer-agent share ledger and daily capitalization records from May 7 through December 31, 2025; and
7. the accounting memorandum supporting equity classification and the later modification accounting for the May warrants used in the October inducement.

VIII. Neutral conclusion

The filed record establishes that the May 2025 offering produced approximately \$7.5 million gross and \$6.690 million net; delivered common shares or pre-funded warrants together with 6,818,180 common warrants; treated the warrants as equity; and valued the common warrants at grant with a Black-Scholes model. The filed record does not provide the separate dollar allocation among the three securities, the detailed valuation workpaper, the transaction-level expense ledger, or the holder-level exercise calculations. Those are accounting records to be obtained and tested, not facts to be presumed.

Source record

1. [Theriva May 7, 2025 pricing release](#)
2. [Theriva May 8, 2025 closing release](#)
3. [Theriva Form 10-Q for the quarter ended June 30, 2025](#)
4. [Theriva Form 10-K for the year ended December 31, 2025](#)

Respectfully submitted,

Dated: _____

Signature: _____

Mark Joseph Nejme, Plaintiff, self-represented

**UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., Defendant.	Case No. 3:26-cv-00705
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SUPPLEMENTAL EXHIBIT SH

Control Provisions and Potential Change of Control Process

Original Part H

The following pages are separately paginated under this supplemental exhibit identifier.

**UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., Defendant.	Case No. 3:26-cv-00705 PROPOSED SUPPLEMENTAL INFORMATION PART H CONTROL PROVISIONS AND POTENTIAL CHANGE OF CONTROL PROCESS
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Part H Control Provisions and Potential Change of Control Process

Purpose. This Part H presents public facts relevant to a narrow question: whether Theriva's financing choices, corporate-control provisions, change-in-control compensation, equity issuances, warrant rights, and strategic-alternatives disclosures warrant limited fact-finding into whether management or directors have evaluated, prepared, negotiated, or facilitated a change-of-control transaction at a time when the company's clinical assets were advancing. The public record reviewed does not establish an insider takeover, an undisclosed counterparty, an executed merger agreement, or an activated shareholder-rights plan. This Part H preserves Parts A through G, does not amend the complaint, adds no party or claim, and asks the Court to distinguish verified facts from a hypothesis that requires records and testimony.

I. Question presented

1. Could the documented sequence and control mechanisms possibly be consistent with management preparing or positioning Theriva for a management-supported business combination, merger, reverse merger, asset transfer, financing, or other change of control at a time when its clinical assets were advancing? If so, do those facts justify narrowly tailored preservation and fact-finding to determine what was considered, whether any transaction was facilitated, what reasonable financing alternatives were evaluated, and how each alternative would affect existing stockholders?
2. Plaintiff asks whether this could be occurring; he does not assert that it is. A management-supported transaction may be lawful and beneficial, harmful, or neutral depending on its terms and process. The facts presently available do not establish intent, breach, manipulation, or an agreement with any private-equity firm or other prospective acquirer.

II. Poison pill and other control provisions

For this Part H, an active poison pill means an adopted shareholder-rights plan with a defined trigger that would permit rights to dilute or otherwise burden an acquiring person. The public sources reviewed do not identify such an active Theriva rights plan. Theriva instead discloses other anti-takeover powers, including blank-check preferred-stock authority. Those powers can affect control but are not, by themselves, proof of an activated poison pill.

Disclosed mechanism	Verified public fact	What the fact does not establish
Blank-check preferred stock	The board may fix the rights, preferences, voting powers, and restrictions of up to 10,000,000 preferred shares in one or more series and authorize issuance, subject to applicable requirements.	That a preferred series has been created for a present takeover or that a rights plan is active.

Disclosed mechanism	Verified public fact	What the fact does not establish
Charter and bylaws	Theriva states that its charter, bylaws, and Nevada law may discourage, delay, or prevent a change in control. The board may alter the bylaws without stockholder approval and fill board vacancies by a majority of directors then in office, even if less than a quorum.	That these provisions have been used for a current transaction or that their use would be lawful in every circumstance.
Common-stock authorization	The company reported 350,000,000 authorized common shares and 45,892,668 outstanding as of March 10, 2026. On August 3, 2026, stockholders approved a further authorized-share amendment, but the board retained discretion whether to file it.	That the amendment has been filed or that any unissued shares are committed to a counterparty.
No preemptive rights	The company's capital-stock description states that common holders do not have preemptive or subscription rights.	That any particular issuance is authorized, fair, or exempt from stockholder or exchange approval.

III. Documented sequence relevant to the hypothesis

Date	Verified event	Limited significance
July to October 2022	The company sold Series C and Series D preferred stock to MSD Credit Opportunity Master Fund for approximately \$3.0 million gross. The Series C purchaser agreed to vote for the name change and increase in authorized common stock from 20,000,000 to 350,000,000. Series D carried 20,000 votes per share on those stockholder items and voted in the same proportion as shares actually voted. The authorized-share increase became effective October 26, 2022.	This is a historical example of specially designated preferred voting rights used for defined corporate proposals. It does not prove current MSD control or a present transaction.
March and April 2025	The 2025 Shallcross employment agreement provided option acceleration at closing of a defined Change in Control and specified severance in certain post-change termination circumstances. On April 28, 2025, he received options for 190,000 shares.	The agreement creates disclosed economic consequences if a change in control occurs. It does not prove that Mr. Shallcross sought or negotiated one.
May and October 2025	The May common warrants included rights in a fundamental transaction, including receipt of transaction consideration and, in specified circumstances, repurchase at Black-Scholes value. Most May common warrants were later exercised in the October inducement addressed in Parts A and G.	Warrant holders have transaction-related contractual rights. Those rights do not establish that warrant holders are acquirers or that a transaction is planned.
January 5, 2026	The Compensation Committee approved options for 475,000 shares for Mr. Shallcross at an exercise price of \$0.241, vesting monthly over 36 months.	The grant affects incentives and the capitalization table but is not evidence of an agreement or improper purpose.
March 2026	The company's ownership table reported all four current officers and directors as a group beneficially owning 351,379 shares, less than 1% of 45,892,668 outstanding shares.	The disclosed insiders did not possess voting control through the reported holdings. Influence, negotiations, or undisclosed arrangements cannot be inferred either way from this table.

Date	Verified event	Limited significance
May through August 2026	Theriva's forward-looking cautionary language repeatedly listed the ability to raise capital and or enter strategic alternatives that may include a business combination, merger, or reverse merger. The August 11 results release also reported first patients dosed in VIRAGE2, \$11.3 million in cash at June 30, and an expected runway into the first quarter of 2027.	This language identifies possible future alternatives while clinical work continues. It is not an announcement that the board authorized a sale process or signed a reverse-merger agreement.
August 3, 2026	Stockholders approved increasing the 2020 Stock Incentive Plan reserve from 4,500,000 to 6,500,000 shares, approved an authorized-share amendment subject to board filing discretion, and approved common-share issuance upon exercise of outstanding warrants.	These approvals expand potential issuances and reduce identified exchange-approval constraints. They do not identify a transaction counterparty or prove that shares will be used in a takeover.

3. The source verified for the merger language is the August 11, 2026 results release, where it appears in the forward-looking-statement discussion. It should not be described as an August 24 announcement of an executed or board-approved reverse merger unless a separate filing is produced that says so.

4. Theriva's predecessor completed a reverse merger in 2006. That history establishes only that a reverse merger is part of the company's corporate history; it does not establish a present plan or intent.

IV. Financing choices and transaction readiness

5. The public record shows several financing methods, including at-the-market sales, registered offerings with warrants, a warrant-inducement transaction, preferred-stock financing, licenses, and government-supported project funding. Describing management as having failed to raise capital properly would require a standard of comparison and evidence of the alternatives available at the time.

6. The relevant inquiry is therefore comparative: whether the board received and evaluated forecasts for an at-the-market program, rights offering, strategic partnership, asset-specific financing, debt, grant funding, ordinary equity without warrants, licensing, sale of assets, merger, or reverse merger; and how each alternative was expected to affect cash runway, voting power, warrant obligations, employee equity, existing stockholders, and the value assigned to VCN-01 and other assets.

7. The timing matters as a fact question because the August 11 release reported both clinical progress and a limited cash runway while also listing merger and reverse-merger alternatives. Those facts may be consistent with ordinary contingency planning. They may also justify asking whether management was positioning the company for a transaction. The public record alone does not choose between those explanations.

V. Facts that support inquiry and facts that limit inference

Public fact	Permissible use	Inference not yet supported
Board authority over preferred stock, bylaws, and vacancies	Shows available control and defensive tools.	An active poison pill or plan to transfer control.
2022 preferred voting structure	Shows that special voting rights were used for specific stockholder items.	Continuing MSD influence, undisclosed coordination, or present control.
Change-in-control compensation and equity grants	Shows financial consequences and incentives that should be disclosed and evaluated for conflicts.	That compensation caused any financing or transaction decision.
Warrant fundamental-transaction provisions	Shows potential transaction economics and records that should be modeled.	That warrant holders are acting together or are intended acquirers.

Public fact	Permissible use	Inference not yet supported
Strategic-alternatives language	Supports asking whether alternatives were formally evaluated and whether an adviser or counterparty was contacted.	A signed deal, identified buyer, or inevitable reverse merger.
Insiders reported below 1% beneficial ownership	Cuts against a claim that disclosed insider shareholdings alone control the vote.	That insiders lack influence or that an undisclosed agreement exists.

VI. Targeted records needed to test the hypothesis

The following records would permit the issue to be tested without assuming motive or wrongdoing:

- (a) all board and committee minutes, written consents, presentations, and financial models concerning strategic alternatives, a sale, business combination, merger, reverse merger, recapitalization, asset transfer, change of control, or shareholder-rights plan from January 1, 2024 forward;
- (b) all engagement letters, fee arrangements, presentations, and communications with A.G.P. or any other investment bank, placement agent, valuation adviser, transaction adviser, or prospective financing source concerning such alternatives;
- (c) all non-disclosure agreements, indications of interest, letters of intent, term sheets, draft agreements, deal logs, data-room access records, and counterparty contact lists for any potential transaction;
- (d) all capitalization tables and transaction models showing common shares, preferred shares, warrants, pre-funded warrants, options, plan reserves, beneficial-ownership blockers, Black-Scholes payment obligations, voting power, and pro forma ownership before and after each alternative;
- (e) all analyses comparing financing without warrants against the May 2025 offering, October 2025 inducement, later at-the-market sales, partnership financing, asset-specific financing, licensing, a merger, or a reverse merger;
- (f) all conflict questionnaires, recusals, related-party analyses, and Audit, Compensation, and Nominating Committee materials concerning management, directors, warrant holders, MSD Credit Opportunity Master Fund, Rasayana Therapeutics, advisers, or prospective counterparties, limited to a financing, asset transfer, or change-of-control process;
- (g) all calculations of management or director benefits upon a transaction, including accelerated vesting, severance, continued benefits, option value, board compensation, and any proposed post-transaction position; and
- (h) preservation of transaction-related email, text messages, messaging applications, board-portal materials, calendars, call logs, and drafts for custodians who participated in any such process.

VII. Narrow sworn clarification requested

If the Court finds the subject relevant to the pleaded warrant-related claims and permits fact-finding, Plaintiff requests a sworn status statement limited to the following matters:

- (a) whether Theriva has adopted, drafted, considered, or activated a shareholder-rights plan and, if so, its date, trigger, duration, rights agent, and present status;
- (b) whether the board or a committee has formally authorized a strategic-alternatives, sale, merger, reverse-merger, recapitalization, or change-of-control process;
- (c) whether any adviser has been retained for that process and whether any non-disclosure agreement, indication of interest, letter of intent, term sheet, or definitive agreement exists;
- (d) whether any counterparty, financing source, warrant holder, director, officer, employee, or related person is expected to receive securities, voting rights, employment, board service, asset rights, or other consideration in a contemplated transaction;
- (e) how outstanding warrants, Black-Scholes provisions, beneficial-ownership blockers, options, plan reserves, contingent consideration, and licensed assets would be treated; and

- (f) what financing and standalone alternatives were evaluated, by whom, with what projected dilution, runway, shareholder consideration, and value assigned to the clinical and intellectual-property assets.

VIII. Limited purpose and requested treatment

8. Plaintiff does not ask the Court, on this Part H alone, to find that an insider takeover exists, that any person acted together with another, or that any financing or corporate transaction was unlawful.
9. Plaintiff asks only that relevant records be preserved and, if the Court determines the subject falls within the existing pleadings and permits discovery, that the status and decision process be clarified through a sworn report and narrowly tailored production. Legitimately confidential transaction material may be addressed through an appropriate protective procedure.
10. If later public filings or authorized discovery produce material evidence, Plaintiff may seek leave to present additional supplemental material. This statement does not alter the existing complaint, emergency motion, jury demand, or requested administrative docket corrections.

IX. Neutral conclusion

The public record establishes available anti-takeover powers, a historical preferred-stock voting structure, transaction rights for warrant holders, change-in-control benefits for the chief executive officer, substantial authorized and reserved share capacity, repeated forward-looking references to merger and reverse-merger alternatives, and clinical progress occurring alongside a limited reported cash runway. It also establishes that disclosed officers and directors held less than 1% as a group and does not identify an active poison pill, an executed transaction, a counterparty, or an insider voting block. The record therefore supports a focused request for preservation and fact-finding, not a conclusion that a takeover from inside has occurred.

Source record

1. [Theriva Form 10-K for the year ended December 31, 2025](#)
2. [Theriva capital-stock and at-the-market prospectus supplement dated October 22, 2025](#)
3. [Theriva Form 8-K reporting the August 3, 2026 annual-meeting results](#)
4. [Theriva May 5, 2026 first-quarter results release](#)
5. [Theriva August 11, 2026 second-quarter results release](#)

Respectfully submitted,

Dated: _____

Signature: _____

Mark Joseph Nejme, Plaintiff, self-represented

**UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., Defendant.	Case No. 3:26-cv-00705
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SUPPLEMENTAL EXHIBIT SI

Debt Obligations and Cash Trace

Original Part I

The following pages are separately paginated under this supplemental exhibit identifier.

Mark Joseph NejmeH
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**UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., a Nevada corporation, Defendant.	Case No. 3:26-cv-00705 PROPOSED SUPPLEMENTAL INFORMATION PART I DISCLOSED DEBT, CONTRACTUAL OBLIGATIONS, AND CASH-RECEIPT TRACE
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PUBLIC-FILING ANALYSIS — FACTS SEPARATED FROM QUESTIONS FOR DISCOVERY

I.A. Purpose, scope, and limits

1. This Part I identifies each material borrowing or debt-like obligation disclosed in Theriva Biologics, Inc.'s public filings, the entity to which it is assigned, whether the filing shows that cash entered the consolidated company, the stated purpose, the amount paid, and the amount remaining. It separates conventional loans from acquisition consideration, lease liabilities, government assistance, and ordinary operating payables.
2. The cutoff is June 30, 2026, the latest balance-sheet date reviewed. Dollar amounts are in U.S. dollars and rounded to the nearest thousand unless otherwise stated. References [S1] through [S3] identify the filed SEC sources.
3. This section does not assert that any loan, payment, accounting treatment, or transfer was improper. It identifies what the public record proves and what the public record does not permit a reader to trace without the underlying agreements, bank records, and general ledger.

I.B. Direct answer

- True loans payable at June 30, 2026: \$1.654 million. Of that amount, \$1.609 million was the THERICEL loan and \$45,000 was NEBT plus RETOS 2015. Other reported liabilities are identified separately below. [S1]
- Only one current borrowing is expressly traceable to a new post-acquisition cash receipt: the €1.3 million THERICEL loan received by Theriva Biologics, S.L. on January 17, 2025. The 2025 cash-flow statement reports \$1.507 million of proceeds from long-term debt. [S1; S2]
- NEBT and RETOS were pre-existing Spanish-government loans acquired with VCN Biosciences, S.L. in March 2022. Theriva's post-acquisition filings show the assumed liabilities and repayments, but do not disclose the original draw dates, original principal by tranche, deposit accounts, or exact project-level uses. [S2; S3]
- A separate VCN CDTI loan caused a \$1.376 million cash payment by Theriva during the first nine months of 2022. The cited filing does not identify the original borrowing date or funded project. [S3]

I.C. Entity map — where each obligation is assigned

Item	Legal or accounting location	Who received or paid cash	Present classification
NEBT	VCN Biosciences, S.L., now Theriva Biologics, S.L.; consolidated by parent	Original VCN receipt not traced in the cited Theriva filings	\$9K current loan
RETOS 2015	VCN Biosciences, S.L., now Theriva Biologics, S.L.; consolidated by parent	Original VCN receipt not traced; \$44K restricted cash remains	\$25K current + \$11K noncurrent loan
THERICEL	Theriva Biologics, S.L., wholly owned Spanish subsidiary	Subsidiary received €1.3M on Jan. 17, 2025	\$1.609M noncurrent loan
VCN CDTI	VCN obligation acquired in 2022	Theriva paid \$1.376M in 2022; original VCN receipt not traced	Repaid item; no separate current balance shown
Grifols purchase obligations	Theriva Biologics, Inc. under VCN purchase agreement	Cash flows from Theriva to Grifols; no cash received	\$5.0M accrued milestone plus \$9.819M contingent consideration
Spain facility lease	Theriva Biologics, S.L.; consolidated lease accounting	Cash rent paid over time; 2026 renewal created noncash ROU/liability entry	Part of \$1.899M consolidated operating-lease liability
Spanish R&D credit	Theriva Biologics, S.L.; consolidated government assistance	\$1.609M cash received in Jan. 2026 for 2024 rebate	\$1.606M deferred credit, not a loan

Name continuity. VCN became a wholly owned subsidiary on March 10, 2022. Effective November 15, 2022, VCN Biosciences, S.L. changed its name to Theriva Biologics, S.L. without another change in corporate structure. Thus a loan described under the VCN name and one later described under the Theriva S.L. name may belong to the same Spanish legal entity. [S2]

I.D. Loan-by-loan cash and balance reconciliation

Loan or transaction	Stated purpose	Cash evidence	June 30, 2026 status
NEBT / ACC10 Generalitat de Catalunya	Not stated at the project level in the cited Theriva loan footnotes	Inherited with VCN; no post-acquisition proceeds line. Original receipt must be obtained from VCN records.	\$9K current; \$0 noncurrent
RETOS 2015 / Spanish science ministry	Not stated at the project level in the cited Theriva loan footnotes	Inherited with VCN; no post-acquisition proceeds line. Restricted cash fell from \$90K in Sept. 2022 to \$44K in June 2026.	\$25K current; \$11K noncurrent
VCN CDTI loan	Not stated in the cited 2022 cash-flow disclosure	Theriva's nine-month 2022 cash-flow statement records a \$1.376M payment, not a receipt.	No separately named balance in latest loan table
THERICEL / National Knowledge Transfer Program	Support Theriva–UAB collaboration adapting the THERICEL suspension-cell platform for clinical manufacture of adenovirus and AAV therapies	€1.3M lump sum received Jan. 17, 2025; 2025 annual cash flow reports \$1.507M long-term-debt proceeds.	\$1.609M noncurrent; 4.015%; seven-year repayment beginning three years after award
Pre-closing Theriva-to-VCN advance	Finance certain VCN R&D costs before closing	Theriva paid \$417K; cash flow classified it as investing. At closing the receivable was 'effectively settled' as purchase consideration.	Not an outside loan payable after consolidation

I.E. Chronology and remaining principal

Date / period	Verified event	Cash or carrying amount
March 10, 2022	Theriva acquired all VCN shares and assumed up to approximately \$2.39M of existing VCN liabilities, including Spanish-government and Catalan-agency loans.	\$2.39M was total assumed liabilities, not all loan principal
Nine months ended Sept. 30, 2022	Pre-closing R&D advance was effectively settled; Theriva paid VCN's CDTI loan; named NEBT and RETOS balances remained.	\$417K advance; \$1.376M CDTI payment; \$254K NEBT/RETOS
Dec. 31, 2024	Named NEBT and RETOS balances before THERICEL borrowing.	\$153K total
Jan. 17, 2025	Theriva S.L. received the THERICEL unsecured loan.	€1.3M (about \$1.4M when announced)
Year ended Dec. 31, 2025	Annual cash flow recorded debt proceeds and repayments; year-end loans included THERICEL.	\$1.507M proceeds; \$70K repayments; \$1.728M ending loans
Six months ended June 30, 2026	No new long-term-debt proceeds; loan repayments continued.	\$60K repayments; \$1.654M ending loans

Reconciliation limitation. The decline from the acquired liabilities to the first named NEBT/RETOS schedule cannot be assigned to repayment, acquisition accounting, foreign exchange, forgiveness, reclassification, or another cause from the cited public filings alone. The executed loan agreements and loan-by-loan rollforward are required before drawing a conclusion.

I.F. Contractual maturity of current loans

Year	Principal due	Year	Principal due
2027	\$34K	2030	\$225K
2028	\$11K	2031	\$235K
2029	\$114K	Thereafter	\$1.035M
TOTAL	\$1.654M		

The maturity table is consolidated. It does not identify change-of-control provisions, acceleration rights, forgiveness terms, grant conversion, offsets, waivers, or project covenants. Those terms require the executed agreements. [S1]

I.G. Material obligations that are not conventional loans

Category at June 30, 2026	Amount	Accounting character and cash direction
Direct loans payable	\$1.654M	Borrowed principal; \$34K current and \$1.620M noncurrent
Operating-lease liabilities	\$1.899M	Present value of future rent; not a cash loan
Contingent consideration	\$9.819M	Probability-weighted VCN purchase liability; \$2.650M current and \$7.169M noncurrent
Deferred Spanish R&D credit	\$1.606M	Government assistance recognized over required R&D/workforce period; not ordinary debt

Category at June 30, 2026	Amount	Accounting character and cash direction
Accounts payable	\$0.984M	Trade payables
Accrued expenses	\$6.418M	Includes separate \$5.000M Grifols milestone, \$0.832M clinical consulting, \$0.332M manufacturing, and \$0.254M vendor accruals
Accrued employee benefits	\$0.290M	Compensation and benefits accruals
TOTAL LIABILITIES	\$22.670M	Agrees to the June 30, 2026 balance sheet

I.H. Grifols purchase obligations and first disclosed deferral

4. At the March 2022 closing, Theriva paid Grifols \$4.7 million and issued shares to the other VCN sellers. Theriva also agreed to milestone-based additional consideration of up to \$70.2 million and assumed approximately \$2.4 million of VCN liabilities. These items were acquisition consideration and assumed liabilities, not loan proceeds received by Theriva. [S2]
5. Theriva later paid Grifols \$3.0 million after the 2022 FDA safe-to-proceed milestone and \$3.25 million following the 2023 first-patient-dosed milestone. A \$6.0 million 2025 milestone was earned; the parties agreed on August 5, 2025 that Theriva would pay \$500,000 in August 2025, \$500,000 in December 2025, and defer the remaining \$5.0 million pending ongoing discussions. [S1; S2]
6. At June 30, 2026, the \$5.0 million earned milestone appeared within accrued expenses. Separately, the balance sheet reported \$9.819 million of contingent consideration at fair value. On the face of the filing these are separate categories; the underlying rollforward should confirm that no amount is duplicated. [S1]
7. The Grifols items represent actual or potential cash outflows. They do not show cash entering Theriva. Theriva stated that paying the deferred \$5.0 million would significantly deplete its cash and could adversely affect liquidity. [S2]

I.I. Lease, laboratory, and government-assistance obligations

8. The June 30, 2026 operating-lease liability was \$1.899 million (\$525,000 current and \$1.374 million noncurrent), with \$2.540 million of undiscounted payments through 2031. The amount combines the Rockville and Spain facilities. No material finance leases were reported. [S1]
9. Theriva S.L. subleases its Parets del Vallès laboratory and office facility from Grifols. During the first half of 2026, the company exercised a five-year renewal and recorded approximately \$1.3 million of additional right-of-use asset and lease liability. That entry is noncash at commencement; rent payments later reduce cash. [S1; S2]
10. The Spanish R&D program reimburses eligible R&D expense but requires certification and specified workforce and R&D spending for 24 months. Theriva received \$1.609 million in January 2026 for the 2024 rebate and reported a \$1.606 million deferred credit at June 30, 2026. The company treats the credit as government assistance and a future reduction of R&D expense, not as a conventional loan. [S1]

I.J. Records needed to complete the cash trace

The following targeted records would answer the remaining questions without presuming misconduct:

1. A complete debt register for Theriva Biologics, Inc., VCN Biosciences, S.L./Theriva Biologics, S.L., and each subsidiary from January 1, 2015 forward, stating lender, borrower, original principal, draw date, cash account, project, interest, maturity, security, guarantor, repayment, forgiveness, waiver, and current balance.
2. Executed NEBT, RETOS 2015, CDTI, and THERICEL agreements, applications, award letters, amendments, project budgets, eligible-cost schedules, bank receipts, compliance reports, covenant calculations, and government correspondence.
3. A loan-by-loan rollforward from original funding through June 30, 2026, separately identifying cash proceeds, repayments, interest accretion, foreign exchange, purchase-accounting adjustments, reclassification, waiver, grant conversion, forgiveness, and writeoff.
4. Bank records and general-ledger entries for the \$417,000 pre-closing VCN advance, the \$1.376 million CDTI repayment, the January 17, 2025 THERICEL receipt, the RETOS restricted-cash account, and every repayment after acquisition.
5. The THERICEL allocation reconciling the announced €2.28 million award to the €1.3 million loan received, including any UAB allocation, nonloan component, invoices, payroll, equipment, subcontractors, and deliverables.
6. The VCN purchase closing statement, assumed-liability schedule, and entries reconciling the approximately \$2.39 million assumed liabilities to the \$254,000 NEBT/RETOS balance reported at September 30, 2022.
7. The complete Grifols milestone ledger, August 5, 2025 deferral agreement and amendments, valuation workpapers, notices, payment records, and a reconciliation of the separate \$5.0 million accrued milestone and \$9.819 million contingent-consideration liability.
8. The Spain sublease, 2026 renewal, payment schedule, ownership and fixed-asset register for laboratory equipment, change-of-control terms, and records allocating the consolidated lease liability between Spain and Rockville.
9. The Spanish R&D credit applications, approvals, receipts, workforce and spending requirements, recapture provisions, and recognition schedules.
10. A certification identifying any debt, guarantee, lien, side agreement, acceleration right, cash-pool balance, due-to/due-from account, or material obligation not separately presented on the consolidated balance sheet, including items eliminated in consolidation.

I.K. Neutral conclusion

11. The latest public balance sheet identifies \$1.654 million of conventional loans, all tied to the Spanish subsidiary's government-supported programs. It also identifies materially larger acquisition, lease, grant-accounting, and operating obligations that must be analyzed separately.
12. The public filings prove that THERICEL proceeds entered the consolidated cash-flow statement and that the company paid the CDTI loan, repaid other loans, paid acquisition milestones, and received R&D reimbursement cash. They do not provide the original cash trail for the inherited NEBT, RETOS, or CDTI loans.
13. The proper next step is a contract-and-ledger reconciliation, not an inference that money is missing. The requested records would show the original purpose, legal borrower, cash destination, spending, repayment, continuing covenants, and effect of any change of control.

14. No conclusion about a proposed buyer, insider transaction, shareholder-rights plan, or takeover follows from this debt schedule alone. Any request for prospective injunctive relief should identify an actual or sufficiently imminent corporate action and address the governing Rule 65 standards separately.

Source record

- [S1] [Theriva Biologics, Inc., Form 10-Q for quarter ended June 30, 2026](#) — filed August 11, 2026; balance sheet, cash flow, government assistance, loans, leases, and contingent consideration
- [S2] [Theriva Biologics, Inc., Form 10-K for year ended December 31, 2025](#) — filed March 12, 2026; acquisition terms, company-name continuity, 2025 cash flow, debt, leases, and Grifols deferral
- [S3] [Theriva Biologics, Inc., Form 10-Q for quarter ended September 30, 2022](#) — filed November 2022; VCN acquisition, pre-closing advance, CDTI payment, first named NEBT/RETOS schedule, and cash flows

Respectfully submitted,

Dated: _____

Signature: _____

Mark Joseph NejmeH, Plaintiff, self-represented

**UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., Defendant.	Case No. 3:26-cv-00705
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SUPPLEMENTAL EXHIBIT SJ

Interim Relief Authorities for Warrants and Rights Plans

Original Part J

The following pages are separately paginated under this supplemental exhibit identifier.

**UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., Defendant.	Case No. 3:26-cv-00705 PROPOSED SUPPLEMENTAL INFORMATION PART J INTERIM RELIEF AUTHORITIES FOR WARRANTS AND RIGHTS PLANS
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Part J Interim Relief Authorities for Warrants and Rights Plans

Submitted as proposed supplemental material, subject to the Court's permission.

Purpose. This Part J identifies the federal and Nevada authorities governing temporary relief directed to the warrant shares already described in the complaint and to any shareholder-rights plan that Theriva might later adopt. It does not represent that a Theriva shareholder-rights plan presently exists; the public materials reviewed did not identify an adopted or triggered plan. It does not add a claim, party, or accusation.

I Terminology and scope

1. The procedurally correct request is a temporary restraining order or preliminary injunction under Federal Rule of Civil Procedure 65, not a stay of an order or proceeding. Any injunction must identify the specific securities and acts to be restrained, the persons bound, and its duration.
2. The operative complaint expressly requests temporary status-quo relief concerning the 16,184,560 New Warrant Shares. A request concerning those identified shares may have the required nexus to the pleaded claims. The complaint reviewed does not plead an adopted shareholder-rights plan or seek final relief against one.

II Controlling federal standard

3. Under *Winter v. Natural Resources Defense Council, Inc.*, 555 U.S. 7, 20 (2008), a movant must establish likely success on the merits, likely irreparable harm without preliminary relief, that the balance of equities favors relief, and that an injunction is in the public interest.
4. *Alliance for the Wild Rockies v. Cottrell*, 632 F.3d 1127, 1131–35 (9th Cir. 2011), permits a serious-questions sliding scale when the balance of hardships tips sharply toward the movant, but likely irreparable harm and the public-interest requirement remain mandatory.
5. *Herb Reed Enterprises, LLC v. Florida Entertainment Management, Inc.*, 736 F.3d 1239, 1250–51 (9th Cir. 2013), holds that conclusory or speculative assertions do not establish irreparable harm. Evidence of imminence and injury is required.
6. *Pacific Radiation Oncology, LLC v. Queen's Medical Center*, 810 F.3d 631, 636–38 (9th Cir. 2015), requires a sufficient nexus between the requested interim relief and the claims in the complaint. The interim relief must be of the same character as relief that could be granted finally.
7. A mandatory order that changes the status quo is especially disfavored and will not issue unless the law and facts clearly favor the moving party. *Garcia v. Google, Inc.*, 786 F.3d 733, 740 (9th Cir. 2015) (en banc).

III Authorities relevant to the identified warrant shares

Authority	Holding relevant here	Permissible use and limitation
<i>Klaus v. Hi-Shear Corp.</i> , 528 F.2d 225, 232–33 (9th Cir. 1975)	A securities-law plaintiff seeking to restrain voting shares still had to prove irreparable injury to protected investors; monetary or remote harm was insufficient.	Binding caution. A claimed statutory violation does not replace proof of imminent, noncompensable harm.
<i>Norlin Corp. v. Rooney, Pace Inc.</i> , 744 F.2d 255, 264–66 (2d Cir. 1984)	Affirmed preliminary restraint on voting shares issued to controlled entities where the record showed probable illegality, no genuine consideration, insider control, and entrenchment.	Persuasive only. It helps only if comparable transaction-specific evidence exists; labels and dilution arithmetic alone are not enough.

Authority	Holding relevant here	Permissible use and limitation
Chicago Stadium Corp. v. Scallen, 530 F.2d 204, 206–08 (8th Cir. 1976)	Affirmed an injunction against issuance and voting after findings supporting probable fiduciary breach and illegality; security was required.	Persuasive only and fact intensive. It does not authorize a categorical restraint on all warrants.
Quinn v. Anvil Corp., 620 F.3d 1005, 1011–17 (9th Cir. 2010)	Affirmed denial of relief attacking a reverse split where fraud and procedural defect were not proved; also enforced derivative-action requirements.	Binding caution. Corporate dissatisfaction or price decline is not itself proof of illegality.

8. A narrowly framed warrant request should be limited to the specifically identified New Warrant Shares and should state whether the requested restraint concerns exercise, issuance, delivery, registration, transfer, or voting. It should not restrain unidentified securities or nonparties beyond Rule 65(d)(2).
9. Before seeking relief, the record should establish by declaration and current issuer or transfer-agent evidence: the number still outstanding; present exercisability; applicable blockers; expiration; whether an exercise, issuance, vote, or registration is imminent; and why later damages, cancellation, or other remedies would be inadequate.
10. A court could also require security under Rule 65(c). The requested duration should be tied to a prompt noticed hearing or a defined event, not an unsupported one-year period.

IV Authorities relevant to a shareholder rights plan

A shareholder-rights plan is commonly called a poison pill. It is distinct from blank-check preferred-stock authority, authorized but unissued common stock, warrant securities, an employee stock plan, or general anti-takeover language. No adopted Theriva rights plan was identified in the public record reviewed.

Authority	What it decided	Effect on a present request
Horwitz v. Southwest Forest Industries, Inc., 604 F. Supp. 1130 (D. Nev. 1985)	Denied a preliminary injunction directed to poison-pill warrants. Possible harm, without probable success or proof of bad faith, self-dealing, or abuse, did not overcome the business-judgment presumption. The court also noted disruption from undoing traded rights.	Closest District of Nevada case and adverse. It must be disclosed. It strongly disfavors relief based only on a possible future plan.
Hilton Hotels Corp. v. ITT Corp., 978 F. Supp. 1342, 1348–52 (D. Nev. 1997)	After an evidentiary record, enjoined a comprehensive defensive restructuring found primarily entrenching and preclusive of a shareholder vote.	Potentially favorable by analogy only. It predates Nevada's 1999 adoption of NRS 78.139 and involved an implemented, documented plan, not a hypothetical future device.
Jewel Companies, Inc. v. Pay Less Drug Stores Northwest, Inc., 741 F.2d 1555, 1560–61 (9th Cir. 1984)	Recognized directors' management role and upheld reasonable, good-faith defensive conduct on the record before it.	Binding caution. Defensive conduct is not unlawful merely because it affects control.

11. Current Nevada law expressly permits plans, rights, options, and instruments that grant or deny rights based on ownership thresholds. NRS 78.195 and NRS 78.378. NRS 78.139's reasonable-threat and reasonable-response test applies to specified control-resisting conduct, but subsection 1 expressly excludes adoption or execution of the threshold-based plans described in NRS 78.195 and NRS 78.378. NRS 78.138 supplies a strong good-faith and informed-decision presumption.
12. Because no adopted plan has been identified and the complaint does not plead such a plan, a blanket order prohibiting any plan for one year would face ripeness, evidentiary, nexus, breadth, and Nevada statutory obstacles. Part H's request for preservation, notice, and sworn clarification is materially safer than a present categorical injunction.
13. If an actual plan is later adopted or imminently proposed, the legally relevant questions would include its text, trigger, duration, exemptions, purpose, process, board record, effect on an actual bid or vote, and whether relief concerning that plan falls within an amended pleading. The Court should receive the instrument, not a label.

V Direct and derivative claim limitations

14. In re Parametric Sound Corp. Shareholders Litigation, 140 Nev. Adv. Op. 36 (2024), holds that claims alleging equity expropriation or dilution are ordinarily derivative unless the plaintiff satisfies Nevada's direct-harm test. Shoen v. SAC Holding Corp., 122 Nev. 621, 137 P.3d 1171 (2006), requires demand or particularized demand-futility allegations for derivative claims.
15. A self-represented litigant may litigate only his own claims and may not represent a corporation or other shareholders. Simon v. Hartford Life, Inc., 546 F.3d 661, 664–65 (9th Cir. 2008); High Country Broadcasting Co. v. United States, 3

F.3d 1244, 1245 (9th Cir. 1993). Any requested relief must therefore be anchored to Plaintiff's own properly pleaded direct claim, not a pro se derivative theory.

VI Filing and emergency-motion requirements

16. District of Nevada Local Rule 7-2(g) bars supplemental briefs, evidence, and authorities without leave of court. Part J should be lodged only through an authorized procedural vehicle or with leave; it should not be filed as a free-standing merits supplement.
17. An emergency motion must comply with Local Rule 7-4, including an emergency declaration, contact information, and a certification describing notice and efforts to resolve the matter. Emergency treatment is rare. Rule 65(b) imposes additional requirements for any request without notice.
18. Rule 65(d) requires a specific order and limits who is bound. Rule 65(c) ordinarily requires security in an amount the Court considers proper. These requirements should be addressed in the proposed order.

VII Evidence and testimony procedure

19. No rule requires every participant to be subpoenaed before a fact can be established. Facts may be proved through authenticated SEC filings, executed agreements, certified records, party admissions, Rule 30(b)(6) testimony, witnesses with personal knowledge, and qualified expert testimony, subject to the Federal Rules of Evidence.
20. Discovery ordinarily may not begin before the Rule 26(f) conference unless a rule, stipulation, or court order permits it. Fed. R. Civ. P. 26(d)(1). Any request for expedited discovery should identify the particular witness, custodian, document category, and urgent issue.
21. A nonparty subpoena under Rule 45 must avoid undue burden, respect privilege, and comply with geographic limits. A court must quash or modify a subpoena that violates those requirements. Targeted entity testimony and records are more defensible than an undifferentiated demand for all participants.

VIII Limited conclusion

The presently pleaded warrant request can be analyzed under Rule 65 because it concerns identified securities and requested final relief of the same general character. Success still depends on admissible, current proof of likely merits, imminent irreparable harm, equities, public interest, specificity, and security. A separate one-year injunction against a hypothetical poison pill is not presently supported by an identified instrument, a pleaded claim, or the closest Nevada authority. Preservation, notice, and fact-specific review if an actual plan appears are the narrower course.

Source record

1. [Federal Rule of Civil Procedure 65](#)
2. [District of Nevada Local Rules 7-2 and 7-4](#)
3. [Alliance for the Wild Rockies v. Cottrell](#)
4. [Herb Reed Enterprises v. Florida Entertainment Management](#)
5. [Pacific Radiation Oncology v. Queen's Medical Center](#)
6. [Garcia v. Google](#)
7. [Nevada Revised Statutes Chapter 78](#)
8. [Horwitz v. Southwest Forest Industries](#)
9. [Hilton Hotels v. ITT](#)
10. [In re Parametric Sound](#)
11. [Shoen v. SAC Holding](#)

Respectfully submitted,

Dated: _____

Signature: _____

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**UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., Defendant.	Case No. 3:26-cv-00705
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SUPPLEMENTAL TO THE ADDENDUM

SUPPLEMENTAL EXHIBIT APPENDIX

VOLUME II

SUPPLEMENTAL EXHIBITS SK THROUGH SP

Organized as a two-volume supplemental exhibit appendix

SUPPLEMENTAL EXHIBIT INDEX - VOLUME II

Nejmeh v. Theriva Biologics, Inc. - Case No. 3:26-cv-00705

ID	Description	Exhibit pages	PDF pages
SK	Price Volume and Company News Record	SK-1 to SK-7	3-10
SL	Valuation Context and Evidence Needed to Evaluate Motive	SL-1 to SL-5	11-16
SM	Insider Led Acquisition Models and Shareholder Protection Inquiry	SM-1 to SM-5	17-22
SN	Plaintiff Internal Disassembly Concern and Theriva Disclosed Sequence	SN-1 to SN-9	23-32
SO	Publicly Identified United States Patent Portfolio and Request for Complete Register	SO-1 to SO-9	33-42
SP	United States Patent No 9404103 B1 Complete Highlighted C difficile Patent	SP-1 to SP-52	43-95

Every supplemental exhibit begins with a descriptive cover sheet. Content pages carry only a bottom-margin identifier.

**UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., Defendant.	Case No. 3:26-cv-00705
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SUPPLEMENTAL EXHIBIT SK

Price Volume and Company News Record

Original Part K

The following pages are separately paginated under this supplemental exhibit identifier.

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**UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., a Nevada corporation, Defendant.	Case No. 3:26-cv-00705 PROPOSED SUPPLEMENTAL INFORMATION PART K PRICE VOLUME AND COMPANY NEWS RECORD 2016 THROUGH SEPTEMBER 22 2026
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DESCRIPTIVE EVENT STUDY WITH FACTS SEPARATED FROM INFERENCE

Purpose and scope

1. This proposed Part K presents a neutral, reproducible comparison of Theriva Biologics, Inc. company news with the price and volume reported for TOVX regular trading sessions. It addresses the timing and magnitude of observable trading activity. It does not repeat the warrant mechanics, financing terms, reverse splits, debt, or related-party subjects addressed in earlier parts.
2. The principal question addressed here is narrower: when the company released news, what happened to regular-session volume and price, and which of the largest volume departures did not have a same-session company press release?
3. This schedule does not identify any buyer, seller, short seller, broker, market maker, order originator, or trading venue. Daily price and volume data cannot prove manipulation, coordination, motive, or causation. Those questions require timestamped trade, quote, routing, and order-level records.

Direct findings

- **Sessions reviewed:** 2,695 regular trading sessions from January 2016 through September 22, 2026.
- **Company news:** 213 trading sessions were matched to one or more English-language company press releases; 2,482 sessions had no same-session company press release.
- **Extreme volume:** 46 sessions traded at least ten times their preceding 20-session average. 18 had a same-session company release and 28 did not.
- **Typical volume:** the median volume multiple was 0.95 times on company-news sessions and 0.64 times on other sessions.
- **Typical price movement:** the median absolute close-to-close move was 3.2% on company-news sessions and 2.7% on other sessions.
- **Largest measured departure:** May 7, 2025 produced approximately 46.0 million regular-session shares, about 2,039 times the preceding 20-session average, and a 40.4 percent close-to-close decline.

Method

4. The company-news source is Theriva's press-release archive. A pre-market or regular-session release was assigned to that trading date. A release posted after 4:00 p.m. Eastern Time was assigned to the next available regular session. Duplicate releases were consolidated so that one trading result was not counted twice.
5. For each session, the study calculated volume divided by the mean volume of the preceding 20 trading sessions, the close-to-close return, the opening gap, the open-to-close return, and the intraday high-low range relative to the prior close. Historical fields were placed on a split-comparable basis for calculations; contemporaneous prices and volume are retained in the detailed workbook.
6. The daily price and volume series was assembled from public historical-data pages. Theriva's SEC filing confirms that TOVX is listed on NYSE American. Nasdaq is used only as a secondary historical-data publisher and is not identified as the listing exchange.

Demonstrative K1 largest volume departures and news timing

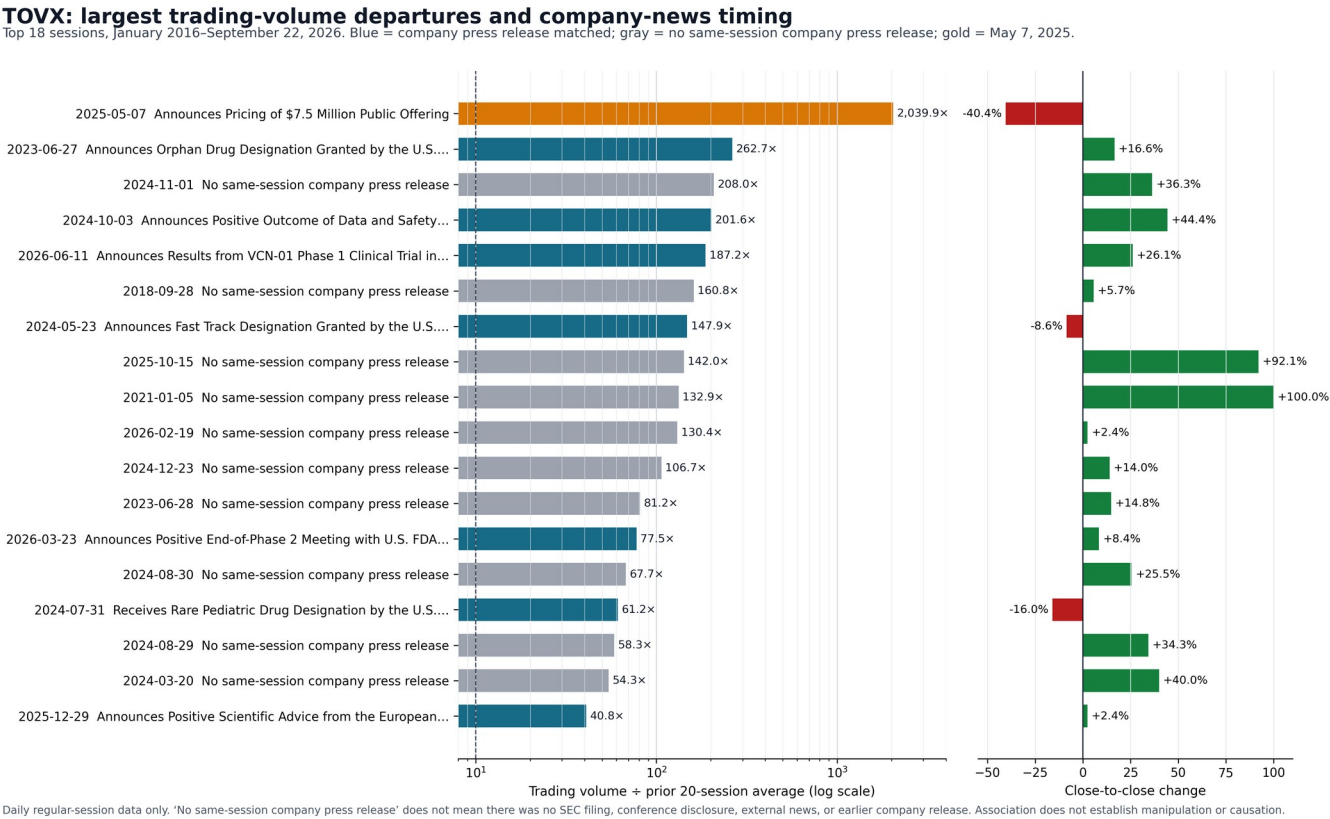


Figure K1 The 18 largest regular-session volume multiples in the reviewed period. Blue denotes a same-session company press release, gray denotes no same-session company press release, and gold identifies May 7, 2025.

The chart uses a logarithmic volume axis because the May 7, 2025 volume multiple is much larger than the other observations. The right panel shows the close-to-close movement on the same sessions. The chart does not attribute a cause to a session that lacks a same-session press release.

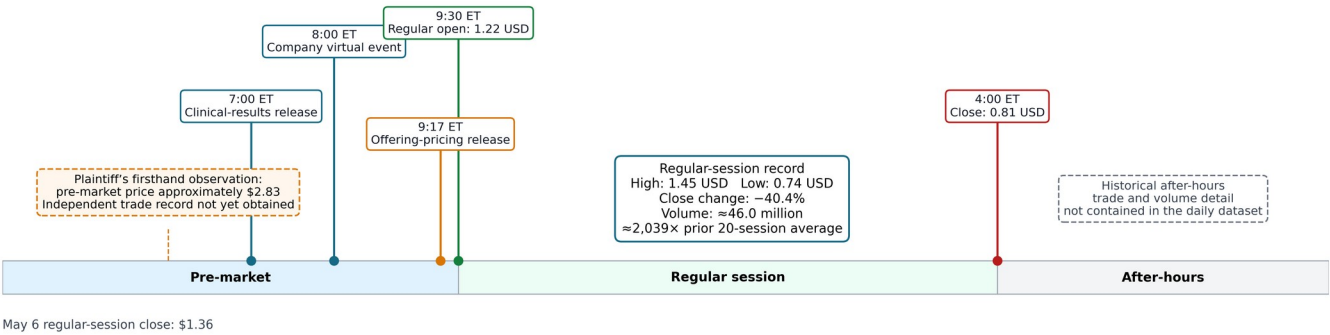
Schedule supporting Demonstrative K1

Session	News status	Company news or status	Volume multiple	Close move
2025-05-07	Company release at 07:00–09:17	Announces Pricing of \$7.5 Million Public Offering	2,039.9x	-40.4%
2023-06-27	Company release at 15:09	Announces Orphan Drug Designation Granted by the U.S. FDA for VCN-01 for the Treatment of P...	262.7x	+16.6%
2024-11-01	No same-session release	No same-session company press release	208.0x	+36.3%
2024-10-03	Company release at 08:00	Announces Positive Outcome of Data and Safety Monitoring Committee (DSMC) Review in Phase 1...	201.6x	+44.4%
2026-06-11	Company release at 08:00	Announces Results from VCN-01 Phase 1 Clinical Trial in Head and Neck Cancer Published in C...	187.2x	+26.1%
2018-09-28	No same-session release	No same-session company press release	160.8x	+5.7%
2024-05-23	Company release at 08:15	Announces Fast Track Designation Granted by the U.S. FDA for VCN-01 for the Treatment of Me...	147.9x	-8.6%
2025-10-15	No same-session release	No same-session company press release	142.0x	+92.1%
2021-01-05	No same-session release	No same-session company press release	132.9x	+100.0%
2026-02-19	No same-session release	No same-session company press release	130.4x	+2.4%
2024-12-23	No same-session release	No same-session company press release	106.7x	+14.0%
2023-06-28	No same-session release	No same-session company press release	81.2x	+14.8%
2026-03-23	Company release at 08:00	Announces Positive End-of-Phase 2 Meeting with U.S. FDA Regarding the Design of a Phase 3 T...	77.5x	+8.4%
2024-08-30	No same-session release	No same-session company press release	67.7x	+25.5%
2024-07-31	Company release at 08:15	Receives Rare Pediatric Drug Designation by the U.S. FDA for VCN-01 for the Treatment of Re...	61.2x	-16.0%
2024-08-29	No same-session release	No same-session company press release	58.3x	+34.3%
2024-03-20	No same-session release	No same-session company press release	54.3x	+40.0%
2025-12-29	Company release at 08:00	Announces Positive Scientific Advice from the European Medicines Agency (EMA) on the Design...	40.8x	+2.4%

A session marked without a same-session company press release may still follow an earlier company release, an SEC filing, conference activity, outside reporting, or another event. The supporting workbook identifies the nearest earlier and later company-news sessions for review.

Demonstrative K2 May 7 2025 session timeline

May 7, 2025: news timing, observed pre-market price, and regular-session outcome



Verified times come from company releases. Regular-session prices and volume come from public daily datasets. The pre-market observation remains expressly unverified pending timestamped trade-and-quote records.

Figure K2 Verified company-news times, the regular-session record, and the separate status of the plaintiff's pre-market observation.

- 7. On May 7, 2025, Theriva issued a clinical-results release at 7:00 a.m. Eastern Time and scheduled a company virtual event for 8:00 a.m. The company issued its offering-pricing release at 9:17 a.m., before the 9:30 a.m. regular-session open. [S2; S3]
- 8. The May 6 regular-session close was \$1.36. On May 7, the regular session opened at \$1.22, reached a high of \$1.45 and a low of \$0.74, and closed at \$0.81. Public daily sources report approximately 46.0 million shares and produce a volume multiple of approximately 2,039 times the preceding 20-session average.
- 9. The plaintiff reports observing an approximate \$2.83 pre-market price. That observation is displayed separately and is not treated as independently verified. The daily datasets do not contain the timestamped pre-market trades or complete after-hours record needed to confirm price, volume, venue, or participant identity.

Data-source precision. The two public daily sources reviewed report 45,957,600 and 46,025,980 shares for May 7, a difference of 68,380 shares, or about 0.15 percent. This schedule therefore states approximately 46.0 million shares. The difference does not materially change the approximately 2,039-times comparison.

Other examples of news and movement

Session	Company news timing	Volume	Volume multiple	Close move
2023-06-27	Orphan Drug Designation release at 3:09 p.m.	7,126,200	262.7x	+16.6%
2024-05-23	Fast Track Designation release at 8:15 a.m.	21,141,750	147.9x	-8.6%
2024-07-31	Rare Pediatric Drug Designation release at 8:15 a.m.	8,145,350	61.2x	-16.0%
2024-10-03	Positive DSMC review release at 8:00 a.m.	76,665,320	201.6x	+44.4%
2026-02-19	No same-session release; SYN-020 license announced prior day	274,439,800	130.4x	+2.4%
2026-06-11	VCN-01 clinical-publication release at 8:00 a.m.	290,080,600	187.2x	+26.1%

These examples show why the chart should be read descriptively. Company science and regulatory news coincided with both positive and negative closing results. A same-day match establishes timing, not causation.

October 2025 sequence

- 10.** Theriva announced upcoming medical-meeting presentations on October 13, 2025, and furnished the disclosure in a Form 8-K filed October 14. [S4]
- 11.** On October 15, no same-session company press release was matched. The stock traded 196,646,100 shares, about 142.0 times the preceding 20-session average, and closed 92.1 percent above the prior close. The October 13 announcement and October 14 filing are relevant context, but the public daily data does not identify the catalyst or market participants.
- 12.** On October 16, the company issued a capital-transaction release at 8:11 a.m. The regular session traded 35,627,010 shares and closed 48.7 percent below October 15. This Part K records the sequence and does not restate the transaction's warrant terms.
- 13.** On October 24, the company issued a response to unusual market action at 2:31 a.m. The session traded 549,590,800 shares, about 38.0 times the preceding 20-session average, and closed 0.9 percent above the prior close after trading between \$0.3416 and \$0.85. [S5]

What the public record can and cannot establish

- 14.** The public record can establish the date and stated time of a company release, the regular-session open, high, low, close, reported volume, and the size of a session relative to a stated historical baseline.
- 15.** The public record used here cannot identify which person or firm initiated, routed, internalized, quoted, shorted, bought, or sold any order. It cannot distinguish customer activity from proprietary activity, identify the market participant identifier attached to an execution, or establish coordination among participants.
- 16.** The event study therefore does not ask the Court to infer unlawful manipulation from volume, price, or timing alone. It identifies sessions for targeted preservation, discovery, and qualified expert review.

Records needed to test the observed sessions

The following records would permit the timing questions to be tested without presuming wrongdoing:

- 1.** Consolidated timestamped trades and quotes for the identified sessions, including pre-market and after-hours activity, sale condition, venue, price, size, and sequence number.
- 2.** NYSE American and applicable regulatory records identifying market participant identifiers, quoting activity, order entry, modification, cancellation, routing, execution, and auction participation.
- 3.** Consolidated Audit Trail or equivalent regulatory order-event data, blue-sheet data, and broker routing records sufficient to connect executions to originating accounts while preserving lawful confidentiality.
- 4.** Broker-dealer locate, borrow, stock-loan, fail-to-deliver, close-out, and short-sale-marking records for the selected event windows.

5. The company's news-distribution records, embargo lists, dissemination timestamps, conference and webcast access logs, and communications with financing or investor-relations participants for the selected event windows.
6. Authenticated broker or platform records supporting the plaintiff's reported May 7, 2025 pre-market observation.
7. An independent market-structure analysis using the produced order-level record, not daily OHLCV alone.

Requested treatment

17. Plaintiff respectfully requests that the Court accept or permit this proposed Part K only to the extent relevant to the pleaded issues and any permitted discovery. This schedule does not add a cause of action and does not ask the Court to decide market manipulation from the demonstratives alone.
18. If the Court permits discovery concerning the identified trading episodes, Plaintiff requests preservation and production sufficient to obtain the timestamped records listed above and to permit independent expert analysis.
19. The full supporting workbook contains the daily calculation schedule, the company-news event schedule, the 100 largest volume departures, extended-hours evidence status, source links, definitions, and limitations. It should be submitted electronically only if permitted by the Clerk or Court.

Source record

- [S1] [Theriva Biologics Form 10-K for 2025](#) - confirms TOVX is registered on NYSE American
- [S2] [Theriva May 7 2025 clinical-results release](#) - 7:00 a.m. Eastern Time release and 8:00 a.m. company virtual event
- [S3] [Theriva May 7 2025 offering-pricing release](#) - 9:17 a.m. Eastern Time company release
- [S4] [Theriva Form 8-K dated October 13 2025](#) - filed October 14, 2025, concerning the expanded VIRAGE data and upcoming presentations
- [S5] [Theriva October 24 2025 response to unusual market action](#) - company press release time and text
- [S6] [Theriva company press-release archive](#) - company-news dates, times, titles, and source links reviewed for the event study
- [S7] [Nasdaq public TOVX historical-data page](#) - secondary publisher of daily historical data; not the listing exchange
- [S8] [Yahoo Finance TOVX historical-data page](#) - secondary public history used as a cross-check and to fill the early 2016 window

Respectfully submitted,

Dated: _____

Signature: _____

Mark Joseph Nejme, Plaintiff, self-represented

Certificate of service

Complete and sign this page only after service has actually been made.

I certify that on _____, I served a true and correct copy of the foregoing Proposed Supplemental Information Part K Price Volume and Company News Record by the following method:

☐ United States mail

☐ Personal service

☐ Electronic service by agreement or court-authorized method

☐ Other: _____

Person or counsel served: _____

Address or authorized electronic destination: _____

Dated: _____

Signature: _____

Mark Joseph Nejme, Plaintiff, self-represented

**UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., Defendant.	Case No. 3:26-cv-00705
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SUPPLEMENTAL EXHIBIT SL

Valuation Context and Evidence Needed to Evaluate Motive

Original Part L

The following pages are separately paginated under this supplemental exhibit identifier.

**UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., Defendant.	Case No. 3:26-cv-00705 PROPOSED SUPPLEMENTAL INFORMATION PART L VALUATION CONTEXT AND EVIDENCE NEEDED TO EVALUATE MOTIVE
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Part L Valuation Context and Evidence Needed to Evaluate Motive

Submitted as proposed supplemental material, subject to the Court's permission.

Purpose. This Part L asks: Could this be a motive? It places the reported VCN-01 survival results beside the reported daraxonrasib results, states the market-valuation disparity, and identifies records needed to determine whether the clinical opportunity, financing choices, or a possible transaction supplied an economic motive. It does not assert that Theriva, its directors, officers, financiers, or any prospective counterparty intend to steal the company or its assets. Public valuation and clinical comparisons can identify a question; they do not establish intent, fair value, wrongdoing, or a transaction.

I Could this be a motive?

1. Could the combination of (a) a VCN-01 subgroup reporting 14.8 months median overall survival, (b) Theriva's extremely low public-market valuation relative to Revolution Medicines, (c) Theriva's financing and dilution history, and (d) a possible strategic or change-of-control process supply an economic motive for a transaction involving Theriva or VCN-01 before a confirmatory Phase 3 trial?
2. That is a question to be tested, not a finding of motive or misconduct. The factual inquiry is whether Theriva's board and management evaluated VCN-01, financing alternatives, partnership alternatives, and possible strategic transactions through a reasonable, informed process directed to corporate and stockholder value. If a merger, reverse merger, asset transfer, financing, change of control, or defensive measure was considered, the relevant evidence includes valuations, conflicts, incentives, counterparties, alternatives, and the contemporaneous board record.

II What can and cannot be compared

Measure	Theriva VCN-01 VIRAGE	Revolution Medicines daraxonrasib	Comparison limit
Setting and phase	Randomized, open-label Phase 2b; first-line newly diagnosed metastatic PDAC; VCN-01 plus gemcitabine/nab-paclitaxel versus chemotherapy.	Randomized Phase 3 RASolute 302; previously treated metastatic PDAC; daraxonrasib versus investigator-choice chemotherapy.	Different lines of therapy, drugs, controls, trial phases, and designs prevent a head-to-head ranking.
Population	96 total patients; 48 per arm.	500 total patients; 248 daraxonrasib and 252 chemotherapy.	The much larger Phase 3 study ordinarily provides more mature evidence.
Requested numerical comparison	Two-dose/at-least-four-cycle subgroup: 14.8 months median overall survival; n=34.	RAS G12 Phase 3 daraxonrasib population: 13.2 months median overall survival; n=228.	The reported VCN-01 subgroup median was numerically 1.6 months higher. This is a numerical fact, not a cross-trial superiority finding.
Median overall survival	10.8 months versus 8.6 months; HR 0.57; two-sided p=0.0546.	RAS G12 population: 13.2 months versus 6.6 months; HR 0.40; p<0.0001.	Absolute medians cannot be compared across different patient populations. Relative effects also arise from different trials.
Other reported endpoints	PFS 7.0 versus 4.6 months, HR 0.55, p=0.0105; duration of response 11.2 versus 5.4 months, HR 0.22, p=0.0035.	PFS in the RAS G12 population 7.3 versus 3.5 months, HR 0.45; p<0.001. Phase 3 primary and key secondary endpoints were reported met.	Both records are clinically important; only daraxonrasib has the cited Phase 3 evidence.
VCN-01 two-dose subgroup	Among patients receiving two VCN-01 doses and at least four chemotherapy cycles: 14.8 versus 11.6 months; HR 0.44; p=0.046; n=34 versus n=29.	Not the same treatment setting or subgroup.	Subgroup evidence can generate a hypothesis but should not be represented as a cross-company superiority finding.

3. The requested numerical fact is correct: the reported median overall survival for the VCN-01 two-dose/at-least-four-cycle subgroup was 14.8 months, compared with 13.2 months for daraxonrasib in the RAS G12 Phase 3 population. The VCN-01 subgroup median was therefore 1.6 months higher.
4. The comparison must remain precisely stated. The 14.8-month VCN-01 figure came from a selected first-line Phase 2b subgroup of 34 patients who received two VCN-01 doses and at least four chemotherapy cycles. The 13.2-month daraxonrasib figure came from 228 previously treated RAS G12 patients in a Phase 3 trial. In the full randomized VCN-01 treatment arm, median overall survival was 10.8 months. Accordingly, the public record proves the numerical 14.8-to-13.2 comparison; it does not prove head-to-head clinical superiority or that one company should have the other's valuation.

III Market-value and capital contrast

Item	Revolution Medicines	Theriva	What it shows
Market snapshot September 22, 2026 close	Approximately \$41.34 billion market capitalization at \$192.86 per share.	Approximately \$10.59 million market capitalization at \$0.2308 per share.	A market-pricing disparity of roughly 3,900 to 1 at one point in time. This is not a fair-value appraisal.
Reported shares	198,173,397 outstanding as of February 20, 2026, before the April offering of 10,563,381 shares and related transactions.	45,892,668 outstanding as of August 7, 2026.	Share counts and later issuances must be normalized before per-share comparisons.
Capital access	At December 31, 2025, approximately \$2.026 billion cash, cash equivalents, and marketable securities; April 2026 common offering reported about \$1.436 billion proceeds before expenses, alongside a convertible-note offering.	June 30, 2026 public financial release reported materially smaller resources and \$12.111 million of current liabilities.	Differences in cash, pipeline breadth, clinical phase, capital access, and liabilities materially affect market value.

Illustrative valuation arithmetic only

Hypothesis	Equity value	TOVX share denominator	Arithmetic result
Apply the cited RVMD market capitalization to TOVX's reported shares	\$41.34 billion	45,892,668 reported outstanding	$\$41.34\text{B} \div 45,892,668 =$ approximately \$900.80 per TOVX share.
Use a rounded \$41.0 billion comparison value	\$41.00 billion	45,892,668 reported outstanding	$\$41.00\text{B} \div 45,892,668 =$ approximately \$893.39 per TOVX share.
Test the proposed \$820 TOVX figure	\$37.632 billion implied	45,892,668 reported outstanding	$\$820 \times 45,892,668 =$ approximately \$37.632 billion.

The per-share prices of RVMD and TOVX cannot be compared directly because the companies have different numbers of shares outstanding. Market capitalization supplies the common denominator. The arithmetic above is a scale illustration, not a valuation opinion, price target, prediction, or evidence that Theriva should trade at RVMD's value. Any fully diluted calculation must separately include outstanding warrants, options, preferred-stock conversion rights, and transaction shares supported by the record.

Authentication note. The September 22, 2026 closing prices and market capitalizations are time-stamped S&P Global Market Intelligence data displayed by StockAnalysis and should be supported by authenticated exchange or vendor records if filed. The remaining figures are drawn from SEC filings and company clinical disclosures.

IV SYN-004 ribaxamase and the documented C difficile economic opportunity

5. SYN-004, also called ribaxamase, is a separate Theriva-controlled clinical asset from VCN-01. It is an orally administered beta-lactamase designed to degrade specified intravenous beta-lactam antibiotics in the upper gastrointestinal tract before those antibiotics disrupt the gut microbiome. The intended clinical purpose includes reducing antibiotic-associated C. difficile infection, vancomycin-resistant Enterococcus colonization, and other consequences of microbiome injury.

Valuation context	Verified public record	What the evidence supports
U.S. disease burden	CDC states that C. difficile causes almost 500,000 infections in the United States each year and can be life-threatening.	The indication addresses a large and continuing public-health need.

Valuation context	Verified public record	What the evidence supports
Published economic burden	Peer-reviewed U.S. analyses reported approximately \$5.4 billion in annual economic cost and approximately \$6.3 billion in CDI-attributable acute-care hospital cost. These are older nominal-dollar estimates using different methods.	The annual economic burden is measured in billions of dollars. These figures are disease-cost estimates, not SYN-004 revenue or asset value.
Phase 2b evidence	A randomized, double-blind, placebo-controlled trial enrolled 412 patients. Seven confirmed CDI cases occurred in the placebo group and two in the SYN-004 group, reported as a 71.4% relative risk reduction with $p=0.045$. The study met its stated primary endpoint.	The program produced positive controlled Phase 2b evidence supporting continued development. It does not establish approval, future sales, or commercial success.
Regulatory and development path	Theriva's 2025 Form 10-K describes a proposed single global event-driven Phase 3 trial of up to approximately 4,000 patients, with estimated completion cost above \$80 million.	Substantial remaining cost, time, regulatory, enrollment, manufacturing, and safety risks must be deducted in any asset valuation.
Current funding decision	Theriva states that it will not fund further SYN-004 development internally and instead intends to seek grant funding, out-licensing, or a development partner. The final Phase 1b/2a cohort will not begin without such funding or a partner.	The board's internal valuation, partner outreach, bids, licensing terms, and reasons for allocating capital away from SYN-004 are material valuation evidence.

6. The most supportable public figure is therefore the value of the disease burden: published U.S. estimates of approximately \$5.4 billion to \$6.3 billion annually, stated in the studies' original nominal dollars. The public record does not support assigning that entire amount to SYN-004. Disease cost, addressable pharmaceutical revenue, and the present value of one unapproved product candidate are different measures.
7. A reliable SYN-004 valuation would require the eligible treated population, expected net price, market penetration, duration of exclusivity, competitive products, gross margin, launch timing, probability of technical and regulatory success, remaining development cost, taxes, discount rate, and any licensing economics. Those inputs are not established by the cited public record. The appropriate inquiry is whether Theriva prepared or received a probability-adjusted valuation and whether it reasonably compared continued development, grants, partnering, licensing, or sale before placing the program on a no-internal-funding path.

Valuation framework. A qualified expert ordinarily would estimate risk-adjusted present value from probability-adjusted future net cash flows and subtract remaining development and commercialization costs. The annual \$5.4-\$6.3 billion disease burden is relevant context and confirms that the indication is economically material, but it is not itself a price tag for SYN-004.

Plaintiff proposed two billion dollar valuation scenario

Hypothesis	Inputs	Arithmetic	Necessary limitation
Assume SYN-004 has a \$2.0 billion equity-value contribution	\$2,000,000,000 assumed value; 45,892,668 reported TOVX shares	$\$2,000,000,000 \div 45,892,668 =$ approximately \$43.58 per share	This is Plaintiff's proposed scenario for testing. It is not a public appraisal, expert opinion, transaction price, or established fair value.
Add the assumed SYN-004 value to the separate \$820-per-share illustration	\$820 base illustration plus \$43.58 per share	Combined illustration = approximately \$863.58 per share; implied combined equity value = approximately \$39.632 billion	The addition is valid only if the \$820 base excludes SYN-004. If the base already values Theriva's entire pipeline, adding \$2.0 billion would double count SYN-004.

This arithmetic gives the proposed valuation hypothesis a defined magnitude. It does not establish that SYN-004 is worth \$2.0 billion or that the combined company is worth \$863.58 per share. It identifies valuation, financing, licensing, and strategic-process records that can confirm or reject the hypothesis.

V Permissible motive inquiry

Observed fact	Question it may support	Conclusion it does not prove
The VCN-01 two-dose subgroup reported 14.8 months median overall survival—1.6 months above the 13.2 months reported for daraxonrasib's RAS G12 Phase 3 population—while Theriva's cited market capitalization was approximately 1/3,900 of RVMD's. At TOVX's reported share count, an RVMD-sized capitalization would arithmetically equal about \$900.80 per share; \$820 would imply about \$37.632 billion.	Could a perceived multi-billion-dollar clinical opportunity, when viewed against Theriva's approximately \$10.59 million cited market capitalization, supply an economic motive for a transaction or asset transfer before confirmatory Phase 3 evidence? What value did the board actually assign to VCN-01?	That TOVX should trade at \$820 or \$900.80, that the trials are directly comparable, that VCN-01 is clinically superior, or that any person intended to capture or transfer value improperly.
VCN-01 reported positive randomized Phase 2b results and Phase 3 regulatory planning.	What internal probability-adjusted value did the board assign to VCN-01, and what financing was required to preserve that opportunity?	That VCN-01 was certain to succeed or worth more than RVMD.
SYN-004 reported a 71.4% relative reduction in CDI in a 412-patient Phase 2b trial, while published estimates place annual U.S. CDI economic burden at approximately \$5.4-\$6.3 billion. Theriva later stated that it would not fund further development internally.	What probability-adjusted value did the board assign to SYN-004, what partner or license interest existed, and why was internal funding rejected?	That SYN-004 is presently worth \$5.4-\$6.3 billion, would obtain approval, or was intentionally suppressed or transferred.
Plaintiff proposes a \$2.0 billion SYN-004 valuation scenario, equal to approximately \$43.58 per reported TOVX share. If added to a separate \$820-per-share base that excludes SYN-004, the combined illustration is approximately \$863.58 per share.	Did any board valuation, partner proposal, licensing analysis, or market check support or reject a value near this scenario? Did the \$820 base include SYN-004, or would the proposed addition avoid double counting?	That SYN-004 is worth \$2.0 billion, that TOVX should trade at \$863.58, that the values may be added without checking their scope, or that any person acted from an improper motive.
Theriva's market capitalization was extremely small relative to RVMD's.	Did the board obtain independent valuation, fairness, strategic, and financing analyses before considering a transaction or asset disposition?	That the market price was manipulated or that insiders intended to appropriate the asset.
Theriva disclosed limited resources, liabilities, warrants, and strategic alternatives.	Were financing and standalone-development alternatives compared against merger, reverse-merger, licensing, sale, or wind-down outcomes?	That any strategic alternative was prearranged, coercive, or unlawful.
Management and directors may have change-in-control or continuing-position incentives.	Were conflicts disclosed, managed, valued, and subject to recusal or independent review?	That compensation caused the decision or establishes disloyal motive.

VI Records needed to evaluate motive rather than speculate

- all board and committee valuations, forecasts, probability-of-success analyses, development budgets, partnering analyses, and risk-adjusted net-present-value models for VCN-01, SYN-004/ribaxamase, SYN-020, and any other material asset;
- all comparisons of standalone development, common-only financing, at-the-market sales, warrant financing, debt, grants, licensing, partnership, asset sale, merger, reverse merger, recapitalization, or dissolution;
- all banker, adviser, consultant, and management presentations identifying potential counterparties, bids, indications of interest, deal structures, exchange ratios, management roles, financing sources, and transaction fees;
- all conflict questionnaires, related-person analyses, recusal records, compensation models, severance calculations, option-acceleration schedules, and post-transaction employment or board discussions;
- all communications concerning the timing of clinical disclosures, financings, warrant exercises, strategic-alternative disclosures, authorized-share changes, employee-plan shares, and any transaction process;
- all VCN-01 regulatory communications and development-cost estimates presented to the board, including the proposed Phase 3 budget, timing, enrollment, manufacturing, and partnership requirements; and
- all independent fairness opinions, valuation reports, market checks, bidder logs, data-room records, and records explaining why any apparently higher-value or less-dilutive alternative was rejected.

VII Targeted witnesses and custodians

The records should be obtained first from the party and entity custodians most likely to possess them. Testimony should then be limited to unresolved facts. Public filings and authenticated records do not require every participant to testify.

Witness or custodian	Factual subject	Why the testimony may matter
Theriva Rule 30(b)(6) designee	VCN-01 valuation, development budget, financing alternatives, warrant treatment, strategic process, and document custody.	Provides binding entity testimony on the decision record and identifies the responsible people and files.
Directors or officers who participated	Materials reviewed, alternatives considered, conflicts disclosed, votes, recusal, and any contemplated post-transaction role.	Tests the process and personal knowledge without assuming an improper motive.
A.G.P. entity designee	Advice provided, alternatives modeled, pricing, fees, transaction structure, and any strategic mandate.	Tests whether less-dilutive or standalone alternatives were presented and how the challenged terms were selected.
Independent valuation or fairness adviser, if one existed	Methods, assumptions, probability adjustments, comparables, and values assigned to clinical assets.	Provides the valuation work actually used, rather than a market-cap inference.
Clinical, regulatory, or health-economics expert	Meaning and limitations of VIRAGE and SYN-004 evidence; Phase 3 requirements; disease burden; addressable market; development cost; and cross-trial comparability.	Separates clinical and economic potential from unsupported claims of superiority, revenue, or fair value.
Relevant counterparty custodian, only if identified by party records	Actual proposal, financing source, asset interest, management role, and communications.	Tests an identified transaction; no nonparty should be accused or burdened based only on suspicion.

Discovery condition. Discovery ordinarily follows the Rule 26(f) conference. Earlier discovery requires authorization by rule, stipulation, or court order. Nonparty subpoenas must be specific, geographically proper, nonprivileged, and not unduly burdensome. Fed. R. Civ. P. 26(d)(1), 45(c)–(d).

VIII Neutral conclusion

The public record establishes that the VCN-01 two-dose subgroup's reported median overall survival of 14.8 months was numerically 1.6 months higher than the 13.2 months reported for daraxonrasib's RAS G12 Phase 3 population. It also shows a very large market-valuation gap. Separately, SYN-004 produced positive controlled Phase 2b evidence in an indication with a documented annual U.S. economic burden measured in billions of dollars, while Theriva has stated that it will not fund further SYN-004 development internally. Plaintiff's \$2.0 billion SYN-004 hypothesis would add approximately \$43.58 to the separate \$820-per-share illustration and produce approximately \$863.58 per share, but only if the \$820 base excludes SYN-004. These facts and calculations do not establish clinical superiority, fair value, future approval, future revenue, that SYN-004 is worth \$2.0 billion or the entire disease burden, that Theriva is more valuable than Revolution Medicines, that the two illustrative values may be added without checking their scope, that management intends to transfer value from stockholders, or that a takeover is occurring from inside. They support the question posed by this Part L—could this be a motive?—and a focused inquiry into what the board believed each clinical asset was worth, what financing, grant, licensing, partnering, and development alternatives were genuinely available, what conflicts existed, and whether any proposed transaction process protected corporate and stockholder value.

Source record

- [1. Theriva May 7, 2025 VIRAGE Phase 2b results](#)
- [2. Theriva March 23, 2026 FDA end-of-Phase-2 meeting release](#)
- [3. NEJM daraxonrasib Phase 3 report](#)
- [4. Revolution Medicines May 31, 2026 Phase 3 release](#)
- [5. Revolution Medicines April 2026 prospectus supplement](#)
- [6. Theriva June 30, 2026 Form 10-Q](#)
- [7. RVMD market-cap and closing-price snapshot](#)
- [8. TOVX market-cap and closing-price snapshot](#)
- [9. CDC About C diff](#)
- [10. Desai et al Epidemiological and economic burden of C difficile in the United States](#)
- [11. Zhang et al Cost of hospital management of C difficile infection in the United States](#)
- [12. Synthetic Biologics January 5, 2017 Phase 2b results filed with the SEC](#)
- [13. Kokai-Kun et al ribaxamase Phase 2b peer-reviewed report](#)
- [14. Theriva 2025 Form 10-K SYN-004 development and funding disclosures](#)

Respectfully submitted,

Dated: _____

Signature: _____

Mark Joseph Nejme, Plaintiff, self-represented

**UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., Defendant.	Case No. 3:26-cv-00705
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SUPPLEMENTAL EXHIBIT SM

Insider Led Acquisition Models and Shareholder Protection Inquiry

Original Part M

The following pages are separately paginated under this supplemental exhibit identifier.

**UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., Defendant.	Case No. 3:26-cv-00705 PROPOSED SUPPLEMENTAL INFORMATION PART M INSIDER-LED ACQUISITION MODELS AND SHAREHOLDER-PROTECTION INQUIRY
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Part M Insider-Led Acquisition Models and Shareholder-Protection Inquiry

Submitted as proposed supplemental material, subject to the Court's permission.

Purpose. This Part M addresses a limited question arising from Part H: whether any officer, director, affiliate, financing counterparty, adviser, or proposed post-transaction manager participated in a process intended to acquire, take private, merge, reverse merge, recapitalize, or transfer material assets of Theriva while also influencing the company's financing, disclosures, operations, capitalization, valuation, or transaction timing. It compares recognized acquisition models taught and studied by leading business schools with the disclosure, conflict, approval, and fairness protections applicable to an actual transaction. It does not allege that a school taught wrongdoing, that any Theriva participant followed a school model, that management intentionally reduced Theriva's value, or that an insider transaction exists.

I. Precise question presented

1. Could the events documented in Parts A through L be separate corporate events, or could some of them form part of a management-supported or affiliate-supported transaction process? If a process existed, who originated it, when did it begin, what alternatives were evaluated, what nonpublic information and management cooperation were available to each potential bidder, what benefits were expected by insiders or affiliates, and what was disclosed to unaffiliated stockholders?
2. Plaintiff's hypothesis is that repeated dilution, depressed market value, limited cash, asset licensing, contingent liabilities, employee equity, warrant rights, and strategic-alternatives language could make a company easier or cheaper to acquire. Those circumstances can also arise from unsuccessful clinical development, ordinary financing need, market risk, or lawful contingency planning. Price decline, dilution, or a possible merger does not by itself establish intent, coordination, breach, manipulation, or an insider bid.
3. The narrower inquiry is therefore whether anyone on the inside acted on both sides of a possible transaction, influenced the conditions under which the company would be valued or sold, expected a continuing role or economic benefit, or restricted a fair market check. That inquiry requires transaction records and testimony; it cannot be resolved from an analogy alone.

II. What the identified schools actually teach or study

The academic record establishes that acquisition, leveraged financing, owner-management, operational restructuring, management participation, and exit are recognized and deliberately taught transaction models. It does not establish that any school teaches concealment, market manipulation, intentional value destruction, or breach of fiduciary duty.

School and source	Verified subject	What it proves—and what it does not
Harvard Business School	HBS publishes a Note on Leveraged Buyouts and multiple cases examining Dell's management-led take-private transaction, including the board and shareholder decision whether to accept the founder-CEO's offer.	HBS teaches and studies leveraged and management-led buyouts of public companies. This does not show that HBS teaches managers to damage a company or conceal a transaction.
Stanford Graduate School of Business	Stanford's 2024 Search Fund Study describes an investment vehicle formed to search for, acquire, and lead a privately held company, ordinarily for six to ten years. It calls the result a path to owner-operator status and tracks later exits.	Stanford documents acquisition followed by inside management and possible exit. The study concerns primarily private-company acquisitions, not an incumbent public-company board preparing its own issuer for a covert takeover.

School and source	Verified subject	What it proves—and what it does not
University of Chicago Booth	Booth teaches entrepreneurship through acquisition and private-equity transactions. Its official materials describe sourcing, diligence, purchase, management, and exit; the buyer commonly joins management after investment and may become CEO. Booth also explains restructuring, cost reduction, repositioning, acquisitions, and management changes after a buyout.	Booth teaches buy, manage, restructure, and exit. It does not say that existing management should first reduce shareholder value or secretly position its own public company for acquisition.
Duke Fuqua	Fuqua's Private Equity Buyouts Lab has student teams identify buyout targets and prepare full acquisition investment theses, including public go-private candidates.	Duke teaches target selection and buyout analysis. It does not prove a Theriva connection or teach unlawful insider conduct.

4. The strongest documented historical link among the schools is limited and precise: Stanford reports that the term search fund originated at Harvard Business School in 1984 and was later popularized at Stanford. The evidence does not support saying that a method of destroying public-company value originated at Harvard, Stanford, Chicago, or Duke.

III. The exact inside-management distinction

Transaction setting	Recognized model	Conflict or protection question
Outside acquisition followed by new management	An outside entrepreneur, search fund, strategic buyer, or private-equity sponsor acquires a company and then installs, retains, or joins management.	Was the seller's process informed, competitive, fairly priced, properly approved, and fully disclosed?
Management-led or affiliate-led buyout	Existing management or an affiliate joins the purchasing group, rolls equity, supplies nonpublic knowledge, expects continued employment, or otherwise participates on the buy side.	Did independent directors control the process? Were conflicts and benefits disclosed? Did other bidders receive cooperation and information? Was there a meaningful market check and fair consideration?
Defensive action or rights plan	A board may use lawful defenses against a change of control, subject to the governing statute and the actual instrument.	Was there a genuine threat? Was the response authorized and proportionate where the statute requires it? Was the tool applied consistently rather than selectively to block an outside bid while favoring an affiliate?
Intentional value impairment alleged	Plaintiff asks whether financing, disclosure, operational, or asset decisions were used to lower the company's market value before a transaction.	This is not established by a falling stock price. It would require contemporaneous communications, models, bidder records, conflict evidence, causation, and proof meeting the applicable legal standard.

Chicago confirmation. The official Booth transcript says the acquisition entrepreneur becomes part of management after the investment and will often take the CEO position. It describes replacement of a retiring founder and later operation of the acquired business. That is management from inside after acquisition. It is not a statement that incumbent management should prepare its own company for a takeover, and Part M does not attribute that proposition to Chicago.

IV. Why a management buyout presents a recognized conflict

5. A Harvard Law Review article by a professor of law and business at Harvard Law School and Harvard Business School uses management buyouts, including Dell, to identify four recurring conditions that can make the bidding field uneven: information asymmetry, management that is valuable to the continuing enterprise, management financial incentives to discourage higher competing bids, and a limited bidding clock.
6. The article explains the structural concern: management may know more than outside bidders, may decide how much cooperation competing bidders receive, may expect to roll equity or continue as management, and may therefore have incentives that differ from those of unaffiliated sellers. It proposes process protections including board control, independent review, meaningful pre-signing competition, management-cooperation commitments, careful treatment of matching rights, and approval by unaffiliated shareholders.
7. This scholarship is not a finding about Theriva and is not binding law. It supplies a recognized framework for asking what records would prove or disprove an insider-supported acquisition hypothesis. It also cuts against the proposition that insider involvement may be inferred merely from a low market price or management's continued employment.

V. Nevada and federal safeguards that would apply to an actual process

Authority	Relevant requirement	Careful application
NRS 78.138	Directors and officers owe duties to exercise power in good faith, on an informed basis, and with a view to the corporation's interests. Nevada presumes compliance; individual damages require rebuttal of the presumption and proof of a breach involving intentional misconduct, fraud, or a knowing legal violation.	An unfavorable outcome is not enough. The statute applies to a change or potential change in control, but the demanding proof standard must be met.
NRS 78.139	For specified actions resisting control that impede shareholder voting, directors must have reasonable grounds for a threat and use a response reasonable in relation to that threat. The section contains exceptions and does not authorize abrogation of statutory or charter rights.	The actual instrument and conduct matter. Part H found no public evidence of an adopted or triggered Theriva rights plan.
NRS 78.140	An interested-director or interested-officer transaction is not automatically void. The statute identifies protections including disclosure of the interest and good-faith approval by disinterested directors or shareholders, lack of knowledge in a specified circumstance, or fairness to the corporation when approved.	Insider participation is not automatically unlawful; disclosure, independent approval, and fairness are central factual questions.
NRS 92A.120	A Nevada merger, conversion, or exchange generally requires board adoption, submission to shareholders, notice containing the plan or a summary, and the required shareholder approval, subject to statutory exceptions.	The transaction structure determines whether an exception applies. A reverse merger or asset transaction must be analyzed from its actual documents.
Exchange Act Rule 13e-3 and Schedule 13E-3	A going-private transaction or series by an issuer or affiliate that has the specified effect triggers filing and disclosure concerning purposes, alternatives, fairness, reports, opinions, funding, and other material facts.	The rule does not cover every strategic review or every merger; issuer or affiliate participation, transaction form, and going-private effect are required.
Exchange Act Rule 14a-9 and Form 8-K	Proxy solicitations may not contain materially false or misleading statements or omit facts needed to make statements not misleading. Form 8-K requires timely disclosure when specified events occur, including certain material definitive agreements, significant asset dispositions, unregistered equity sales, and completed changes of control.	Not every preliminary thought or discussion must be announced immediately. Once a reporting trigger occurs, the filing must accurately describe the required event and relationships.

8. Federal securities law also imposes limits on pleading. *Santa Fe Industries, Inc. v. Green* holds that corporate mismanagement or fiduciary breach without deception or manipulation is not itself a Rule 10b-5 claim. *Macquarie Infrastructure Corp. v. Moab Partners, L.P.* holds that a pure omission is not actionable under Rule 10b-5(b) unless the omission makes an affirmative statement misleading. *Basic Inc. v. Levinson* rejects a rigid merger-discussion rule and assesses materiality from probability and magnitude. Part M therefore asks for facts and does not convert suspicion into a federal fraud conclusion.

VI. Theriva-specific bridge without repeating Parts A through L

9. Part H already identifies the relevant public sequence: financing and warrant terms, equity and plan capacity, change-in-control compensation, anti-takeover powers, the August 11, 2026 forward-looking reference to possible business combinations, mergers, or reverse mergers, and the absence of a publicly identified executed agreement, counterparty, insider voting block, or active shareholder-rights plan. Part M incorporates that verified record rather than repeating it.
10. The university materials do not connect any Theriva officer, director, lawyer, adviser, warrant holder, or counterparty to a particular course or playbook. The relevant evidentiary question is functional, not academic pedigree: did any person with inside influence participate on the prospective buy side, expect a post-transaction benefit, limit competing bidders, time disclosures or financing decisions to affect price, or use defensive tools selectively?
11. If no insider or affiliate process existed, board minutes, adviser mandates, conflict questionnaires, capitalization models, and counterparty logs should tend to show that. If one existed, those same records should identify its beginning, participants, purposes, alternatives, projected ownership, asset treatment, benefits, valuation, approvals, and disclosure analysis.

VII. Targeted records needed to test—not assume—the hypothesis

- a dated chronology of every strategic-alternatives, merger, reverse-merger, go-private, recapitalization, asset-sale, or change-of-control proposal considered from January 1, 2024 forward, identifying who first proposed it and every officer, director, affiliate, adviser, financier, and counterparty involved;
- all board and committee minutes, written consents, presentations, banker books, valuation materials, fairness analyses, market checks, bidder lists, data-room logs, nondisclosure agreements, indications of interest, letters of intent, term sheets, draft agreements, and abandoned proposals;
- all records of management rollover equity, continuing employment, board seats, consulting arrangements, severance, accelerated vesting, option or warrant treatment, transaction bonuses, indemnification, or other personal benefit;
- all documents showing whether management agreed to cooperate with all bidders, only one bidder, or no competing bidder, including information access, management presentations, forecasts, clinical data, intellectual-property data, and financing support;
- all analyses of how warrant exercise, employee-plan issuances, authorized shares, preferred shares, beneficial-ownership blockers, reverse splits, contingent consideration, debt, and licensed assets would affect pre- and post-transaction voting power and economic ownership;
- all projections and board analyses comparing continued independence and less-dilutive financing with merger, reverse merger, private-equity financing, licensing, asset sale, wind-down, or going-private alternatives;
- all conflict questionnaires, related-person analyses, recusals, special-committee records, independent-counsel engagements, and records identifying which directors were treated as disinterested; and
- all analyses under Rule 13e-3, Schedule 13E-3, Rule 14a-9, Form 8-K, NYSE American shareholder-approval requirements, NRS 78.138 through 78.140, and NRS Chapter 92A for any actual or contemplated structure.

VIII. Narrow requested treatment

12. Plaintiff asks the Court to treat the school materials only as background showing that acquisition, inside operation, restructuring, management participation, and exit are recognized transaction models. Plaintiff does not ask the Court to infer that Theriva followed a model merely because the model is taught.
13. If the Court finds the issue relevant to the existing warrant- and disclosure-related claims and permits discovery, Plaintiff requests preservation and limited production of the records in Section VII, followed by entity testimony limited to unresolved transaction-process facts. Legitimately confidential deal material may be handled under an appropriate protective order.
14. This Part M seeks no present finding that an insider takeover, management buyout, poison pill, manipulation, fiduciary breach, conspiracy, or criminal conduct occurred. It preserves Parts A through L, the complaint, the existing emergency motion, the jury demand, and the requested administrative docket corrections. It adds no party or claim. Any later request for additional relief must rest on admissible evidence and an authorized procedural vehicle.

IX. Neutral conclusion

Leading business schools teach and study acquisition, leveraged buyouts, management participation, post-acquisition operation, restructuring, and exit. Stanford traces the search-fund concept to Harvard and describes acquisition followed by owner-management; Chicago expressly teaches sourcing, diligence, purchase, management, restructuring, and exit; Duke students develop buyout theses; and Harvard studies management-led public-company buyouts and their conflicts. The record does not establish that these institutions teach managers to destroy their own companies or that any Theriva participant used a university-derived method. It does establish that management-supported acquisitions create recognized information, incentive, valuation, and competition issues. The appropriate question is whether such a process existed at Theriva and, if so, whether it was independently controlled, fairly valued, properly disclosed, lawfully approved, and structured to protect the corporation and unaffiliated stockholders.

Source record

1. [Harvard Business School, Note on Leveraged Buyouts](#)
2. [Harvard Business School, Taking Dell Private](#)
3. [Harvard Business School, The Management Buyout of Dell, Inc.](#)
4. [Guhan Subramanian, Deal Process Design in Management Buyouts, Harvard Law Review](#)
5. [Stanford GSB, 2024 Search Fund Study](#)
6. [Chicago Booth, Models of Entrepreneurship Through Acquisition](#)
7. [Chicago Booth, MBA Career Paths in Private Equity](#)
8. [Chicago Booth Review, Going Private](#)

9. [Duke Innovation & Entrepreneurship, Private Equity Buyouts Lab presentations](#)
10. [SEC Form 8-K](#)
11. [17 C.F.R. § 240.13e-3, Going-private transactions by issuers or affiliates](#)
12. [Schedule 13E-3](#)
13. [17 C.F.R. § 240.14a-9, False or misleading proxy statements](#)
14. [Nevada Revised Statutes Chapter 78](#)
15. [Nevada Revised Statutes Chapter 92A](#)
16. [Basic Inc. v. Levinson, 485 U.S. 224 \(1988\)](#)
17. [Santa Fe Industries, Inc. v. Green, 430 U.S. 462 \(1977\)](#)
18. [Macquarie Infrastructure Corp. v. Moab Partners, L.P., 601 U.S. 257 \(2024\)](#)
19. [Theriva August 11, 2026 second-quarter results release](#)

Respectfully submitted,

Dated: _____

Signature: _____

Mark Joseph Nejme, Plaintiff, self-represented

**UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., Defendant.	Case No. 3:26-cv-00705
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SUPPLEMENTAL EXHIBIT SN

Plaintiff Internal Disassembly Concern and Theriva Disclosed Sequence

Original Part N

The following pages are separately paginated under this supplemental exhibit identifier.

**UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., Defendant.	Case No. 3:26-cv-00705 PROPOSED SUPPLEMENTAL INFORMATION PART N PLAINTIFF'S INTERNAL-DISASSEMBLY CONCERN AND THERIVA'S DISCLOSED SEQUENCE
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Part N Plaintiff's Internal-Disassembly Concern and Theriva's Disclosed Sequence

Submitted as proposed supplemental material, subject to the Court's permission.

Preliminary statement and relief requested. Plaintiff assembled Parts A through N to place in one organized record Theriva's publicly disclosed reverse splits, warrant and equity financings, capitalization, cash use, debt and contingent obligations, patent and clinical assets, asset licensing, governance and related-person questions, price-volume and news chronology, strategic review, and potential change-of-control mechanics. Plaintiff does not ask the Court to accept his interpretation. He asks the Court to review the following record, weigh the competing explanations, and decide for itself.

Preservation requested. Plaintiff contends that further exercise, issuance, delivery, registration, transfer, or voting of the specifically identified New Warrant Shares could materially alter economic and voting interests before a hearing, and that implementation of any actual shareholder-rights plan could affect whether a transaction proceeds on equal terms. Subject to the findings required by Federal Rule of Civil Procedure 65, Plaintiff therefore requests a temporary stay of those New Warrant mechanics and, if a shareholder-rights plan actually exists or is adopted, a temporary stay of its implementation, triggering, distribution, exercise, or selective waiver pending prompt notice and hearing. These are requests to preserve the status quo while the record is tested, not findings that a rights plan, insider transaction, or wrongful purpose exists.

We now see how premier graduate schools teach leveraged buyouts. But one thing we could not find in the reviewed materials is any instruction teaching incumbent management to disassemble its own public company in preparation for a leveraged buyout or reverse merger.

Purpose. Part M addresses the academic record. This separate Part N states Plaintiff's Theriva-specific concern: could the Company's disclosed sequence reflect internal preparation for a transaction that reduces the stand-alone company and the relative position of existing stockholders? The word "disassemble" is Plaintiff's characterization and inference. It is not a fact established by the present public record. This Part therefore identifies the proved sequence, competing explanations, facts that cut against the inference, and focused records needed to test it.

Scope. This Part incorporates Parts A through M without repeating their complete financing, patent, accounting, relationship, price-volume, debt, valuation, and legal analyses. It adds no party or cause of action, makes no criminal accusation, does not identify an inside buyer, and does not assert that a merger, reverse merger, going-private transaction, asset diversion, active shareholder-rights plan, or agreement to exclude existing stockholders presently exists.

Question to the Court—and, on issues properly tried, the jury

Does the disclosed sequence, viewed as a whole, suggest that Theriva was being disassembled from within in preparation for a possible merger or reverse merger? Theriva said its strategic alternatives may include a reverse merger after it had: (a) formally resized and restructured the Company; (b) reduced its workforce and research spending; (c) licensed SYN-020 and its covered intellectual property to Rasayana for \$300,000 up front plus contingent consideration, even though a 2025 patent concerns alkaline-phosphatase-based oncology treatment and identifies gemcitabine and cancer treatment; (d) stated that it would not fund further SYN-004 development internally without grant or partner funding after a Phase 2b trial that the Company reported met its primary endpoint, while retaining a patent portfolio; and (e) used financings that substantially expanded the share count and warrant overhang. Parts E and G separately model financing alternatives that Plaintiff believes may have raised capital with less dilution. Is this combination only a response to genuine financial constraints, or does it reflect internal preparation for a transaction that may transfer value or control away from existing stockholders? Plaintiff asks the Court—and, if the evidence reaches trial on properly triable issues, the jury—to decide from admissible evidence, not from Plaintiff's belief alone.

Scientific limitation. U.S. Patent No. 12,318,434 concerns alkaline-phosphatase-based oncology treatment and identifies gemcitabine; it does not establish an early-detection diagnostic. The SYN-004 patent family concerns beta-lactamase formulations and treating or preventing antibiotic-related adverse effects, including prevention of *C. difficile* overgrowth or infection. U.S. Patent No. 9,404,103 B1, application No. 15/160,669, was filed May 20, 2016 and issued August 2, 2016. Its summary states that the improved beta-lactamases have therapeutic uses “including the prevention or treatment of CDI and/or a *C. difficile*-associated disease.” Prevention is itself a clinically meaningful therapeutic objective: the disclosed approach is intended to prevent or treat CDI and associated disease before or while toxin-driven colon injury develops. *C. difficile* is a bacterium and CDI is an infection that can cause colitis; CDI is not itself an ulcer. The complete patent attached as Exhibit P-7 highlights the patent's exact language. The patent does not use “cure” to characterize SYN-004 or CDI; the word appears in the title of a cited article concerning a duodenal ulcer and *Helicobacter pylori*, not as a statement about the claimed therapy. The Company's July 12, 2016 release announced a Notice of Allowance for that same application before the patent issued, although the reviewed sources did not locate a separate post-issuance company release naming the final patent number. The Company's SEC-filed January 5, 2017 release reported that SYN-004 met the primary endpoint in a randomized, double-blind, placebo-controlled, 412-patient Phase 2b trial: seven confirmed CDI cases occurred in the placebo group and two in the ribaxamase group, a reported 71.4% relative-risk reduction ($p=0.045$). The Company's May 11, 2017 release states that FDA granted Breakthrough Therapy Designation for prevention of CDI based on that trial. Part N uses those precise patent, reported-trial, and regulatory descriptions; it does not characterize SYN-004 as an approved cure. Patent issuance does not itself establish FDA approval or clinical efficacy. The alternative-financing models are counterfactual analyses, not proof that the modeled capital was actually available on the assumed terms. These limitations are part of the question presented.

Potential-value question. A patent grant establishes enforceable patent rights if valid; it does not establish clinical efficacy, a cure, or a dollar valuation. Here, however, the patent estate did not stand alone: the Company also reported a successful Phase 2b primary endpoint and an FDA Breakthrough Therapy Designation for prevention of CDI. Those combined facts permit a focused valuation inquiry. What contemporaneous fair-value analyses, licensing proposals, partnering discussions, market assessments, patent-maintenance decisions, and development budgets existed for SYN-004/ribaxamase? The public record reviewed for this supplement does not establish a reliable dollar value. Published observational research reporting increased CDI risk among patients with pancreatic disease supports a clinical-risk association only; it does not establish that CDI causes pancreatic cancer or that SYN-004 treats pancreatic cancer.

I. Question presented and Plaintiff's inference

- 1. Could the combined sequence—three reverse stock splits since 2018; restoration of 350,000,000 authorized common shares after the 2024 split; repeated equity and warrant financings; a 2025 workforce restructuring; reduced research spending; licensing SYN-020; declining cash and working capital; a large exercisable warrant overhang; increased employee-plan capacity; change-in-control compensation keyed in part to a private-company transaction; and an announced strategic review that expressly includes a merger or reverse merger—reflect deliberate preparation for a transaction rather than only ordinary financing and survival measures?
- 2. Plaintiff believes the sequence could reflect an internal process of reducing the Company as a stand-alone operating enterprise while preserving a public-company platform, clinical assets, authorized equity capacity, and transaction flexibility for a later counterparty. Plaintiff asks whether current stockholders would retain fair economic participation in the existing assets, cash, intellectual property, and any later combined company.
- 3. The same events may also have ordinary explanations: a clinical-stage company can lose money, reduce staff, license a program, issue discounted securities, increase equity compensation, and explore strategic alternatives because it lacks sufficient capital and faces substantial clinical and regulatory risk. The Court should not choose between those explanations from chronology alone. Contemporaneous records, ownership evidence, transaction terms, and testimony are required.

II. Verified disclosed sequence

Date	Verified disclosure	What the fact permits the Court to ask
June 30–Aug. 5, 2021	The Company reported \$74.291 million of cash and cash equivalents at June 30, 2021 and described that amount as cash available to fund clinical programs. Its later 2021 Form 10-K reported a distinct figure: approximately \$79.2 million of public float at June 30, 2021, based on a \$0.60 NYSE American closing price. The same Form 10-K checked “Non-accelerated Filer” and “Smaller Reporting Company,” and the Section 404(b) auditor-attestation box was unchecked.	Cash, market capitalization, and public float are different measures. The \$74.3 million cash balance was not proof that market value or public float was being held below \$75 million. What capital-allocation, public-float, shelf-capacity, and program-funding analyses did the board receive?

Date	Verified disclosure	What the fact permits the Court to ask
Mar. 1–Aug. 2, 2016	Synthetic Biologics announced three USPTO Notices of Allowance covering SYN-004 and prevention of CDI and antibiotic-associated diarrhea. On July 12 it announced U.S. Patent No. 9,376,673 and a Notice of Allowance for application No. 15/160,669. That application issued August 2 as U.S. Patent No. 9,404,103, whose summary refers to “the prevention or treatment of CDI and/or a C. difficile-associated disease.”	What contemporaneous valuation and development plans did the patent estate support, and how were those plans later reassessed? The July release publicized the allowed application before issuance; the reviewed sources did not locate a separate post-issuance release naming No. 9,404,103. A patent documents claimed inventions, not approval or commercial value.
Jan. 5–May 11, 2017	The Company reported that its randomized, double-blind, placebo-controlled 412-patient Phase 2b trial met its primary endpoint, with seven confirmed CDI cases in the placebo group and two in the ribaxamase group—a reported 71.4% relative-risk reduction ($p=0.045$). It later announced FDA Breakthrough Therapy Designation for prevention of CDI based on that trial.	What development, financing, partnering, and valuation plans followed the reported trial result and regulatory designation, and why did those plans change? These events document reported results and regulatory status, not approval or guaranteed efficacy.
Mar. 11–June 30, 2022	The Company completed the VCN acquisition by paying \$4.7 million to Grifols, issuing 26,395,303 pre-split common shares representing 19.99% of the Dec. 14, 2021 outstanding shares, and agreeing to an approximately \$27.8 million VCN research-and-development budgetary commitment plus contingent milestones. The 2022 Form 10-K later reported approximately \$32.7 million of public float at June 30, 2022, based on a reverse-split-adjusted \$2.10 closing price.	What forecasts compared continued SYN-004/SYN-020 development, the VCN acquisition, and other uses of cash? The chronology permits discovery into allocation and valuation; it does not prove that VCN was acquired to reduce public float or market value.
2018, 2022, and Aug. 26, 2024	Theriva's 2025 Form 10-K lists reverse splits of 1-for-35, 1-for-10, and 1-for-25. The 2024 split reduced 25,131,230 outstanding shares to 1,005,249 and reduced authorized common shares from 350,000,000 to 14,000,000; stockholders later approved restoration to 350,000,000.	Why was each split needed, what financing followed, and what alternatives were modeled? Restoration of authorized shares creates capacity; it does not itself prove an issuance plan.
May 8, 2025	The Company sold 1,990,900 common shares, pre-funded warrants for 4,827,280 shares, and accompanying common warrants for 6,818,180 shares. The combined public price was \$1.10 per common share and accompanying common warrant, or \$1.099 per pre-funded warrant and accompanying common warrant. Gross proceeds were about \$7.5 million.	What values were assigned to the common shares and warrants, what less-dilutive alternatives were considered, and what did the board expect the full capitalization effect to be?
Sept. 28–30, 2025	The board approved a plan to “resize and restructure” the Company around business development and licensing, clinical-trial planning, exploratory manufacturing scale-up, limited preclinical work, and regulatory interactions. The Company reduced its workforce by seven employees, or 32%, recorded about \$520,000 in charges, and projected about \$1.8 million in annual compensation-and-benefit savings.	Was the plan solely a cash-conservation measure, or did it also prepare a smaller organization for a transaction? The filing states the Company's rationale but does not answer that broader process question.
Oct. 16–17, 2025	Holders exercised warrants for 8,092,280 shares at a reduced \$0.54 exercise price. In consideration, the Company issued New Warrants for 16,184,560 shares—200% of the exercised amount—at \$0.54. The Company later reported about \$3.9 million of net exercise proceeds.	Who held the securities, what ownership and hedging information was known, why was the 200% replacement selected, and what post-exercise ownership did the board model?
Year ended Dec. 31, 2025	The Company reported approximately \$6.8 million of net ATM proceeds, \$6.7 million of net May-offering proceeds, \$3.9 million of net inducement exercise proceeds, a \$1.7 million Spanish R&D rebate, and a \$1.4 million THERICEL project loan. It ended 2025 with 35,688,350 common shares outstanding.	How were financing proceeds allocated by program and payee, and what cash runway, ownership, and clinical-development alternatives were compared?
Feb. 18, 2026	Theriva granted Rasayana an exclusive worldwide, sublicensable license covering SYN-020 and licensed intellectual property. Rasayana assumed development and commercialization costs. Theriva received \$300,000 up front and retained contingent milestones, royalties, and a percentage of sublicense revenue; Theriva said it did not anticipate additional funds under the agreement during the following twelve months.	What valuation, diligence, funding evidence, conflicts review, development plan, and alternative bids supported the license? Contingent amounts are not cash already received.
Six months ended June 30, 2026	The Company sold 10,204,319 ATM shares for approximately \$2.303 million net. Outstanding shares increased from 35,688,350 at Dec. 31, 2025 to 45,892,668 at June 30, 2026. It used \$5.541 million of operating cash, which it said was primarily related to VCN-01 development.	What price, timing, volume, and alternative-financing analyses governed the ATM sales, and how much financing value reached the operating programs?

Date	Verified disclosure	What the fact permits the Court to ask
Aug. 3, 2026	Stockholders approved issuance of up to 16,184,560 shares upon exercise of the New Warrants and increased the 2020 Stock Plan's share authority from 4,500,000 to 6,500,000. The New Warrants became exercisable at \$0.54 and expire Aug. 3, 2031.	What amounts have since been exercised or granted, who beneficially owns them, and how would exercise or awards affect voting and economic ownership in a transaction? Approval and exercisability do not establish actual issuance.
Aug. 6–11, 2026	Theriva announced that the first patients had been dosed in VIRAGE2. Its Form 10-Q stated that the Company was devoting substantial time and resources to a strategic review that may include a business combination, merger, or reverse merger, but that no definitive timeline or board-approved course of action existed.	What process existed, when did it begin, which alternatives and counterparties were considered, and how would the legacy assets and current stockholders be treated? Continued dosing is evidence against a conclusion that all development stopped.
Aug. 10, 2026	The amended chief-executive employment agreement provided change-in-control vesting and severance. Its definition includes execution of a definitive agreement with a non-listed private operating company for a reverse merger or similar transaction intended to shift board or voting control.	When and why was that transaction-specific language selected, what advice supported it, and were any actual discussions underway? A contingency clause does not prove that such a transaction was agreed.

III. Capitalization and liquidity map

The \$75 million threshold must be applied to the correct measure. The SEC provisions discussed below use public float—the market value of voting and non-voting common equity held by non-affiliates—not cash, total assets, stockholders' equity, or unrestricted market capitalization. Synthetic Biologics reported \$74.291 million of cash but approximately \$79.2 million of public float on the same June 30, 2021 measurement date.

Rule or measure	Below \$75 million public float	\$75 million or more public float	Theriva/Synthetic record and limitation
Cash versus public float	A cash balance below \$75 million does not establish public float below \$75 million.	A cash balance above \$75 million does not establish public float above \$75 million.	At June 30, 2021: cash was \$74.291 million; reported public float was approximately \$79.2 million. They cannot be substituted for one another.
Form S-3 primary shelf capacity	If otherwise eligible, General Instruction I.B.6 limits primary sales in a rolling 12-month period to no more than one-third of public float; derivative securities are included through their underlying shares.	General Instruction I.B.1 permits primary offerings without the I.B.6 one-third cap, subject to the Form's other eligibility and transaction requirements.	Crossing below \$75 million generally narrows—not expands—primary shelf capacity. It does not bar registered financing altogether.
Accelerated-filer status	Public float below \$75 million is one route to non-accelerated status.	From \$75 million to below \$700 million, accelerated status also requires the other Rule 12b-2 conditions and excludes issuers eligible for the low-revenue smaller-reporting-company test.	The 2021 Form 10-K reported \$79.2 million public float but checked non-accelerated filer because the low-revenue/SRC rules can apply above \$75 million.
Smaller reporting company	A company below \$250 million public float can qualify for scaled disclosure; a separate revenue test can extend eligibility to public float below \$700 million.	\$75 million is not the current SRC ceiling. It was the historical public-float threshold before the SEC's amendments effective Sept. 10, 2018.	Synthetic checked SRC in its 2021 Form 10-K. The present record does not show that keeping public float below \$75 million was necessary to retain SRC status.
SOX §404(b) auditor attestation and filing cadence	Non-accelerated filers generally do not provide the Section 404(b) auditor attestation and receive the non-accelerated filing schedule, while management remains responsible for internal-control assessment.	An issuer can be accelerated above \$75 million if all conditions apply, but a low-revenue SRC can remain non-accelerated.	Synthetic's 2021 10-K left the 404(b) attestation box unchecked even though reported public float exceeded \$75 million. No avoidance inference follows from the cash figure alone.

Measure	Filed figure or transparent calculation	Required limitation
Outstanding common shares	2,811,259 at Dec. 31, 2024; 35,688,350 at Dec. 31, 2025; 45,892,668 at June 30 and Aug. 7, 2026. The Dec. 2024-to-Aug. 2026 increase is approximately 16.3 times.	This is share-count expansion, not by itself a measure of loss, manipulation, insider ownership, or proceeds retained.
New Warrant overhang	16,184,560 shares equal 35.3% of the 45,892,668 then-outstanding shares. If all were newly issued with no other change, those warrant shares would be 26.1% of the resulting 62,077,228 shares.	The warrants are material but do not alone constitute majority control. Current exercises, holders, blockers, transfers, and voting arrangements must be proved.
Employee-plan increase	The plan authority increased by 2,000,000 shares, from 4,500,000 to 6,500,000. The incremental 2,000,000 equals about 4.4% of the 45,892,668-share base.	Plan authority is not the same as grants, vesting, exercise, issuance, beneficial ownership, or voting control.

Measure	Filed figure or transparent calculation	Required limitation
Authorized capacity	350,000,000 authorized common shares less 45,892,668 outstanding shares left approximately 304.1 million authorized but unissued shares before accounting for reservations and other commitments.	Authorized capacity is not an issued security, an active poison pill, or proof that the board intends to use all capacity.
Cash and working capital	Cash was \$13.056 million at Dec. 31, 2025, \$11.337 million at June 30, 2026, and approximately \$9.6 million in early Aug. 2026. Working capital declined from about \$7.5 million to \$1.6 million during the first six months of 2026.	Cash burn and a going-concern disclosure show financial pressure; they do not establish that management manufactured the pressure.
Equity and accumulated deficit	Total stockholders' equity declined from \$15.383 million at Dec. 31, 2025 to \$12.187 million at June 30, 2026. Accumulated deficit increased from \$358.710 million to \$363.971 million.	An accumulated deficit is not a cash ledger and cannot identify who received funds. Parts D, E, and I address the cash and balance-sheet trace.
Grifols contingent consideration	At June 30, 2026 the balance sheet reported \$2.650 million current and \$7.169 million non-current contingent consideration, or \$9.819 million in total. The Company separately stated that \$5.0 million had been earned but deferred pending discussions with Grifols.	The \$5.0 million disclosure may overlap the recorded liability and must not be added again without the rollforward and contract. The entire liability was not stated to be immediately payable.

IV. What may support the inference—and what cuts against it

Disclosed circumstance	Plaintiff's permissible question	Competing record or explanation
Repeated splits followed by substantial share issuance and warrant capacity	Was capitalization managed to create a transaction-ready public platform while existing ownership percentages repeatedly contracted?	Splits can preserve exchange listing; equity and warrants can finance a non-revenue clinical-stage company. No public record reviewed proves a hidden ownership agreement.
Workforce reduction and lower R&D	Was the operating organization intentionally reduced before a strategic transaction?	The Company expressly described cash conservation and a narrower development plan. It continued VIRAGE2 and reported that first patients were dosed.
SYN-020 license and no internally funded SYN-004 development	Were existing assets fairly valued, preserved for current holders, and exposed to a genuine market check?	Licensing or partnering can conserve cash and transfer development risk. SYN-020 rights revert upon specified termination events, and Theriva retained contingent economics.
Strategic-review language and transaction-specific change-in-control terms	Were officers preparing for or negotiating an insider-supported transaction while influencing capitalization and asset decisions?	The Form 10-Q expressly said no definitive timeline or board-approved course existed. Compensation provisions commonly address possible transactions and do not establish one.
Cash, working-capital, and contingent-liability pressure	Did earlier financing choices unnecessarily create urgency that could reduce negotiating leverage?	Clinical development consumes cash, outcomes are uncertain, and the Company disclosed substantial going-concern risk. Poor results or poor judgment are not proof of intentional value impairment.
Counterfactual financing analysis in Parts E and G	Could common-only, differently timed, or otherwise less-dilutive financing have preserved more ownership and supplied more development capital?	A what-if model depends on price, volume, shelf capacity, purchaser demand, transaction costs, and market impact. It identifies a discovery question; it does not prove that management could have completed the modeled financing.

V. Findings the present record does not support

- The public materials reviewed do not identify an executed merger or reverse-merger agreement, a buyer, an insider purchasing group, a voting agreement transferring control, a commitment to take the Company private, or an adopted or triggered shareholder-rights plan. The strategic review and employment provisions create questions; they are not substitutes for those missing facts.
- The New Warrants are economically and legally material, but the filed arithmetic does not show that they alone deliver majority control. They equal approximately 35.3% of the then-outstanding share count and approximately 26.1% of the enlarged total if fully exercised with no other change. Ownership blockers, current exercises, holder coordination, other securities, and voting arrangements must be proved rather than assumed.
- The record does not establish intentional value destruction, market manipulation, short selling by any identified holder, asset diversion, a related-party Rasayana transaction, commingling with another issuer, or a criminal or fraudulent

agreement. Those conclusions require admissible evidence of participants, communications, conduct, causation, state of mind where required, and resulting harm.

VI. Focused records needed to test the internal-disassembly concern

Patent-register disclosure gap. Theriva's 2025 Form 10-K reports more than 135 U.S. and foreign patents and more than 50 pending applications, and states that VCN controls more than 50 patents through assignment or exclusive license. The filing names examples rather than a complete patent-by-patent register. The accompanying proposed patent exhibit schedule identifies 53 U.S. issued-grant records located under the Theriva, predecessor Synthetic Biologics, and VCN assignee names, but it is not a current title opinion and does not establish the complete worldwide portfolio. The difference between an aggregate count and an itemized, current register is a disclosure gap to be tested with records; it is not, by itself, proof that any patent was concealed, abandoned, transferred, encumbered, or omitted improperly.

Specific \$74.3 million limitation. The reviewed record does not prove that management selected, preserved, or depleted the June 30, 2021 cash balance to remain below a regulatory market-value threshold, or that VCN was acquired for that purpose. The same-date public-float figure reported in the 2021 Form 10-K was approximately \$79.2 million. Plaintiff requests the contemporaneous analyses below so that purpose, alternatives, and effects can be tested rather than assumed.

- a complete, verified patent-and-application register from January 1, 2016 to the present, organized by program and patent family, identifying every jurisdiction, application and patent number, legal and beneficial owner, chain of title, current status, expiration date, maintenance or annuity status, abandonment or lapse, security interest, assignment, exclusive or nonexclusive license, sublicense, field and territory restriction, reversion right, and transaction in which the right was valued, transferred, or proposed to be transferred;
- the complete patent schedules and chain-of-title materials delivered with the Rasayana license, the VCN acquisition, and the IDIBELL/ICO/Hospital Sant Joan de Deu licenses, together with any later amendments, notices, consents, valuations, maintenance decisions, or proposed dispositions;
- the Company's patent-announcement and investor-relations chronology from January 1, 2016 forward, including the criteria used to issue or withhold a release after allowance, issuance, abandonment, lapse, assignment, licensing, or other material change, and the records concerning application No. 15/160,669, the July 12, 2016 Notice-of-Allowance release, and its August 2, 2016 issuance as U.S. Patent No. 9,404,103;
- all board and committee minutes, written consents, banker materials, management presentations, forecasts, capitalization models, and cash-runway analyses concerning the 2025 restructuring and every financing, warrant inducement, equity-plan increase, asset license, and strategic alternative from January 1, 2024 forward;
- a transaction-process chronology identifying every proposed acquisition, merger, reverse merger, recapitalization, asset sale, license, liquidation, or private-company combination, including when it arose, who proposed it, each counterparty, present status, and reasons for acceptance or rejection;
- all comparisons of common-only ATM financing, warrant financing, debt, grants, licensing, partnering, sale, merger, reverse merger, wind-down, and continued stand-alone development, including projected dilution, voting power, cash runway, fees, and program funding;
- all calculations and presentations from January 1, 2021 through December 31, 2022 concerning cash, market capitalization, public float, Form S-3 General Instructions I.B.1 and I.B.6, accelerated-filer or smaller-reporting-company status, Section 404(b), exchange-listing thresholds, and the expected effect of the VCN acquisition and related share issuance on each measure;
- current transfer-agent and warrant-agent records showing exercises, issuances, beneficial holders, 4.99% or 9.99% blockers, notices to increase blockers, transfers, registration status, and any voting or disposition agreements concerning the New Warrants;
- all valuations, bids, diligence, funding verification, conflict review, development plans, milestone tracking, and sublicense communications concerning Rasayana and SYN-020, together with any alternative proposals;
- all Grifols communications and accounting work papers concerning the earned-but-deferred \$5.0 million milestone, the current and non-current contingent consideration, payment timing, deferral consideration, amendments, waivers, and treatment in any strategic transaction;
- all records concerning treatment of VCN-01, SYN-004, SYN-020 economics, cash, debt, contingent liabilities, patents, licenses, tax credits, and existing stockholders in any proposed transaction, including exchange ratios, rollover interests, contingent-value rights, and post-transaction ownership;
- all conflict questionnaires, related-person analyses, recusals, special-committee records, independent-counsel or valuation engagements, severance calculations, option-acceleration schedules, continuing-employment discussions, and proposed post-transaction board seats; and

- any actual shareholder-rights agreement, board resolution, preferred-stock designation, transfer-agent instruction, or other instrument claimed to operate as a poison pill. If none exists, a sworn statement identifying that fact and the authorized decisionmaker.

VII. Witness sequence and procedural limits

7. Records should come first. A Theriva Rule 30(b)(6) witness can then identify the decisionmakers, custodians, transaction process, capitalization analyses, asset valuations, and financing alternatives. Individual directors, officers, advisers, warrant holders, Rasayana, Grifols, scientists, auditors, or other nonparties should be examined only on specific unresolved subjects for which they possess personal knowledge.
8. Discovery ordinarily follows the Rule 26(f) conference unless a rule, stipulation, or court order authorizes earlier discovery. Any expedited request should identify the urgent issue, specific custodian, narrow time period, and documents needed. Nonparty subpoenas remain subject to Rule 45's relevance, privilege, geographic, and undue-burden protections.
9. This Part is proposed supplemental information, not a self-executing discovery demand or a new emergency motion. District of Nevada Local Rule 7-2(g) requires leave for supplemental briefs, evidence, or authorities. Any injunction must satisfy Rule 65, address the claims actually pleaded, identify a presently threatened act, and be specific in scope and duration.

VIII. Narrow requested treatment

10. If the Court permits this supplemental record and finds it relevant to the existing warrant- and disclosure-related claims, Plaintiff respectfully asks the Court to grant the Rule 65 temporary restraining and preliminary-injunction relief properly requested concerning exercise, issuance, delivery, registration, transfer, or voting of the specifically identified New Warrant Shares, pending a prompt noticed hearing, but only if the Court finds the governing Rule 65 factors satisfied on current evidence of imminent and irreparable harm.
11. Plaintiff also requests preservation of the Section VI records and a sworn status disclosure identifying: (a) New Warrant exercises and shares issued; (b) current outstanding shares; (c) any executed or pending strategic-transaction agreement; (d) whether any shareholder-rights plan, sometimes called a poison pill, exists or is imminent; and (e) the proposed treatment of current stockholders and material legacy assets in any signed transaction, subject to lawful confidentiality protections. If an actual rights-plan instrument is identified, Plaintiff asks the Court to restrain implementation, triggering, distribution, exercise, or selective waiver only through the prompt hearing and only upon the required Rule 65 showing. If no such plan exists, Plaintiff requests preservation and advance notice rather than an injunction against a hypothetical device.
12. Plaintiff does not ask the Court in this Part to find that Theriva has disassembled itself, that an insider takeover exists, or that lawful financing or strategic review is inherently wrongful. Plaintiff asks that the disclosed sequence be tested with the records capable of proving or disproving the inference before irreversible ownership or asset changes occur.

IX. Conclusion

Part M shows that premier business schools teach leveraged buyouts, acquisition, management participation, restructuring, and exit; it did not identify teaching that incumbent managers should secretly impair or disassemble their own public company for a later transaction. Part N identifies why Plaintiff believes Theriva's disclosed sequence may present that different concern. The Company itself disclosed repeated capitalization changes, a formal resizing and restructuring, a 32% workforce reduction, substantial share-count expansion, a 200% replacement-warrant issuance, licensing of SYN-020, no internally funded further development of SYN-004, declining liquidity, transaction-specific change-in-control protection, and an active review that includes a merger or reverse merger. The same record also shows continued VCN-01 development, first patients dosed in VIRAGE2, financing need, going-concern risk, no named buyer, no executed transaction, and no identified active rights plan. The present record therefore supports a focused factual inquiry—not a verdict—into whether the sequence was ordinary corporate survival, reasonable preparation for alternatives, or an internally influenced process that failed to protect the corporation and existing stockholders. Plaintiff respectfully asks the Court to grant only the properly supported, specific Rule 65 relief stated above concerning the identified New Warrant Shares and any actual rights-plan instrument, and respectfully thanks the Court for its consideration.

Source record

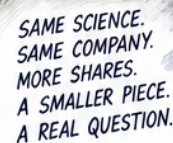
1. [Synthetic Biologics Form 10-K for the year ended Dec. 31, 2021 \(public float, filer status, and 404\(b\) check box\)](#)
2. [Theriva Form 10-K for the year ended Dec. 31, 2022 \(June 30, 2022 public float\)](#)
3. [Synthetic Biologics Aug. 5, 2021 second-quarter release \(\\$74.291 million cash\)](#)
4. [Synthetic Biologics Mar. 11, 2022 VCN acquisition release](#)
5. [SEC Form S-3, General Instructions I.B.1 and I.B.6](#)
6. [SEC Accelerated Filer and Large Accelerated Filer Definitions compliance guide](#)
7. [SEC Amendments to the Smaller Reporting Company Definition compliance guide](#)
8. [Theriva Form 10-K for the year ended Dec. 31, 2025](#)
9. [Theriva Form 10-Q for the quarter ended June 30, 2026](#)
10. [Theriva Aug. 3, 2026 annual-meeting Form 8-K](#)
11. [Theriva Aug. 11, 2026 operational and financial release](#)
12. [Theriva May 7, 2025 VIRAGE Phase 2b results](#)
13. [U.S. Patent No. 12,318,434, Alkaline phosphate-based oncology treatments](#)
14. [U.S. Patent No. 9,404,103 B1, Beta-lactamases with improved properties for therapy](#)
15. [U.S. Patent No. 10,041,056, Modified beta-lactamases and methods and uses related thereto](#)
16. [U.S. Patent No. 12,274,740, Microbiome protection from oral antibiotics](#)
17. [Synthetic Biologics Jan. 5, 2017 SEC-filed SYN-004 Phase 2b results release](#)
18. [Synthetic Biologics Mar. 1, 2016 SYN-004 patent-allowance release](#)
19. [Synthetic Biologics July 12, 2016 ribaxamase patent-grant release](#)
20. [Synthetic Biologics May 11, 2017 FDA Breakthrough Therapy Designation release](#)
21. [Vaishnavi et al., 2019, pancreatic disease and observed CDI risk \(peer-reviewed article\)](#)
22. [Federal Rule of Civil Procedure 65](#)
23. [Federal Rules of Civil Procedure 26 and 45](#)
24. [District of Nevada Local Rules](#)

Respectfully submitted,

Dated: _____

Signature: _____

Mark Joseph Nejme, Plaintiff, self-represented



GREED IS GOOD?
NOT WHEN PEOPLE ARE DYING.

AM I IMAGINING THIS?

ILLUSTRATIVE QUESTION — NOT A FINDING OF INTENT

**UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., Defendant.	Case No. 3:26-cv-00705
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SUPPLEMENTAL EXHIBIT SO

**Publicly Identified United States Patent Portfolio and Request for
Complete Register**

Part N Patent Schedule

The following pages are separately paginated under this supplemental exhibit identifier.

**UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., Defendant.	Case No. 3:26-cv-00705 PROPOSED SUPPLEMENTAL INFORMATION PART N—EXHIBITS PUBLICLY IDENTIFIED U.S. PATENT PORTFOLIO AND REQUEST FOR COMPLETE REGISTER
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Proposed Patent Exhibit Schedule and Disclosure-Gap Request

Organizational appendix; not a patent-title opinion.

Purpose and limitation. This appendix organizes the U.S. issued-grant records located in the reviewed public sources under Theriva Biologics, Inc., its predecessor name Synthetic Biologics, Inc., and its subsidiary VCN Biosciences, S.L. It identifies 53 U.S. issued grants. It does not purport to be the complete worldwide portfolio, a current chain-of-title report, or proof that every listed right remains owned, enforceable, maintained, unencumbered, or commercially valuable. It also does not equate patent language with FDA approval or clinical efficacy.

Why a complete register is still required. Theriva's 2025 Form 10-K reports more than 135 U.S. and foreign patents and more than 50 pending applications and states that VCN controls more than 50 patents through assignment or exclusive license, but it names only examples. Public assignee indexes are useful locating tools, not certified title evidence. A complete answer therefore requires the Company's program-by-program register, assignment records, maintenance records, and every governing license. The gap between aggregate counts and an itemized current register is a question for production and testimony, not proof of concealment.

Ownership categories. An assigned patent and an exclusively licensed patent are different interests. The SYN-004 record is described by Theriva as assigned to Theriva. The SYN-020 record is also described as assigned to Theriva, but the 10-K states that all such patents were exclusively licensed to Rasayana in February 2026. VCN's portfolio includes patents assigned to VCN and patents exclusively licensed from third parties. Those categories are kept separate below.

Exhibit P-1 — What Theriva's 2025 Form 10-K discloses

Portfolio statement	Named examples	What remains unverified publicly
Theriva reports more than 135 U.S. and foreign patents and more than 50 pending applications.	The filing names nine U.S. SYN-004 patents as examples: 8,894,994; 9,587,234; 9,301,995; 9,301,996; 9,290,754; 9,376,673; 9,404,103; 9,464,280; and 9,695,409.	Every patent/application number, jurisdiction, family, current owner, status, expiration, maintenance history, encumbrance, and license term.
SYN-020 is supported by U.S. and foreign patents and applications assigned to Theriva.	The filing describes formulations, medical uses, and manufacture, with expected expiration in 2038–2040, and states all such patents were licensed to Rasayana in February 2026.	The complete Rasayana patent schedule, present chain of title, retained rights, sublicenses, maintenance duties, security interests, valuation, and any later amendment.
VCN controls more than 50 U.S. and foreign patents through assignment or exclusive license and more than 15 pending applications.	The filing names U.S. 10,316,065 as an example and identifies licenses from IDIBELL, ICO, and Hospital Sant Joan de Deu.	Which rights are assigned to VCN versus third-party-owned and licensed; the complete family schedule; current license compliance; royalties; reversion; maintenance; and change-of-control provisions.

Exhibit P-2 — U.S. grants located under Theriva Biologics, Inc. (14)

These are public issued-grant records located under the current company name. Their appearance in an assignee index does not replace an official current title and maintenance review. The 2025 Form 10-K states that the SYN-020 patents were licensed to Rasayana in February 2026.

Ex.	Patent and title	Dates	Program, record name, and inventors
P-2.1	U.S. 12,648,987 Alkaline phosphatase agents for treatment of radiation disorders	Filed: Apr. 14, 2023 Issued: June 9, 2026	Program: SYN-020 / IAP Record: Theriva Biologics assignee record Inventor(s): Michael Kaleko; Christian Furlan Freguia; Sheila Connelly Link: https://patents.google.com/patent/US12648987
P-2.2	U.S. 12,508,304 Alkaline phosphatase formulations	Filed: June 17, 2024 Issued: Dec. 30, 2025	Program: SYN-020 / IAP Record: Theriva Biologics assignee record Inventor(s): Vincent Wachter; J. Blair West; Michael Kaleko; Christian Furlan Freguia Link: https://patents.google.com/patent/US12508304
P-2.3	U.S. 12,472,147 Intestinal alkaline phosphatase formulations	Filed: Mar. 17, 2023 Issued: Nov. 18, 2025	Program: SYN-020 / IAP Record: Theriva Biologics assignee record Inventor(s): Andrew Bristol; Ray Stapleton; Michael Kaleko; Christian Furlan Freguia; Steven Hubert; Cristina Freire; James Gubbins Link: https://patents.google.com/patent/US12472147
P-2.4	U.S. 12,318,434 Alkaline phosphate-based oncology treatments	Filed: May 5, 2020 Issued: June 3, 2025	Program: SYN-020 / IAP Record: Theriva Biologics assignee record Inventor(s): Christian Furlan Freguia; Michael Kaleko Link: https://patents.google.com/patent/US12318434
P-2.5	U.S. 12,274,740 Microbiome protection from oral antibiotics	Filed: Dec. 11, 2023 Issued: Apr. 15, 2025	Program: SYN-004 / microbiome Record: Theriva Biologics assignee record Inventor(s): Michael Kaleko; Sheila Connelly Link: https://patents.google.com/patent/US12274740
P-2.6	U.S. 12,178,917 Beta-lactamase formulations and uses thereof	Filed: Sept. 28, 2020 Issued: Dec. 31, 2024	Program: SYN-004 / ribaxamase Record: Theriva Biologics assignee record Inventor(s): Andrew Bristol; Michael Kaleko; Sheila Connelly Link: https://patents.google.com/patent/US12178917
P-2.7	U.S. 12,048,735 Alkaline phosphatase formulations	Filed: Mar. 29, 2021 Issued: July 30, 2024	Program: SYN-020 / IAP Record: Theriva Biologics assignee record Inventor(s): Vincent Wachter; J. Blair West; Michael Kaleko; Christian Furlan Freguia Link: https://patents.google.com/patent/US12048735
P-2.8	U.S. 11,981,899 E. coli-based production of beta-lactamase	Filed: Dec. 1, 2022 Issued: May 14, 2024	Program: SYN-004 / manufacture Record: Theriva Biologics assignee record Inventor(s): Andrew Bristol; Michael Kaleko; Steven Hubert Link: https://patents.google.com/patent/US11981899

Ex.	Patent and title	Dates	Program, record name, and inventors
P-2.9	U.S. 11,944,669 Method and compositions for inhibiting or preventing adverse effects of oral antibiotics	Filed: Feb. 2, 2023 Issued: Apr. 2, 2024	Program: SYN-004 / microbiome Record: Theriva Biologics assignee record Inventor(s): Michael Kaleko; Sheila Connelly; Vincent John Wachter Link: https://patents.google.com/patent/US11944669
P-2.10	U.S. 11,883,474 Microbiome protection from oral antibiotics	Filed: Oct. 26, 2022 Issued: Jan. 30, 2024	Program: SYN-004 / microbiome Record: Theriva Biologics assignee record Inventor(s): Michael Kaleko; Sheila Connelly Link: https://patents.google.com/patent/US11883474
P-2.11	U.S. 11,872,268 Safe and effective beta-lactamase dosing for microbiome protection	Filed: Jan. 10, 2022 Issued: Jan. 16, 2024	Program: SYN-004 / dosing Record: Theriva Biologics assignee record Inventor(s): Joseph Sliman Link: https://patents.google.com/patent/US11872268
P-2.12	U.S. 11,654,184 Alkaline phosphatase agents for treatment of radiation disorders	Filed: Mar. 20, 2019 Issued: May 23, 2023	Program: SYN-020 / IAP Record: Theriva Biologics assignee record Inventor(s): Michael Kaleko; Christian Furlan Freguia; Sheila Connelly Link: https://patents.google.com/patent/US11654184
P-2.13	U.S. 11,638,699 Intestinal alkaline phosphatase formulations	Filed: Mar. 20, 2019 Issued: May 2, 2023	Program: SYN-020 / IAP Record: Theriva Biologics assignee record Inventor(s): Andrew Bristol; Ray Stapleton; Michael Kaleko; Christian Furlan Freguia; Steven Hubert; Cristina Freire; James Gubbins Link: https://patents.google.com/patent/US11638699
P-2.14	U.S. 11,608,494 Beta-lactamases with improved properties for therapy	Filed: Dec. 8, 2021 Issued: Mar. 21, 2023	Program: SYN-004 / ribaxamase Record: Theriva Biologics assignee record Inventor(s): Michael Kaleko; Sheila Connelly Link: https://patents.google.com/patent/US11608494

Exhibit P-3 — U.S. grants located under Synthetic Biologics or otherwise tied to the predecessor record (38)

Synthetic Biologics changed its name to Theriva Biologics in October 2022. The entries below preserve the name shown in the public locating source. Some are co-assigned with another institution. No conclusion about current ownership, exclusivity, lapse, abandonment, or encumbrance should be drawn without the official chain of title and maintenance record.

Ex.	Patent and title	Dates	Program, public record, and inventors
P-3.1	U.S. 11,542,510 E. coli-based production of beta-lactamase	Filed: May 7, 2021 Issued: Jan. 3, 2023	Program: SYN-004 / manufacture Record: Synthetic Biologics record Inventor(s): Andrew Bristol; Michael Kaleko; Steven Hubert Link: https://patents.google.com/patent/US11542510
P-3.2	U.S. 11,517,614 Microbiome protection from oral antibiotics	Filed: June 28, 2017 Issued: Dec. 6, 2022	Program: SYN-004 / microbiome Record: Synthetic Biologics record Inventor(s): Michael Kaleko; Sheila Connelly Link: https://patents.google.com/patent/US11517614

Ex.	Patent and title	Dates	Program, public record, and inventors
P-3.3	U.S. 11,338,020 Alkaline phosphatase agents for treatment of neurodevelopmental disorders	Filed: Jan. 8, 2019 Issued: May 24, 2022	Program: SYN-020 / IAP Record: Synthetic Biologics record Inventor(s): Michael Kaleko; Christian Furlan Freguia; Sheila Connelly Link: https://patents.google.com/patent/US11338020
P-3.4	U.S. 11,253,577 Safe and effective beta-lactamase dosing for microbiome protection	Filed: June 4, 2020 Issued: Feb. 22, 2022	Program: SYN-004 / dosing Record: Synthetic Biologics record Inventor(s): Joseph Sliman Link: https://patents.google.com/patent/US11253577
P-3.5	U.S. 11,236,319 Beta-lactamases with improved properties for therapy	Filed: July 24, 2020 Issued: Feb. 1, 2022	Program: SYN-004 / ribaxamase Record: Google patent record; 2023 name-change entry Inventor(s): Michael Kaleko; Sheila Connelly Link: https://patents.google.com/patent/US11236319
P-3.6	U.S. 11,034,966 E. coli-based production of beta-lactamase	Filed: Aug. 27, 2015 Issued: June 15, 2021	Program: SYN-004 / manufacture Record: Synthetic Biologics record Inventor(s): Andrew Bristol; Michael Kaleko; Steven Hubert Link: https://patents.google.com/patent/US11034966
P-3.7	U.S. 10,987,410 Alkaline phosphatase formulations	Filed: July 17, 2018 Issued: Apr. 27, 2021	Program: SYN-020 / IAP Record: Synthetic Biologics record Inventor(s): Vincent Wachter; J. Blair West; Michael Kaleko; Christian Furlan Freguia Link: https://patents.google.com/patent/US10987410
P-3.8	U.S. 10,828,260 Beta-lactamase formulations and uses thereof	Filed: Aug. 24, 2018 Issued: Nov. 10, 2020	Program: SYN-004 / ribaxamase Record: Synthetic Biologics record Inventor(s): Andrew Bristol; Michael Kaleko; Sheila Connelly Link: https://patents.google.com/patent/US10828260
P-3.9	U.S. 10,792,346 Method and compositions for inhibiting or preventing adverse effects of oral antibiotics	Filed: July 10, 2018 Issued: Oct. 6, 2020	Program: SYN-004 / microbiome Record: Synthetic Biologics record Inventor(s): Michael Kaleko; Sheila Connelly; Vincent John Wachter Link: https://patents.google.com/patent/US10792346
P-3.10	U.S. 10,767,171 Beta-lactamases with improved properties for therapy	Filed: Jan. 24, 2020 Issued: Sept. 8, 2020	Program: SYN-004 / ribaxamase Record: Synthetic Biologics record Inventor(s): Michael Kaleko; Sheila Connelly Link: https://patents.google.com/patent/US10767171
P-3.11	U.S. 10,736,871 Anti-methanogenic lovastatin analogs or derivatives and uses thereof	Filed: Mar. 31, 2016 Issued: Aug. 11, 2020	Program: Legacy / IBS Record: Cedars-Sinai and Synthetic Biologics co-assignee record Inventor(s): Mark Pimentel; Vince Wachter Link: https://patents.google.com/patent/US10736871
P-3.12	U.S. 10,709,773 Safe and effective beta-lactamase dosing for microbiome protection	Filed: Mar. 7, 2016 Issued: July 14, 2020	Program: SYN-004 / dosing Record: Synthetic Biologics record Inventor(s): Joseph Sliman Link: https://patents.google.com/patent/US10709773
P-3.13	U.S. 10,584,326 Beta-lactamases with improved properties for therapy	Filed: May 16, 2019 Issued: Mar. 10, 2020	Program: SYN-004 / ribaxamase Record: Synthetic Biologics record Inventor(s): Michael Kaleko; Sheila Connelly Link: https://patents.google.com/patent/US10584326

Ex.	Patent and title	Dates	Program, public record, and inventors
P-3.14	U.S. 10,548,955 Carbapenemases for use with antibiotics for the protection of the intestinal microbiome	Filed: Jan. 9, 2018 Issued: Feb. 4, 2020	Program: Microbiome / beta-lactamase Record: Synthetic Biologics record Inventor(s): Michael Kaleko; Sheila Connelly Link: https://patents.google.com/patent/US10548955
P-3.15	U.S. 10,526,400 Humanized pertussis antibodies and uses thereof	Filed: June 25, 2018 Issued: Jan. 7, 2020	Program: Legacy / pertussis Record: University of Texas System and Synthetic Biologics co-assignee record Inventor(s): Jennifer Maynard; Annalee Nguyen; Eduardo Padlan; Ellen Wagner Link: https://patents.google.com/patent/US10526400
P-3.16	U.S. 10,336,995 Beta-lactamases with improved properties for therapy	Filed: Aug. 24, 2018 Issued: July 2, 2019	Program: SYN-004 / ribaxamase Record: Synthetic Biologics record Inventor(s): Michael Kaleko; Sheila Connelly Link: https://patents.google.com/patent/US10336995
P-3.17	U.S. 10,253,306 Modified beta-lactamases and methods and uses related thereto	Filed: July 27, 2017 Issued: Apr. 9, 2019	Program: SYN-004 / ribaxamase Record: Synthetic Biologics record Inventor(s): Pertti Koski; Ulla Airaksinen; Katja Valimaki Link: https://patents.google.com/patent/US10253306
P-3.18	U.S. 10,105,322 Beta-lactamase formulations and uses thereof	Filed: Oct. 8, 2015 Issued: Oct. 23, 2018	Program: SYN-004 / ribaxamase Record: Synthetic Biologics record Inventor(s): Andrew Bristol; Michael Kaleko; Sheila Connelly Link: https://patents.google.com/patent/US10105322
P-3.19	U.S. 10,087,433 Beta-lactamases with improved properties for therapy	Filed: May 30, 2018 Issued: Oct. 2, 2018	Program: SYN-004 / ribaxamase Record: Synthetic Biologics record Inventor(s): Michael Kaleko; Sheila Connelly Link: https://patents.google.com/patent/US10087433
P-3.20	U.S. 10,046,035 Methods and compositions for inhibiting or preventing adverse effects of oral antibiotics	Filed: July 5, 2017 Issued: Aug. 14, 2018	Program: SYN-004 / microbiome Record: Synthetic Biologics record Inventor(s): Michael Kaleko; Sheila Connelly; Vincent John Wachter Link: https://patents.google.com/patent/US10046035
P-3.21	U.S. 10,041,056 Modified beta-lactamases and methods and uses related thereto	Filed: Jan. 27, 2017 Issued: Aug. 7, 2018	Program: SYN-004 / ribaxamase Record: Synthetic Biologics record Inventor(s): Pertti Koski; Ulla Airaksinen; Katja Valimaki Link: https://patents.google.com/patent/US10041056
P-3.22	U.S. 10,035,846 Humanized pertussis antibodies and uses thereof	Filed: Nov. 18, 2016 Issued: July 31, 2018	Program: Legacy / pertussis Record: University of Texas System and Synthetic Biologics co-assignee record Inventor(s): Jennifer Maynard; Annalee Nguyen; Eduardo Padlan; Ellen Wagner Link: https://patents.google.com/patent/US10035846
P-3.23	U.S. 10,011,824 Beta-lactamases with improved properties for therapy	Filed: Aug. 10, 2017 Issued: July 3, 2018	Program: SYN-004 / ribaxamase Record: Synthetic Biologics record Inventor(s): Michael Kaleko; Sheila Connelly Link: https://patents.google.com/patent/US10011824

Ex.	Patent and title	Dates	Program, public record, and inventors
P-3.24	U.S. 9,956,292 Anti-methanogenic compositions and uses thereof	Filed: Aug. 13, 2015 Issued: May 1, 2018	Program: Legacy / IBS Record: Cedars-Sinai and Synthetic Biologics co-assignee record Inventor(s): Mark Pimentel; Ruchi Mathur; Steve Kanzer; Vince Wachter Link: https://patents.google.com/patent/US9956292
P-3.25	U.S. 9,783,797 Beta-lactamases with improved properties for therapy	Filed: June 2, 2017 Issued: Oct. 10, 2017	Program: SYN-004 / ribaxamase Record: Synthetic Biologics record Inventor(s): Michael Kaleko; Sheila Connelly Link: https://patents.google.com/patent/US9783797
P-3.26	U.S. 9,765,320 Modified beta-lactamases and methods and uses related thereto	Filed: Apr. 26, 2016 Issued: Sept. 19, 2017	Program: SYN-004 / ribaxamase Record: Synthetic Biologics record Inventor(s): Pertti Koski; Ulla Airaksinen; Katja Valimaki Link: https://patents.google.com/patent/US9765320
P-3.27	U.S. 9,744,221 Method and compositions for inhibiting or preventing adverse effects of oral antibiotics	Filed: Dec. 23, 2015 Issued: Aug. 29, 2017	Program: SYN-004 / microbiome Record: Synthetic Biologics record Inventor(s): Michael Kaleko; Sheila Connelly; Vincent John Wachter Link: https://patents.google.com/patent/US9744221
P-3.28	U.S. 9,695,409 Beta-lactamases with improved properties for therapy	Filed: Aug. 24, 2016 Issued: July 4, 2017	Program: SYN-004 / ribaxamase Record: Synthetic Biologics record Inventor(s): Michael Kaleko; Sheila Connelly Link: https://patents.google.com/patent/US9695409
P-3.29	U.S. 9,587,234 Modified beta-lactamases and methods and uses related thereto	Filed: Feb. 26, 2016 Issued: Mar. 7, 2017	Program: SYN-004 / ribaxamase Record: Synthetic Biologics record Inventor(s): Pertti Koski; Ulla Airaksinen; Katja Valimaki Link: https://patents.google.com/patent/US9587234
P-3.30	U.S. 9,512,204 Humanized pertussis antibodies and uses thereof	Filed: Mar. 31, 2015 Issued: Dec. 6, 2016	Program: Legacy / pertussis Record: University of Texas System and Synthetic Biologics co-assignee record Inventor(s): Jennifer Maynard; Annalee Nguyen; Eduardo Padlan; Ellen Wagner Link: https://patents.google.com/patent/US9512204
P-3.31	U.S. 9,464,280 Beta-lactamases with improved properties for therapy	Filed: July 1, 2016 Issued: Oct. 11, 2016	Program: SYN-004 / ribaxamase Record: Synthetic Biologics record Inventor(s): Michael Kaleko; Sheila Connelly Link: https://patents.google.com/patent/US9464280
P-3.32	U.S. 9,404,103 Beta-lactamases with improved properties for therapy	Filed: May 20, 2016 Issued: Aug. 2, 2016	Program: SYN-004 / ribaxamase Record: Synthetic Biologics record Inventor(s): Michael Kaleko; Sheila Connelly Link: https://patents.google.com/patent/US9404103
P-3.33	U.S. 9,376,673 Beta-lactamases with improved properties for therapy	Filed: Feb. 9, 2016 Issued: June 28, 2016	Program: SYN-004 / ribaxamase Record: Synthetic Biologics record Inventor(s): Michael Kaleko; Sheila Connelly Link: https://patents.google.com/patent/US9376673
P-3.34	U.S. 9,301,996 Modified beta-lactamases and methods and uses related thereto	Filed: Apr. 1, 2015 Issued: Apr. 5, 2016	Program: SYN-004 / ribaxamase Record: Synthetic Biologics record Inventor(s): Pertti Koski; Ulla Airaksinen; Katja Valimaki Link: https://patents.google.com/patent/US9301996

Ex.	Patent and title	Dates	Program, public record, and inventors
P-3.35	U.S. 9,301,995 Modified beta-lactamases and methods and uses related thereto	Filed: Oct. 17, 2014 Issued: Apr. 5, 2016	Program: SYN-004 / ribaxamase Record: Synthetic Biologics record Inventor(s): Pertti Koski; Ulla Airaksinen; Katja Valimaki Link: https://patents.google.com/patent/US9301995
P-3.36	U.S. 9,290,754 Beta-lactamases with improved properties for therapy	Filed: Apr. 17, 2015 Issued: Mar. 22, 2016	Program: SYN-004 / ribaxamase Record: Form 10-K and Google patent record; 2023 name-change entry Inventor(s): Michael Kaleko; Sheila Connelly Link: https://patents.google.com/patent/US9290754
P-3.37	U.S. 9,034,602 Modified beta-lactamases and methods and uses related thereto	Filed: May 17, 2011 Issued: May 19, 2015	Program: SYN-004 / ribaxamase Record: Synthetic Biologics record Inventor(s): Pertti Koski; Ulla Airaksinen; Katja Valimaki Link: https://patents.google.com/patent/US9034602
P-3.38	U.S. 8,894,994 Modified beta-lactamases and methods and uses related thereto	Filed: Oct. 7, 2013 Issued: Nov. 25, 2014	Program: SYN-004 / ribaxamase Record: Synthetic Biologics record Inventor(s): Pertti Koski; Ulla Airaksinen; Katja Valimaki Link: https://patents.google.com/patent/US8894994

Exhibit P-4 — U.S. grant located under VCN Biosciences, S.L. (1)

Ex.	Patent and title	Dates	Program, public record, and inventors
P-4.1	U.S. 11,000,560 Oncolytic adenoviruses with mutations in immunodominant adenovirus epitopes and their use in cancer treatment	Filed: May 4, 2016 Issued: May 11, 2021	Program: VCN oncolytic-virus platform Record: VCN Biosciences, S.L. assignee record Inventor(s): Ramon Alemany; Raul Gil; Miriam Bazan Link: https://patents.google.com/patent/US1100560

Exhibit P-5 — Disclosed controlled or licensed rights not represented as Theriva-owned

Right	Publicly disclosed ownership/control	Necessary limitation
U.S. Patent No. 10,316,065, Oncolytic adenoviruses for treating cancer	The public patent record identifies IDIBELL and Institut Catala d'Oncologia as assignees. Theriva's 10-K identifies the patent as an example within VCN's assigned-or-exclusively-licensed estate.	Treat as third-party-owned and licensed/controlled unless a later recorded assignment proves otherwise.
Spanish patent P200901201, Oncolytic adenoviruses for treating cancer	The 10-K states VCN obtained an exclusive license from IDIBELL in 2010 and that the patent is co-owned by IDIBELL and ICO.	The license includes royalty, termination, utilization, and maintenance provisions; the full agreement and current compliance must be reviewed.
Spanish patent P200700665 and EP-family priority application 14 38 2162.7	The 10-K describes additional VCN licenses from ICO and IDIBELL/ICO.	Patent-family members, present owners, territories, status, payment duties, and change-of-control terms require the complete schedules and agreements.

Exhibit P-6 — C. difficile patent and company-announcement chronology

Date	Documented event	What it establishes—and does not
Mar. 1, 2016	Synthetic Biologics announced three USPTO Notices of Allowance covering SYN-004-related composition and methods, including prevention of CDI and antibiotic-associated diarrhea.	The Company publicized patent allowances. A notice of allowance is not an issued patent, FDA approval, or proof of value.
July 12, 2016	The Company announced U.S. Patent No. 9,376,673 and a Notice of Allowance for U.S. application No. 15/160,669.	The release identified the allowed application before issuance. It could not yet name the later final patent number.

Date	Documented event	What it establishes—and does not
Aug. 2, 2016	Application No. 15/160,669 issued as U.S. Patent No. 9,404,103 B1. Its summary states that the improved beta-lactamases have uses “including the prevention or treatment of CDI and/or a C. difficile-associated disease.”	This is patent language describing claimed technology. Prevention is a clinically meaningful therapeutic objective, but CDI is an infection that can cause colitis; it is not itself an ulcer. The patent does not call SYN-004 or CDI a “cure.” Its only “Cure” reference is the title of a cited article about a duodenal ulcer and Helicobacter pylori. Patent language is not FDA approval or proof of clinical efficacy.
Post-issuance search	The reviewed Theriva/Synthetic release archive and web sources did not locate a separate company release after August 2, 2016 that names final U.S. Patent No. 9,404,103.	Absence from the reviewed sources is not proof that no communication existed. The Company should identify any release, investor communication, or decision record concerning post-issuance publicity.
Jan. 5 and May 11, 2017	The Company later reported a 412-patient Phase 2b result and FDA Breakthrough Therapy Designation for prevention of CDI.	Those later clinical/regulatory disclosures are separate from patent issuance and do not transform the patent itself into approval.

Exhibit P-7 — Complete C. difficile patent attached separately

The separate Exhibit P-7 PDF reproduces all 52 pages of U.S. Patent No. 9,404,103 B1. Yellow highlighting is limited to the two-line statement on patent page 2 that the improved beta-lactamases have therapeutic uses “including the prevention or treatment of CDI and/or a C. difficile-associated disease.” The exhibit cover explains that the patent does not use “cure” to describe SYN-004 or CDI and that patent language is not a regulatory approval or efficacy finding. No other complete patent is attached; the remaining patent records are supplied by number and public link in this schedule.

Exhibit P-8 — Focused production needed to close the public-record gaps

- a complete patent-and-application register from January 1, 2016 to the present, organized by program and patent family, listing every jurisdiction, application and patent number, owner, beneficial owner, chain of title, inventors, priority, filing, issue, expiration, maintenance or annuity status, terminal disclaimer, lapse, abandonment, revival, security interest, assignment, license, sublicense, field restriction, territory, reversion right, and enforcement or challenge;
- the complete patent schedules, assignment instruments, recorded title documents, maintenance records, and amendments for the Rasayana license, the VCN acquisition, and every IDIBELL, ICO, Hospital Sant Joan de Deu, Cedars-Sinai, and University of Texas agreement reflected in the public record;
- all patent-specific valuations, impairment analyses, sale or license proposals, bidder or counterparty diligence, board or committee materials, and banker presentations used in any financing, restructuring, strategic review, merger, reverse merger, asset transfer, or license analysis;
- a program-by-program reconciliation showing which patents support SYN-004, SYN-020, VCN-01, Albumin Shield, or a discontinued legacy program, and identifying who paid prosecution and maintenance costs and where those costs were recorded; and
- the investor-relations and disclosure-decision record for patent allowances, issuances, lapses, abandonments, assignments, licenses, and sublicenses, including application No. 15/160,669 and U.S. Patent No. 9,404,103.

Requested treatment and evidentiary guardrails

1. Use this schedule only as an organizational index to identified public records. A patent document may be capable of judicial notice for what it says and when it issued, but disputed ownership, maintenance, commercial value, materiality, and scientific effect require proper evidence and, where disputed, testimony.
2. If the Court finds the subject proportional and relevant to the pleaded warrant- and disclosure-related issues, Plaintiff requests production of the focused records above through the ordinary Rule 26 process or a specifically supported expedited order. This appendix is not itself a subpoena, title opinion, or proof of wrongdoing.
3. No adverse inference should be drawn merely because a patent number was not individually named in a Form 10-K, a press release was not located, a patent belongs to a continuation family, or a company used an exclusive license instead of ownership. The requested records are intended to determine the facts.

Source record

1. [Theriva Form 10-K for the year ended Dec. 31, 2025, Intellectual Property and collaboration disclosures](#)
2. [Public assignee index — Theriva Biologics, Inc.](#)
3. [Public assignee index — Synthetic Biologics, Inc., page 1](#)
4. [Public assignee index — Synthetic Biologics, Inc., page 2](#)
5. [Public assignee index — Synthetic Biologics, Inc., page 3](#)
6. [Public assignee index — VCN Biosciences, S.L.](#)
7. [U.S. Patent No. 9,290,754 public patent record](#)
8. [U.S. Patent No. 11,236,319 public patent record](#)
9. [U.S. Patent No. 10,316,065 public patent record](#)
10. [USPTO MPEP § 301, ownership and assignability](#)
11. [U.S. Patent No. 9,404,103 B1 public patent text and downloadable PDF](#)
12. [Synthetic Biologics Mar. 1, 2016 SYN-004 patent-allowance release](#)
13. [Synthetic Biologics July 12, 2016 ribaxamase patent release](#)
14. [Synthetic Biologics Jan. 5, 2017 SEC-filed SYN-004 Phase 2b results release](#)
15. [Synthetic Biologics May 11, 2017 FDA Breakthrough Therapy Designation release](#)

Respectfully submitted,

Dated: _____

Signature: _____

Mark Joseph Nejme, Plaintiff, self-represented

**UNITED STATES DISTRICT COURT
DISTRICT OF NEVADA**

MARK JOSEPH NEJMEH, Plaintiff, v. THERIVA BIOLOGICS, INC., Defendant.	Case No. 3:26-cv-00705
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SUPPLEMENTAL EXHIBIT SP

**United States Patent No 9404103 B1 Complete Highlighted C difficile
Patent**

Former Exhibit P 7

The following pages are separately paginated under this supplemental exhibit identifier.



US009404103B1

(12) **United States Patent**
Kaleko et al.

(10) **Patent No.:** **US 9,404,103 B1**
(45) **Date of Patent:** ***Aug. 2, 2016**

(54) **BETA-LACTAMASES WITH IMPROVED PROPERTIES FOR THERAPY**

FOREIGN PATENT DOCUMENTS

(71) Applicant: **Synthetic Biologics, Inc.**, Rockville, MD (US)

(72) Inventors: **Michael Kaleko**, Rockville, MD (US);
Sheila Connelly, Rockville, MD (US)

(73) Assignee: **Synthetic Biologics, Inc.**, Rockville, MD (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

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WO	WO 2008/065247	A1	6/2008
WO	2011148041		12/2011

(21) Appl. No.: **15/160,669**

(22) Filed: **May 20, 2016**

Related U.S. Application Data

(63) Continuation of application No. 15/019,474, filed on Feb. 9, 2016, now Pat. No. 9,376,673, which is a continuation of application No. 14/689,877, filed on Apr. 17, 2015, now Pat. No. 9,290,754.

(60) Provisional application No. 61/980,844, filed on Apr. 17, 2014, provisional application No. 62/046,627, filed on Sep. 5, 2014.

(51) **Int. Cl.**
C12N 9/86 (2006.01)
A61K 45/06 (2006.01)
A61K 9/00 (2006.01)
A61K 31/54 (2006.01)

(52) **U.S. Cl.**
CPC **C12N 9/86** (2013.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

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(Continued)

Primary Examiner — Suzanne M Noakes

(74) Attorney, Agent, or Firm — Morgan, Lewis & Bockius LLP

(57) **ABSTRACT**

This invention relates to, in part, compositions of beta-lactamases and methods of using these enzymes in, for example, gastrointestinal tract (GI tract) disorders such as *C. difficile* infection (CDI).

29 Claims, 8 Drawing Sheets

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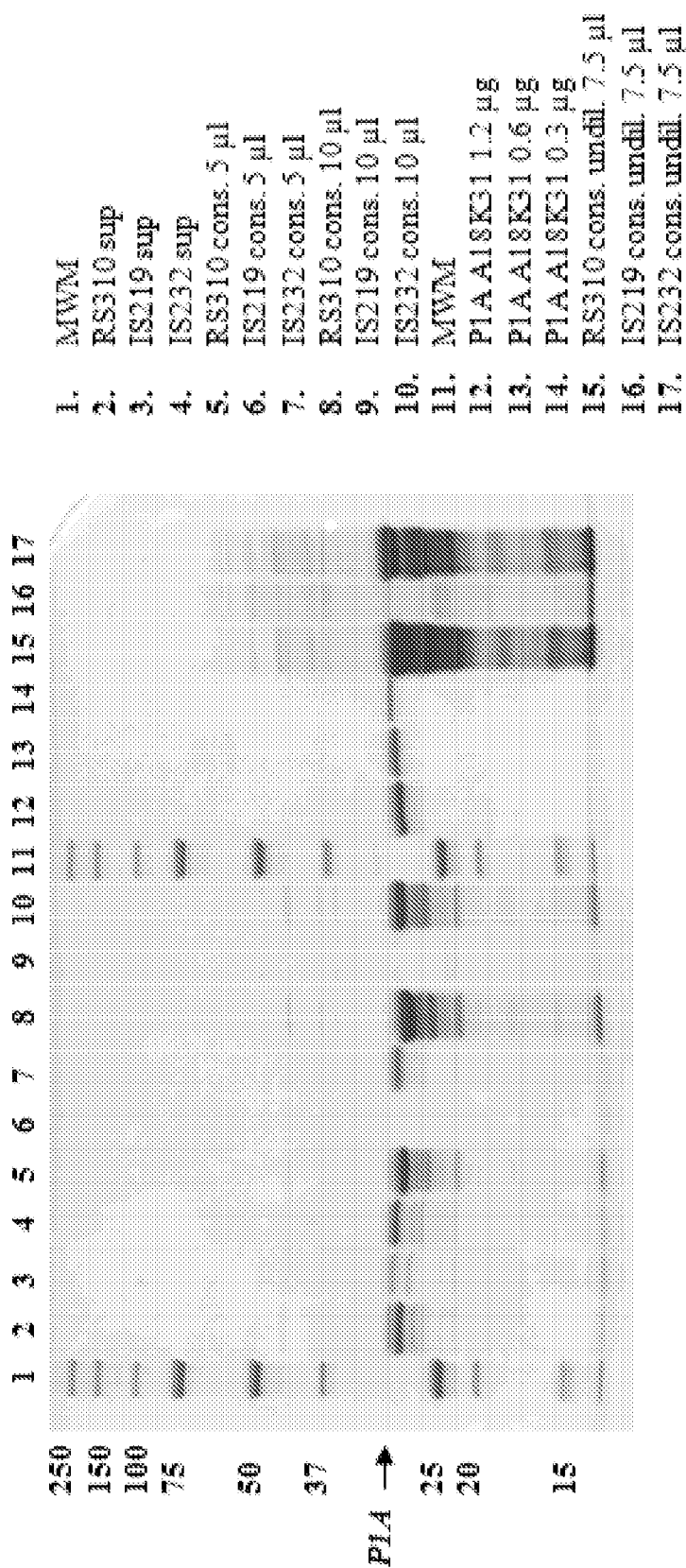


FIG. 1

1. MWM
2. RS310 9.5h
3. IS219 9.5h
4. IS232 9.5h
5. RS310 10.5h
6. IS219 10.5h
7. IS232 10.5h
8. RS310 11h
9. IS219 11h
10. IS232 11h
11. RS310 25h
12. IS219 25h
13. IS232 25h
14. RS310 27.5h
15. IS219 27.5h
16. IS232 27.5h
17. MWM

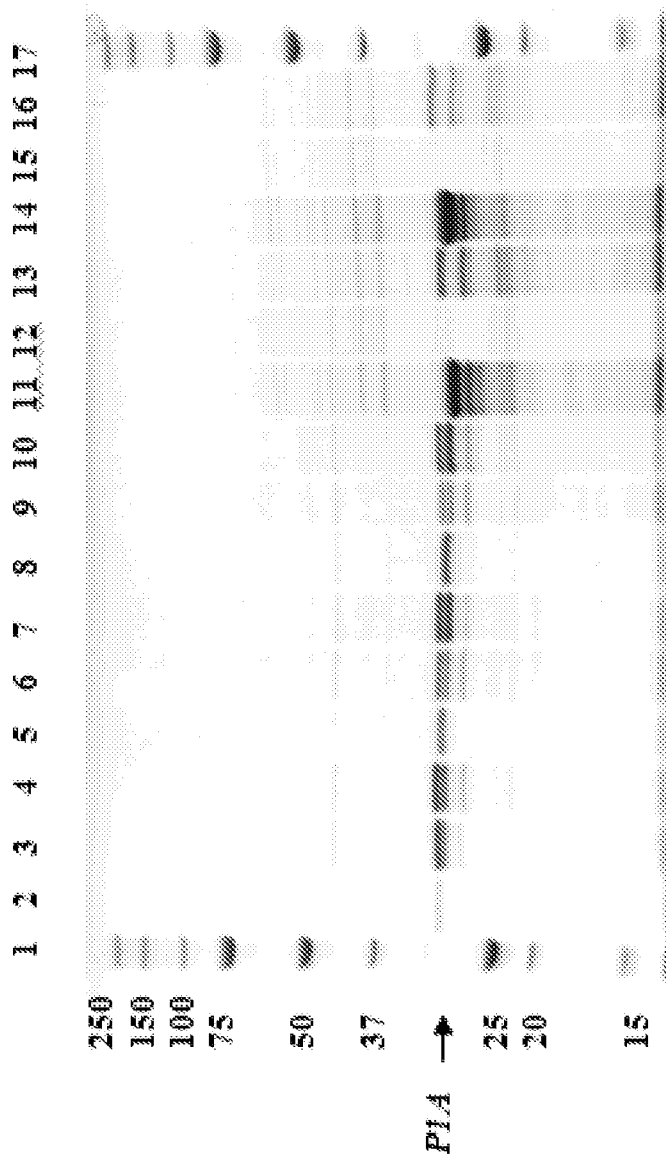


FIG. 2

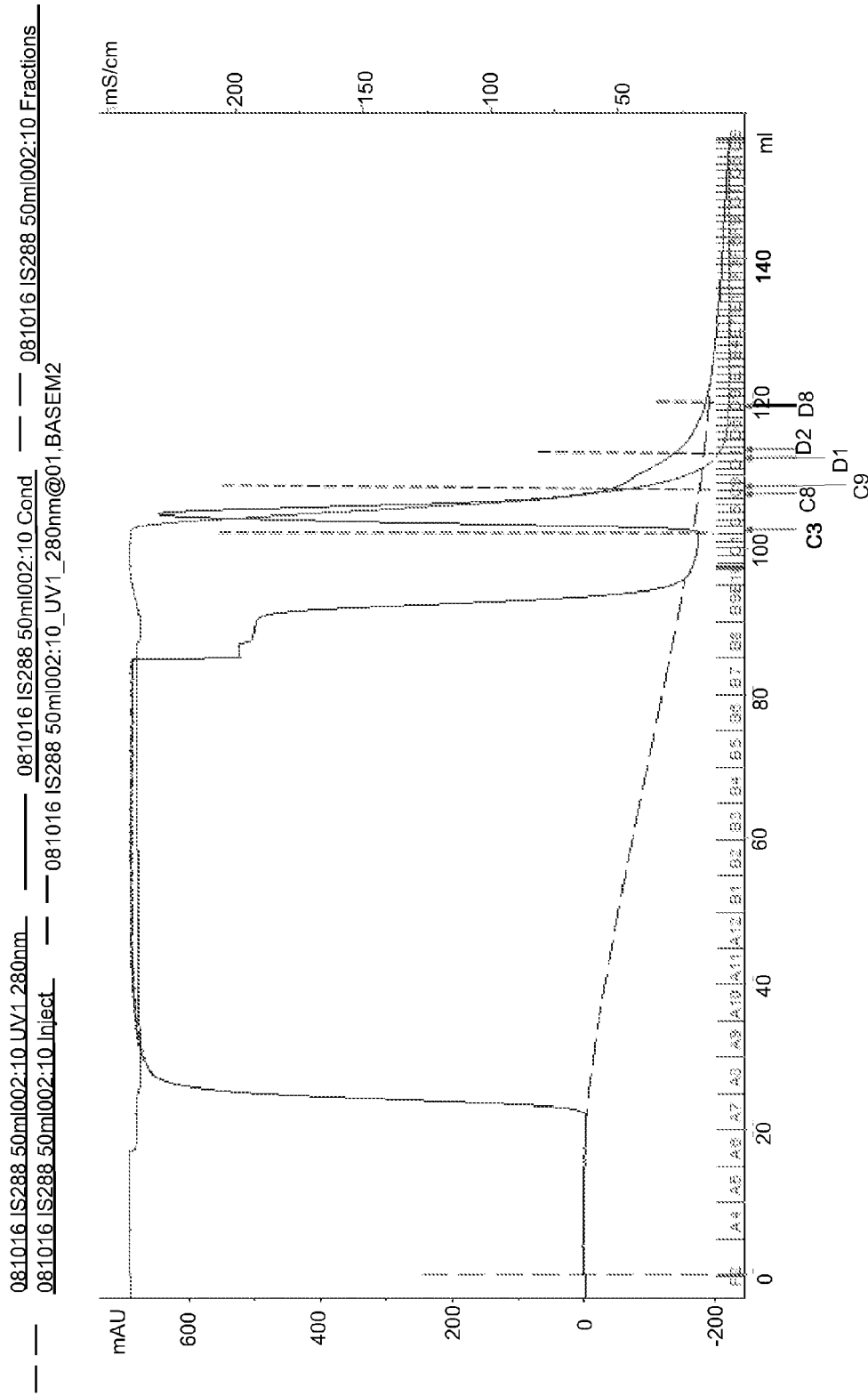


FIG. 3

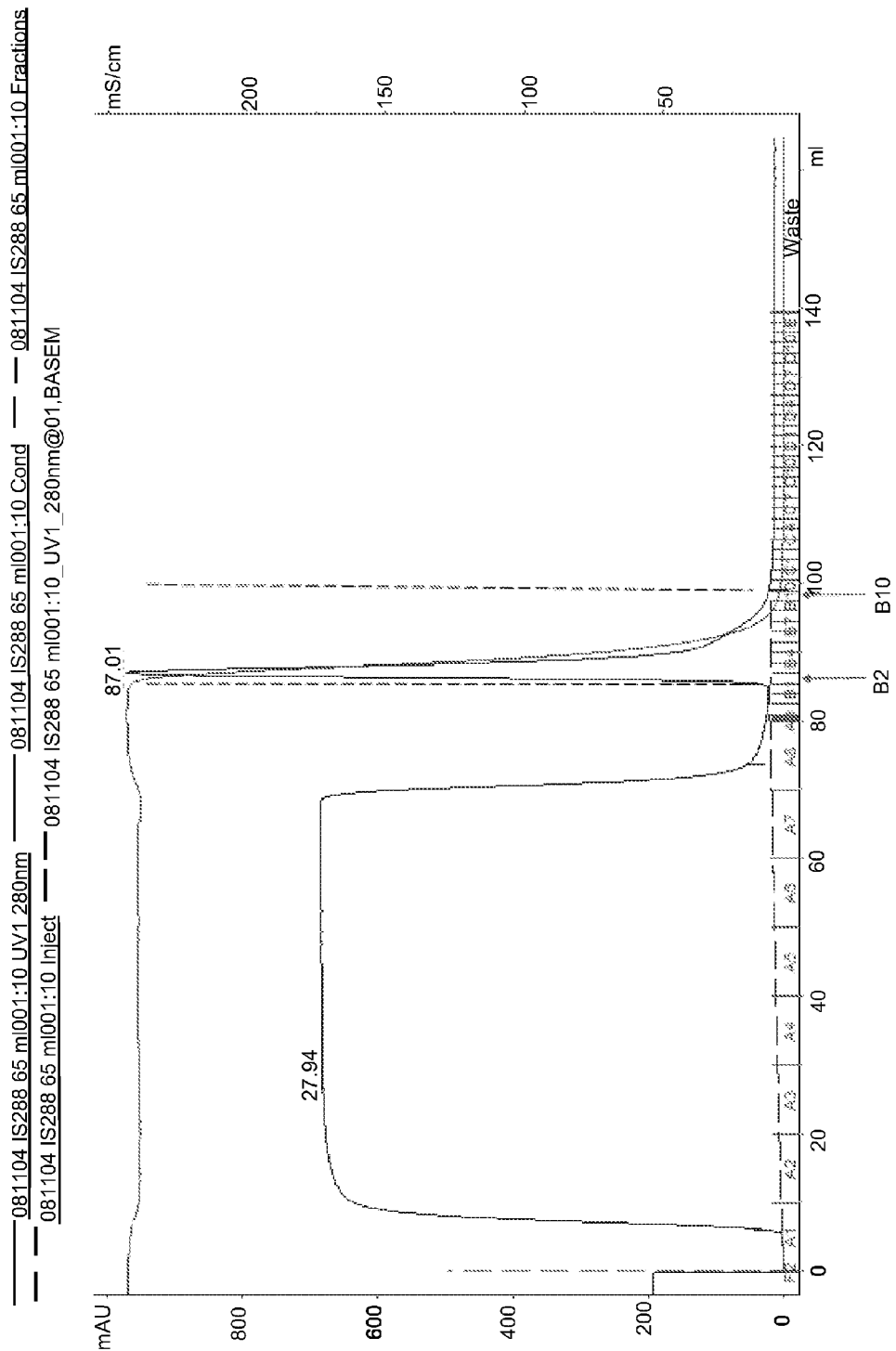


FIG. 4

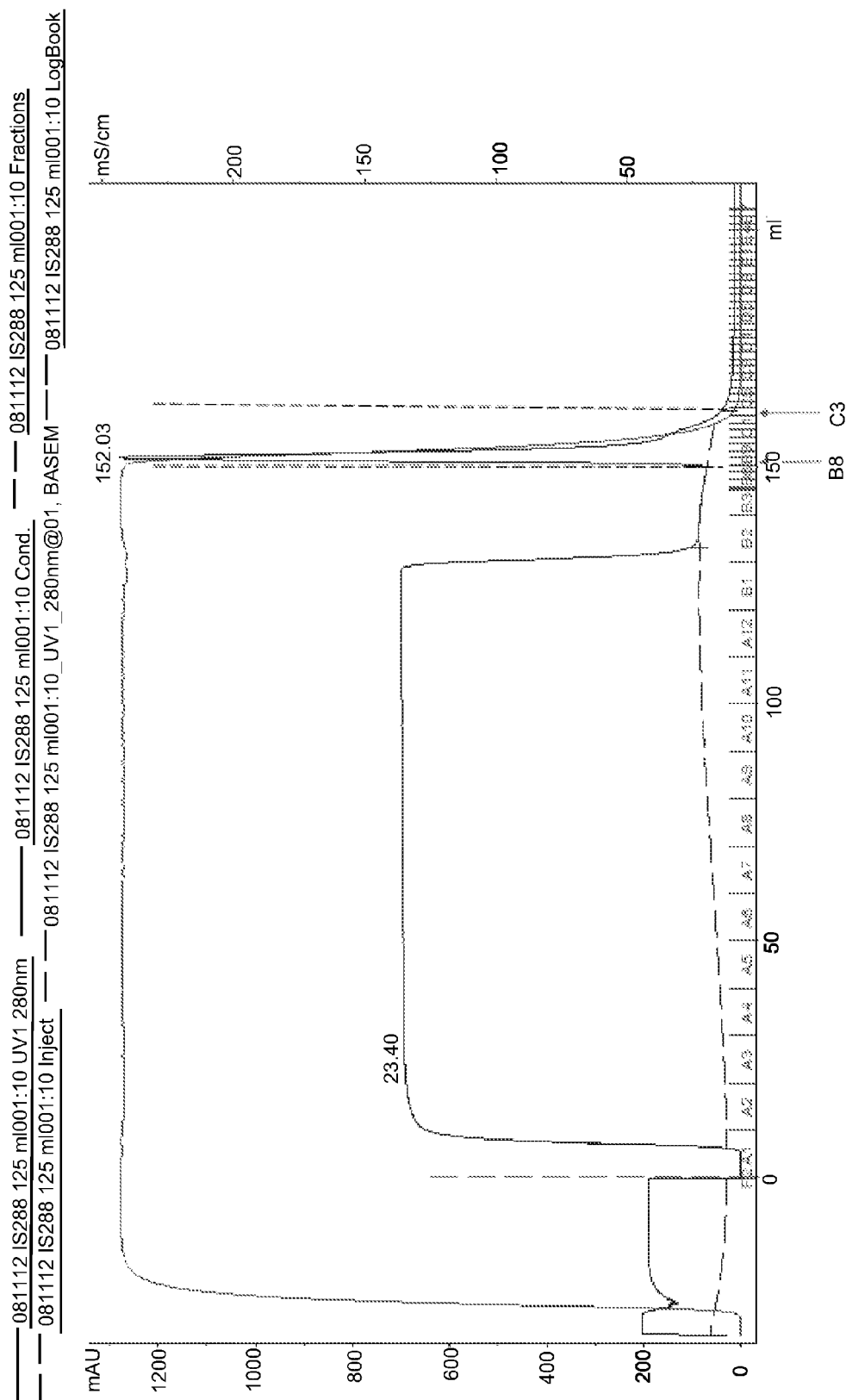


FIG. 5

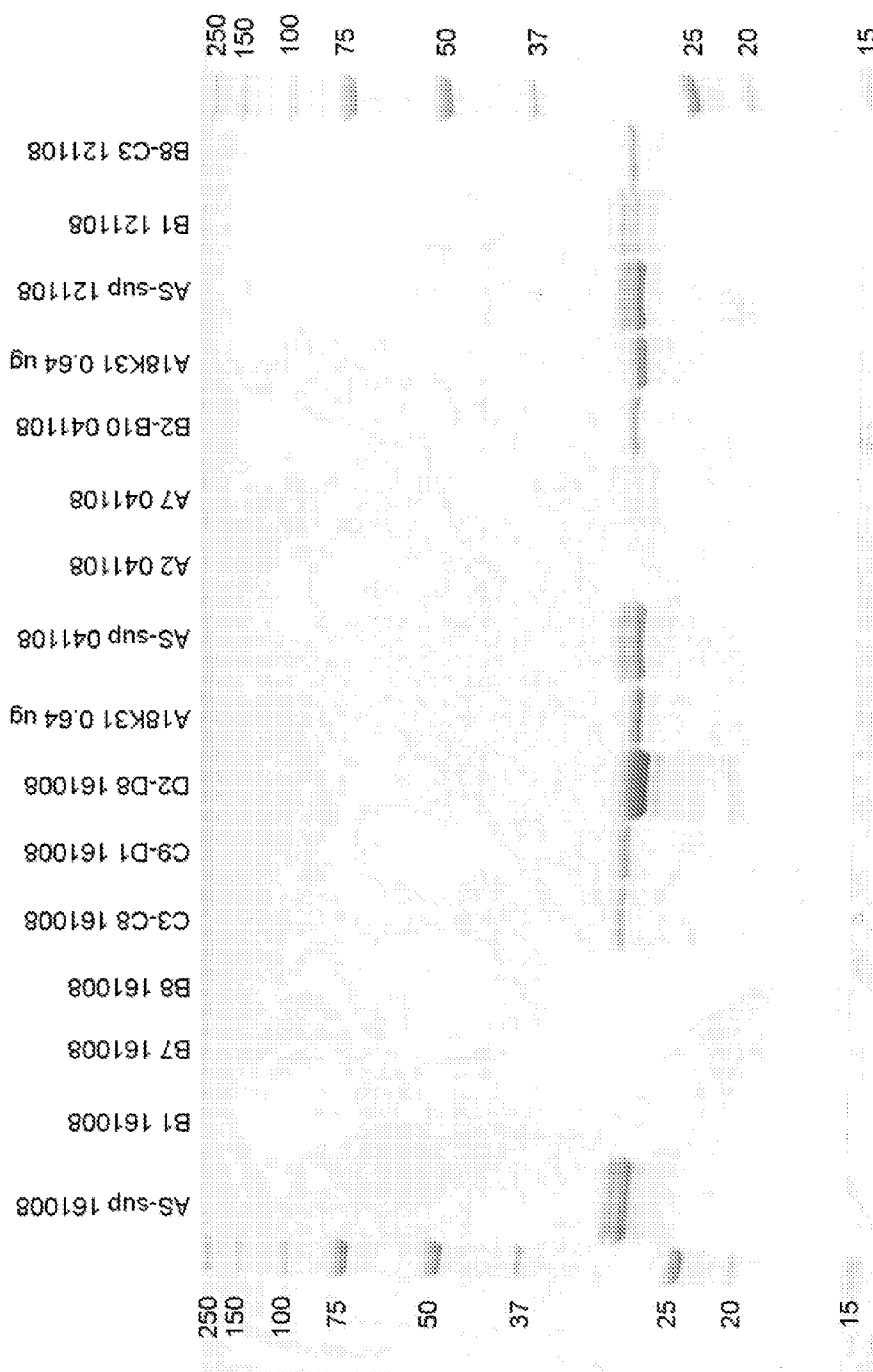


FIG. 6

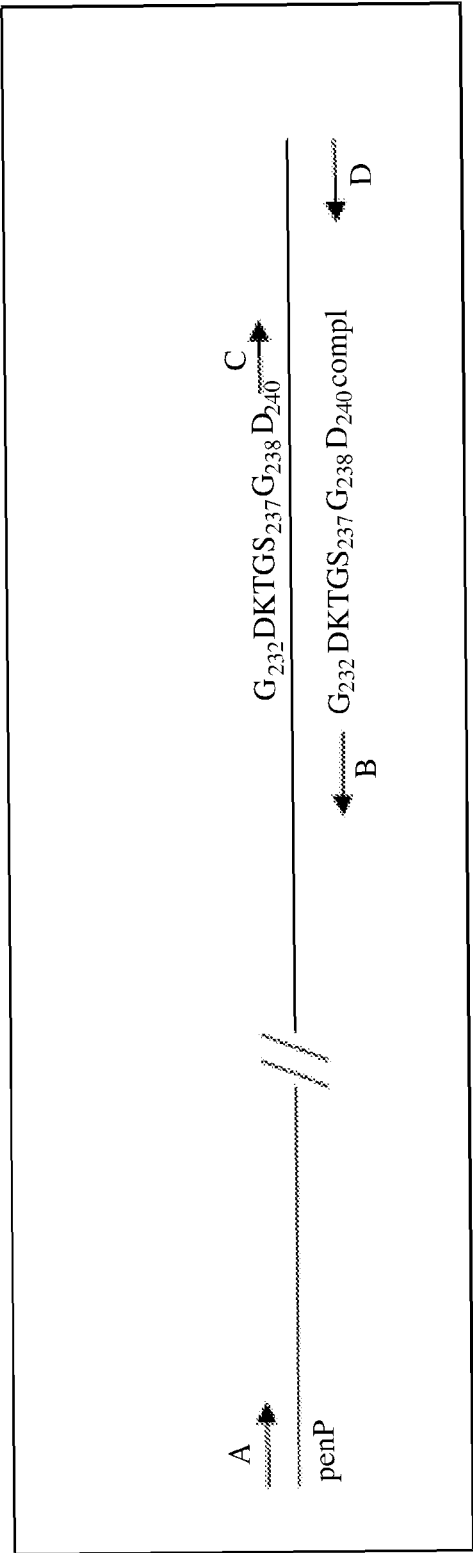


FIG. 7

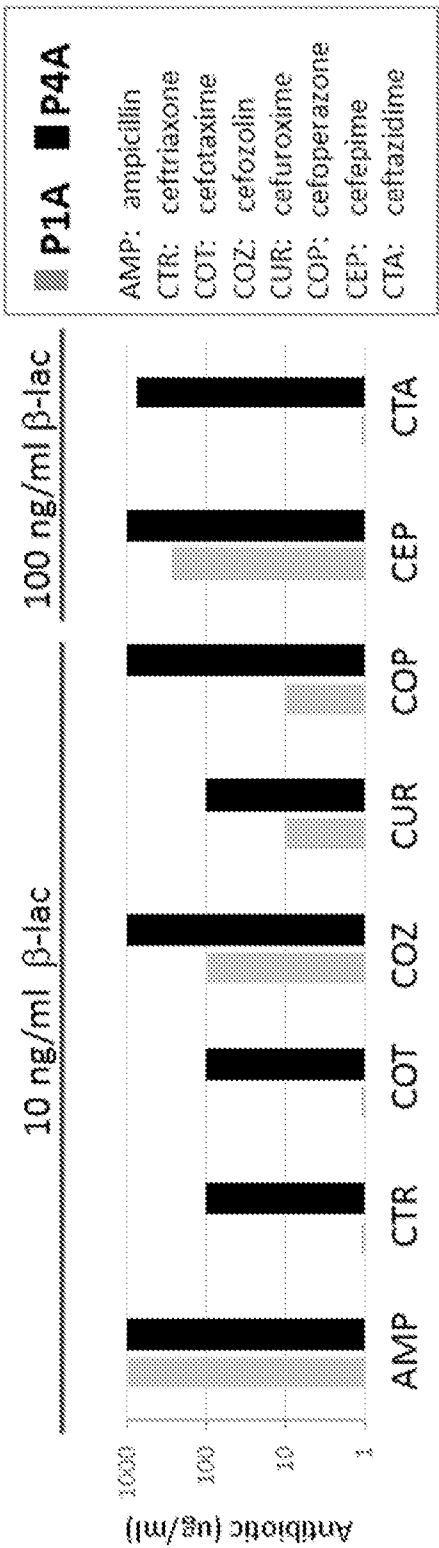


FIG. 8

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BETA-LACTAMASES WITH IMPROVED PROPERTIES FOR THERAPY

RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 15/019,474, filed Feb. 9, 2016 and U.S. patent application Ser. No. 14/689,877, filed Apr. 17, 2015 (now U.S. Pat. No. 9,290,754) and claims the benefit of U.S. Provisional Patent Application No. 61/980,844, filed Apr. 17, 2014, and U.S. Provisional Patent Application No. 62/046,627, filed Sep. 5, 2014, both of which are hereby incorporated by reference in their entirety.

FIELD OF THE INVENTION

This invention relates to, in part, compositions of beta-lactamases and methods for using these enzymes in, for example, gastrointestinal tract (GI tract) disorders such as *C. difficile* infection (CDI).

BACKGROUND

Humans may be considered to be a “superorganism” which is a conglomerate of mammalian and microbial cells, with the latter estimated to outnumber the former by ten to one. This microbial component, and its microbial genetic repertoire, the microbiome, is roughly 100-times greater than that of the human host. Strikingly, despite this enormous diversity of foreign organisms, the human immune system generally maintains a state of synergy. This is particularly true of the distal GI tract, which houses up to 1000 distinct bacterial species and an estimated excess of 1×10^{14} microorganisms, and appears to be central in defining human host health status. Loss of the careful balance in the microbiome, especially in the GI tract, can lead to various diseases.

However, antibiotic medical treatments, which are needed to treat certain aspects of disease, can induce disruption in the microbiome, including in the GI tract, and lead to further disease. For instance, certain parentally administered beta-lactams like ampicillin, ceftriaxone, cefoperazone, and piperacillin are, in part, eliminated via biliary excretion into the proximal part of the small intestine (duodenum). Residual unabsorbed beta-lactams in the intestinal tract may cause an undesirable effect on the ecological balance of normal intestinal microbiota resulting in, for example, CDI, antibiotic-associated diarrhea, overgrowth of pathogenic bacteria such as vancomycin resistant enterococci (VRE), extended-spectrum beta-lactamase producing Gram-negative bacilli (ESBL), and fungi, and selection of antibiotic-resistance strains among both normal intestinal microbiota and potential pathogen bacteria.

One approach for avoiding or rebalancing the ecological balance of normal intestinal microbiota is the therapeutic use of beta-lactamases, for example, by inactivating excreted or unabsorbed antibiotics in the GI tract, thereby maintaining a normal intestinal microbiota and preventing its overgrowth with potentially pathogenic micro-organisms.

There remains a need for agents and medicaments for the treatment of gastrointestinal tract (GI tract) disorders such as *C. difficile* infection (CDI) which have enzymatic properties that are best suited for these uses.

SUMMARY OF THE INVENTION

Accordingly, the present invention provides for beta-lactamases comprising an amino acid sequence that is based on the

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P1A beta-lactamase but containing advantageous mutations that, for example, change substrate specificity and/or activity and provide for therapeutic advantages. For example, such mutants can have at least 70% sequence identity with SEQ ID NO: 1 and have one or more mutations at the following Ambler classification positions F33, Q135, G156, A232, A237, A238, S240, T243, R244, S266, and D276. Such beta-lactamases have the ability to hydrolyze both penicillins and cephalosporins and do so with enzymatically desirable characteristics, such as low K_M s and/or high V_{max} s for select antibiotic substrates.

These improved beta-lactamases find uses in a number of therapies, including the prevention or treatment of CDI and/or a *C. difficile*-associated disease or other antibiotic-induced adverse effects in the GI tract. For example, the beta-lactamases find use in allowing a patient to undergo antibiotic therapy while being protected against disease that could result from excess antibiotics negatively affecting the microbiome. Such use does not interfere with the systemic utility of the antibiotic. Rather, the beta-lactamases clear out excess antibiotic that may populate parts of the GI tract and, in doing so, prevent the disruption of the microbiota that is linked to the various disease states described herein.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 shows a 12% Bis-Tris Criterion XT SDS-PAGE (BioRad) gel with concentrated samples (see labels of the samples to the right of the gel) from a 10 g/l glucose *B. subtilis* growth. MWM=molecular weight marker Precision Plus (BioRad), RS310=P1A *B. subtilis* strain and P1A A18K31=P1A reference material. “P1A→” denotes the right size of P1A and the mutant proteins.

FIG. 2 shows a 12% Bis-Tris Criterion XT SDS-PAGE (BioRad) gel with supernatant samples (see labels of the samples to the right of the gel) from a 5 g/l glucose *B. subtilis* growth. 14 μ l of each supernatant was applied on the gel. MWM=molecular weight marker Precision Plus (BioRad), RS310=P1A *B. subtilis* strain and “P1A” denotes the right size of P1A and the mutant proteins.

FIG. 3 shows an elution diagram from a HIC run. The places of collected elution fractions are marked with arrows and black dashed lines (dashed lines farthest to the right) both perpendicular to the X-axis. Flow-through fractions A7-B8 can be easily seen on the diagram. The A_{280} graph is represented with a solid line (top line), conductivity of the eluted solution by a solid line (second line from the top). Sample injection is marked with a dashed line (dashed line farthest to the left) and fractions with multiple solid lines both perpendicular to the X-axis.

FIG. 4 shows an elution diagram from a HIC run. The places of collected elution fractions are marked with arrows and black dashed lines (dashed lines farthest to the right) both perpendicular to the X-axis. Flow-through fractions A1-A8 can be easily seen on the diagram. The A_{280} graph is represented with a solid line (second line from the top), conductivity of the eluted solution by a solid line (top line). Sample injection is marked with a dashed line (dashed line farthest to the left) and fractions with multiple solid lines both perpendicular to the X-axis.

FIG. 5 shows an elution diagram from a HIC run. The places of collected elution fractions are marked with arrows and black dashed lines (dashed lines farthest to the right) both perpendicular to the X-axis. Flow-through fractions A1-B1 can be easily seen on the diagram. The A_{280} graph is represented with a solid line (second line from the top), conductivity of the eluted solution by a solid line (top line). Sample

injection is marked with a dashed line (dashed line farthest to the left) and fractions with multiple solid lines both perpendicular to the X-axis. The section before zero at the X-axis represents the column equilibration with buffer A.

FIG. 6 shows a SDS-PAGE gel of the fractions of the three HIC purifications shown in FIGS. 3-5. AS-sup=ammonium sulphate filtrate used in the particular purification, i.e. starting material for purification. B1 161008, B7 161008, B8 161008, A2 041108, A7 041108, and B1 121108: flow through fractions of each of purifications. C3-C8 161008, C9-D1 161008, D2-D8 161008, B2-B10 041108, and B8-C3 121108: eluted protein peak, pooled fractions. A18K31 P1A: reference material, 0.64 µg/lane.

FIG. 7 demonstrates the SOE (spliced overlap extension) technique. The target amino acids of the B3 β-strand of P1A are so closely situated to each other that all four amino acids can be exchanged using the SOE technique. The region encompassing the exchanges contains 24 nucleotides (8 amino acid codons) and is included in the overlap extension part of two PCR primers. Mutated penP gene is first amplified by PCR in two parts, using primers A+B and C+D. The two PCR products are then combined and amplified using primers A and D, complementary to the 5' and 3' parts of the penP gene. Cloning sites are included in primers A and D. Once the construct is ready, further mutations can be added using the Multi kit. The amino acid sequences encoded by the overlapping primers (B and C) are shown.

FIG. 8 demonstrate antibiotic degradation activity of P1A (left bars) and P4A (right bars). Concentrations of 10 or 100 ng/ml of enzyme are used against the indicated antibiotics. The data are plotted as the highest antibiotic concentration that is degraded by the indicated beta-lactamase (the antibiotic concentrations used were 10, 100, or 1000 µg/ml).

DETAILED DESCRIPTION

The present invention is based, in part, on the discovery of certain mutant beta-lactamase enzymes that have desirable characteristics for use in therapy, including, for example, an ability to hydrolyze one or more of penicillins and cephalosporins.

In some aspects, the present invention provides for beta-lactamases, and pharmaceutical compositions of these beta-lactamases. In various aspects, the beta-lactamases and/or pharmaceutical compositions are derived by mutation from SEQ ID NO: 1 (*Bacillus licheniformis* PenP, i.e. P1A). Accordingly, in various aspects, the present invention provides for beta-lactamases, and/or pharmaceutical compositions which are P1A derivatives.

P1A was designed to inactivate parenterally administered penicillin group beta-lactams (e.g. penicillin, amoxicillin, ampicillin and piperacillin) with or without beta-lactamase inhibitors (e.g. tazobactam, sulbactam, clavulanic acid) excreted via biliary system (see International Patent Publications WO 1993/013795 and WO 2008/065247; Tarkkanen, A. M. et al., *Antimicrob Agents Chemother.* 53: 2455, the contents of which are hereby incorporated by reference in their entirety). The P1A enzyme is a recombinant form of *Bacillus licheniformis* 749/C small exo beta-lactamase (WO 2008/065247) which belongs to class A and is grouped to subgroup 2a in functional classification. *B. licheniformis* beta-lactamase and its P1A derivative are considered as penicillinases which have high hydrolytic capacity to degrade e.g. penicillin, ampicillin, amoxicillin or piperacillin and they are generally inhibited by active site-directed beta-lactamase inhibitors such as clavulanic acid, sulbactam or tazobactam. However, the P1A enzyme has a limited capacity to inactivate

beta-lactam antibiotics that belong to the cephalosporin or the carbapenem group. Because the employed beta-lactamases possess poor activity to cephalosporins, they cannot be applied in conjunction with parenteral cephalosporin therapy for inactivation of unabsorbed beta-lactam in the small intestinal tract. Therefore, in some embodiments, the beta-lactamases and/or pharmaceutical compositions of the present invention improve upon the properties of P1A, as described herein.

In various embodiments, the inventive beta-lactamases possess desirable characteristics, including, for example, having an ability to efficiently target a broad spectra of antibiotics. In various embodiments, the inventive beta-lactamases possess desirable enzyme kinetic characteristics. For example, in some embodiments, the beta-lactamases possess a low K_M for at least one cephalosporin, including, for example, a K_M of less than about 500 µM, or about 100 µM, or about 10 µM, or about 1 µM, or about 0.1 µM (100 nM), or about 0.01 µM (10 nM), or about 1 nM. For example, in some embodiments, the beta-lactamases possess a low K_M for at least one penicillin, including, for example, a K_M of less than about 500 µM, or about 100 µM, or about 10 µM, or about 1 µM, or about 0.1 µM (100 nM), or about 0.01 µM (10 nM), or about 1 nM. In various embodiments, the inventive beta-lactamases possess a high V_{max} for at least one cephalosporin, including, for example, V_{max} which is greater than about 100 s⁻¹, or about 1000 s⁻¹, or about 10000 s⁻¹, or about 100000 s⁻¹, or about 1000000 s⁻¹. In various embodiments, the inventive beta-lactamases possess a high V_{max} for at least one penicillin, including, for example, V_{max} which is greater than about 100 s⁻¹, or about 1000 s⁻¹, or about 10000 s⁻¹, or about 100000 s⁻¹, or about 1000000 s⁻¹. In various embodiments, the inventive beta-lactamases possess catalytic efficiency is greater than about 10⁶ M⁻¹s⁻¹ for at least one cephalosporin. In various embodiments, the inventive beta-lactamases possess catalytic efficiency is greater than about 10⁶ M⁻¹s⁻¹ for at least one penicillin. In various embodiments, the inventive beta-lactamases possess the desirable enzyme kinetic characteristics for at least one of either or both of cephalosporins and penicillins. In various embodiments, specific antibiotics (e.g. cephalosporins and penicillins are described herein).

Provided herein is the 263 amino acid sequence of the P1A enzyme (after removal of a 31 amino acid signal sequence and the QASKT (Gln-Ala-Ser-Lys-Thr) pentapeptide at the N terminus, see SEQ ID NO: 3). As described herein, mutations may be made to this sequence to generate the inventive beta-lactamases.

SEQ ID NO: 1

Glu Met Lys Asp Asp Phe Ala Lys Leu Glu Glu Gln
Phe Asp Ala Lys Leu Gly Ile Phe Ala Leu Asp Thr
Gly Thr Asn Arg Thr Val Ala Tyr Arg Pro Asp Glu
Arg Phe Ala Phe Ala Ser Thr Ile Lys Ala Leu Thr
Val Gly Val Leu Leu Gln Gln Lys Ser Ile Glu Asp
Leu Asn Gln Arg Ile Thr Tyr Thr Arg Asp Asp Leu
Val Asn Tyr Asn Pro Ile Thr Glu Lys His Val Asp
Thr Gly Met Thr Leu Lys Glu Leu Ala Asp Ala Ser
Leu Arg Tyr Ser Asp Asn Ala Ala Gln Asn Leu Ile
Leu Lys Gln Ile Gly Gly Pro Glu Ser Leu Lys Lys

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Glu Leu Arg Lys Ile Gly Asp Glu Val Thr Asn Pro
 Glu Arg Phe Glu Pro Glu Leu Asn Glu Val Asn Pro
 Gly Glu Thr Gln Asp Thr Ser Thr Ala Arg Ala Leu
 Val Thr Ser Leu Arg Ala Phe Ala Leu Glu Asp Lys
 Leu Pro Ser Glu Lys Arg Glu Leu Leu Ile Asp Trp
 Met Lys Arg Asn Thr Thr Gly Asp Ala Leu Ile Arg
 Ala Gly Val Pro Asp Gly Trp Glu Val Ala Asp Lys
 Thr Gly Ala Ala Ser Tyr Gly Thr Arg Asn Asp Ile
 Ala Ile Ile Trp Pro Pro Lys Gly Asp Pro Val Val
 Leu Ala Val Leu Ser Ser Arg Asp Lys Lys Asp Ala
 Lys Tyr Asp Asp Lys Leu Ile Ala Glu Ala Thr Lys
 Val Val Met Lys Ala Leu Asn Met Asn Gly Lys.

In some embodiments, SEQ ID NO: 1 may have a Met and/or Thr preceeding the first residue of the sequence. In various embodiments, the Met may be cleaved. As described herein, mutations may be made to the sequence comprising the Met and/or Thr preceeding the first residue to generate the inventive beta-lactamases.

Also provided herein is the 299 amino acid sequence of the P1A enzyme before removal of a 31 amino acid signal sequence and the QASKT (Gln-Ala-Ser-Lys-Thr) pentapeptide at the N terminus as SEQ ID NO: 3:

SEQ ID NO: 3

Met Ile Gln Lys Arg Lys Arg Thr Val Ser Phe Arg
 Leu Val Leu Met Cys Thr Leu Leu Phe Val Ser Leu
 Pro Ile Thr Lys Thr Ser Ala Gln Ala Ser Lys Thr
 Glu Met Lys Asp Asp Phe Ala Lys Leu Glu Glu Gln
 Phe Asp Ala Lys Leu Gly Ile Phe Ala Leu Asp Thr
 Gly Thr Asn Arg Thr Val Ala Tyr Arg Pro Asp Glu
 Arg Phe Ala Phe Ala Ser Thr Ile Lys Ala Leu Thr
 Val Gly Val Leu Leu Gln Gln Lys Ser Ile Glu Asp
 Leu Asn Gln Arg Ile Thr Tyr Thr Arg Asp Asp Leu
 Val Asn Tyr Asn Pro Ile Thr Glu Lys His Val Asp
 Thr Gly Met Thr Leu Lys Glu Leu Ala Asp Ala Ser
 Leu Arg Tyr Ser Asp Asn Ala Ala Gln Asn Leu Ile
 Leu Lys Gln Ile Gly Gly Pro Glu Ser Leu Lys Lys
 Glu Leu Arg Lys Ile Gly Asp Glu Val Thr Asn Pro
 Glu Arg Phe Glu Pro Glu Leu Asn Glu Val Asn Pro
 Gly Glu Thr Gln Asp Thr Ser Thr Ala Arg Ala Leu
 Val Thr Ser Leu Arg Ala Phe Ala Leu Glu Asp Lys
 Leu Pro Ser Glu Lys Arg Glu Leu Leu Ile Asp Trp
 Met Lys Arg Asn Thr Thr Gly Asp Ala Leu Ile Arg
 Ala Gly Val Pro Asp Gly Trp Glu Val Ala Asp Lys
 Thr Gly Ala Ala Ser Tyr Gly Thr Arg Asn Asp Ile

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Ala Ile Ile Trp Pro Pro Lys Gly Asp Pro Val Val
 Leu Ala Val Leu Ser Ser Arg Asp Lys Lys Asp Ala
 Lys Tyr Asp Asp Lys Leu Ile Ala Glu Ala Thr Lys
 Val Val Met Lys Ala Leu Asn Met Asn Gly Lys

Further, the present invention also provides for additional upstream residues from the first residue of SEQ ID NO: 1 (see, e.g., *JBC* 258 (18): 11211, 1983, the contents of which are hereby incorporated by reference—including the exo-large and exo-small versions of penP and penP1). Further, the present invention also provides for additional downstream residues from the last residue of SEQ ID NO: 1.

The polynucleotide sequence of P1A (after removal of a 31 amino acid signal sequence and the QAKST pentapeptide at the N terminus) is also provided as SEQ ID NO: 2. As described herein, mutations may be made to this sequence to generate the inventive beta-lactamases (including, taking into account degeneracy of the genetic code).

SEQ ID NO: 2

gagatgaaagatgattttgcaaaacttgaggacaatttgatgcaaaact
 cgggatctttgcattggatacaggtacaaacggacggttagcgtatcggc
 cggatgagcgttttgcctttgcttcgacgattaaggctttaactgtaggc
 gtgcttttgcaacagaaatcaatagaagatctgaaccagagaataacata
 tacacgtgatgatcttgtaaactacaacccgattacggaagacacgttg
 atacgggaatgacgctcaaagagcttgccgatgcttcgcttcgatatagt
 gacaatgcggcacagaatctcattcttaacaaattggcggacctgaaag
 tttgaaaaaggaaactgaggaagattggtgatgaggtacaaatcccgaaac
 gattcgaaaccagagttaaatgaagtgaatccgggtgaaactcaggatacc
 agtacagcaagagcacttgtcacaagccttcgagcctttgctcttgaaaga
 taaacttccaagtgaataacgcgagcttttaacgattggatgaaacgaa
 ataccactggagacgccttaacccgtgcccgggtgcccggacggttgggaa
 gtggctgataaaactggagcggcatcatatggaacccggaatgacattgc
 catcatttggccgcaaaaggagatcctgtcgttcttgacgtattatcca
 gcagggataaaaaggacgccaagtatgatgataaacttattgcagaggca
 acaaagggtggaatgaaagccttaaacatgaacggcaataa

Also provided is the polynucleotide sequence of P1A before the removal of a 31 amino acid signal sequence and the QASKT pentapeptide at the N terminus as SEQ ID NO: 4. As described herein, mutations may be made to this sequence to generate the inventive beta-lactamases (including, taking into account degeneracy of the genetic code).

SEQ ID NO: 4

atgattcaaaaacgaaagcggacagtttcgttcagacttgcttattgtg
 cacgctgttatttgcagtttgccgattacaaaaacatcagcgcaagctt
 ccaagacggagatgaaagatgattttgcaaaacttgaggacaatttgat
 gcaaaactcgggatctttgcattggatacaggtacaaacggacggttagc
 gtatcgccggatgagcgttttgcctttgcttcgacgattaaggctttaa

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ctgtaggcgtgcttttgcacagaaatcaatagaagatctgaaccagaga
 ataacatatacacgtgatgatcttgtaaaactacaacccgattacggaaaa
 gcacgttgatacgggaatgacgctcaaagagcttgccgatgcttcgcttc
 gatatagtgacaaatgcggcacagaatctcattcttaacaaattggcgga
 cctgaaagtgtgaaaaaggaactgaggaagattggtgatgaggttacaaa
 tcccgaaacgattcgaaccagagttaaatgaagtgaatccgggtgaaactc
 aggataccagtagcagaagagcacttggtcacaaagccttcgagccttgc
 ctggaagataaaactccaagtgaacacgcgagcttttaacgattggat
 gaaacgaaataaccactggagacgccttaacccgtgccggtgtgccggacg
 gttgggaagtggctgataaaactggagcggcatcatatggaacccggaat
 gacattgccatcatattggccgcaaaaaggagatcctgtcgttcttgcagt
 attatccagcaggataaaaaaggacgccaagtatgatgat aaacttatt
 gcagaggcaacaaaggtggtaatgaaagccttaaacatgaacggcgaata
 a

In some embodiments, the invention pertains to mutagenesis of a beta-lactamase (e.g. a class A beta-lactamase) to derive advantageous enzymes (e.g. those that can target a broad spectra of antibiotics). In some embodiments, the invention includes site-directed mutagenesis, random mutagenesis, and/or directed evolution approaches. In some embodiments, mutation design is based on, inter alia, structural data (e.g. crystal structure data, homolog models, etc.) of the following: P1A crystal structure (Knox and Moews, *J. Mol Biol.*, 220, 435-455 (1991)), CTX-M-44 (1BZA (Ibuka et al. *Journal of Molecular Biology* Volume 285, Issue 5 2079-2087 (1999)), 1IYS (Ibuka et al. *Biochemistry*, 2003, 42 (36): 10634-43), 1IYO, 1IYP and 1IYQ (Shimamura et al. 2002 *J. Biol. Chem.* 277:46601-08), *Proteus vulgaris* K1 (1HZO, Nugaka et al. *J Mol Biol.* 2002 Mar. 15;317(1):109-17) and *Proteus penneri* Huga (Liassine et al. *Antimicrob Agents Chemother.* 2002 January;46(1):216-9. 2002), and reviewed in Bonnet, *Antimicrob. Agents Chemother* 48(1): 1-14 (2004) (for CTM-X), the contents of all of these documents are hereby incorporated by reference in their entirety). In some embodiments, the present mutations are informed by analysis of structural data (e.g. crystal structure data, homolog models, etc.) of any one of the following beta-lactamases: P1A (see, e.g. U.S. Pat. No. 5,607,671, the contents of which are hereby incorporated by reference), P2A (see, e.g., WO 2007/147945, the contents of which are hereby incorporated by reference), P3A (see, e.g., WO 2011/148041, the contents of which are hereby incorporated by reference), CTX-M-3, CTX-M-4, CTX-M-5, CTX-M-9, CTX-M-10, CTX-M-14, CTX-M-15, CTX-M-16, CTX-M-18, CTX-M-19, CTX-M-25, CTX-M-26, CTX-M-27, CTX-M-32, CTX-M-44, CTX-M-45, and CTX-M-54. Such information is available to one skilled in the art at known databases, for example, Swiss-Prot Protein Sequence Data Bank, NCBI, and PDB.

In some embodiments, the invention pertains to one or more (e.g. about 1, or about 2, or about 3, or about 4, or about 5, or about 6, or about 7, or about 8, or about 9, or about 10, or about 15, or about 20, or about 30, or about 40, or about 50, or about 60, or about 70, or about 80, or about 90, or about 100, or about 110, or about 120, or about 130, or about 140, or about 150) mutations to SEQ ID NO: 1 or SEQ ID NO: 3 or

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a sequence with at least 30, 35, 40, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 99.5, 99.8, 99.9% identity to SEQ ID NO: 1 or SEQ ID NO: 3 (or about 70%, or about 75%, or about 80%, or about 85%, or about 90, or about 95%, or about 96%, or about 97%, or about 98%, or about 99% identity to SEQ ID NO: 1 or SEQ ID NO: 3). In various embodiments, one or more amino acid of SEQ ID NO: 1 or SEQ ID NO: 3 is substituted with a naturally occurring amino acid, such as a hydrophilic amino acid (e.g. a polar and positively charged hydrophilic amino acid, such as arginine (R) or lysine (K); a polar and neutral of charge hydrophilic amino acid, such as asparagine (N), glutamine (Q), serine (S), threonine (T), proline (P), and cysteine (C), a polar and negatively charged hydrophilic amino acid, such as aspartate (D) or glutamate (E), or an aromatic, polar and positively charged hydrophilic amino acid, such as histidine (H)) or a hydrophobic amino acid (e.g. a hydrophobic, aliphatic amino acid such as glycine (G), alanine (A), leucine (L), isoleucine (I), methionine (M), or valine (V), a hydrophobic, aromatic amino acid, such as phenylalanine (F), tryptophan (W), or tyrosine (Y) or a non-classical amino acid (e.g. selenocysteine, pyrrolysine, N-formylmethionine β -alanine, GABA and δ -Aminolevulinic acid. 4-Aminobenzoic acid (PABA), D-isomers of the common amino acids, 2,4-diaminobutyric acid, α -amino isobutyric acid, 4-aminobutyric acid, Abu, 2-amino butyric acid, γ -Abu, ϵ -Ahx, 6-amino hexanoic acid, Aib, 2-amino isobutyric acid, 3-amino propionic acid, ornithine, norleucine, norvaline, hydroxyproline, sarcosine, citrulline, homocitrulline, cysteine acid, t-butylglycine, t-butylalanine, phenylglycine, cyclohexylalanine, β -alanine, fluoramino acids, designer amino acids such as β methyl amino acids, C α -methyl amino acids, N α -methyl amino acids, and amino acid analogs in general).

In illustrative embodiments, inventive mutations include, but are not limited to one or more (e.g. about 1, or about 2, or about 3, or about 4, or about 5, or about 6, or about 7, or about 8, or about 9, or about 10, or about 15, or about 20, or about 30, or about 40, or about 50, or about 60, or about 70, or about 80, or about 90, or about 100, or about 110, or about 120, or about 130, or about 140, or about 150) of the following mutations to SEQ ID NO: 1 or SEQ ID NO: 3 or a sequence with at least 30, 35, 40, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 99.5, 99.8, 99.9% identity to SEQ ID NO: 1 or SEQ ID NO: 3 (or about 70%, or about 75%, or about 80%, or about 85%, or about 90, or about 95%, or about 96%, or about 97%, or about 98%, or about 99% identity to SEQ ID NO: 1 or SEQ ID NO: 3): Glu1A1a; Glu1Cys; Glu1Asp; Glu1Phe; Glu1Gly; Glu1His; Glu1Ile; Met1Lys; Glu1Leu; Glu1Met; Glu1Asn; Glu1Pro; Glu1Gln; Glu1Arg; Glu1Ser; Glu1Thr; Glu1Val; Glu1Trp; Glu1Tyr; Met2Ala; Met2Cys; Met2Asp; Met2Glu; Met2Phe; Met2Gly; Met2His; Met2Ile; Met1Lys; Met2Leu; Met2Asn; Met2Pro; Met2Gln; Met2Arg; Met2Ser; Met2Thr; Met2Val; Met2Trp; Met2Tyr; Lys3Ala; Lys3Cys; Lys3Asp; Lys3Glu; Lys3Phe; Lys3Gly; Lys3His; Lys3Ile; Lys3Leu; Lys3Met; Lys3Asn; Lys3Pro; Lys3Gln; Lys3Arg; Lys3Ser; Lys3Thr; Lys3Val; Lys3Trp; Lys3Tyr; Asp4Ala; Asp4Cys; Asp4Glu; Asp4Phe; Asp4Gly; Asp4His; Asp4Ile; Asp4Lys; Asp4Leu; Asp4Met; Asp4Asn; Asp4Pro; Asp4Gln; Asp4Arg; Asp4Ser; Asp4Thr; Asp4Val; Asp4Trp; Asp4Tyr; Asp5Ala; Asp5Cys; Asp5Glu; Asp5Phe; Asp5Gly; Asp5His; Asp5Ile; Asp5Lys; Asp5Leu; Asp5Met; Asp5Asn; Asp5Pro; Asp5Gln; Asp5Arg; Asp5Ser; Asp5Thr; Asp5Val; Asp5Trp; Asp5Tyr; Phe6Ala; Phe6Cys; Phe6Asp;

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Phe6Glu; Phe6Gly; Phe6His; Phe6Ile; Phe6Lys; Phe6Leu;
 Phe6Met; Phe6Asn; Phe6Pro; Phe6Gln; Phe6Arg; Phe6Ser;
 Phe6Thr; Phe6Val; Phe6Trp; Phe6Tyr; Ala7Cys; Ala7Asp;
 Ala7Glu; Ala7Phe; Ala7Gly; Ala7His; Ala7Ile; Ala7Lys;
 Ala7Leu; Ala7Met; Ala7Asn; Ala7Pro; Ala7Gln; Ala7Arg;
 Ala7Ser; Ala7Thr; Ala7Val; Ala7Trp; Ala7Tyr; Lys8Ala;
 Lys8Cys; Lys8Asp; Lys8Glu; Lys8Phe; Lys8Gly; Lys8His;
 Lys8Ile; Lys8Leu; Lys8Met; Lys8Asn; Lys8Pro; Lys8Gln;
 Lys8Arg; Lys8Ser; Lys8Thr; Lys8Val; Lys8Trp; Lys8Tyr;
 Leu9Ala; Leu9Cys; Leu9Asp; Leu9Glu; Leu9Phe; Leu9Gly;
 Leu9His; Leu9Ile; Leu9Lys; Leu9Met; Leu9Asn; Leu9Pro;
 Leu9Gln; Leu9Arg; Leu9Ser; Leu9Thr; Leu9Val; Leu9Trp;
 Leu9Tyr; Glu10Ala; Glu10Cys; Glu10Asp; Glu10Phe;
 Glu10Gly; Glu10His; Glu10Ile; Glu10Lys; Glu10Leu;
 Glu10Met; Glu10Asn; Glu10Pro; Glu10Gln; Glu10Arg;
 Glu10Ser; Glu10Thr; Glu10Val; Glu10Trp; Glu10Tyr;
 Glu11Ala; Glu11Cys; Glu11Asp; Glu11Phe; Glu11Gly;
 Glu11His; Glu11Ile; Glu11Lys; Glu11Leu; Glu11Met;
 Glu11Asn; Glu11Pro; Glu11Gln; Glu11Arg; Glu11Ser;
 Glu11Thr; Glu11Val; Glu11Trp; Glu11Tyr; Gln12Ala;
 Gln12Cys; Gln12Asp; Gln12Glu; Gln12Phe; Gln12Gly;
 Gln12His; Gln12Ile; Gln12Lys; Gln12Leu; Gln12Met;
 Gln12Asn; Gln12Pro; Gln12Arg; Gln12Ser; Gln12Thr;
 Gln12Val; Gln12Trp; Gln12Tyr; Phe13Ala; Phe13Cys;
 Phe13Asp; Phe13Glu; Phe13Gly; Phe13His; Phe13Ile;
 Phe13Lys; Phe13Leu; Phe13Met; Phe13Asn; Phe13Pro;
 Phe13Gln; Phe13Arg; Phe13Ser; Phe13Thr; Phe13Val;
 Phe13Trp; Phe13Tyr; Asp14Ala; Asp14Cys; Asp14Glu;
 Asp14Phe; Asp14Gly; Asp14His; Asp14Ile; Asp14Lys;
 Asp14Leu; Asp14Met; Asp14Asn; Asp14Pro; Asp14Gln;
 Asp14Arg; Asp14Ser; Asp14Thr; Asp14Val; Asp14Trp;
 Asp14Tyr; Ala15Cys; Ala15Asp; Ala15Glu; Ala15Phe;
 Ala15Gly; Ala15His; Ala15Ile; Ala15Lys; Ala15Leu;
 Ala15Met; Ala15Asn; Ala15Pro; Ala15Gln; Ala15Arg;
 Ala15Ser; Ala15Thr; Ala15Val; Ala15Trp; Ala15Tyr;
 Lys16Ala; Lys16Cys; Lys16Asp; Lys16Glu; Lys16Phe;
 Lys16Gly; Lys16His; Lys16Ile; Lys16Leu; Lys16Met;
 Lys16Asn; Lys16Pro; Lys16Gln; Lys16Arg; Lys16Ser;
 Lys16Thr; Lys16Val; Lys16Trp; Lys16Tyr; Leu17Ala;
 Leu17Cys; Leu17Asp; Leu17Glu; Leu17Phe; Leu17Gly;
 Leu17His; Leu17Ile; Leu17Lys; Leu17Met; Leu17Asn;
 Leu17Pro; Leu17Gln; Leu17Arg; Leu17Ser; Leu17Thr;
 Leu17Val; Leu17Trp; Leu17Tyr; Gly18Ala; Gly18Cys;
 Gly18Asp; Gly18Glu; Gly18Phe; Gly18His; Gly18Ile;
 Gly18Lys; Gly18Leu; Gly18Met; Gly18Asn; Gly18Pro;
 Gly18Gln; Gly18Arg; Gly18Ser; Gly18Thr; Gly18Val;
 Gly18Trp; Gly18Tyr; Ile19Ala; Ile19Cys; Ile19Asp;
 Ile19Glu; Ile19Phe; Ile19Gly; Ile19His; Ile19Lys; Ile19Leu;
 Ile19Met; Ile19Asn; Ile19Pro; Ile19Gln; Ile19Arg; Ile19Ser;
 Ile19Thr; Ile19Val; Ile19Trp; Ile19Tyr; Phe20Ala;
 Phe20Cys; Phe20Asp; Phe20Glu; Phe20Gly; Phe20His;
 Phe20Ile; Phe20Lys; Phe20Leu; Phe20Met; Phe20Asn;
 Phe20Pro; Phe20Gln; Phe20Arg; Phe20Ser; Phe20Thr;
 Phe20Val; Phe20Trp; Phe20Tyr; Ala21Cys; Ala21Asp;
 Ala21Glu; Ala21Phe; Ala21Gly; Ala21His; Ala21Ile;
 Ala21Lys; Ala21Leu; Ala21Met; Ala21Asn; Ala21Pro;
 Ala21Gln; Ala21Arg; Ala21Ser; Ala21Thr; Ala21Val;
 Ala21Trp; Ala21Tyr; Leu22Ala; Leu22Cys; Leu22Asp;
 Leu22Glu; Leu22Phe; Leu22Gly; Leu22His; Leu22Ile;
 Leu22Lys; Leu22Met; Leu22Asn; Leu22Pro; Leu22Gln;
 Leu22Arg; Leu22Ser; Leu22Thr; Leu22Val; Leu22Trp;
 Leu22Tyr; Asp23Ala; Asp23Cys; Asp23Glu; Asp23Phe;
 Asp23Gly; Asp23His; Asp23Ile; Asp23Lys; Asp23Leu;
 Asp23Met; Asp23Asn; Asp23Pro; Asp23Gln; Asp23Arg;
 Asp23Ser; Asp23Thr; Asp23Val; Asp23Trp; Asp23Tyr;
 Thr24Ala; Thr24Cys; Thr24Asp; Thr24Glu; Thr24Phe;
 Thr24Gly; Thr24His; Thr24Ile; Thr24Lys; Thr24Leu;

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Thr24Met; Thr24Asn; Thr24Pro; Thr24Gln; Thr24Arg;
 Thr24Ser; Thr24Val; Thr24Trp; Thr24Tyr; Gly25Ala;
 Gly25Cys; Gly25Asp; Gly25Glu; Gly25Phe; Gly25His;
 Gly25Ile; Gly25Lys; Gly25Leu; Gly25Met; Gly25Asn;
 Gly25Pro; Gly25Gln; Gly25Arg; Gly25Ser; Gly25Thr;
 Gly25Val; Gly25Trp; Gly25Tyr; Thr26Ala; Thr26Cys;
 Thr26Asp; Thr26Glu; Thr26Phe; Thr26Gly; Thr26His;
 Thr26Ile; Thr26Lys; Thr26Leu; Thr26Met; Thr26Asn;
 Thr26Pro; Thr26Gln; Thr26Arg; Thr26Ser; Thr26Val;
 Thr26Trp; Thr26Tyr; Asn27Ala; Asn27Cys; Asn27Asp;
 Asn27Glu; Asn27Phe; Asn27Gly; Asn27His; Asn27Ile;
 Asn27Lys; Asn27Leu; Asn27Met; Asn27Pro; Asn27Gln;
 Asn27Arg; Asn27Ser; Asn27Thr; Asn27Val; Asn27Trp;
 Asn27Tyr; Arg28Ala; Arg28Cys; Arg28Asp; Arg28Glu;
 Arg28Phe; Arg28Gly; Arg28His; Arg28Ile; Arg28Lys;
 Arg28Leu; Arg28Met; Arg28Asn; Arg28Pro; Arg28Gln;
 Arg28Ser; Arg28Thr; Arg28Val; Arg28Trp; Arg28Tyr;
 Thr29Ala; Thr29Cys; Thr29Asp; Thr29Glu; Thr29Phe;
 Thr29Gly; Thr29His; Thr29Ile; Thr29Lys; Thr29Leu;
 Thr29Met; Thr29Asn; Thr29Pro; Thr29Gln; Thr29Arg;
 Thr29Ser; Thr29Val; Thr29Trp; Thr29Tyr; Val30Ala;
 Val30Cys; Val30Asp; Val30Glu; Val30Phe; Val30Gly;
 Val30His; Val30Ile; Val30Lys; Val30Leu; Val30Met;
 Val30Asn; Val30Pro; Val30Gln; Val30Arg; Val30Ser;
 Val30Thr; Val30Trp; Val30Tyr; Ala31Ala; Ala31Cys;
 Ala31Asp; Ala31Glu; Ala31Phe; Ala31Gly; Ala31His;
 Ala31Ile; Ala31Lys; Ala31Leu; Ala31Met; Ala31Asn;
 Ala31Pro; Ala31Gln; Ala31Arg; Ala31Ser; Ala31Thr;
 Ala31Val; Ala31Trp; Ala31Tyr; Tyr32Ala; Tyr32Cys;
 Tyr32Asp; Tyr32Glu; Tyr32Phe; Tyr32Gly; Tyr32His;
 Tyr32Ile; Tyr32Lys; Tyr32Leu; Tyr32Met; Tyr32Asn;
 Tyr32Pro; Tyr32Gln; Tyr32Arg; Tyr32Ser; Tyr32Thr;
 Tyr32Val; Tyr32Trp; Arg33Ala; Arg33Cys; Arg33Asp;
 Arg33Glu; Arg33Phe; Arg33Gly; Arg33His; Arg33Ile;
 Arg33Lys; Arg33Leu; Arg33Met; Arg33Asn; Arg33Pro;
 Arg33Gln; Arg33Ser; Arg33Thr; Arg33Val; Arg33Trp;
 Arg33Tyr; Pro34Ala; Pro34Cys; Pro34Asp; Pro34Glu;
 Pro34Phe; Pro34Gly; Pro34His; Pro34Ile; Pro34Lys;
 Pro34Leu; Pro34Met; Pro34Asn; Pro34Gln; Pro34Arg;
 Pro34Ser; Pro34Thr; Pro34Val; Pro34Trp; Pro34Tyr;
 Asp35Ala; Asp35Cys; Asp35Glu; Asp35Phe; Asp35Gly;
 Asp35His; Asp35Ile; Asp35Lys; Asp35Leu; Asp35Met;
 Asp35Asn; Asp35Pro; Asp35Gln; Asp35Arg; Asp35Ser;
 Asp35Thr; Asp35Val; Asp35Trp; Asp35Tyr; Glu36Ala;
 Glu36Cys; Glu36Asp; Glu36Phe; Glu36Gly; Glu36His;
 Glu36Ile; Glu36Lys; Glu36Leu; Glu36Met; Glu36Asn;
 Glu36Pro; Glu36Gln; Glu36Arg; Glu36Ser; Glu36Thr;
 Glu36Val; Glu36Trp; Glu36Tyr; Arg37Ala; Arg37Cys;
 Arg37Asp; Arg37Glu; Arg37Phe; Arg37Gly; Arg37His;
 Arg37Ile; Arg37Lys; Arg37Leu; Arg37Met; Arg37Asn;
 Arg37Pro; Arg37Gln; Arg37Ser; Arg37Thr; Arg37Val;
 Arg37Trp; Arg37Tyr; Phe38Ala; Phe38Cys; Phe38Asp;
 Phe38Glu; Phe38Gly; Phe38His; Phe38Ile; Phe38Lys;
 Phe38Leu; Phe38Met; Phe38Asn; Phe38Pro; Phe38Gln;
 Phe38Arg; Phe38Ser; Phe38Thr; Phe38Val; Phe38Trp;
 Phe38Tyr; Ala39Cys; Ala39Asp; Ala39Glu; Ala39Phe;
 Ala39Gly; Ala39His; Ala39Ile; Ala39Lys; Ala39Leu;
 Ala39Met; Ala39Asn; Ala39Pro; Ala39Gln; Ala39Arg;
 Ala39Ser; Ala39Thr; Ala39Val; Ala39Trp; Ala39Tyr;
 Phe40Ala; Phe40Cys; Phe40Asp; Phe40Glu; Phe40Gly;
 Phe40His; Phe40Ile; Phe40Lys; Phe40Leu; Phe40Met;
 Phe40Asn; Phe40Pro; Phe40Gln; Phe40Arg; Phe40Ser;
 Phe40Thr; Phe40Val; Phe40Trp; Phe40Tyr; Ala41Cys;
 Ala41Asp; Ala41Glu; Ala41Phe; Ala41Gly; Ala41His;
 Ala41Ile; Ala41Lys; Ala41Leu; Ala41Met; Ala41Asn;
 Ala41Pro; Ala41Gln; Ala41Arg; Ala41Ser; Ala41Thr;
 Ala41Val; Ala41Trp; Ala41Tyr; Ser42Ala; Ser42Cys;

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Ser42Asp; Ser42Glu; Ser42Phe; Ser42Gly; Ser42His;
 Ser42Ile; Ser42Lys; Ser42Leu; Ser42Met; Ser42Asn;
 Ser42Pro; Ser42Gln; Ser42Arg; Ser42Thr; Ser42Val;
 Ser42Trp; Ser42Tyr; Thr43Ala; Thr43Cys; Thr43Asp;
 Thr43Glu; Thr43Phe; Thr43Gly; Thr43His; Thr43Ile;
 Thr43Lys; Thr43Leu; Thr43Met; Thr43Asn; Thr43Pro;
 Thr43Gln; Thr43Arg; Thr43Ser; Thr43Val; Thr43Trp;
 Thr43Tyr; Ile44Ala; Ile44Cys; Ile44Asp; Ile44Glu;
 Ile44Phe; Ile44Gly; Ile44His; Ile44Lys; Ile44Leu; Ile44Met;
 Ile44Asn; Ile44Pro; Ile44Gln; Ile44Arg; Ile44Ser; Ile44Thr;
 Ile44Val; Ile44Trp; Ile44Tyr; Lys45Ala; Lys45Cys;
 Lys45Asp; Lys45Glu; Lys45Phe; Lys45Gly; Lys45His;
 Lys45Ile; Lys45Leu; Lys45Met; Lys45Asn; Lys45Pro;
 Lys45Gln; Lys45Arg; Lys45Ser; Lys45Thr; Lys45Val;
 Lys45Trp; Lys45Tyr; Ala46Cys; Ala46Asp; Ala46Glu;
 Ala46Phe; Ala46Gly; Ala46His; Ala46Ile; Ala46Lys;
 Ala46Leu; Ala46Met; Ala46Asn; Ala46Pro; Ala46Gln;
 Ala46Arg; Ala46Ser; Ala46Thr; Ala46Val; Ala46Trp;
 Ala46Tyr; Leu47Ala; Leu47Cys; Leu47Asp; Leu47Glu;
 Leu47Phe; Leu47Gly; Leu47His; Leu47Ile; Leu47Lys;
 Leu47Met; Leu47Asn; Leu47Pro; Leu47Gln; Leu47Arg;
 Leu47Ser; Leu47Thr; Leu47Val; Leu47Trp; Leu47Tyr;
 Thr48Ala; Thr48Cys; Thr48Asp; Thr48Glu; Thr48Phe;
 Thr48Gly; Thr48His; Thr48Ile; Thr48Lys; Thr48Leu;
 Thr48Met; Thr48Asn; Thr48Pro; Thr48Gln; Thr48Arg;
 Thr48Ser; Thr48Val; Thr48Trp; Thr48Tyr; Val49Ala;
 Val49Cys; Val49Asp; Val49Glu; Val49Phe; Val49Gly;
 Val49His; Val49Ile; Val49Lys; Val49Leu; Val49Met;
 Val49Asn; Val49Pro; Val49Gln; Val49Arg; Val49Ser;
 Val49Thr; Val49Trp; Val49Tyr; Gly50Ala; Gly50Cys;
 Gly50Asp; Gly50Glu; Gly50Phe; Gly50His; Gly50Ile;
 Gly50Lys; Gly50Leu; Gly50Met; Gly50Asn; Gly50Pro;
 Gly50Gln; Gly50Arg; Gly50Ser; Gly50Thr; Gly50Val;
 Gly50Trp; Gly50Tyr; Val51Ala; Val51Cys; Val51Asp;
 Val51Glu; Val51Phe; Val51Gly; Val51His; Val51Ile;
 Val51Lys; Val51Leu; Val51Met; Val51Asn; Val51Pro;
 Val51Gln; Val51Arg; Val51Ser; Val51Thr; Val51Trp;
 Val51Tyr; Leu52Ala; Leu52Cys; Leu52Asp; Leu52Glu;
 Leu52Phe; Leu52Gly; Leu52His; Leu52Ile; Leu52Lys;
 Leu52Met; Leu52Asn; Leu52Pro; Leu52Gln; Leu52Arg;
 Leu52Ser; Leu52Thr; Leu52Val; Leu52Trp; Leu52Tyr;
 Leu53Ala; Leu53Cys; Leu53Asp; Leu53Glu; Leu53Phe;
 Leu53Gly; Leu53His; Leu53Ile; Leu53Lys; Leu53Met;
 Leu53Asn; Leu53Pro; Leu53Gln; Leu53Arg; Leu53Ser;
 Leu53Thr; Leu53Val; Leu53Trp; Leu53Tyr; Gln54Ala;
 Gln54Cys; Gln54Asp; Gln54Glu; Gln54Phe; Gln54Gly;
 Gln54His; Gln54Ile; Gln54Lys; Gln54Leu; Gln54Met;
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 Gln54Val; Gln54Trp; Gln54Tyr; Gln55Ala; Gln55Cys;
 Gln55Asp; Gln55Glu; Gln55Phe; Gln55Gly; Gln55His;
 Gln55Ile; Gln55Lys; Gln55Leu; Gln55Met; Gln55Asn;
 Gln55Pro; Gln55Arg; Gln55Ser; Gln55Thr; Gln55Val;
 Gln55Trp; Gln55Tyr; Lys56Ala; Lys56Cys; Lys56Asp;
 Lys56Glu; Lys56Phe; Lys56Gly; Lys56His; Lys56Ile;
 Lys56Leu; Lys56Met; Lys56Asn; Lys56Pro; Lys56Gln;
 Lys56Arg; Lys56Ser; Lys56Thr; Lys56Val; Lys56Trp;
 Lys56Tyr; Ser57Ala; Ser57Cys; Ser57Asp; Ser57Lys;
 Ser57Phe; Ser57Gly; Ser57His; Ser57Ile; Ser57Glu;
 Ser57Leu; Ser57Met; Ser57Asn; Ser57Pro; Ser57Gln;
 Ser57Arg; Ser57Thr; Ser57Val; Ser57Trp; Ser57Tyr;
 Ile58Ala; Ile58Cys; Ile58Asp; Ile58Glu; Ile58Phe; Ile58Gly;
 Ile58His; Ile58Lys; Ile58Leu; Ile58Met; Ile58Asn; Ile58Pro;
 Ile58Gln; Ile58Ile; Ile58Ser; Ile58Thr; Ile58Val; Ile58Trp;
 Ile58Tyr; Glu59Ala; Glu59Cys; Glu59Asp; Glu59Phe;
 Glu59Gly; Glu59His; Glu59Ile; Glu59Lys; Glu59Leu;
 Glu59Met; Glu59Asn; Glu59Pro; Glu59Gln; Glu59Arg;
 Glu59Ser; Glu59Thr; Glu59Val; Glu59Trp; Glu59Tyr;

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Asp60Ala; Asp60Cys; Asp60Glu; Asp60Phe; Asp60Gly;
 Asp60His; Asp60Ile; Asp60Lys; Asp60Leu; Asp60Met;
 Asp60Asn; Asp60Pro; Asp60Gln; Asp60Arg; Asp60Ser;
 Asp60Thr; Asp60Val; Asp60Trp; Asp60Tyr; Leu61Ala;
 Leu61Cys; Leu61Asp; Leu61Glu; Leu61Phe; Leu61Gly;
 Leu61His; Leu61Ile; Leu61Lys; Leu61Met; Leu61Asn;
 Leu61Pro; Leu61Gln; Leu61Arg; Leu61Ser; Leu61Thr;
 Leu61Val; Leu61Trp; Leu61Tyr; Asn62Ala; Asn62Cys;
 Asn62Asp; Asn62Glu; Asn62Phe; Asn62Gly; Asn62His;
 Asn62Ile; Asn62Lys; Asn62Leu; Asn62Met; Asn62Pro;
 Asn62Gln; Asn62Arg; Asn62Ser; Asn62Thr; Asn62Val;
 Asn62Trp; Asn62Tyr; Gln63Ala; Gln63Cys; Gln63Asp;
 Gln63Glu; Gln63Phe; Gln63Gly; Gln63His; Gln63Ile;
 Gln63Lys; Gln63Leu; Gln63Met; Gln63Asn; Gln63Pro;
 Gln63Arg; Gln63Ser; Gln63Thr; Gln63Val; Gln63Trp;
 Arg64Ala; Arg64Cys; Arg64Asp; Arg64Glu;
 Arg64Phe; Arg64Gly; Arg64His; Arg64Ile; Arg64Lys;
 Arg64Leu; Arg64Met; Arg64Asn; Arg64Pro; Arg64Gln;
 Arg64Ser; Arg64Thr; Arg64Val; Arg64Trp; Arg64Tyr;
 Ile65Ala; Ile65Cys; Ile65Asp; Ile65Glu; Ile65Phe; Ile65Gly;
 Ile65His; Ile65Lys; Ile65Leu; Ile65Met; Ile65Asn; Ile65Pro;
 Ile65Gln; Ile65Arg; Ile65Ser; Ile65Thr; Ile65Val; Ile65Trp;
 Ile65Tyr; Thr66Ala; Thr66Cys; Thr66Asp; Thr66Glu;
 Thr66Phe; Thr66Gly; Thr66His; Thr66Ile; Thr66Lys;
 Thr66Leu; Thr66Met; Thr66Asn; Thr66Pro; Thr66Gln;
 Thr66Arg; Thr66Ser; Thr66Val; Thr66Trp; Thr66Tyr;
 Tyr67Ala; Tyr67Cys; Tyr67Asp; Tyr67Glu; Tyr67Phe;
 Tyr67Gly; Tyr67His; Tyr67Ile; Tyr67Lys; Tyr67Leu;
 Tyr67Met; Tyr67Asn; Tyr67Pro; Tyr67Gln; Tyr67Arg;
 Tyr67Ser; Tyr67Thr; Tyr67Val; Tyr67Trp; Thr68Ala;
 Thr68Cys; Thr68Asp; Thr68Glu; Thr68Phe; Thr68Gly;
 Thr68His; Thr68Ile; Thr68Lys; Thr68Leu; Thr68Met;
 Thr68Asn; Thr68Pro; Thr68Gln; Thr68Arg; Thr68Ser;
 Thr68Val; Thr68Trp; Thr68Tyr; Arg69Ala; Arg69Cys;
 Arg69Asp; Arg69Glu; Arg69Phe; Arg69Gly; Arg69His;
 Arg69Ile; Arg69Lys; Arg69Leu; Arg69Met; Arg69Asn;
 Arg69Pro; Arg69Gln; Arg69Ser; Arg69Thr; Arg69Val;
 Arg69Trp; Arg69Tyr; Asp70Ala; Asp70Cys; Asp70Glu;
 Asp70Phe; Asp70Gly; Asp70His; Asp70Ile; Asp70Lys;
 Asp70Leu; Asp70Met; Asp70Asn; Asp70Pro; Asp70Gln;
 Asp70Arg; Asp70Ser; Asp70Thr; Asp70Val; Asp70Trp;
 Asp70Tyr; Asp71Ala; Asp71Cys; Asp71Glu; Asp71Phe;
 Asp71Gly; Asp71His; Asp71Ile; Asp71Lys; Asp71Leu;
 Asp71Met; Asp71Asn; Asp71Pro; Asp71Gln; Asp71Arg;
 Asp71Ser; Asp71Thr; Asp71Val; Asp71Trp; Asp71Tyr;
 Leu72Ala; Leu72Cys; Leu72Asp; Leu72Glu; Leu72Phe;
 Leu72Gly; Leu72His; Leu72Ile; Leu72Lys; Leu72Met;
 Leu72Asn; Leu72Pro; Leu72Gln; Leu72Arg; Leu72Ser;
 Leu72Thr; Leu72Val; Leu72Trp; Leu72Tyr; Val73Ala;
 Val73Cys; Val73Asp; Val73Glu; Val73Phe; Val73Gly;
 Val73His; Val73Ile; Val73Lys; Val73Leu; Val73Met;
 Val73Asn; Val73Pro; Val73Gln; Val73Arg; Val73Ser;
 Val73Thr; Val73Trp; Val73Tyr; Asn74Ala; Asn74Cys;
 Asn74Asp; Asn74Glu; Asn74Phe; Asn74Gly; Asn74His;
 Asn74Ile; Asn74Lys; Asn74Leu; Asn74Met; Asn74Pro;
 Asn74Gln; Asn74Arg; Asn74Ser; Asn74Thr; Asn74Val;
 Asn74Trp; Asn74Tyr; Tyr75Ala; Tyr75Cys; Tyr75Asp;
 Tyr75Glu; Tyr75Phe; Tyr75Gly; Tyr75His; Tyr75Ile;
 Tyr75Lys; Tyr75Leu; Tyr75Met; Tyr75Asn; Tyr75Pro;
 Tyr75Gln; Tyr75Arg; Tyr75Ser; Tyr75Thr; Tyr75Val;
 Tyr75Trp; Asn76Ala; Asn76Cys; Asn76Asp; Asn76Glu;
 Asn76Phe; Asn76Gly; Asn76His; Asn76Ile; Asn76Lys;
 Asn76Leu; Asn76Met; Asn76Pro; Asn76Gln; Asn76Arg;
 Asn76Ser; Asn76Thr; Asn76Val; Asn76Trp; Asn76Tyr;
 Pro77Ala; Pro77Cys; Pro77Asp; Pro77Glu; Pro77Phe;
 Pro77Gly; Pro77His; Pro77Ile; Pro77Lys; Pro77Leu;
 Pro77Met; Pro77Asn; Pro77Gln; Pro77Arg; Pro77Ser;

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Pro77Thr; Pro77Val; Pro77Trp; Pro77Tyr; Ile78Ala; Ile78Cys; Ile78Asp; Ile78Glu; Ile78Phe; Ile78Gly; Ile78His; Ile78Lys; Ile78Leu; Ile78Met; Ile78Asn; Ile78Pro; Ile78Gln; Ile78Arg; Ile78Ser; Ile78Thr; Ile78Val; Ile78Trp; Ile78Tyr; Thr79Ala; Thr79Cys; Thr79Asp; Thr79Glu; Thr79Phe; Thr79Gly; Thr79His; Thr79Ile; Thr79Lys; Thr79Leu; Thr79Met; Thr79Asn; Thr79Pro; Thr79Gln; Thr79Arg; Thr79Ser; Thr79Val; Thr79Trp; Thr79Tyr; Glu80Ala; Glu80Cys; Glu80Asp; Glu80Phe; Glu80Gly; Glu80His; Glu80Ile; Glu80Lys; Glu80Leu; Glu80Met; Glu80Asn; Glu80Pro; Glu80Gln; Glu80Arg; Glu80Ser; Glu80Thr; Glu80Val; Glu80Trp; Glu80Tyr; Lys81Ala; Lys81Cys; Lys81Asp; Lys81Glu; Lys81Phe; Lys81Gly; Lys81His; Lys81Ile; Lys81Leu; Lys81Met; Lys81Asn; Lys81Pro; Lys81Gln; Lys81Arg; Lys81Ser; Lys81Thr; Lys81Val; Lys81Trp; Lys81Tyr; His82Ala; His82Cys; His82Glu; His82Phe; His82Gly; His82Ile; His82Leu; His82Met; His82Asn; His82Pro; His82Gln; His82Arg; His82Ser; His82Thr; His82Val; His82Trp; His82Tyr; Val83Ala; Val83Cys; Val83Asp; Val83Glu; Val83Phe; Val83Gly; Val83His; Val83Ile; Val83Lys; Val83Leu; Val83Met; Val83Asn; Val83Pro; Val83Gln; Val83Arg; Val83Ser; Val83Thr; Val83Trp; Val83Tyr; Asp84Ala; Asp84Cys; Asp84Glu; Asp84Phe; Asp84Gly; Asp84His; Asp84Ile; Asp84Lys; Asp84Leu; Asp84Met; Asp84Asn; Asp84Pro; Asp84Gln; Asp84Arg; Asp84Ser; Asp84Thr; Asp84Val; Asp84Trp; Asp84Tyr; Thr85Ala; Thr85Cys; Thr85Asp; Thr85Glu; Thr85Phe; Thr85Gly; Thr85His; Thr85Ile; Thr85Lys; Thr85Leu; Thr85Met; Thr85Asn; Thr85Pro; Thr85Gln; Thr85Arg; Thr85Ser; Thr85Val; Thr85Trp; Thr85Tyr; Gly86Ala; Gly86Cys; Gly86Asp; Gly86Glu; Gly86Phe; Gly86His; Gly86Ile; Gly86Lys; Gly86Leu; Gly86Met; Gly86Asn; Gly86Pro; Gly86Gln; Gly86Arg; Gly86Ser; Gly86Thr; Gly86Val; Gly86Trp; Gly86Tyr; Met87Ala; Met87Cys; Met87Asp; Met87Glu; Met87Phe; Met87Gly; Met87His; Met87Ile; Met87Lys; Met87Leu; Met87Asn; Met87Pro; Met87Gln; Met87Arg; Met87Ser; Met87Thr; Met87Val; Met87Trp; Met87Tyr; Thr88Ala; Thr88Cys; Thr88Asp; Thr88Glu; Thr88Phe; Thr88Gly; Thr88His; Thr88Ile; Thr88Lys; Thr88Leu; Thr88Met; Thr88Asn; Thr88Pro; Thr88Gln; Thr88Arg; Thr88Ser; Thr88Val; Thr88Trp; Thr88Tyr; Leu89Ala; Leu89Cys; Leu89Asp; Leu89Glu; Leu89Phe; Leu89Gly; Leu89His; Leu89Ile; Leu89Lys; Leu89Met; Leu89Asn; Leu89Pro; Leu89Gln; Leu89Arg; Leu89Ser; Leu89Thr; Leu89Val; Leu89Tyr; Lys90Cys; Lys90Asp; Lys90Glu; Lys90Phe; Lys90Gly; Lys90His; Lys90Ile; Lys90Leu; Lys90Met; Lys90Asn; Lys90Pro; Lys90Gln; Lys90Arg; Lys90Ser; Lys90Thr; Lys90Val; Lys90Trp; Lys90Tyr; Glu91Ala; Glu91Cys; Glu91Asp; Glu91Phe; Glu91Gly; Glu91His; Glu91Ile; Glu91Lys; Glu91Leu; Glu91Met; Glu91Asn; Glu91Pro; Glu91Gln; Glu91Arg; Glu91Ser; Glu91Thr; Glu91Val; Glu91Trp; Glu91Tyr; Leu92Ala; Leu92Cys; Leu92Asp; Leu92Glu; Leu92Phe; Leu92Gly; Leu92His; Leu92Ile; Leu92Lys; Leu92Met; Leu92Asn; Leu92Pro; Leu92Gln; Leu92Arg; Leu92Ser; Leu92Thr; Leu92Val; Leu92Trp; Leu92Tyr; Ala93Cys; Ala93Asp; Ala93Glu; Ala93Phe; Ala93Gly; Ala93His; Ala93Ile; Ala93Lys; Ala93Leu; Ala93Met; Ala93Asn; Ala93Pro; Ala93Gln; Ala93Arg; Ala93Ser; Ala93Thr; Ala93Val; Ala93Trp; Ala93Tyr; Asp94Ala; Asp94Cys; Asp94Glu; Asp94Phe; Asp94Gly; Asp94His; Asp94Ile; Asp94Lys; Asp94Leu; Asp94Met; Asp94Asn; Asp94Pro; Asp94Gln; Asp94Arg; Asp94Ser; Asp94Thr; Asp94Val; Asp94Trp; Asp94Tyr; Ala95Cys; Ala95Asp; Ala95Glu; Ala95Phe; Ala95Gly; Ala95His; Ala95Ile; Ala95Lys; Ala95Leu; Ala95Met; Ala95Asn;

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Ala95Pro; Ala95Gln; Ala95Arg; Ala95Ser; Ala95Thr; Ala95Val; Ala95Trp; Ala95Tyr; Ser96Ala; Ser96Cys; Ser96Asp; Ser96Glu; Ser96Phe; Ser96Gly; Ser96His; Ser96Ile; Ser96Lys; Ser96Leu; Ser96Met; Ser96Asn; Ser96Pro; Ser96Gln; Ser96Arg; Ser96Thr; Ser96Val; Ser96Trp; Ser96Tyr; Leu97Ala; Leu97Cys; Leu97Asp; Leu97Glu; Leu97Phe; Leu97Gly; Leu97His; Leu97Ile; Leu97Lys; Leu97Met; Leu97Asn; Leu97Pro; Leu97Gln; Leu97Arg; Leu97Ser; Leu97Thr; Leu97Val; Leu97Trp; Leu97Tyr; Arg98Ala; Arg98Cys; Arg98Glu; Arg98Phe; Arg98Gly; Arg98His; Arg98Ile; Arg98Lys; Arg98Leu; Arg98Met; Arg98Asn; Arg98Pro; Arg98Gln; Arg98Ser; Arg98Thr; Arg98Val; Arg98Trp; Arg98Tyr; Tyr99Ala; Tyr99Cys; Tyr99Asp; Tyr99Glu; Tyr99Phe; Tyr99Gly; Tyr99His; Tyr99Ile; Tyr99Lys; Tyr99Leu; Tyr99Met; Tyr99Asn; Tyr99Pro; Tyr99Gln; Tyr99Arg; Tyr99Ser; Tyr99Thr; Tyr99Val; Tyr99Trp; Ser100Ala; Ser100Cys; Ser100Asp; Ser100Glu; Ser100Phe; Ser100Gly; Ser100His; Ser100Ile; Ser100Lys; Ser100Leu; Ser100Met; Ser100Asn; Ser100Pro; Ser100Gln; Ser100Arg; Ser100Thr; Ser100Val; Ser100Trp; Ser100Tyr; Asp101Ala; Asp101Cys; Asp101Glu; Asp101Phe; Asp101Gly; Asp101His; Asp101Ile; Asp101Lys; Asp101Leu; Asp101Met; Asp101Asn; Asp101Pro; Asp101Gln; Asp101Arg; Asp101Ser; Asp101Thr; Asp101Val; Asp101Trp; Asp101Tyr; Asn102Ala; Asn102Cys; Asn102Asp; Asn102Glu; Asn102Phe; Asn102Gly; Asn102His; Asn102Ile; Asn102Lys; Asn102Leu; Asn102Met; Asn102Pro; Asn102Gln; Asn102Arg; Asn102Ser; Asn102Thr; Asn102Val; Asn102Trp; Asn102Tyr; Ala103Cys; Ala103Asp; Ala103Glu; Ala103Phe; Ala103Gly; Ala103His; Ala103Ile; Ala103Lys; Ala103Leu; Ala103Met; Ala103Asn; Ala103Pro; Ala103Gln; Ala103Arg; Ala103Ser; Ala103Thr; Ala103Val; Ala103Trp; Ala103Tyr; Ala104Cys; Ala104Asp; Ala104Glu; Ala104Phe; Ala104Gly; Ala104His; Ala104Ile; Ala104Lys; Ala104Leu; Ala104Met; Ala104Asn; Ala104Pro; Ala104Gln; Ala104Arg; Ala104Ser; Ala104Thr; Ala104Val; Ala104Trp; Ala104Tyr; Gln105Ala; Gln105Cys; Gln105Asp; Gln105Glu; Gln105Phe; Gln105Gly; Gln105His; Gln105Ile; Gln105Lys; Gln105Leu; Gln105Met; Gln105Asn; Gln105Pro; Gln105Arg; Gln105Ser; Gln105Thr; Gln105Val; Gln105Trp; Gln105Tyr; Asn106Ala; Asn106Cys; Asn106Asp; Asn106Glu; Asn106Phe; Asn106Gly; Asn106His; Asn106Ile; Asn106Lys; Asn106Leu; Asn106Met; Asn106Pro; Asn106Gln; Asn106Arg; Asn106Ser; Asn106Thr; Asn106Val; Asn106Trp; Asn106Tyr; Leu107Ala; Leu107Cys; Leu107Asp; Leu107Glu; Leu107Phe; Leu107Gly; Leu107His; Leu107Ile; Leu107Lys; Leu107Met; Leu107Asn; Leu107Pro; Leu107Gln; Leu107Arg; Leu107Ser; Leu107Thr; Leu107Val; Leu107Trp; Leu107Tyr; Ile108Ala; Ile108Cys; Ile108Asp; Ile108Glu; Ile108Phe; Ile108Gly; Ile108His; Ile108Lys; Ile108Leu; Ile108Met; Ile108Asn; Ile108Pro; Ile108Gln; Ile108Arg; Ile108Ser; Ile108Thr; Ile108Val; Ile108Trp; Ile108Tyr; Leu109Ala; Leu109Cys; Leu109Asp; Leu109Glu; Leu109Phe; Leu109Gly; Leu109His; Leu109Ile; Leu109Lys; Leu109Met; Leu109Asn; Leu109Pro; Leu109Gln; Leu109Arg; Leu109Ser; Leu109Thr; Leu109Val; Leu109Trp; Leu109Tyr; Lys110Ala; Lys110Cys; Lys110Asp; Lys110Glu; Lys110Phe; Lys110Gly; Lys110His; Lys110Ile; Lys110Leu; Lys110Met; Lys110Asn; Lys110Pro; Lys110Gln; Lys110Arg; Lys110Ser; Lys110Thr; Lys110Val; Lys110Trp; Lys110Tyr; Gln111Ala; Gln111Cys; Gln111Asp; Gln111Glu; Gln111Phe; Gln111Gly; Gln111His; Gln111Ile;

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Gln111Lys; Gln111Leu; Gln111Met; Gln111Asn;
 Gln111Pro; Gln111Arg; Gln111Ser; Gln111Thr;
 Gln111Val; Gln111Trp; Gln111Tyr; Ile112Ala; Ile112Cys;
 Ile112Asp; Ile112Glu; Ile112Phe; Ile112Gly; Ile112His;
 Ile112Lys; Ile112Leu; Ile112Met; Ile112Asn; Ile112Pro;
 Ile112Gln; Ile112Arg; Ile112Ser; Ile112Thr; Ile112Val;
 Ile112Trp; Ile112Tyr; Gly113Ala; Gly113Cys; Gly113Asp;
 Gly113Glu; Gly113Phe; Gly113His; Gly113Ile; Gly113Lys;
 Gly113Leu; Gly113Met; Gly113Asn; Gly113Pro;
 Gly113Gln; Gly113Arg; Gly113Ser; Gly113Thr;
 Gly113Val; Gly113Trp; Gly113Tyr; Gly114Ala;
 Gly114Cys; Gly114Asp; Gly114Glu; Gly114Phe;
 Gly114His; Gly114Ile; Gly114Lys; Gly114Leu;
 Gly114Met; Gly114Asn; Gly114Pro; Gly114Gln;
 Gly114Arg; Gly114Ser; Gly114Thr; Gly114Val;
 Gly114Trp; Gly114Tyr; Pro115A; Pro115Cys;
 Pro115Asp; Pro115Glu; Pro115Phe; Pro115Gly; Pro115His;
 Pro115Ile; Pro115Lys; Pro115Leu; Pro115Met; Pro115Asn;
 Pro115Gln; Pro115Arg; Pro115Ser; Pro115Thr; Pro115Val;
 Pro115Trp; Pro115Tyr; Glu116Ala; Glu116Cys;
 Glu116Asp; Glu116Phe; Glu116Gly; Glu116His;
 Glu116Ile; Glu116Lys; Glu116Leu; Glu116Met;
 Glu116Asn; Glu116Pro; Glu116Gln; Glu116Arg;
 Glu116Ser; Glu116Thr; Glu116Val; Glu116Trp; Glu116Tyr;
 Ser117Ala; Ser117Cys; Ser117Asp; Ser117Glu; Ser117Phe;
 Ser117Gly; Ser117His; Ser117Ile; Ser117Lys; Ser117Leu;
 Ser117Met; Ser117Asn; Ser117Pro; Ser117Gln; Ser117Arg;
 Ser117Thr; Ser117Val; Ser117Trp; Ser117Tyr; Leu118Ala;
 Leu118Cys; Leu118Asp; Leu118Glu; Leu118Phe;
 Leu118Gly; Leu118His; Leu118Ile; Leu118Lys;
 Leu118Met; Leu118Asn; Leu118Pro; Leu118Gln;
 Leu118Arg; Leu118Ser; Leu118Thr; Leu118Val;
 Leu118Trp; Leu118Tyr; Lys119Ala; Lys119Cys;
 Lys119Asp; Lys119Glu; Lys119Phe; Lys119Gly;
 Lys119His; Lys119Ile; Lys119Leu; Lys119Met; Lys119Asn;
 Lys119Pro; Lys119Gln; Lys119Arg; Lys119Ser; Lys119Thr;
 Lys119Val; Lys119Trp; Lys119Tyr; Lys120Ala; Lys120Cys;
 Lys120Asp; Lys120Glu; Lys120Phe; Lys120Gly;
 Lys120His; Lys120Ile; Lys120Leu; Lys120Met; Lys120Asn;
 Lys120Pro; Lys120Gln; Lys120Arg; Lys120Ser; Lys120Thr;
 Lys120Val; Lys120Trp; Lys120Tyr; Glu121Ala; Glu121Cys;
 Glu121Asp; Glu121Phe; Glu121Gly; Glu121His;
 Glu121Ile; Glu121Lys; Glu121Leu; Glu121Met;
 Glu121Asn; Glu121Pro; Glu121Gln; Glu121Arg;
 Glu121Ser; Glu121Thr; Glu121Val; Glu121Trp; Glu121Tyr;
 Leu122Ala; Leu122Cys; Leu122Asp; Leu122Glu;
 Leu122Phe; Leu122Gly; Leu122His; Leu122Ile;
 Leu122Lys; Leu122Met; Leu122Asn; Leu122Pro;
 Leu122Gln; Leu122Arg; Leu122Ser; Leu122Thr;
 Leu122Val; Leu122Trp; Leu122Tyr; Arg123Ala;
 Arg123Cys; Arg123Asp; Arg123Glu; Arg123Phe;
 Arg123Gly; Arg123His; Arg123Ile; Arg123Lys; Arg123Leu;
 Arg123Met; Arg123Asn; Arg123Pro; Arg123Gln;
 Arg123Ser; Arg123Thr; Arg123Val; Arg123Trp; Arg123Tyr;
 Lys124Ala; Lys124Cys; Lys124Asp; Lys124Glu;
 Lys124Phe; Lys124Gly; Lys124His; Lys124Ile; Lys124Leu;
 Lys124Met; Lys124Asn; Lys124Pro; Lys124Gln;
 Lys124Arg; Lys124Ser; Lys124Thr; Lys124Val; Lys124Trp;
 Lys124Tyr; Ile125Ala; Ile125Cys; Ile125Asp; Ile125Glu;
 Ile125Phe; Ile125Gly; Ile125His; Ile125Lys; Ile125Leu;
 Ile125Met; Ile125Asn; Ile125Pro; Ile125Gln; Ile125Arg;
 Ile125Ser; Ile125Thr; Ile125Val; Ile125Trp; Ile125Tyr;
 Gly126Ala; Gly126Cys; Gly126Asp; Gly126Glu;
 Gly126Phe; Gly126His; Gly126Ile; Gly126Lys; Gly126Leu;
 Gly126Met; Gly126Asn; Gly126Pro; Gly126Gln;
 Gly126Arg; Gly126Ser; Gly126Thr; Gly126Val;
 Gly126Trp; Gly126Tyr; Asp127Ala; Asp127Cys;

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Asp127Glu; Asp127Phe; Asp127Gly; Asp127His;
 Asp127Ile; Asp127Lys; Asp127Leu; Asp127Met;
 Asp127Asn; Asp127Pro; Asp127Gln; Asp127Arg;
 Asp127Ser; Asp127Thr; Asp127Val; Asp127Trp;
 5 Asp127Tyr; Glu128Ala; Glu128Cys; Glu128Asp;
 Glu128Phe; Glu128Gly; Glu128His; Glu128Ile; Glu128Lys;
 Glu128Leu; Glu128Met; Glu128Asn; Glu128Pro;
 Glu128Gln; Glu128Arg; Glu128Ser; Glu128Thr;
 Glu128Val; Glu128Trp; Glu128Tyr; Val129Ala; Val129Cys;
 10 Val129Asp; Val129Glu; Val129Phe; Val129Gly; Val129His;
 Val129Ile; Val129Lys; Val129Leu; Val129Met; Val129Asn;
 Val129Pro; Val129Gln; Val129Arg; Val129Ser; Val129Thr;
 Val129Trp; Val129Tyr; Thr130Ala; Thr130Cys; Thr130Asp;
 Thr130Glu; Thr130Phe; Thr130Gly; Thr130His; Thr130Ile;
 15 Thr130Lys; Thr130Leu; Thr130Met; Thr130Asn;
 Thr130Pro; Thr130Gln; Thr130Arg; Thr130Ser; Thr130Val;
 Thr130Trp; Thr130Tyr; Asn131Ala; Asn131Cys;
 Asn131Asp; Asn131Glu; Asn131Phe; Asn131Gly;
 Asn131His; Asn131Ile; Asn131Lys; Asn131Leu;
 20 Asn131Met; Asn131Pro; Asn131Gln; Asn131Arg;
 Asn131Ser; Asn131Thr; Asn131Val; Asn131Trp;
 Asn131Tyr; Pro132Ala; Pro132Cys; Pro132Asp;
 Pro132Glu; Pro132Phe; Pro132Gly; Pro132His; Pro132Ile;
 Pro132Lys; Pro132Leu; Pro132Met; Pro132Asn;
 25 Pro132Gln; Pro132Arg; Pro132Ser; Pro132Thr; Pro132Val;
 Pro132Trp; Pro132Tyr; Glu133Ala; Glu133Cys;
 Glu133Asp; Glu133Phe; Glu133Gly; Glu133His;
 Glu133Ile; Glu133Lys; Glu133Leu; Glu133Met;
 Glu133Asn; Glu133Pro; Glu133Gln; Glu133Arg;
 30 Glu133Ser; Glu133Thr; Glu133Val; Glu133Trp; Glu133Tyr;
 Arg134Ala; Arg134Cys; Arg134Asp; Arg134Glu;
 Arg134Phe; Arg134Gly; Arg134His; Arg134Ile; Arg134Lys;
 Arg134Leu; Arg134Met; Arg134Asn; Arg134Trp;
 Arg134Gln; Arg134Ser; Arg134Thr; Arg134Val; Arg134Trp;
 35 Arg134Tyr; Phe135Ala; Phe135Cys; Phe135Asp;
 Phe135Glu; Phe135Gly; Phe135His; Phe135Ile; Phe135Lys;
 Phe135Leu; Phe135Met; Phe135Asn; Phe135Pro;
 Phe135Gln; Phe135Arg; Phe135Ser; Phe135Thr;
 Phe135Val; Phe135Trp; Phe135Tyr; Glu136Ala;
 40 Glu136Cys; Glu136Asp; Glu136Phe; Glu136Gly;
 Glu136His; Glu136Ile; Glu136Lys; Glu136Leu;
 Glu136Met; Glu136Asn; Glu136Pro; Glu136Gln;
 Glu136Arg; Glu136Ser; Glu136Thr; Glu136Val;
 Glu136Trp; Glu136Tyr; Pro137Ala; Pro137Cys;
 45 Pro137Asp; Pro137Glu; Pro137Phe; Pro137Gly; Pro137His;
 Pro137Ile; Pro137Lys; Pro137Leu; Pro137Met; Pro137Asn;
 Pro137Gln; Pro137Arg; Pro137Ser; Pro137Thr; Pro137Val;
 Pro137Trp; Pro137Tyr; Glu138Ala; Glu138Cys;
 Glu138Asp; Glu138Phe; Glu138Gly; Glu138His;
 50 Glu138Ile; Glu138Lys; Glu138Leu; Glu138Met;
 Glu138Asn; Glu138Pro; Glu138Gln; Glu138Arg;
 Glu138Ser; Glu138Thr; Glu138Val; Glu138Trp; Glu138Tyr;
 Leu139Ala; Leu139Cys; Leu139Asp; Leu139Glu;
 Leu139Phe; Leu139Gly; Leu139His; Leu139Ile;
 55 Leu139Lys; Leu139Met; Leu139Asn; Leu139Pro;
 Leu139Gln; Leu139Arg; Leu139Ser; Leu139Thr;
 Leu139Val; Leu139Trp; Leu139Tyr; Asn140Ala;
 Asn140Cys; Asn140Asp; Asn140Glu; Asn140Phe;
 Asn140Gly; Asn140His; Asn140Ile; Asn140Lys;
 60 Asn140Leu; Asn140Met; Asn140Pro; Asn140Gln;
 Asn140Arg; Asn140Ser; Asn140Thr; Asn140Val;
 Asn140Trp; Asn140Tyr; Glu141Ala; Glu141Cys;
 Glu141Asp; Glu141Phe; Glu141Gly; Glu141His;
 Glu141Ile; Glu141Lys; Glu141Leu; Glu141Met;
 65 Glu141Asn; Glu141Pro; Glu141Gln; Glu141Arg;
 Glu141Ser; Glu141Thr; Glu141Val; Glu141Trp; Glu141Tyr;
 Val142Ala; Val142Cys; Val142Asp; Val142Glu; Val142Phe;

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Val142Gly; Val142His; Val142Ile; Val142Lys; Val142Leu;
 Val142Met; Val142Asn; Val142Pro; Val142Gln; Val142Arg;
 Val142Ser; Val142Thr; Val142Trp; Val142Tyr; Asn143Ala;
 Asn143Cys; Asn143Asp; Asn143Glu; Asn143Phe;
 Asn143Gly; Asn143His; Asn143Ile; Asn143Lys;
 Asn143Leu; Asn143Met; Asn143Pro; Asn143Gln;
 Asn143Arg; Asn143Ser; Asn143Thr; Asn143Val;
 Asn143Trp; Asn143Tyr; Pro144Ala; Pro144Cys;
 Pro144Asp; Pro144Glu; Pro144Phe; Pro144Gly; Pro144His;
 Pro144Ile; Pro144Lys; Pro144Leu; Pro144Met; Pro144Asn;
 Pro144Gln; Pro144Arg; Pro144Ser; Pro144Thr; Pro144Val;
 Pro144Trp; Pro144Tyr; Gly145Ala; Gly145Cys;
 Gly145Asp; Gly145Glu; Gly145Phe; Gly145His;
 Gly145Ile; Gly145Lys; Gly145Leu; Gly145Met;
 Gly145Asn; Gly145Pro; Gly145Gln; Gly145Arg;
 Gly145Ser; Gly145Thr; Gly145Val; Gly145Trp; Gly145Tyr;
 Glu146Ala; Glu146Cys; Glu146Asp; Glu146Phe;
 Glu146Gly; Glu146His; Glu146Ile; Glu146Lys; Glu146Leu;
 Glu146Met; Glu146Asn; Glu146Pro; Glu146Gln;
 Glu146Arg; Glu146Ser; Glu146Thr; Glu146Val;
 Glu146Trp; Glu146Tyr; Thr147Ala; Thr147Cys;
 Thr147Asp; Thr147Glu; Thr147Phe; Thr147Gly;
 Thr147His; Thr147Ile; Thr147Lys; Thr147Leu; Thr147Met;
 Thr147Asn; Thr147Pro; Thr147Gln; Thr147Arg;
 Thr147Ser; Thr147Val; Thr147Trp; Thr147Tyr; Gln148Ala;
 Gln148Cys; Gln148Asp; Gln148Glu; Gln148Phe;
 Gln148Gly; Gln148His; Gln148Ile; Gln148Lys; Gln148Leu;
 Gln148Met; Gln148Asn; Gln148Pro; Gln148Arg;
 Gln148Ser; Gln148Thr; Gln148Val; Gln148Trp; Gln148Tyr;
 Asp149Ala; Asp149Cys; Asp149Glu; Asp149Phe;
 Asp149Gly; Asp149His; Asp149Ile; Asp149Lys;
 Asp149Leu; Asp149Met; Asp149Asn; Asp149Pro;
 Asp149Gln; Asp149Arg; Asp149Ser; Asp149Thr;
 Asp149Val; Asp149Trp; Asp149Tyr; Thr150Ala;
 Thr150Cys; Thr150Asp; Thr150Glu; Thr150Phe;
 Thr150Gly; Thr150His; Thr150Ile; Thr150Lys; Thr150Leu;
 Thr150Met; Thr150Asn; Thr150Pro; Thr150Gln;
 Thr150Arg; Thr150Ser; Thr150Val; Thr150Trp; Thr150Tyr;
 Ser151Ala; Ser151Cys; Ser151Asp; Ser151Glu; Ser151Phe;
 Ser151Gly; Ser151His; Ser151Ile; Ser151Lys; Ser151Leu;
 Ser151Met; Ser151Asn; Ser151Pro; Ser151Gln; Ser151Arg;
 Ser151Thr; Ser151Val; Ser151Trp; Ser151Tyr; Thr152Ala;
 Thr152Cys; Thr152Asp; Thr152Glu; Thr152Phe;
 Thr152Gly; Thr152His; Thr152Ile; Thr152Lys; Thr152Leu;
 Thr152Met; Thr152Asn; Thr152Pro; Thr152Gln;
 Thr152Arg; Thr152Ser; Thr152Val; Thr152Trp; Thr152Tyr;
 Ala153Cys; Ala153Asp; Ala153Glu; Ala153Phe;
 Ala153Gly; Ala153His; Ala153Ile; Ala153Lys; Ala153Leu;
 Ala153Met; Ala153Asn; Ala153Pro; Ala153Gln;
 Ala153Arg; Ala153Ser; Ala153Thr; Ala153Val; Ala153Trp;
 Ala153Tyr; Arg154Ala; Arg154Cys; Arg154Asp;
 Arg154Glu; Arg154Phe; Arg154Gly; Arg154His; Arg154Ile;
 Arg154Lys; Arg154Leu; Arg154Met; Arg154Asn;
 Arg154Pro; Arg154Gln; Arg154Ser; Arg154Thr; Arg154Val;
 Arg154Trp; Arg154Tyr; Ala155Cys; Ala155Asp;
 Ala155Glu; Ala155Phe; Ala155Gly; Ala155His; Ala155Ile;
 Ala155Lys; Ala155Leu; Ala155Met; Ala155Asn;
 Ala155Pro; Ala155Gln; Ala155Arg; Ala155Ser; Ala155Thr;
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 Leu156His; Leu156Ile; Leu156Lys; Leu156Met;
 Leu156Asn; Leu156Pro; Leu156Gln; Leu156Arg;
 Leu156Ser; Leu156Thr; Leu156Val; Leu156Trp;
 Leu156Tyr; Val157Ala; Val157Cys; Val157Asp; Val157Glu;
 Val157Phe; Val157Gly; Val157His; Val157Ile; Val157Lys;
 Val157Leu; Val157Met; Val157Asn; Val157Pro; Val157Gln;
 Val157Arg; Val157Ser; Val157Thr; Val157Trp; Val157Tyr;

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Thr158Ala; Thr158Cys; Thr158Asp; Thr158Glu;
 Thr158Phe; Thr158Gly; Thr158His; Thr158Ile; Thr158Lys;
 Thr158Leu; Thr158Met; Thr158Asn; Thr158Pro;
 Thr158Gln; Thr158Arg; Thr158Ser; Thr158Val; Thr158Trp;
 Thr158Tyr; Ser159Ala; Ser159Cys; Ser159Asp; Ser159Glu;
 Ser159Phe; Ser159Gly; Ser159His; Ser159Ile; Ser159Lys;
 Ser159Leu; Ser159Met; Ser159Asn; Ser159Pro; Ser159Gln;
 Ser159Arg; Ser159Thr; Ser159Val; Ser159Trp; Ser159Tyr;
 Leu160Ala; Leu160Cys; Leu160Asp; Leu160Glu;
 Leu160Phe; Leu160Gly; Leu160His; Leu160Ile;
 Leu160Lys; Leu160Met; Leu160Asn; Leu160Pro;
 Leu160Gln; Leu160Arg; Leu160Ser; Leu160Thr;
 Leu160Val; Leu160Trp; Leu160Tyr; Arg161Ala;
 Arg161Cys; Arg161Asp; Arg161Glu; Arg161Phe;
 Arg161Gly; Arg161His; Arg161Ile; Arg161Lys; Arg161Leu;
 Arg161Met; Arg161Asn; Arg161Pro; Arg161Gln;
 Arg161Ser; Arg161Thr; Arg161Val; Arg161Trp; Arg161Tyr;
 Ala162Cys; Ala162Asp; Ala162Glu; Ala162Phe;
 Ala162Gly; Ala162His; Ala162Ile; Ala162Lys; Ala162Leu;
 Ala162Met; Ala162Asn; Ala162Pro; Ala162Gln;
 Ala162Arg; Ala162Ser; Ala162Thr; Ala162Val; Ala162Trp;
 Ala162Tyr; Phe163Ala; Phe163Cys; Phe163Asp;
 Phe163Glu; Phe163Gly; Phe163His; Phe163Ile; Phe163Lys;
 Phe163Leu; Phe163Met; Phe163Asn; Phe163Pro;
 Phe163Gln; Phe163Arg; Phe163Ser; Phe163Thr;
 Phe163Val; Phe163Trp; Phe163Tyr; Ala164Cys;
 Ala164Asp; Ala164Glu; Ala164Phe; Ala164Gly; Ala164His;
 Ala164Ile; Ala164Lys; Ala164Leu; Ala164Met; Ala164Asn;
 Ala164Pro; Ala164Gln; Ala164Arg; Ala164Ser; Ala164Thr;
 Ala164Val; Ala164Trp; Ala164Tyr; Leu165Ala; Leu165Cys;
 Leu165Asp; Leu165Glu; Leu165Phe; Leu165Gly;
 Leu165His; Leu165Ile; Leu165Lys; Leu165Met;
 Leu165Asn; Leu165Pro; Leu165Gln; Leu165Arg;
 Leu165Ser; Leu165Thr; Leu165Val; Leu165Tyr;
 Glu166Ala; Glu166Cys; Glu166Asp;
 Glu166Phe; Glu166Gly; Glu166His; Glu166Ile; Glu166Lys;
 Glu166Leu; Glu166Met; Glu166Asn; Glu166Pro;
 Glu166Gln; Glu166Arg; Glu166Ser; Glu166Thr;
 Glu166Val; Glu166Trp; Glu166Tyr; Asp167Ala;
 Asp167Cys; Asp167Glu; Asp167Phe; Asp167Gly;
 Asp167His; Asp167Ile; Asp167Lys; Asp167Leu;
 Asp167Met; Asp167Asn; Asp167Pro; Asp167Gln;
 Asp167Arg; Asp167Ser; Asp167Thr; Asp167Val;
 Asp167Trp; Asp167Tyr; Lys168Ala; Lys168Cys;
 Lys168Asp; Lys168Glu; Lys168Phe; Lys168Gly;
 Lys168His; Lys168Ile; Lys168Leu; Lys168Met; Lys168Asn;
 Lys168Pro; Lys168Gln; Lys168Arg; Lys168Ser; Lys168Thr;
 Lys168Val; Lys168Trp; Lys168Tyr; Leu169Ala;
 Leu169Cys; Leu169Asp; Leu169Glu; Leu169Phe;
 Leu169Gly; Leu169His; Leu169Ile; Leu169Lys;
 Leu169Met; Leu169Asn; Leu169Pro; Leu169Gln;
 Leu169Arg; Leu169Ser; Leu169Thr; Leu169Val;
 Leu169Trp; Leu169Tyr; Pro170Ala; Pro170Cys;
 Pro170Asp; Pro170Glu; Pro170Phe; Pro170Gly; Pro170His;
 Pro170Ile; Pro170Lys; Pro170Leu; Pro170Met; Pro170Asn;
 Pro170Gln; Pro170Arg; Pro170Ser; Pro170Thr; Pro170Val;
 Pro170Trp; Pro170Tyr; Ser171Ala; Ser171Cys; Ser171Asp;
 Ser171Glu; Ser171Phe; Ser171Gly; Ser171His; Ser171Ile;
 Ser171Lys; Ser171Leu; Ser171Met; Ser171Asn; Ser171Pro;
 Ser171Gln; Ser171Arg; Ser171Thr; Ser171Val; Ser171Trp;
 Ser171Tyr; Glu172Ala; Glu172Cys; Glu172Asp;
 Glu172Phe; Glu172Gly; Glu172His; Glu172Ile; Glu172Lys;
 Glu172Leu; Glu172Met; Glu172Asn; Glu172Pro;
 Glu172Gln; Glu172Arg; Glu172Ser; Glu172Thr;
 Glu172Val; Glu172Trp; Glu172Tyr; Lys173Ala; Lys173Cys;
 Lys173Asp; Lys173Glu; Lys173Phe; Lys173Gly;
 Lys173His; Lys173Ile; Lys173Leu; Lys173Met; Lys173Asn;

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Lys173Pro; Lys173Gln; Lys173Arg; Lys173Ser; Lys173Thr;
 Lys173Val; Lys173Trp; Lys173Tyr; Arg174Ala; Arg174Cys;
 Arg174Asp; Arg174Glu; Arg174Phe; Arg174Gly;
 Arg174His; Arg174Ile; Arg174Lys; Arg174Leu;
 Arg174Met; Arg174Asn; Arg174Pro; Arg174Gln;
 Arg174Ser; Arg174Thr; Arg174Val; Arg174Trp; Arg174Tyr;
 Glu175Ala; Glu175Cys; Glu175Asp; Glu175Phe;
 Glu175Gly; Glu175His; Glu175Ile; Glu175Lys; Glu175Leu;
 Glu175Met; Glu175Asn; Glu175Pro; Glu175Gln;
 Glu175Arg; Glu175Ser; Glu175Thr; Glu175Val;
 Glu175Trp; Glu175Tyr; Leu176Ala; Leu176Cys;
 Leu176Asp; Leu176Glu; Leu176Phe; Leu176Gly;
 Leu176His; Leu176Ile; Leu176Lys; Leu176Met;
 Leu176Asn; Leu176Pro; Leu176Gln; Leu176Arg;
 Leu176Ser; Leu176Thr; Leu176Val; Leu176Trp;
 Leu177Arg; Leu177Cys; Leu177Asp; Leu177Phe;
 Leu177Gly; Leu177His; Leu177Ile; Leu177Lys;
 Leu177Met; Leu177Asn; Leu177Pro; Leu177Gln;
 Leu177Arg; Leu177Ser; Leu177Thr; Leu177Val;
 Leu177Trp; Leu177Tyr; Ile178Ala; Ile178Cys;
 Ile178Asp; Ile178Glu; Ile178Phe; Ile178Gly;
 Ile178His; Ile178Lys; Ile178Leu; Ile178Met; Ile178Asn;
 Ile178Pro; Ile178Gln; Ile178Arg; Ile178Ser; Ile178Thr;
 Ile178Val; Ile178Trp; Ile178Tyr; Asp179Ala; Asp179Cys;
 Asp179Glu; Asp179Phe; Asp179Gly; Asp179His;
 Asp179Ile; Asp179Lys; Asp179Leu; Asp179Met;
 Asp179Asn; Asp179Pro; Asp179Gln; Asp179Arg;
 Asp179Ser; Asp179Thr; Asp179Val; Asp179Trp;
 Asp179Tyr; Trp180Ala; Trp180Cys; Trp180Asp;
 Trp180Glu; Trp180Phe; Trp180Gly; Trp180His; Trp180Ile;
 Trp180Lys; Trp180Leu; Trp180Met; Trp180Asn;
 Trp180Pro; Trp180Gln; Trp180Arg; Trp180Ser; Trp180Thr;
 Trp180Val; Trp180Tyr; Met181Ala; Met181Cys;
 Met181Asp; Met181Glu; Met181Phe; Met181Gly;
 Met181His; Met181Ile; Met181Lys; Met181Leu;
 Met181Asn; Met181Pro; Met181Gln; Met181Arg;
 Met181Ser; Met181Thr; Met181Val; Met181Trp;
 Met181Tyr; Lys182Ala; Lys182Cys; Lys182Asp;
 Lys182Glu; Lys182Phe; Lys182Gly; Lys182His; Lys182Ile;
 Lys182Leu; Lys182Met; Lys182Asn; Lys182Pro;
 Lys182Gln; Lys182Arg; Lys182Ser; Lys182Thr; Lys182Val;
 Lys182Trp; Lys182Tyr; Arg183Ala; Arg183Cys;
 Arg183Asp; Arg183Glu; Arg183Phe; Arg183Gly;
 Arg183His; Arg183Ile; Arg183Lys; Arg183Leu;
 Arg183Met; Arg183Asn; Arg183Pro; Arg183Gln;
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 Asn184Ala; Asn184Cys; Asn184Asp; Asn184Glu;
 Asn184Phe; Asn184Gly; Asn184His; Asn184Ile;
 Asn184Lys; Asn184Leu; Asn184Met; Asn184Pro;
 Asn184Gln; Asn184Arg; Asn184Ser; Asn184Thr;
 Asn184Val; Asn184Trp; Asn184Tyr; Thr185Ala;
 Thr185Cys; Thr185Asp; Thr185Glu; Thr185Phe;
 Thr185Gly; Thr185His; Thr185Ile; Thr185Lys;
 Thr185Leu; Thr185Met; Thr185Asn; Thr185Pro;
 Thr185Gln; Thr185Arg; Thr185Ser; Thr185Val;
 Thr185Trp; Thr185Tyr; Thr186Ala; Thr186Cys;
 Thr186Asp; Thr186Glu; Thr186Phe; Thr186Gly;
 Thr186His; Thr186Ile; Thr186Lys; Thr186Leu;
 Thr186Met; Thr186Asn; Thr186Pro; Thr186Gln;
 Thr186Arg; Thr186Ser; Thr186Val; Thr186Trp;
 Thr186Tyr; Gly187Ala; Gly187Cys; Gly187Asp;
 Gly187Glu; Gly187Phe; Gly187His; Gly187Ile;
 Gly187Lys; Gly187Leu; Gly187Met; Gly187Asn;
 Gly187Pro; Gly187Gln; Gly187Arg; Gly187Ser;
 Gly187Thr; Gly187Val; Gly187Trp; Gly187Tyr;
 Asp188Cys; Asp188Glu; Asp188Phe; Asp188Gly;
 Asp188His; Asp188Ile; Asp188Lys; Asp188Leu;
 Asp188Met; Asp188Asn; Asp188Pro; Asp188Gln;

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Asp188Arg; Asp188Ser; Asp188Thr; Asp188Val;
 Asp188Trp; Asp188Tyr; Ala189Cys; Ala189Asp;
 Ala189Glu; Ala189Phe; Ala189Gly; Ala189His; Ala189Ile;
 Ala189Lys; Ala189Leu; Ala189Met; Ala189Asn;
 Ala189Pro; Ala189Gln; Ala189Arg; Ala189Ser; Ala189Thr;
 Ala189Val; Ala189Trp; Ala189Tyr; Leu190Ala; Leu190Cys;
 Leu190Asp; Leu190Glu; Leu190Phe; Leu190Gly;
 Leu190His; Leu190Ile; Leu190Lys; Leu190Met;
 Leu190Asn; Leu190Pro; Leu190Gln; Leu190Arg;
 Leu190Ser; Leu190Thr; Leu190Val; Leu190Trp;
 Leu190Tyr; Ile191Ala; Ile191Cys; Ile191Asp; Ile191Glu;
 Ile191Phe; Ile191Gly; Ile191His; Ile191Lys; Ile191Leu;
 Ile191Met; Ile191Asn; Ile191Pro; Ile191Gln; Ile191Arg;
 Ile191Ser; Ile191Thr; Ile191Val; Ile191Trp; Ile191Tyr;
 Arg192Ala; Arg192Cys; Arg192Asp; Arg192Glu;
 Arg192Phe; Arg192Gly; Arg192His; Arg192Ile; Arg192Lys;
 Arg192Leu; Arg192Met; Arg192Asn; Arg192Pro;
 Arg192Gln; Arg192Ser; Arg192Thr; Arg192Val; Arg192Trp;
 Arg192Tyr; Ala193Cys; Ala193Asp; Ala193Glu;
 Ala193Phe; Ala193Gly; Ala193His; Ala193Ile; Ala193Lys;
 Ala193Leu; Ala193Met; Ala193Asn; Ala193Pro;
 Ala193Gln; Ala193Arg; Ala193Ser; Ala193Thr; Ala193Val;
 Ala193Trp; Ala193Tyr; Gly194Ala; Gly194Cys;
 Gly194Asp; Gly194Glu; Gly194Phe; Gly194His;
 Gly194Ile; Gly194Lys; Gly194Leu; Gly194Met;
 Gly194Asn; Gly194Pro; Gly194Gln; Gly194Arg;
 Gly194Ser; Gly194Thr; Gly194Val; Gly194Trp; Gly194Tyr;
 Val195Ala; Val195Cys; Val195Asp; Val195Glu; Val195Phe;
 Val195Gly; Val195His; Val195Ile; Val195Lys; Val195Leu;
 Val195Met; Val195Asn; Val195Pro; Val195Gln; Val195Arg;
 Val195Ser; Val195Thr; Val195Trp; Val195Tyr; Pro196Ala;
 Pro196Cys; Pro196Asp; Pro196Glu; Pro196Phe;
 Pro196Gly; Pro196His; Pro196Ile; Pro196Lys; Pro196Leu;
 Pro196Met; Pro196Asn; Pro196Gln; Pro196Arg;
 Pro196Ser; Pro196Thr; Pro196Val; Pro196Trp; Pro196Tyr;
 Asp197Ala; Asp197Cys; Asp197Glu; Asp197Phe;
 Asp197Gly; Asp197His; Asp197Ile; Asp197Lys;
 Asp197Leu; Asp197Met; Asp197Asn; Asp197Pro;
 Asp197Gln; Asp197Arg; Asp197Ser; Asp197Thr;
 Asp197Val; Asp197Trp; Asp197Tyr; Gly198Ala;
 Gly198Cys; Gly198Asp; Gly198Glu; Gly198Phe;
 Gly198His; Gly198Ile; Gly198Lys; Gly198Leu;
 Gly198Met; Gly198Asn; Gly198Pro; Gly198Gln;
 Gly198Arg; Gly198Ser; Gly198Thr; Gly198Val;
 Gly198Trp; Gly198Tyr; Thr199Ala; Thr199Cys;
 Thr199Asp; Thr199Glu; Thr199Phe; Thr199Gly; Thr199His;
 Thr199Ile; Thr199Lys; Thr199Leu; Thr199Met; Thr199Asn;
 Thr199Pro; Thr199Gln; Thr199Arg; Thr199Ser; Thr199Thr;
 Thr199Val; Thr199Tyr; Glu200Ala; Glu200Cys;
 Glu200Asp; Glu200Phe; Glu200Gly; Glu200His;
 Glu200Ile; Glu200Lys; Glu200Leu; Glu200Met;
 Glu200Asn; Glu200Pro; Glu200Gln; Glu200Arg;
 Glu200Ser; Glu200Thr; Glu200Val; Glu200Trp; Glu200Tyr;
 Val201Ala; Val201Cys; Val201Asp; Val201Glu; Val201Phe;
 Val201Gly; Val201His; Val201Ile; Val201Lys; Val201Leu;
 Val201Met; Val201Asn; Val201Pro; Val201Gln; Val201Arg;
 Val201Ser; Val201Thr; Val201Trp; Val201Tyr; Ala202Cys;
 Ala202Asp; Ala202Glu; Ala202Phe; Ala202Gly; Ala202His;
 Ala202Ile; Ala202Lys; Ala202Leu; Ala202Met; Ala202Asn;
 Ala202Pro; Ala202Gln; Ala202Arg; Ala202Ser; Ala202Thr;
 Ala202Val; Ala202Trp; Ala202Tyr; Asp203Ala; Asp203Cys;
 Asp203Glu; Asp203Phe; Asp203Gly; Asp203His;
 Asp203Ile; Asp203Lys; Asp203Leu; Asp203Met;
 Asp203Asn; Asp203Pro; Asp203Gln; Asp203Arg;
 Asp203Ser; Asp203Thr; Asp203Val; Asp203Trp;
 Asp203Tyr; Lys204Ala; Lys204Cys; Lys204Asp;
 Lys204Glu; Lys204Phe; Lys204Gly; Lys204His; Lys204Ile;

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Lys204Leu; Lys204Met; Lys204Asn; Lys204Pro;
 Lys204Gln; Lys204Arg; Lys204Ser; Lys204Thr; Lys204Val;
 Lys204Trp; Lys204Tyr; Thr205Ala; Thr205Cys;
 Thr205Asp; Thr205Glu; Thr205Phe; Thr205Gly;
 Thr205His; Thr205Ile; Thr205Lys; Thr205Leu; Thr205Met;
 Thr205Asn; Thr205Pro; Thr205Gln; Thr205Arg;
 Thr205Ser; Thr205Val; Thr205Trp; Thr205Tyr; Gly206Ala;
 Gly206Cys; Gly206Asp; Gly206Glu; Gly206Phe;
 Gly206His; Gly206Ile; Gly206Lys; Gly206Leu;
 Gly206Met; Gly206Asn; Gly206Pro; Gly206Gln;
 Gly206Arg; Gly206Ser; Gly206Thr; Gly206Val;
 Gly206Trp; Gly206Tyr; Ala207Cys; Ala207Asp;
 Ala207Glu; Ala207Phe; Ala207Gly; Ala207His; Ala207Ile;
 Ala207Lys; Ala207Leu; Ala207Met; Ala207Asn;
 Ala207Pro; Ala207Gln; Ala207Arg; Ala207Ser; Ala207Thr;
 Ala207Val; Ala207Trp; Ala207Tyr; Ala208Cys; Ala208Asp;
 Ala208Glu; Ala208Phe; Ala208Gly; Ala208His; Ala208Ile;
 Ala208Lys; Ala208Leu; Ala208Met; Ala208Asn;
 Ala208Pro; Ala208Gln; Ala208Arg; Ala208Ser; Ala208Thr;
 Ala208Val; Ala208Trp; Ala208Tyr; Ser209Ala; Ser209Cys;
 Ser209Asp; Ser209Glu; Ser209Phe; Ser209Gly; Ser209His;
 Ser209Ile; Ser209Lys; Ser209Leu; Ser209Met; Ser209Asn;
 Ser209Pro; Ser209Gln; Ser209Arg; Ser209Thr; Ser209Val;
 Ser209Trp; Ser209Tyr; Tyr210Ala; Tyr210Cys; Tyr210Asp;
 Tyr210Glu; Tyr210Phe; Tyr210Gly; Tyr210His; Tyr210Ile;
 Tyr210Lys; Tyr210Leu; Tyr210Met; Tyr210Asn; Tyr210Pro;
 Tyr210Gln; Tyr210Arg; Tyr210Ser; Tyr210Thr; Tyr210Val;
 Tyr210Trp; Gly211Ala; Gly211Cys; Gly211Asn;
 Gly211Glu; Gly211Phe; Gly211His; Gly211Ile; Gly211Lys;
 Gly211Leu; Gly211Met; Gly211Asn; Gly211Pro;
 Gly211Gln; Gly211Arg; Gly211Ser; Gly211Thr;
 Gly211Val; Gly211Trp; Gly211Tyr; Thr212Ala; Thr212Cys;
 Thr212Asp; Thr212Glu; Thr212Phe; Thr212Gly;
 Thr212His; Thr212Ile; Thr212Lys; Thr212Leu; Thr212Met;
 Thr212Asn; Thr212Pro; Thr212Gln; Thr212Arg;
 Thr212Ser; Thr212Val; Thr212Trp; Thr212Tyr; Arg213Ala;
 Arg213Cys; Arg213Asp; Arg213Glu; Arg213Phe;
 Arg213Gly; Arg213His; Arg213Ile; Arg213Lys; Arg213Leu;
 Arg213Met; Arg213Asn; Arg213Pro; Arg213Gln;
 Arg213Ser; Arg213Thr; Arg213Val; Arg213Trp; Arg213Tyr;
 Asn214Ala; Asn214Cys; Asn214Asp; Asn214Glu;
 Asn214Phe; Asn214Gly; Asn214His; Asn214Ile;
 Asn214Lys; Asn214Leu; Asn214Met; Asn214Pro;
 Asn214Gln; Asn214Arg; Asn214Ser; Asn214Thr;
 Asn214Val; Asn214Trp; Asn214Tyr; Asp215Ala;
 Asp215Cys; Asp215Glu; Asp215Phe; Asp215Gly;
 Asp215His; Asp215Ile; Asp215Lys; Asp215Leu;
 Asp215Met; Asp215Asn; Asp215Pro; Asp215Gln;
 Asp215Arg; Asp215Ser; Asp215Thr; Asp215Val;
 Asp215Trp; Asp215Tyr; Ile216Ala; Ile216Cys; Ile216Asp;
 Ile216Glu; Ile216Phe; Ile216Gly; Ile216His; Ile216Lys;
 Ile216Leu; Ile216Met; Ile216Asn; Ile216Pro; Ile216Gln;
 Ile216Arg; Ile216Ser; Ile216Thr; Ile216Val; Ile216Trp;
 Ile216Tyr; Ala217Cys; Ala217Asp; Ala217Glu; Ala217Phe;
 Ala217Gly; Ala217His; Ala217Ile; Ala217Lys; Ala217Leu;
 Ala217Met; Ala217Asn; Ala217Pro; Ala217Gln;
 Ala217Arg; Ala217Ser; Ala217Thr; Ala217Val; Ala217Trp;
 Ala217Tyr; Ile218Ala; Ile218Cys; Ile218Asp; Ile218Glu;
 Ile218Phe; Ile218Gly; Ile218His; Ile218Lys; Ile218Leu;
 Ile218Met; Ile218Asn; Ile218Pro; Ile218Gln; Ile218Arg;
 Ile218Ser; Ile218Thr; Ile218Val; Ile218Trp; Ile218Tyr;
 Ile219Ala; Ile219Cys; Ile219Asp; Ile219Glu; Ile219Phe;
 Ile219Gly; Ile219His; Ile219Lys; Ile219Leu; Ile219Met;
 Ile219Asn; Ile219Pro; Ile219Gln; Ile219Arg; Ile219Ser;
 Ile219Thr; Ile219Val; Ile219Trp; Ile219Tyr; Trp220Ala;
 Trp220Cys; Trp220Asp; Trp220Glu; Trp220Phe;
 Trp220Gly; Trp220His; Trp220Ile; Trp220Lys; Trp220Leu;

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Trp220Met; Trp220Asn; Trp220Pro; Trp220Gln;
 Trp220Arg; Trp220Ser; Trp220Thr; Trp220Val; Trp220Tyr;
 Pro221Ala; Pro221Cys; Pro221Asp; Pro221Glu;
 Pro221Phe; Pro221Gly; Pro221His; Pro221Ile; Pro221Lys;
 Pro221Leu; Pro221Met; Pro221Asn; Pro221Gln;
 Pro221Arg; Pro221Ser; Pro221Thr; Pro221Val; Pro221Trp;
 Pro221Tyr; Pro222Ala; Pro222Cys; Pro222Asp; Pro222Glu;
 Pro222Phe; Pro222Gly; Pro222His; Pro222Ile; Pro222Lys;
 Pro222Leu; Pro222Met; Pro222Asn; Pro222Gln;
 Pro222Arg; Pro222Ser; Pro222Thr; Pro222Val; Pro222Trp;
 Pro222Tyr; Lys223Ala; Lys223Cys; Lys223Asp;
 Lys223Glu; Lys223Phe; Lys223Gly; Lys223His; Lys223Ile;
 Lys223Leu; Lys223Met; Lys223Asn; Lys223Pro;
 Lys223Gln; Lys223Arg; Lys223Ser; Lys223Thr; Lys223Val;
 Lys223Trp; Lys223Tyr; Gly224Ala; Gly224Cys;
 Gly224Asp; Gly224Glu; Gly224Phe; Gly224His;
 Gly224Ile; Gly224Lys; Gly224Leu; Gly224Met;
 Gly224Asn; Gly224Pro; Gly224Gln; Gly224Arg;
 Gly224Ser; Gly224Thr; Gly224Val; Gly224Trp; Gly224Tyr;
 Asp225Ala; Asp225Cys; Asp225Glu; Asp225Phe;
 Asp225Gly; Asp225His; Asp225Ile; Asp225Lys;
 Asp225Leu; Asp225Met; Asp225Asn; Asp225Pro;
 Asp225Gln; Asp225Arg; Asp225Ser; Asp225Thr;
 Asp225Val; Asp225Trp; Asp225Tyr; Pro226Ala;
 Pro226Cys; Pro226Asp; Pro226Glu; Pro226Phe;
 Pro226Gly; Pro226His; Pro226Ile; Pro226Lys; Pro226Leu;
 Pro226Met; Pro226Asn; Pro226Gln; Pro226Arg;
 Pro226Ser; Pro226Thr; Pro226Val; Pro226Trp; Pro226Tyr;
 Val227Ala; Val227Cys; Val227Asp; Val227Glu; Val227Phe;
 Val227Gly; Val227His; Val227Ile; Val227Lys; Val227Leu;
 Val227Met; Val227Asn; Val227Pro; Val227Gln; Val227Arg;
 Val227Ser; Val227Thr; Val227Trp; Val227Tyr; Val228Ala;
 Val228Cys; Val228Asp; Val228Glu; Val228Phe; Val228Gly;
 Val228His; Val228Ile; Val228Lys; Val228Leu; Val228Met;
 Val228Asn; Val228Pro; Val228Gln; Val228Arg; Val228Ser;
 Val228Thr; Val228Trp; Val228Tyr; Leu229Ala; Leu229Cys;
 Leu229Asp; Leu229Glu; Leu229Phe; Leu229Gly;
 Leu229His; Leu229Ile; Leu229Lys; Leu229Met;
 Leu229Asn; Leu229Pro; Leu229Gln; Leu229Arg;
 Leu229Ser; Leu229Thr; Leu229Val; Leu229Trp;
 Leu229Tyr; Ala230Cys; Ala230Asp; Ala230Glu;
 Ala230Phe; Ala230Gly; Ala230His; Ala230Ile; Ala230Lys;
 Ala230Leu; Ala230Met; Ala230Asn; Ala230Pro;
 Ala230Gln; Ala230Arg; Ala230Ser; Ala230Thr; Ala230Val;
 Ala230Trp; Ala230Tyr; Val231Ala; Val231Cys; Val231Asp;
 Val231Glu; Val231Phe; Val231Gly; Val231His; Val231Ile;
 Val231Lys; Val231Leu; Val231Met; Val231Asn; Val231Pro;
 Val231Gln; Val231Arg; Val231Ser; Val231Thr; Val231Trp;
 Val231Tyr; Leu232Ala; Leu232Cys; Leu232Asp;
 Leu232Glu; Leu232Phe; Leu232Gly; Leu232His;
 Leu232Ile; Leu232Lys; Leu232Met; Leu232Asn;
 Leu232Pro; Leu232Gln; Leu232Arg; Leu232Ser;
 Leu232Thr; Leu232Val; Leu232Trp; Leu232Tyr;
 Ser233Ala; Ser233Cys; Ser233Asp; Ser233Glu; Ser233Phe;
 Ser233Gly; Ser233His; Ser233Ile; Ser233Lys; Ser233Leu;
 Ser233Met; Ser233Asn; Ser233Pro; Ser233Gln; Ser233Arg;
 Ser233Thr; Ser233Val; Ser233Trp; Ser233Tyr; Ser234Ala;
 Ser234Cys; Ser234Asp; Ser234Glu; Ser234Phe; Ser234Gly;
 Ser234His; Ser234Ile; Ser234Lys; Ser234Leu; Ser234Met;
 Ser234Asn; Ser234Pro; Ser234Gln; Ser234Arg; Ser234Thr;
 Ser234Val; Ser234Trp; Ser234Tyr; Arg235Ala; Arg235Cys;
 Arg235Asp; Arg235Glu; Arg235Phe; Arg235Gly;
 Arg235His; Arg235Ile; Arg235Lys; Arg235Leu;
 Arg235Met; Arg235Asn; Arg235Ser; Arg235Pro; Arg235Gln;
 Arg235Ser; Arg235Thr; Arg235Val; Arg235Trp; Arg235Tyr;
 Asp236Ala; Asp236Cys; Asp236Glu; Asp236Phe;
 Asp236Gly; Asp236His; Asp236Ile; Asp236Lys;

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Asp236Leu; Asp236Met; Asp236Asn; Asp236Pro;
 Asp236Gln; Asp236Arg; Asp236Ser; Asp236Thr;
 Asp236Val; Asp236Trp; Asp236Tyr; Lys237Ala;
 Lys237Cys; Lys237Asp; Lys237Glu; Lys237Phe;
 Lys237Gly; Lys237His; Lys237Ile; Lys237Leu; Lys237Met;
 Lys237Asn; Lys237Pro; Lys237Gln; Lys237Arg;
 Lys237Ser; Lys237Thr; Lys237Val; Lys237Trp; Lys237Tyr;
 Lys238Ala; Lys238Cys; Lys238Asp; Lys238Glu;
 Lys238Phe; Lys238Gly; Lys238His; Lys238Ile; Lys238Leu;
 Lys238Met; Lys238Asn; Lys238Pro; Lys238Gln;
 Lys238Arg; Lys238Ser; Lys238Thr; Lys238Val; Lys238Trp;
 Lys238Tyr; Asp239Ala; Asp239Cys; Asp239Glu;
 Asp239Phe; Asp239Gly; Asp239His; Asp239Ile;
 Asp239Lys; Asp239Leu; Asp239Met; Asp239Asn;
 Asp239Pro; Asp239Gln; Asp239Arg; Asp239Ser;
 Asp239Thr; Asp239Val; Asp239Trp; Asp239Tyr;
 Ala240Cys; Ala240Asp; Ala240Glu; Ala240Phe;
 Ala240Gly; Ala240His; Ala240Ile; Ala240Lys; Ala240Leu;
 Ala240Met; Ala240Asn; Ala240Pro; Ala240Gln;
 Ala240Arg; Ala240Ser; Ala240Thr; Ala240Val; Ala240Trp;
 Ala240Tyr; Lys241Ala; Lys241Cys; Lys241Asn;
 Lys241Glu; Lys241Phe; Lys241Gly; Lys241His; Lys241Ile;
 Lys241Leu; Lys241Met; Lys241Asn; Lys241Pro;
 Lys241Gln; Lys241Arg; Lys241Ser; Lys241Thr; Lys241Val;
 Lys241Trp; Lys241Tyr; Tyr242Ala; Tyr242Cys; Tyr242Asp;
 Tyr242Glu; Tyr242Phe; Tyr242Gly; Tyr242His; Tyr242Ile;
 Tyr242Lys; Tyr242Leu; Tyr242Met; Tyr242Asn; Tyr242Pro;
 Tyr242Gln; Tyr242Arg; Tyr242Ser; Tyr242Thr; Tyr242Val;
 Tyr242Trp; Asp243Ala; Asp243Cys; Asp243Glu;
 Asp243Phe; Asp243Gly; Asp243His; Asp243Ile;
 Asp243Lys; Asp243Leu; Asp243Met; Asp243Asn;
 Asp243Pro; Asp243Gln; Asp243Arg; Asp243Ser;
 Asp243Thr; Asp243Val; Asp243Trp; Asp243Tyr;
 Asp244Ala; Asp244Cys; Asp244Glu; Asp244Phe;
 Asp244Gly; Asp244His; Asp244Ile; Asp244Lys;
 Asp244Leu; Asp244Met; Asp244Asn; Asp244Pro;
 Asp244Gln; Asp244Arg; Asp244Ser; Asp244Thr;
 Asp244Val; Asp244Trp; Asp244Tyr; Lys245Ala;
 Lys245Cys; Lys245Asp; Lys245Glu; Lys245Phe;
 Lys245Gly; Lys245His; Lys245Ile; Lys245Leu; Lys245Met;
 Lys245Asn; Lys245Pro; Lys245Gln; Lys245Arg;
 Lys245Ser; Lys245Thr; Lys245Val; Lys245Trp; Lys245Tyr;
 Leu246Ala; Leu246Cys; Leu246Asp; Leu246Glu;
 Leu246Phe; Leu246Gly; Leu246His; Leu246Ile;
 Leu246Lys; Leu246Met; Leu246Asn; Leu246Pro;
 Leu246Gln; Leu246Arg; Leu246Ser; Leu246Thr;
 Leu246Val; Leu246Trp; Leu246Tyr; Ile247Ala; Ile247Cys;
 Ile247Asp; Ile247Glu; Ile247Phe; Ile247Gly; Ile247His;
 Ile247Lys; Ile247Leu; Ile247Met; Ile247Asn; Ile247Pro;
 Ile247Gln; Ile247Arg; Ile247Ser; Ile247Thr; Ile247Val;
 Ile247Trp; Ile247Tyr; Ala248Cys; Ala248Asp; Ala248Glu;
 Ala248Phe; Ala248Gly; Ala248His; Ala248Ile; Ala248Lys;
 Ala248Leu; Ala248Met; Ala248Asn; Ala248Pro;
 Ala248Gln; Ala248Arg; Ala248Ser; Ala248Thr; Ala248Val;
 Ala248Trp; Ala248Tyr; Glu249Ala; Glu249Cys;
 Glu249Asp; Glu249Phe; Glu249Gly; Glu249His;
 Glu249Ile; Glu249Lys; Glu249Leu; Glu249Met;
 Glu249Asn; Glu249Pro; Glu249Gln; Glu249Arg;
 Glu249Ser; Glu249Thr; Glu249Val; Glu249Trp; Glu249Tyr;
 Ala250Cys; Ala250Asp; Ala250Glu; Ala250Phe;
 Ala250Gly; Ala250His; Ala250Ile; Ala250Lys; Ala250Leu;
 Ala250Met; Ala250Asn; Ala250Pro; Ala250Gln;
 Ala250Arg; Ala250Ser; Ala250Thr; Ala250Val; Ala250Trp;
 Ala250Tyr; Thr251Ala; Thr251Cys; Thr251Asp;
 Thr251Glu; Thr251Phe; Thr251Gly; Thr251His; Thr251Ile;
 Thr251Lys; Thr251Leu; Thr251Met; Thr251Asn;
 Thr251Pro; Thr251Gln; Thr251Arg; Thr251Ser; Thr251Val;

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Thr251Trp; Thr251Tyr; Lys252Ala; Lys252Cys;
 Lys252Asp; Lys252Glu; Lys252Phe; Lys252Gly;
 Lys252His; Lys252Ile; Lys252Leu; Lys252Met; Lys252Asn;
 Lys252Pro; Lys252Gln; Lys252Arg; Lys252Ser; Lys252Thr;
 Lys252Val; Lys252Trp; Lys252Tyr; Val253Ala; Val253Cys;
 Val253Asp; Val253Glu; Val253Phe; Val253Gly; Val253His;
 Val253Ile; Val253Lys; Val253Leu; Val253Met; Val253Asn;
 Val253Pro; Val253Gln; Val253Arg; Val253Ser; Val253Thr;
 Val253Trp; Val253Tyr; Val254Ala; Val254Cys; Val254Asp;
 Val254Glu; Val254Phe; Val254Gly; Val254His; Val254Ile;
 Val254Lys; Val254Leu; Val254Met; Val254Asn; Val254Pro;
 Val254Gln; Val254Arg; Val254Ser; Val254Thr; Val254Trp;
 Val254Tyr; Met255Ala; Met255Cys; Met255Asp;
 Met255Glu; Met255Phe; Met255Gly; Met255His;
 Met255Ile; Met255Lys; Met255Leu; Met255Asn;
 Met255Pro; Met255Gln; Met255Arg; Met255Ser;
 Met255Thr; Met255Val; Met255Trp; Met255Tyr;
 Lys256Ala; Lys256Cys; Lys256Asp; Lys256Glu;
 Lys256Phe; Lys256Gly; Lys256His; Lys256Ile; Lys256Leu;
 Lys256Met; Lys256Asn; Lys256Pro; Lys256Gln;
 Lys256Arg; Lys256Ser; Lys256Thr; Lys256Val; Lys256Trp;
 Lys256Tyr; Ala257Cys; Ala257Asp; Ala257Glu;
 Ala257Phe; Ala257Gly; Ala257His; Ala257Ile; Ala257Lys;
 Ala257Leu; Ala257Met; Ala257Asn; Ala257Pro;
 Ala257Gln; Ala257Arg; Ala257Ser; Ala257Thr; Ala257Val;
 Ala257Trp; Ala257Tyr; Leu258Ala; Leu258Cys;
 Leu258Asp; Leu258Glu; Leu258Phe; Leu258Gly;
 Leu258His; Leu258Ile; Leu258Lys; Leu258Met;
 Leu258Asn; Leu258Pro; Leu258Gln; Leu258Arg;
 Leu258Ser; Leu258Thr; Leu258Val; Leu258Trp;
 Leu258Tyr; Asn259Ala; Asn259Cys; Asn259Asp;
 Asn259Glu; Asn259Phe; Asn259Gly; Asn259His;
 Asn259Ile; Asn259Lys; Asn259Leu; Asn259Met;
 Asn259Pro; Asn259Gln; Asn259Arg; Asn259Ser;
 Asn259Thr; Asn259Val; Asn259Trp; Asn259Tyr;
 Met260Ala; Met260Cys; Met260Asp; Met260Glu;
 Met260Phe; Met260Gly; Met260His; Met260Ile;
 Met260Lys; Met260Leu; Met260Asn; Met260Pro;
 Met260Gln; Met260Arg; Met260Ser; Met260Thr;
 Met260Val; Met260Trp; Met260Tyr; Asn261Ala;
 Asn261Cys; Asn261Asp; Asn261Glu; Asn261Phe;
 Asn261Gly; Asn261His; Asn261Ile; Asn261Lys;
 Asn261Leu; Asn261Met; Asn261Pro; Asn261Gln;
 Asn261Arg; Asn261Ser; Asn261Thr; Asn261Val;
 Asn261Trp; Asn261Tyr; Gly262Ala; Gly262Cys;
 Gly262Asp; Gly262Glu; Gly262Phe; Gly262His;
 Gly262Ile; Gly262Lys; Gly262Leu; Gly262Met;
 Gly262Asn; Gly262Pro; Gly262Gln; Gly262Arg;
 Gly262Ser; Gly262Thr; Gly262Val; Gly262Trp; Gly262Tyr;
 Lys263Ala; Lys263Cys; Lys263Asp; Lys263Glu;
 Lys263Phe; Lys263Gly; Lys263His; Lys263Ile; Lys263Leu;
 Lys263Met; Lys263Asn; Lys263Pro; Lys263Gln;
 Lys263Arg; Lys263Ser; Lys263Thr; Lys263Val; Lys263Trp;
 Lys263Tyr; Met 264Ala; Met 264Cys; Met 264Asp; Met
 264Glu; Met 264Phe; Met 264Gly; Met 264His; Met 264Ile;
 Met 264Lys; Met 264Leu; Met 264Asn; Met 264Pro; Met
 264Gln; Met 264Arg; Met 264Ser; Met 264Thr; Met 264Val;
 Met 264Trp; Met 264Tyr; Asn 265Ala; Asn 265Cys; Asn
 265Asp; Asn 265Glu; Asn 265Phe; Asn 265Gly; Asn 265His;
 Asn 265Ile; Asn 265Lys; Asn 265Leu; Asn 265Met; ; Asn
 265Pro; Asn 265Gln; Asn 265Arg; Asn 265Ser; Asn 265Thr;
 Asn 265Val; Asn 265Trp; Asn 265Tyr; Gly 266Ala; Gly
 266Cys; Gly 266Asp; Gly 266Glu; Gly 266Phe; ; Gly
 266His; Gly 266Ile; Gly 266Lys; Gly 266Leu; Gly 266Met;
 Gly 266Asn; Gly 266Pro; Gly 266Gln; Gly 266Arg; Gly
 266Ser; Gly 266Thr; Gly 266Val; Gly 266Trp; Gly 266Tyr;
 Lys267Ala; Lys267Cys; Lys267Asp; Lys267Glu;

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Lys267Phe; Lys267Gly; Lys267His; Lys267Ile; Lys267Leu; Lys267Met; Lys267Asn; Lys267Pro; Lys267Gln; Lys267Arg; Lys267Ser; Lys267Thr; Lys267Val; Lys267Trp; and Lys267Tyr. In some embodiments, SEQ ID NO: 1 may have a Met and/or Thr preceeding the first residue of the sequence. These residues may be similarly mutated as above.

In all of these mutants, the numbering of residues corresponds to SEQ ID NO: 1. These residue numbers may be converted to Ambler numbers (Ambler et al., 1991, A standard numbering scheme for the Class A β -lactamases, *Biochem. J.* 276:269-272, the contents of which are hereby incorporated by reference) through use of any conventional bioinformatic method, for example by using BLAST (Basic Local Alignment Search Tools) or FASTA (FAST-All). For example, residue 244 corresponds to Ambler 276. For example, the following conversions may be used:

Ambler Classification No.	SEQ ID NO: 1 Residue
F33	F6
I72	I44
Q135	Q105
G156	G126
T160	T130
A232	A202
A237	A207
A238	A208
S240	S209
T243	T212
R244	R213
S266	S234
D276	D244

Furthermore, percent identity may also be assessed with these conventional bioinformatic methods.

In one aspect, the present invention pertains to beta-lactamases and/or pharmaceutical compositions comprising an amino acid sequence having at least 70% (e.g. about 70%, or about 71%, or about 72%, or about 73%, or about 74%, or about 75%, or about 76%, or about 77%, or about 78%, or about 79%, or about 80%, or about 81%, or about 82%, or about 83%, or about 84%, or about 85%, or about 86%, or about 87%, or about 88%, or about 89%, or about 90%, or about 91%, or about 92%, or about 93%, or about 94%, or about 95%, or about 96%, or about 97%, or about 98%, or about 99%) sequence identity with SEQ ID NO: 1 or SEQ ID NO: 3 and one or more of the following mutations of Ambler classification: F33X, Q135X, G156X, A232X, A237X, A238X, S240X, T243X, R244X, S266X, and D276X, wherein X is any naturally-occurring amino acid and with the proviso that D276X is not present in the context of a single mutant. In some embodiments, X is a naturally occurring hydrophilic or hydrophobic amino acid residue or a non-classical amino acid.

In another aspect, the present invention pertains to beta-lactamases and/or pharmaceutical compositions comprising an amino acid sequence having at least 70% sequence identity with SEQ ID NO: 1 or SEQ ID NO: 3 and one or more of the following mutations of Ambler classification: a hydrophobic residue other than phenylalanine (F) at position 33; a hydrophobic residue other than glutamine (Q) at position 135; a hydrophilic residue other than glycine (G) at position 156; a hydrophobic residue other than alanine (A) at position 232; a hydrophilic residue other than alanine (A) at position 237; a hydrophobic or hydrophilic residue other than alanine (A) at position 238; a hydrophilic residue other than serine (S) at position 240; a hydrophobic residue other than threonine (T) at position 243; a hydrophobic residue other than arginine (R)

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at position 244; a hydrophilic residue other than serine (S) at position 266; and a hydrophilic residue other than aspartate (D) at position 276, with the proviso that hydrophilic amino acid residue other than aspartic acid (D) at a position corresponding to position 276 is not present in the context of a single mutant.

As used throughout, a hydrophilic amino acid residue may include a polar and positively charged hydrophilic residue selected from arginine (R) and lysine (K), a polar and neutral of charge hydrophilic residue selected from asparagine (N), glutamine (Q), serine (S), threonine (T), proline (P), and cysteine (C), a polar and negatively charged hydrophilic residue selected from aspartate (D) and glutamate (E), or an aromatic, polar and positively charged hydrophilic including histidine (H). As used throughout, a hydrophobic amino acid residue may include a hydrophobic, aliphatic amino acid selected from glycine (G), alanine (A), leucine (L), isoleucine (I), methionine (M), or valine (V) or a hydrophobic, aromatic amino acid selected from phenylalanine (F), tryptophan (W), or tyrosine (Y).

Mutations may be made to the gene sequence of a beta-lactamase (e.g. SEQ ID NOs: 3 and 4) by reference to the genetic code, including taking into account codon degeneracy.

In some embodiments, the beta-lactamases and/or pharmaceutical compositions comprise one or more of the following mutations at positions of Ambler classification: F33Y, Q135M, G156R, A232G, A237S, A238G or T, S240P or D, T243I, R244T, S266N, D276N or R or K, provided that D276N or R or K is not in the context of a single mutant. In one embodiment, the beta-lactamases and/or pharmaceutical compositions comprise Q135M. In another embodiment, the beta-lactamases and/or pharmaceutical compositions comprise G156R and A238T. In another embodiment, the beta-lactamases and/or pharmaceutical compositions comprise F33Y and D276N. In still another embodiment, the beta-lactamases and/or pharmaceutical compositions comprise F33Y, S240P, and D276N. In one embodiment, the beta-lactamases and/or pharmaceutical compositions comprise F33Y, A238T, and D276N. In another embodiment, the beta-lactamases and/or pharmaceutical compositions comprise A232G, A237S, A238G, and S240D. In a further embodiment, the beta-lactamases and/or pharmaceutical compositions comprise A232G, A237S, A238G, S240D, and R244T. In another embodiment, the beta-lactamases and/or pharmaceutical compositions comprise A232G, A237S, A238G, S240D, and D276R. In one embodiment, the beta-lactamases and/or pharmaceutical compositions comprise A232G, A237S, A238G, S240D, and D276K. In one embodiment, the beta-lactamases and/or pharmaceutical compositions comprise A232G, A237S, A238G, S240D, and Q135M. In one embodiment, the beta-lactamases and/or pharmaceutical compositions comprise A238T. In one embodiment, the beta-lactamases and/or pharmaceutical compositions comprise T243I, S266N, and D276N. In one embodiment, the beta-lactamases and/or pharmaceutical compositions comprise A232G, A237S, A238G, S240D, and D276N.

In other embodiments, the beta-lactamases and/or pharmaceutical compositions comprise an amino acid sequence having at least 70% sequence identity with SEQ ID NO: 2 and the following of Ambler classification: a hydrophobic residue other than alanine (A) at position 232; a hydrophilic residue other than alanine (A) at position 237; a hydrophobic residue other than alanine (A) at position 238; a hydrophilic residue other than serine (S) at position 240; and a hydrophilic residue other than aspartate (D) at position 276. In some embodiments, the hydrophobic residue other than alanine (A) at

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position 232 is glycine (G). In some embodiments, the hydrophilic residue other than alanine (A) at position 237 is serine (S). In some embodiments, the hydrophobic residue other than alanine (A) at position 238 is glycine (G). In some embodiments, the hydrophilic residue other than serine (S) at position 240 is aspartate (D). In some embodiments, the other than aspartate (D) at position 276 is asparagine (N). In some embodiments, the beta-lactamase and/or pharmaceutical composition comprises one or more of A232G, A237S, A238G, S240D, and D276N. In some embodiments, the beta-lactamase and/or pharmaceutical composition comprises all of A232G, A237S, A238G, S240D, and D276N, the sequence of which may be SEQ ID NO:5:

SEQ ID NO: 5

EMKDDFAKLEEQFADKLGI FALDTGTNRTVAYRPDERFAFASTIKALTVG
VLLQOKSIEDLNQRITTRDDL VNYNPITEKHVDGMTLKLADASLRYSD
NAAQNLIKQIGGPESLKKELRKIGDEV TNP EPELNEVNPGETQDTS
TARALVTS LRAFALEDKL PSEKRELLIDWMKRNTTG DALIRAGVPDGEV
GDKTGS G DYGTRNDIAI IWPPKGPVVLAVLSSRDKKDAKYDNKLIAEAT
KVVMKALNMNGK
or

SEQ ID NO: 6: SEQ ID NO: 5, including the signal and the addition of the QASKT amino acids (the coding region is underlined):

M I Q K R K R T V S F R L V L M C T L L F V S L P
I T K T S A Q A S K T E M K D D F A K L E E Q F D
A K L G I F A L D T G T N R T V A Y R P D E R F A
F A S T I K A L T V G V L L Q Q K S I E D L N Q R
I T Y T R D D L V N Y N P I T E K H V D T G M T L
K E L A D A S L R Y S D N A A Q N L I L K Q I G G
P E S L K K E L R K I G D E V T N P E R F E P E L
N E V N P G E T Q D T S T A R A L V T S L R A F A
L E D K L P S E K R E L L I D W M K R N T T G D A
L I R A G V P D G W E V G D K T G S G D Y G T R N
D I A I I W P P K G D P V V L A V L S S R D K K D
A K Y D N K L I A E A T K V V M K A L N M N G K

The invention also provides for polynucleotides encoding any of the beta-lactamases of the invention, including, for example, vectors, comprising such polynucleotides. Such polynucleotides may further comprise, in addition to sequences encoding the beta-lactamases of the invention, one or more expression control elements. For example, the polynucleotide, may comprise one or more promoters or transcriptional enhancers, ribosomal binding sites, transcription termination signals, and polyadenylation signals, as expression control elements. The polynucleotide may be inserted within any suitable vector, which may be contained within any suitable host cell for expression. An illustrative polynucleotide of the invention is SEQ ID NO: 7:

SEQ ID NO: 7

Full nucleotide sequence of A232G, A237S, A238G, S240D, and D276N mutant, Hind III site (AAGCTT-in bold) and additional K and T amino acids. The leader and additional

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nucleotides (Hind III site and K and T amino acids—for the addition of the amino acid sequence QASKT) are underlined.

5 atgattcaaaaacgaaagcggacagtttcggttcagacttgcgttatgtg
cacgctgttatttgcagtttgccgattacaaaacatcagcgcaagcctt
ccaagacggagatgaaagatgattttgcaaaacttgaggaaacatttgat
10 gcaaaactcgggatctttgcattggatacaggtacaaacggacggttagc
gtatcgccggatgagcggtttgcttttgccttcgacgattaaggctttaa
ctgtaggcggtgcttttgcaacagaaatcaatagaagatctgaaccagaga
15 ataacatatcacgtgatgatcttgtaactacaacccgattacggaaaa
gcacgttgatacgggaatgacgctcaaagagcttgccggtgcttcgcttc
gatatagtgcacatgcggcacagatctcattcttaacaaattggcgga
cctgaaagtgttgaagaaaggaaactgaggaagattggtgatgaggttacaaa
20 tcccgaacgattcgaaccagagttaaatgaagtgaatccgggtgaaactc
aggataccagtcacagcaagagcacttgctcacaagccttcgagcctttgct
cttgaagataaaacttccaagtgaagaaacgcgagcttttaacgattggat
25 gaaacgaaataccactggagacgccttaacccgtgcgggtgtgccggacg
gttgggaagtgggtgataaaactggaagcggagattatggaaccgggaat
gacattgccatcatttgccgcccagagagatcctgctgttcttgcagt
30 attatccagcagggataaaaaggacgccaagtatgataataaacttattg
cagaggcaacaaggtggtaatgaaagccttaacatgaacggcaataa

In some embodiments, the vector can remain episomal or become chromosomally integrated, as long as the insert encoding the therapeutic agent can be transcribed. Vectors can be constructed by standard recombinant DNA technology. Vectors can be, for example, plasmids, phages, cosmids, phagemids, viruses, or any other types known in the art, which are used for replication and expression in prokaryotic or eukaryotic cells (e.g. an adenovirus; a retrovirus; a lentivirus; an scAAV; pGEX vector; pET vector; and pHT vector). It will be appreciated by one of skill in the art that a wide variety of components known in the art (such as expression control elements) may be included in such vectors, including a wide variety of transcription signals, such as promoters and other sequences that regulate the binding of RNA polymerase onto the promoter. Any promoter known to be effective in the cells in which the vector will be expressed can be used to initiate expression of the therapeutic agent. Suitable promoters may be inducible or constitutive. Examples of suitable promoters include the pET system (INVTROGEN), lac promoter, tac, trc, T7, T7A3 promoter, PhoA, lux, and Phage lambda pR, lambda pL promoter (see, e.g. *J Ind Microbiol Biotechnol* (2012) 39:383-399; *Curr Opin Biotech* 2001, 12: 195, the contents of which are hereby incorporated by reference). The promoter may be inducible (e.g. via IPTG, metabolites, temperature). Examples of suitable promoters include the SV40 early promoter region, the promoter contained in the 3' long terminal repeat of Rous sarcoma virus, the HSV-1 thymidine kinase promoter, the regulatory sequences of the metallothionein gene, etc., as well as the following animal transcriptional control regions, which exhibit tissue specificity and have been utilized in transgenic animals: elastase I gene control region which is active in pancreatic acinar cells; insulin gene control region which is active in pancreatic beta cells, immunoglobulin gene control region which is active in lymphoid cells,

mouse mammary tumor virus control region which is active in testicular, breast, lymphoid and mast cells, albumin gene control region which is active in liver, alpha-fetoprotein gene control region which is active in liver, alpha 1-antitrypsin gene control region which is active in the liver, beta-globin gene control region which is active in erythroid cells, myelin basic protein gene control region which is active in oligodendrocyte cells in the brain, myosin light chain-2 gene control region which is active in skeletal muscle, and gonadotropin releasing hormone gene control region which is active in the hypothalamus.

The invention provides polynucleotides, such as polynucleotides comprising a nucleotide sequence encoding a beta-lactamase of the invention. For example, the nucleotide sequence that encodes a beta-lactamase of the invention. Such polynucleotides may further comprise additional control element(s) operably linked to the nucleotide sequence, such as promoter elements and/or other transcription or expression-related signals. The polynucleotide may be inserted into various vectors, which may be useful for production of the therapeutic agent in host cells, including, for example, bacterial and eukaryotic host cells. The beta-lactamase of the invention can be prepared by known recombinant expression techniques. For example, to recombinantly produce the beta-lactamase of the invention, a nucleic acid sequence encoding the respective gene is operatively linked to a suitable promoter sequence such that the nucleic acid sequence encoding such beta-lactamase of the invention will be transcribed and/or translated in the host cells. Illustrative promoters are those useful for expression in various systems such as the T7 promoter, Pspac, PgroES, Pgsi and Plux and amyQ promoter and/or amyQ signal peptide from *Bacillus amyloliquefaciens* (by way of non-limiting example Gen Bank ID No. J01542.1, the contents of which are hereby incorporated by reference). Any commonly used expression system may be used, including eukaryotic or prokaryotic systems. Specific examples include yeast (e.g., *Saccharomyces* spp., *Pichia* spp.), baculovirus, mammalian, and bacterial systems, such as *E. coli*, *B. subtilis*, and *Caulobacter*.

Host cells of the present invention include, for example, prokaryotic, eukaryotic, bacterial, yeast, algal, plant, insect, and/or mammalian cells. In some embodiments, the invention encompasses any type of cell that recombinantly expresses genes associated with the invention, including prokaryotic and eukaryotic cells. In some embodiments the cell is a bacterial cell, such as, for example, *Escherichia* spp., *Streptomyces* spp., *Zyomonas* spp., *Acetobacter* spp., *Citrobacter* spp., *Synechocystis* spp., *Rhizobium* spp., *Clostridium* spp., *Corynebacterium* spp., *Streptococcus* spp., *Xanthomonas* spp., *Lactobacillus* spp., *Lactococcus* spp., *Bacillus* spp., *Alcaligenes* spp., *Pseudomonas* spp., *Aeromonas* spp., *Azotobacter* spp., *Comamonas* spp., *Mycobacterium* spp., *Rhodococcus* spp., *Gluconobacter* spp., *Ralstonia* spp., *Acidithiobacillus* spp., *Microlunatus* spp., *Geobacter* spp., *Geobacillus* spp., *Arthrobacter* spp., *Flavobacterium* spp., *Serratia* spp., *Saccharopolyspora* spp., *Thermus* spp., *Stenotrophomonas* spp., *Chromobacterium* spp., *Sinorhizobium* spp., *Saccharopolyspora* spp., *Agrobacterium* spp. and *Pantoea* spp. The bacterial cell can be a Gram-negative cell such as an *Escherichia coli* (*E. coli*) cell, or a Gram-positive cell such as a species of *Bacillus*. In other embodiments, the cell is a fungal cell such as, for example, a yeast cell, e.g., *Saccharomyces* spp., *Schizosaccharomyces* spp., *Pichia* spp., *Paffia* spp., *Kluyveromyces* spp., *Candida* spp., *Talaromyces* spp., *Brettanomyces* spp., *Pachysolen* spp., *Debaryomyces* spp., *Yarrowia* spp., and industrial polyploid yeast strains. The yeast strain may be a *S. cerevisiae* strain or a *Yarrowia*

spp. strain. Other examples of fungi include, for example, *Aspergillus* spp., *Pennicillium* spp., *Fusarium* spp., *Rhizopus* spp., *Acremonium* spp., *Neurospora* spp., *Sordaria* spp., *Magnaporthe* spp., *Allomyces* spp., *Ustilago* spp., *Botrytis* spp., and *Trichoderma* spp. In other embodiments, the cell is an algal cell or a plant cell (e.g., *A. thaliana*, *C. reinhardtii*, *Arthrospira*, *P. tricomutum*, *T. suecica*, *P. carterae*, *P. tricomutum*, *Chlorella* spp., such as *Chlorella vulgaris*). Target cells can include transgenic and recombinant cell lines. In addition, heterologous cell lines can be used, such as Chinese Hamster Ovary cells (CHO). Host cells may be unicellular host cells or multicellular host cells.

In various embodiments, the beta-lactamases of the invention possess functional characteristics that make them desirable for a variety of uses, including therapeutic uses. Methods of characterizing beta-lactamases are known in the art (e.g. nitrocefin assay as described by O'Callaghan, et al. *Antimicrob. Agents Chemother.* 1 :283-288; the various methods of Viswanatha et al. *Methods Mol Med.* 2008;142:239-60).

In one embodiment, the inventive beta-lactamases hydrolyze one or more of penicillins and cephalosporins. As used throughout, penicillins include, for example, Amoxicillin (e.g. NOVAMOX, AMOXIL); Ampicillin (e.g. PRINCIPEN); Azlocillin; Carbenicillin (e.g. GEOCILLIN); Cloxacillin (e.g. TEGOPEN); Dicloxacillin (e.g. DYNAPEN); Flucloxacillin (e.g. FLOXAPEN); Mezlocillin (e.g. MEZLIN); Methicillin (e.g. STAPHICILLIN); Nafcillin (e.g. UNIPEN); Oxacillin (e.g. PROSTAPHLIN); Penicillin G (e.g. PENTIDS or PFIZERPEN); Penicillin V (e.g. VEETIDS (PEN-VEE-K)); Piperacillin (e.g. PIPRACIL); Temocillin (e.g. NEGABAN); and Ticarcillin (e.g. TICAR). As used throughout, cephalosporins include, for example, a first generation cephalosporin (e.g. Cefadroxil (e.g. DURICEF); Cefazolin (e.g. ANCEF); Cefotiozane, Cefalotin/Cefalothin (e.g. KEFLIN); Cefalexin (e.g. KEFLEX); a second generation cephalosporin (e.g. Cefaclor (e.g. DISTACLOX); Cefamandole (e.g. MANDOL); Cefoxitin (e.g. MEFOXIN); Cefprozil (e.g. CEFZIL); Cefuroxime (e.g. CEFTIN, ZINNAT)); a third generation cephalosporin (e.g. Cefixime (e.g. SUPRAX); Cefdinir (e.g. OMNICEF, CEFDIEL); Cefditoren (e.g. SPECTRACEF); Cefoperazone (e.g. CEFOTID); Cefotaxime (e.g. CLAFORAN); Cefpodoxime (e.g. VANTIN); Ceftriaxime (e.g. FORTAZ); Cefibuten (e.g. CEDAX) Cefizoxime (e.g. CEFIZOX); and Ceftriaxone (e.g. ROCEPHIN)); a fourth generation cephalosporin (e.g. Cefepime (e.g. MAXIPIME)); or a fifth generation cephalosporin (e.g. Ceftriaxone fosamil (e.g. TEFLARO); Cefotibiprole (e.g. ZEFTERA)). In a specific embodiment, cephalosporins include, for example, cefoperazone, ceftriaxone or cefazolin. In a specific embodiment, the inventive beta-lactamases have improved catalytic efficiency against cephalosporins as compared to SEQ ID NO: 1.

In various embodiments, the inventive beta-lactamases possess desirable enzyme kinetic characteristics. For example, in some embodiments, the beta-lactamases possess a low K_M for at least one cephalosporin, including, for example, a K_M of less than about 500 μM , or about 100 μM , or about 10 μM , or about 1 μM , or about 0.1 μM (100 nM), or about 0.01 μM (10 nM), or about 1 nM. For example, in some embodiments, the beta-lactamases possess a low K_M for at least one penicillin, including, for example, a K_M of less than about 500 μM , or about 100 μM , or about 10 μM , or about 1 μM , or about 0.1 μM (100 nM), or about 0.01 μM (10 nM), or about 1 nM. In various embodiments, the inventive beta-lactamases possess a high V_{max} for at least one cephalosporin, including, for example, V_{max} which is greater than about 100 s⁻¹, or about 1000 s⁻¹, or about 10000 s⁻¹, or about 100000

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s-1, or about 1000000 s-1. In various embodiments, the inventive beta-lactamases possess a high V_{max} for at least one penicillin, including, for example, V_{max} which is greater than about 100 s-1, or about 1000 s-1, or about 10000 s-1, or about 100000 s-1, or about 1000000 s-1. In various embodiments, the inventive beta-lactamases possess catalytic efficiency is greater than about $10^6 \text{ M}^{-1} \text{ s}^{-1}$ for at least one cephalosporin. In various embodiments, the inventive beta-lactamases possess catalytic efficiency is greater than about $10^6 \text{ M}^{-1} \text{ s}^{-1}$ for at least one penicillin. In various embodiments, the inventive beta-lactamases possess the desirable enzyme kinetic characteristics for at least one of either or both of cephalosporins and penicillins.

In various embodiments, the inventive beta-lactamases are stable and/or active in the GI tract, e.g. in one or more of the mouth, esophagus, stomach, duodenum, small intestine, duodenum, jejunum, ileum, large intestine, colon transversum, colon descendens, colon ascendens, colon sigmoideum, cecum, and rectum. In a specific embodiment, the beta-lactamase is stable in the large intestine, optionally selected from one or more of colon transversum, colon descendens, colon ascendens, colon sigmoideum and cecum. In a specific embodiment, the beta-lactamase is stable in the small intestine, optionally selected from one or more of duodenum, jejunum, and ileum. In some embodiments, the beta-lactamase is resistant to proteases in the GI tract, including for example, the small intestine. In some embodiments, the beta-lactamase is substantially active at a pH of about 6.0 to about 7.5, e.g. about 6.0, or about 6.1, or about 6.2, or about 6.3, or about 6.4, or about 6.5, or about 6.6, or about 6.7, or about 6.8, or about 6.9, or about 7.0, or about 7.1, or about 7.2, or about 7.3, or about 7.4, or about 7.5 (including, for example, via formulation, as described herein). In various embodiments, the beta-lactamases of the present invention are resistant to one or more beta-lactamase inhibitors, optionally selected from avibactam, tazobactam, sulbactam, and clavulanic acid. In some embodiments, stable refers to an enzyme that has a long enough half life and maintains enough activity for therapeutic effectiveness.

In various embodiments, the present invention pertains to pharmaceutical compositions comprising a beta-lactamase described herein and one or more of and a pharmaceutically acceptable carrier or excipient as described herein. In a specific embodiment, the pharmaceutical composition is formulated for oral administration, e.g. as a tablet or multi-particulate sprinkle or a multi-particulate capsule. However, as described herein, other administration routes and formulations are also provided.

In some embodiments, the pharmaceutical compositions may be used in conjunction with or be co-formulations with an additional agent. In some embodiments, the additional agent is an additional antibiotic degradation enzyme, such as, for example, a beta-lactamase of class EC 3.5.2.6. In some embodiments, the antibiotic degradation enzyme is selected from a functional Group 1, Group 2, Group 3, or a Group 4 beta-lactamase (see, e.g., Bush et al., *Antimicrob. Agents Chemother.* 39: 1211, the contents of which are hereby incorporated by reference); without wishing to be bound by theory, Group 1 consists of cephalosporinases that are not well inhibited by clavulanic acid; Group 2 consists of penicillinases, cephalosporinases and broad-spectrum beta-lactamases that are generally inhibited by active site-directed beta-lactamase inhibitors; Group 3 consists of metallo-beta-lactamases that hydrolyze penicillins, cephalosporins and carbapenems, and that are poorly inhibited by almost all beta-lactam-containing molecules; and Group 4 consists of penicillinases that are not well inhibited by clavulanic acid) and/or a molecular/Ambler

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class A, or class B, or class C, or class D beta-lactamase (see, e.g., Ambler 1980, *Philos Trans R Soc Lond B Biol Sci.* 289: 321 the contents of which are hereby incorporated by reference), without wishing to be bound by theory: Classes A, C, and D gather evolutionarily distinct groups of serine beta-lactamase enzymes, and class B the zinc-dependent ("EDTA-inhibited") beta-lactamase enzymes (see Ambler R. P. et al., 1991, *Biochem J.* 276: 269-270, the contents of which are hereby incorporated by reference). In some embodiments, the antibiotic degradation enzyme is a serine beta-lactamase or a zinc-dependent (EDTA-inhibited) beta-lactamase. For example, in some embodiments, the beta-lactamase is one or more of P1A, P2A, or P3A. Further, the beta-lactamase may be an extended-spectrum beta-lactamase (ESBL), optionally selected from a TEM, SHV, CTX-M, OXA, PER, VEB, GES, and IBC beta-lactamase. Further, the beta-lactamase may be an inhibitor-resistant β -lactamase, optionally selected from an AmpC-type β -lactamases, Carbapenemase, IMP-type carbapenemases (metallo- β -lactamases), VIM (Verona integron-encoded metallo- β -lactamase), OXA (oxacillinase) group of β -lactamases, KPC (*K. pneumoniae* carbapenemase), CMY (Class C), SME, IMI, NMC and CcrA, and a NDM (New Delhi metallo- β -lactamase, e.g. NDM-1) beta-lactamase.

In some embodiments, the additional agent is an adjunctive therapy that is used in, for example, the treatment of CDI as described herein. In some embodiments, the additional agent is metronidazole (e.g. FLAGYL), fidaxomicin (e.g. DIFI-CID), or vancomycin (e.g. Vancocin), rifaximin, fecal bacteriotherapy, charcoal-based binders (e.g. DAV132), probiotic therapy (see, e.g., *Intnat'l J Inf Dis.* 16 (11): e786, the contents of which are hereby incorporated by reference, illustrative probiotics include *Saccharomyces boulardii*; *Lactobacillus rhamnosus* GG; *Lactobacillus plantarum* 299v; *Clostridium butyricum* M588; *Clostridium difficile* VP20621 (non-toxigenic *C. difficile* strain); combination of *Lactobacillus casei*, *Lactobacillus acidophilus* (Bio-K+CL1285); combination of *Lactobacillus casei*, *Lactobacillus bulgaricus*, *Streptococcus thermophilus* (Actimel); combination of *Lactobacillus acidophilus*, *Bifidobacterium bifidum* (Florajen3); combination of *Lactobacillus acidophilus*, *Lactobacillus bulgaricus delbrueckii* subsp. *bulgaricus*, *Lactobacillus bulgaricus casei*, *Lactobacillus bulgaricus plantarum*, *Bifidobacterium longum*, *Bifidobacterium infantis*, *Bifidobacterium breve*, *Streptococcus salivarius* subsp. *thermophilus* (VSL#3)) and antibody or other biologic therapy (e.g. monoclonal antibodies against *C. difficile* toxins A and B as described in *N Engl J Med.* 2010;362(3):197, the content of which are hereby incorporated by reference in their entirety; neutralizing binding proteins, for example, arranged as multimers, which are directed to one or more of SEQ ID NOs. recited in United States Patent Publication No. 2013/0058962 (e.g. one or more of SEQ ID Nos.: 59, 60, 95, 67, 68, and 87), the contents of which are hereby incorporated by reference); or any neutralizing binding protein directed against *C. difficile* binary toxin. In some embodiments, any of the penicillins and cephalosporins described herein may be the additional agent.

For all additional agent compositions and methods, targeting to various parts of the GI tract may be employed as described herein.

In some embodiments, the inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) described herein, include derivatives that are modified, i.e., by the covalent attachment of any type of molecule to the composition such that covalent attachment does not prevent the activity of the composition. For example, but not by way of limitation, derivatives include composition that have been

modified by, inter alia, glycosylation, lipidation, acetylation, pegylation, phosphorylation, amidation, derivatization by known protecting/blocking groups, proteolytic cleavage, linkage to a cellular ligand or other protein, etc. Any of numerous chemical modifications can be carried out by known techniques, including, but not limited to specific chemical cleavage, acetylation, formylation, metabolic synthesis of turicamycin, etc. Additionally, the derivative can contain one or more non-classical amino acids.

In still other embodiments, the inventive beta-lactamases and/or pharmaceutical compositions described herein may be modified to add effector moieties such as chemical linkers, detectable moieties such as for example fluorescent dyes, enzymes, substrates, bioluminescent materials, radioactive materials, and chemiluminescent moieties, or functional moieties such as for example streptavidin, avidin, biotin, a cytotoxin, a cytotoxic agent, and radioactive materials.

The inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) described herein can possess a sufficiently basic functional group, which can react with an inorganic or organic acid, or a carboxyl group, which can react with an inorganic or organic base, to form a pharmaceutically acceptable salt. A pharmaceutically acceptable acid addition salt is formed from a pharmaceutically acceptable acid, as is well known in the art. Such salts include the pharmaceutically acceptable salts listed in, for example, *Journal of Pharmaceutical Science*, 66, 2-19 (1977) and *The Handbook of Pharmaceutical Salts; Properties, Selection, and Use*. P. H. Stahl and C. G. Wermuth (eds.), Verlag, Zurich (Switzerland) 2002, which are hereby incorporated by reference in their entirety.

Pharmaceutically acceptable salts include, by way of non-limiting example, sulfate, citrate, acetate, oxalate, chloride, bromide, iodide, nitrate, bisulfate, phosphate, acid phosphate, isonicotinate, lactate, salicylate, acid citrate, tartrate, oleate, tannate, pantothenate, bitartrate, ascorbate, succinate, maleate, gentisinate, fumarate, gluconate, glucuronate, saccharate, formate, benzoate, glutamate, methanesulfonate, ethanesulfonate, benzenesulfonate, p-toluenesulfonate, camphorsulfonate, pamoate, phenylacetate, trifluoroacetate, acrylate, chlorobenzoate, dinitrobenzoate, hydroxybenzoate, methoxybenzoate, methylbenzoate, o-acetoxybenzoate, naphthalene-2-benzoate, isobutyrate, phenylbutyrate, α -hydroxybutyrate, butyne-1,4-dicarboxylate, hexyne-1,4-dicarboxylate, caprate, caprylate, cinnamate, glycollate, heptanoate, hippurate, malate, hydroxymaleate, malonate, mandelate, mesylate, nicotinate, phthalate, teraphthalate, propiolate, propionate, phenylpropionate, sebacate, suberate, p-bromobenzenesulfonate, chlorobenzenesulfonate, ethylsulfonate, 2-hydroxyethylsulfonate, methylsulfonate, naphthalene-1-sulfonate, naphthalene-2-sulfonate, naphthalene-1,5-sulfonate, xylenesulfonate, and tartarate salts.

The term "pharmaceutically acceptable salt" also refers to a salt of the compositions of the present invention having an acidic functional group, such as a carboxylic acid functional group, and a base. Suitable bases include, but are not limited to, hydroxides of alkali metals such as sodium, potassium, and lithium; hydroxides of alkaline earth metal such as calcium and magnesium; hydroxides of other metals, such as aluminum and zinc; ammonia, and organic amines, such as unsubstituted or hydroxy-substituted mono-, di-, or tri-alkylamines, dicyclohexylamine; tributyl amine; pyridine; N-methyl, N-ethylamine; diethylamine; triethylamine; mono-, bis-, or tris-(2-OH-lower alkylamines), such as mono-, bis-, or tris-(2-hydroxyethyl)amine, 2-hydroxy-tert-butylamine, or tris-(hydroxymethyl)methylamine, N,N-di-lower alkyl-N-(hydroxyl-lower alkyl)-amines, such as N,N-dimethyl-N-(2-

hydroxyethyl)amine or tri-(2-hydroxyethyl)amine; N-methyl-D-glucamine; and amino acids such as arginine, lysine, and the like.

In some embodiments, the compositions described herein are in the form of a pharmaceutically acceptable salt.

Further, any inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) described herein can be administered to a subject as a component of a composition that comprises a pharmaceutically acceptable carrier or vehicle. Such compositions can optionally comprise a suitable amount of a pharmaceutically acceptable excipient so as to provide the form for proper administration.

Pharmaceutical excipients can be liquids, such as water and oils, including those of petroleum, animal, vegetable, or synthetic origin, such as peanut oil, soybean oil, mineral oil, sesame oil and the like. The pharmaceutical excipients can be, for example, saline, gum acacia, gelatin, starch paste, talc, keratin, colloidal silica, urea and the like. In addition, auxiliary, stabilizing, thickening, lubricating, and coloring agents can be used. In one embodiment, the pharmaceutically acceptable excipients are sterile when administered to a subject. Water is a useful excipient when any agent described herein is administered intravenously. Saline solutions and aqueous dextrose and glycerol solutions can also be employed as liquid excipients, specifically for injectable solutions. Suitable pharmaceutical excipients also include starch, glucose, lactose, sucrose, gelatin, malt, rice, flour, chalk, silica gel, sodium stearate, glycerol monostearate, talc, sodium chloride, dried skim milk, glycerol, propylene glycol, water, ethanol and the like. Any agent described herein, if desired, can also comprise minor amounts of wetting or emulsifying agents, or pH buffering agents.

The present invention includes the described inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) in various formulations. Any inventive beta-lactamase and/or pharmaceutical composition (and/or additional agents) described herein can take the form of solutions, suspensions, emulsion, drops, tablets, pills, pellets, capsules, capsules containing liquids, powders, sustained-release formulations, suppositories, emulsions, aerosols, sprays, suspensions, or any other form suitable for use. In one embodiment, the composition is in the form of a capsule (see, e.g., U.S. Pat. No. 5,698,155). Other examples of suitable pharmaceutical excipients are described in *Remington's Pharmaceutical Sciences* 1447-1676 (Alfonso R. Gennaro eds., 19th ed. 1995), incorporated herein by reference.

Where necessary, the inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) can also include a solubilizing agent. Also, the agents can be delivered with a suitable vehicle or delivery device as known in the art. Compositions for administration can optionally include a local anesthetic such as, for example, lignocaine to lessen pain at the site of the injection. Combination therapies outlined herein can be co-delivered in a single delivery vehicle or delivery device.

The formulations comprising the inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) of the present invention may conveniently be presented in unit dosage forms and may be prepared by any of the methods well known in the art of pharmacy. Such methods generally include the step of bringing the therapeutic agents into association with a carrier, which constitutes one or more accessory ingredients. Typically, the formulations are prepared by uniformly and intimately bringing the therapeutic agent into association with a liquid carrier, a finely divided solid carrier, or both, and then, if necessary, shaping the product into dosage forms of the desired formulation (e.g.,

wet or dry granulation, powder blends, etc., followed by tableting using conventional methods known in the art)

In one embodiment, any inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) described herein is formulated in accordance with routine procedures as a composition adapted for a mode of administration described herein.

In some embodiments, the administration of any inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) is any one of oral, intravenous, and parenteral. In some embodiments, the administration of any inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) is not intravenous in order to, for example, prevent interference with an antibiotic administered systemically. In other embodiments, routes of administration include, for example: oral, intradermal, intramuscular, intraperitoneal, intravenous, subcutaneous, intranasal, epidural, sublingual, intranasal, intracerebral, intravaginal, transdermal, rectally, by inhalation, or topically, particularly to the ears, nose, eyes, or skin. In some embodiments, the administering is effected orally or by parenteral injection. The mode of administration can be left to the discretion of the practitioner, and depends in-part upon the site of the medical condition. In most instances, administration results in the release of any agent described herein into the bloodstream.

Any inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) described herein can be administered orally. Such inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) can also be administered by any other convenient route, for example, by intravenous infusion or bolus injection, by absorption through epithelial or mucocutaneous linings (e.g., oral mucosa, rectal and intestinal mucosa, etc.) and can be administered together with an additional agent. Administration can be systemic or local. In some embodiments, administration is not at the site of infection to avoid, for example, hydrolysis of an antibiotic at the site of infection. Various delivery systems are known, e.g., encapsulation in liposomes, microparticles, microcapsules, capsules, etc., and can be used to administer.

In specific embodiments, it may be desirable to administer locally to the area in need of treatment.

In one embodiment, any inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) described herein is formulated in accordance with routine procedures as a composition adapted for oral administration to humans. Compositions for oral delivery can be in the form of tablets, lozenges, aqueous or oily suspensions, granules, powders, sprinkles, emulsions, capsules, syrups, or elixirs, for example. Orally administered compositions can comprise one or more agents, for example, sweetening agents such as fructose, aspartame or saccharin; flavoring agents such as peppermint, oil of wintergreen, or cherry; coloring agents; and preserving agents, to provide a pharmaceutically palatable preparation. Moreover, where in tablet or pill form, the compositions can be coated to delay disintegration to provide a sustained action over an extended period of time. Selectively permeable membranes surrounding an osmotically active agent driving any inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) described herein are also suitable for orally administered compositions. In these latter platforms, fluid from the environment surrounding the capsule is imbibed by the driving compound, which swells to displace the agent or agent composition through an aperture. These delivery platforms can provide an essentially zero order delivery profile as opposed to the spiked profiles of immediate release formulations. A

time-delay material such as glycerol monostearate or glycerol stearate can also be useful. Oral compositions can include standard excipients such as mannitol, lactose, starch, magnesium stearate, sodium saccharin, cellulose, ethacrylic acid and derivative polymers thereof, and magnesium carbonate. In one embodiment, the excipients are of pharmaceutical grade. Suspensions, in addition to the active compounds, may contain suspending agents such as, for example, ethoxylated isostearyl alcohols, polyoxyethylene sorbitol and sorbitan esters, microcrystalline cellulose, aluminum metahydroxide, bentonite, agar-agar, tragacanth, etc., and mixtures thereof.

Dosage forms suitable for parenteral administration (e.g. intravenous, intramuscular, intraperitoneal, subcutaneous and intra-articular injection and infusion) include, for example, solutions, suspensions, dispersions, emulsions, and the like. They may also be manufactured in the form of sterile solid compositions (e.g. lyophilized composition), which can be dissolved or suspended in sterile injectable medium immediately before use. They may contain, for example, suspending or dispersing agents known in the art.

Inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) may be administered to a subject, by, for example, directly or indirectly contacting the mucosal tissues of the gastrointestinal tract. The gastrointestinal tract includes organs of the digestive system such as mouth, esophagus, stomach, duodenum, small intestine, large intestine and rectum and includes all subsections thereof (e.g. the small intestine may include the duodenum, jejunum and ileum; the large intestine may include the colon transversum, colon descendens, colon ascendens, colon sigmoidum and cecum). Various methods may be used to formulate and/or deliver the agents described herein to a location of interest. For example, the inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) described herein may be formulated for delivery to one or more of the stomach, small intestine, large intestine and rectum and includes all subsections thereof (e.g. duodenum, jejunum and ileum, colon transversum, colon descendens, colon ascendens, colon sigmoidum and cecum). In some embodiments, the compositions described herein may be formulated to deliver to the upper or lower GI tract.

In some embodiments, the compositions of the invention are formulated for enteric delivery. For example, the compositions may be formulated as capsules or tablets for oral delivery, and may comprise a delayed-release coating containing one or more enteric agents. A delayed-release coating is substantially stable in gastric fluid and substantially unstable (e.g., dissolves rapidly or is physically unstable) in intestinal fluid, thus providing for substantial release of the active agent from the composition in the affected region of the small intestine, e.g., the duodenum, the jejunum, and/or the ileum or large intestine, e.g. the colon transversum, colon descendens, colon ascendens, colon sigmoidum and cecum.

The inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) are generally stable in gastric fluid or simulated intestinal fluid, that is, the compositions are stable in acidic environments. Thus, the compositions release less than 30% by weight of the active agent in gastric fluid with a pH of 5 or less, or simulated gastric fluid with a pH of 5 or less, in approximately 120 minutes. Compositions of the invention may release from about 0% to about 25%, or from about 0% to about 10% by weight of the active agent in gastric fluid with a pH of 5 or less, or simulated gastric fluid with a pH of 5 or less, in approximately 120 minutes. Compositions of the invention in certain embodiments release no more than about 1%, about 2%, about 3%,

about 4%, about 5%, about 6%, about 7%, about 8%, about 9%, or about 10% by weight of the active agent in gastric fluid, or simulated gastric fluid with a pH of 5 or less, in approximately 120 minutes. The pharmaceutical compositions generally release active agent to act locally at regions of the small or large intestine. In certain embodiments, the composition of the invention release about 70% or more by weight of the active agent in the small or large intestine within about 120 minutes. In certain embodiments, the composition releases 80% or more, or 90% or more, of the active agent in small or large intestine, within about ninety minutes or within about 120 minutes. In certain embodiments this release in the small or large intestine is mediated by pH of gastric fluid or simulated gastric fluid—for example, a release when the pH of about 5 or greater (e.g. pH of about 5.5–6.5 for release in the duodenum, pH of about 6–7 for release in the colon ascendens or jejunum, pH of about 6.5–7 for release in the ileum, pH of about 7–7.5 for release in the colon descendens).

The pharmaceutical composition may control intestinal release of the active agent through one or more delayed-release coating(s), which remain essentially intact, and/or which may be essentially insoluble, in gastric fluid. The stability of the delayed-release coating can be pH dependent. Delayed-release coatings that are pH dependent will be substantially stable in acidic environments (pH 5 or less), and substantially unstable in near neutral to alkaline environments (pH greater than 5). For example, the delayed-release coating may essentially disintegrate or dissolve in near neutral to alkaline environments, such as are found in the small intestine or large intestine, to thereby release active agent locally to diseased or affected tissue.

Examples of simulated gastric fluid and simulated intestinal fluid include, but are not limited to, those disclosed in the 2005 Pharmacopeia 23NF/28USP in Test Solutions at page 2858 and/or other simulated gastric fluids and simulated intestinal fluids known to those of skill in the art, for example, simulated gastric fluid and/or intestinal fluid prepared without enzymes.

Alternatively, the stability of the delayed-release coating may be enzyme-dependent. Delayed-release coatings that are enzyme dependent will be substantially stable in fluid that does not contain a particular enzyme and substantially unstable in fluid containing the enzyme. The delayed-release coating will essentially disintegrate or dissolve in fluid containing the appropriate enzyme. Enzyme-dependent control can be brought about, for example, by using materials which release the active ingredient only on exposure to enzymes in the intestine, such as galactomannans.

In some embodiments, the target organ for the release of the active agent from the compositions of the invention, which include gastric resistant capsules or tablets, is the small intestine, such as the duodenum and the jejunum or the large intestine, e.g. the colon transversum, colon descendens, colon ascendens, colon sigmoidum and cecum. See Remington's Pharmaceutical Sciences, 16th Ed., Eds. Osol, Mack Publishing Co., Chapter 89 (1980); Digenis et al., *J. Pharm. Sci.*, 83:915-921 (1994); Vantini et al., *Clinica Terapeutica*, 145: 445-451 (1993); Yoshitomi et al., *Chem. Pharm. Bull.*, 40:1902-1905 (1992); Thoma et al., *Pharmazie*, 46:331-336 (1991); Morishita et al., *Drug Design and Delivery*, 7:309-319 (1991); and Lin et al., *Pharmaceutical Res.*, 8:919-924 (1991) for examples of the preparation of such tablets or capsules (the contents of the above are hereby incorporated by reference in their entirety).

In some embodiments, the compositions of the present invention may be formulated using the EUDRAGIT system,

as known in the art and described in *Pharma Polymer* No. 7, October 2000, the contents of which are hereby incorporated by reference in their entirety.

The dosage of any inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) described herein as well as the dosing schedule can depend on various parameters, including, but not limited to, the disease being treated, the subject's general health, and the administering physician's discretion. Any agent described herein, can be administered prior to (e.g., 5 minutes, 15 minutes, 30 minutes, 45 minutes, 1 hour, 2 hours, 4 hours, 6 hours, 12 hours, 24 hours, 48 hours, 72 hours, 96 hours, 1 week, 2 weeks, 3 weeks, 4 weeks, 5 weeks, 6 weeks, 8 weeks, or 12 weeks before), concurrently with, or subsequent to (e.g., 5 minutes, 15 minutes, 30 minutes, 45 minutes, 1 hour, 2 hours, 4 hours, 6 hours, 12 hours, 24 hours, 48 hours, 72 hours, 96 hours, 1 week, 2 weeks, 3 weeks, 4 weeks, 5 weeks, 6 weeks, 8 weeks, or 12 weeks after) the administration of an additional therapeutic agent, to a subject in need thereof. In various embodiments any agent described herein is administered 1 minute apart, 10 minutes apart, 30 minutes apart, less than 1 hour apart, 1 hour apart, 1 hour to 2 hours apart, 2 hours to 3 hours apart, 3 hours to 4 hours apart, 4 hours to 5 hours apart, 5 hours to 6 hours apart, 6 hours to 7 hours apart, 7 hours to 8 hours apart, 8 hours to 9 hours apart, 9 hours to 10 hours apart, 10 hours to 11 hours apart, 11 hours to 12 hours apart, no more than 24 hours apart or no more than 48 hours apart.

The amount of any inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) described herein that is admixed with the carrier materials to produce a single dosage can vary depending upon the subject being treated and the particular mode of administration. In vitro or in vivo assays can be employed to help identify optimal dosage ranges.

In general, the doses that are useful are known to those in the art. For example, doses may be determined with reference *Physicians' Desk Reference*, 66th Edition, PDR Network; 2012 Edition (Dec. 27, 2011), the contents of which are incorporated by reference in its entirety. In some embodiment, the present invention allows a patient to receive doses that exceed those determined with reference *Physicians' Desk Reference*.

The dosage of any inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) described herein can depend on several factors including the severity of the condition, whether the condition is to be treated or prevented, and the age, weight, and health of the subject to be treated. Additionally, pharmacogenomic (the effect of genotype on the pharmacokinetic, pharmacodynamic or efficacy profile of a therapeutic) information about a particular subject may affect dosage used. Furthermore, the exact individual dosages can be adjusted somewhat depending on a variety of factors, including the specific combination of the agents being administered, the time of administration, the route of administration, the nature of the formulation, the rate of excretion, the particular disease being treated, the severity of the disorder, and the anatomical location of the disorder. Some variations in the dosage can be expected.

In specific embodiments, the concentration of any inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) is about 0.2 to about 20 g/L when administered orally or about 4 g/mL when administered intravenously.

In some embodiments, when orally administered to a mammal, the dosage of any inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents)

described herein may be about 0.001 mg/kg/day to about 100 mg/kg/day, about 0.01 mg/kg/day to about 50 mg/kg/day, or about 0.1 mg/kg/day to about 10 mg/kg/day. When orally administered to a human, the dosage of any agent described herein is normally about 0.001 mg to 1000 mg per day, about 1 mg to about 600 mg per day, or about 5 mg to about 30 mg per day. In one embodiment, 100 mg per day of the inventive beta-lactamases and/or pharmaceutical compositions is administered orally to a human. For administration of any inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) described herein by parenteral injection, the dosage is normally about 0.1 mg to about 250 mg per day, about 1 mg to about 20 mg per day, or about 3 mg to about 5 mg per day. Injections may be given up to four times daily. Generally, when orally or parenterally administered, the dosage of any agent described herein is normally about 0.1 mg to about 1500 mg per day, or about 0.5 mg to about 10 mg per day, or about 0.5 mg to about 5 mg per day. A dosage of up to about 3000 mg per day can be administered. In some embodiments, the dose may be about 1000 mg per day. In some embodiments, the following dose regimen can be used: 100 mg, four times a day.

In various embodiments, the dosage of any inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) described herein is administered to achieve an intestinal concentration of from about 1 ng/ml to about 10,000 ng/ml, about 1 ng/ml to about 5,000 ng/ml, about 1 ng/ml to about 2,500 ng/ml, about 1 ng/ml to about 1,000 ng/ml, about 1 ng/ml to about 500 ng/ml, about 1 ng/ml to about 250 ng/ml, about 1 ng/ml to about 100 ng/ml, about 1 ng/ml to about 50 ng/ml, about 1 ng/ml to about 25 ng/ml, about 1 ng/ml to about 10 ng/ml, about 10 ng/ml to about 10,000 ng/ml, about 10 ng/ml to about 5,000 ng/ml, about 10 ng/ml to about 2,500 ng/ml, about 10 ng/ml to about 1,000 ng/ml, about 10 ng/ml to about 500 ng/ml, about 10 ng/ml to about 250 ng/ml, about 10 ng/ml to about 100 ng/ml, about 10 ng/ml to about 50 ng/ml, about 10 ng/ml to about 25 ng/ml, about 100 ng/ml to about 10,000 ng/ml, about 100 ng/ml to about 5,000 ng/ml, about 100 ng/ml to about 2,500 ng/ml, about 100 ng/ml to about 1,000 ng/ml, about 100 ng/ml to about 500 ng/ml, or about 100 ng/ml to about 250 ng/ml. In various embodiments, the intestinal concentration of any inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) described herein is 10 ng/ml, 100 ng/ml, or 1,000 ng/ml.

In another embodiment, delivery can be in a vesicle, in particular a liposome (see Langer, 1990, *Science* 249:1527-1533; Treat et al., in *Liposomes in the Therapy of Infectious Disease and Cancer*, Lopez-Berestein and Fidler (eds.), Liss, New York, pp. 353-365 (1989).

Any inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) described herein can be administered by controlled-release or sustained-release means or by delivery devices that are well known to those of ordinary skill in the art. Examples include, but are not limited to, those described in U.S. Pat. Nos. 3,845,770; 3,916,899; 3,536,809; 3,598,123; 4,008,719; 5,674,533; 5,059,595; 5,591,767; 5,120,548; 5,073,543; 5,639,476; 5,354,556; and 5,733,556, each of which is incorporated herein by reference in its entirety. Such dosage forms can be useful for providing controlled- or sustained-release of one or more active ingredients using, for example, hydropropyl cellulose, hydropropylmethyl cellulose, polyvinylpyrrolidone, other polymer matrices, gels, permeable membranes, osmotic systems, multilayer coatings, microparticles, liposomes, microspheres, or a combination thereof to provide the desired release profile in varying proportions. Suitable controlled- or sustained-release

formulations known to those skilled in the art, including those described herein, can be readily selected for use with the active ingredients of the agents described herein. The invention thus provides single unit dosage forms suitable for oral administration such as, but not limited to, tablets, capsules, gels, and caplets that are adapted for controlled- or sustained-release.

Controlled- or sustained-release of an active ingredient can be stimulated by various conditions, including but not limited to, changes in pH, changes in temperature, stimulation by an appropriate wavelength of light, concentration or availability of enzymes, concentration or availability of water, or other physiological conditions or compounds.

In another embodiment, polymeric materials can be used (see *Medical Applications of Controlled Release*, Langer and Wise (eds.), CRC Pres., Boca Raton, Fla. (1974); *Controlled Drug Bioavailability, Drug Product Design and Performance*, Smolen and Ball (eds.), Wiley, New York (1984); Ranger and Peppas, 1983, *J. Macromol. Sci. Rev. Macromol. Chem.* 23:61; see also Levy et al., 1985, *Science* 228:190; During et al., 1989, *Ann. Neurol.* 25:351; Howard et al., 1989, *J. Neurosurg.* 71:105).

In another embodiment, a controlled-release system can be placed in proximity of the target area to be treated, thus requiring only a fraction of the systemic dose (see, e.g., Goodson, in *Medical Applications of Controlled Release*, supra, vol. 2, pp. 115-138 (1984)). Other controlled-release systems discussed in the review by Langer, 1990, *Science* 249:1527-1533) may be used.

Administration of any inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) described herein can, independently, be one to six times daily (e.g. 1, or 2, or 3, or 4, or 5, or 6 times per day) or one to four times per month or one to six times per year or once every two, three, four or five years. Administration can be for the duration of one day or one month, two months, three months, six months, one year, two years, three years, and may even be for the life of the subject. Chronic, long-term administration will be indicated in many cases. The dosage may be administered as a single dose or divided into multiple doses. In general, the desired dosage should be administered at set intervals for a prolonged period, usually at least over several weeks or months, although longer periods of administration of several months or years or more may be needed.

The dosage regimen utilizing any inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) described herein can be selected in accordance with a variety of factors including type, species, age, weight, sex and medical condition of the subject; the severity of the condition to be treated; the route of administration; the renal or hepatic function of the subject; the pharmacogenomic makeup of the individual; and the specific compound of the invention employed. Any inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) described herein can be administered in a single daily dose, or the total daily dosage can be administered in divided doses of two, three or four times daily. Furthermore, any inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) described herein can be administered continuously rather than intermittently throughout the dosage regimen.

In various aspects, the present invention provides methods for treating or preventing an antibiotic-induced adverse effect in the GI tract, comprising administering an effective amount of a beta-lactamase and/or pharmaceutical composition (and/or additional agent) described herein to a patient in need thereof. In one aspect, the present invention provides methods

for preventing an antibiotic-induced adverse effect in the GI tract, comprising administering an effective amount of a beta-lactamase and/or pharmaceutical composition (and/or additional agent) described herein to a patient in need thereof (by way of non-limiting example, a patient that is being administered or will be administered an antibiotic, including those described herein). In various aspects, the present invention provides methods for treating or preventing *C. difficile* infection (CDI) and/or a *C. difficile*-associated disease, comprising administering an effective amount of a beta-lactamase and/or pharmaceutical composition (and/or additional agent) described herein to a patient in need thereof. In one aspect, the present invention provides methods for preventing *C. difficile* infection (CDI) and/or a *C. difficile*-associated disease, comprising administering an effective amount of a beta-lactamase and/or pharmaceutical composition (and/or additional agent) described herein to a patient in need thereof (by way of non-limiting example, a patient that is being administered or will be administered an antibiotic, including those described herein).

In various embodiments, the antibiotic-induced adverse effect and/or CDI or *C. difficile*-associated disease is one or more of: antibiotic-associated diarrhea, *C. difficile* diarrhea (CDD), *C. difficile* intestinal inflammatory disease, colitis, pseudomembranous colitis, fever, abdominal pain, dehydration and disturbances in electrolytes, megacolon, peritonitis, and perforation and/or rupture of the colon.

In various aspects, the present invention provides methods for protecting a subject's gastrointestinal microbiome, comprising administering an effective amount of a beta-lactamase and/or pharmaceutical composition (and/or additional agent) described herein. In various embodiments, the subject is undergoing treatment or has recently undergone treatment with an antibiotic. In various embodiments, the beta-lactamase and/or pharmaceutical composition (and/or additional agent) described herein is capable of degrading or inactivating the antibiotic.

In various embodiments, the subjects include, but are not limited to, subjects that are at a particular risk for a microbiome-mediated disorder, such as, by way of non-limiting example, those undergoing treatment or having recently undergone treatment with an antibiotic. For example, the subject may have taken an antibiotic during the past about 30 or so days and/or have an immune system that is weak (e.g. from a chronic illness) and/or is a woman and/or is elderly (e.g. over about 65 years old) and/or is an elderly woman and/or is undergoing (or has undergone) treatment with for heartburn or stomach acid disorders (e.g. with agents such as PREVACID, TAGAMET, PRILOSEC, or NEXIUM and related drugs) and/or has recently been in the hospital, including in an intensive care unit, or lives in a nursing home. Accordingly, in some embodiments, the methods and uses of the present invention treat or prevent a nosocomial infection and/or a secondary emergent infection and/or a hospital acquired infection (HAI).

In various embodiments, the methods of the invention comprise treating or preventing a microbiome-mediated disorder. Illustrative microbiome-mediated disorder includes, but are not limited to, for example, those found in Table 3 of WO 2014/121298, the entire contents of which are incorporated herein by reference. For example, the microbiome-mediated disorder may be selected from an antibiotic-induced adverse effect, a *C. difficile* infection (CDI), a *C. difficile*-associated disease, ulcerative colitis, Crohn's disease, and irritable bowel syndrome. In various embodiments, the microbiome-mediated disorder is an antibiotic-induced adverse effect, a *C. difficile* infection (CDI), or a *C. difficile*-

associated disease. In an embodiment, the present invention provides methods for treating an antibiotic-induced adverse effect in the GI tract, comprising administering an effective amount of a beta-lactamase and/or pharmaceutical composition (and/or additional agent) described herein to a subject who is undergoing treatment or has recently undergone treatment with an antibiotic. In another embodiment, the present invention provides methods for preventing an antibiotic-induced adverse effect in the GI tract, comprising administering an effective amount of a beta-lactamase and/or pharmaceutical composition (and/or additional agent) described herein to a subject who is undergoing treatment or has recently undergone treatment with an antibiotic.

In various embodiments, the present uses and methods pertain to co-treatment (simultaneously or sequentially) with the beta-lactamases and/or pharmaceutical compositions of the present invention and any additional agent described herein and/or any initial and/or adjunctive therapy, or treatment with a co-formulation of the beta-lactamases and/or pharmaceutical compositions of the present invention and any additional agent described herein and/or any initial and/or adjunctive therapy for treatment of the various diseases described herein, or methods of treating the various diseases described herein in a patient undergoing treatment with any additional agent described herein and/or any initial and/or adjunctive therapy described herein by administering a beta-lactamases and/or pharmaceutical compositions of the present invention to the patient.

In various embodiments, the CDI and/or *C. difficile* associated disease is treated or prevented in the context of initial onset or relapse/recurrence (e.g. due to continued or restarted antibiotic therapy). For example, in a patient that has previously suffered from CDI, the present beta-lactamases and/or pharmaceutical compositions (and/or additional agents) may be administered upon the first symptoms of recurrence. By way of non-limiting example, symptoms of recurrence include, in a mild case, about 5 to about 10 watery bowel movements per day, no significant fever, and only mild abdominal cramps while blood tests may show a mild rise in the white blood cell count up to about 15,000 (normal levels are up to about 10,000), and, in a severe case, more than about 10 watery stools per day, nausea, vomiting, high fever (e.g. about 102-104° F.), rectal bleeding, severe abdominal pain (e.g. with tenderness), abdominal distention, and a high white blood count (e.g. of about 15,000 to about 40,000).

Regardless of initial onset or relapse/recurrence, CDI and/or *C. difficile* associated disease may be diagnosed via any of the symptoms described herein (e.g. watery diarrhea about 3 or more times a day for about 2 days or more, mild to bad cramping and pain in the belly, fever, blood or pus in the stool, nausea, dehydration, loss of appetite, loss of weight, etc.). Regardless of initial onset or relapse/recurrence, CDI and/or *C. difficile* associated disease may also be diagnosed via enzyme immunoassays e.g. to detect the *C. difficile* toxin A or B antigen and/or glutamine dehydrogenase (GDH), which is produced by *C. difficile* organisms), polymerase chain reaction (e.g. to detect the *C. difficile* toxin A or B gene or a portion thereof (e.g. tcdA or tcdB), including the ILLUMIGENE LAMP assay), a cell cytotoxicity assay. For example, any one of the following tests may be used may be used: Meridian ImmunoCard Toxins A/B; Wampole Toxin A/B Quik Chek; Wampole *C. diff* Quik Chek Complete; Remel Xpect *Clostridium difficile* Toxin A/B; Meridian Premier Toxins A/B; Wampole *C. difficile* Tox A/B II; Remel Prospect Toxin A/B EIA; Biomerieux Vidas *C. difficile* Toxin A&B; BD

Geneohm *C. diff*; Prodesse Progestro CD; and Cepheid Xpert *C. diff*. In various embodiments, the clinical sample is a patient stool sample.

Also a flexible sigmoidoscopy “scope” test and/or an abdominal X-ray and/or a computerized tomography (CT) scan, which provides images of your colon, may be used in assessing a patient (e.g. looking for characteristic creamy white or yellow plaques adherent to the wall of the colon). Further, biopsies (e.g. of any region of the GI tract) may be used to assess a potential CDI and/or *C. difficile* associated disease patient.

Furthermore, the patients of the invention include, but are not limited to, patients that are at a particular risk for CDI and/or *C. difficile* associated disease, such as those which have been taking an antibiotic during the past 30 or so days and/or have an immune system that is weak (e.g. from a chronic illness) and/or are women and/or are elderly (e.g. over about 65 years old) and/or are elderly woman and/or undergo treatment with for heartburn or stomach acid disorders (e.g. with agents such as PREVACID, TAGAMET, PRILOSEC, or NEXIUM and related drugs) and/or have recently been in the hospital, including in an intensive care unit, or live in a nursing home. Accordingly, in some embodiments, the methods and uses of the present invention treat or prevent a nosocomial infection and/or a secondary emergent infection and/or a hospital acquired infection (HAI).

In some embodiments, the methods of the invention treat or prevent a ceftriaxone-associated adverse effect (e.g. diarrhea, nausea, vomiting, dysgeusia, and pseudomembranous colitis disease and/or symptoms).

In some embodiments, the methods and uses of the present invention relate to a patient is undergoing treatment or has recently undergone treatment with one or more primary antibiotic. A “primary antibiotic” refers to an antibiotic that is administered to a patient and which may result in CDI and/or *C. difficile* associated disease. These include the antibiotics that most often lead to CDI and/or *C. difficile* associated disease: fluoroquinolones, cephalosporins, clindamycin and penicillins.

In some embodiments, the methods and uses of the present invention relate to the inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) which hydrolyze a primary antibiotic before it enters the large intestine. In some embodiments, the methods and uses of the present invention relate to the inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) which hydrolyze excess antibiotic residue in the GI tract. In some embodiments, methods and uses of the present invention relate to the inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) which maintain a normal intestinal microbiota and/or prevents the overgrowth of one or more pathogenic microorganisms in the GI tract of a patient. In various embodiments, the beta-lactamases and/or pharmaceutical compositions (and/or additional agents) do not substantially interfere with plasma levels of a primary antibiotic. For example, the beta-lactamases and/or pharmaceutical compositions (and/or additional agents) of the present invention allow for a patient to receive a primary antibiotic that might be required for an infection and do not interfere with the systemic utility of the antibiotic. Rather, the beta-lactamases and/or pharmaceutical compositions (and/or additional agents) inactivate excess antibiotic that may populate parts of the GI tract and in doing so, prevent the disruption of the microbiota that is linked to the various disease states described herein.

In various embodiments, the inventive beta-lactamases and/or pharmaceutical compositions (and/or additional

agents) are not systemically absorbed. In various embodiments, the inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) do not substantially interfere with the activity of systemically administered antibiotics. In various embodiments, the inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) function to eliminate antibiotics from interfering with the microbiota of a microbiome (e.g. the gut, including the large intestine). In some embodiments, the inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) do not interfere with the antibiotic absorption from the gut and/or enterohepatically sufficiently to alter the half-lives of antibiotic circulation. In some embodiments, the inventive beta-lactamases and/or pharmaceutical compositions (and/or additional agents) do not interfere with the antibiotic absorption from the gut and/or enterohepatically enough to be clinically important.

In some embodiments, the methods and uses of the present invention include those in which an initial and/or adjunctive therapy is administered to a patient. Initial and/or adjunctive therapy indicates therapy that is used to treat CDI and/or *C. difficile* associated disease upon detection of such disease. In some embodiments, the initial and/or adjunctive therapy is one or more of metronidazole, vancomycin, fidaxomicin, rifaximin, fecal bacteriotherapy, probiotic therapy, and antibody therapy, as described herein. In various embodiments, the methods and uses of the present invention include use of the inventive beta-lactamases as an adjuvant to any of these initial and/or adjunctive therapies (including co-administration or sequential administration). In various embodiments, the methods and uses of the present invention include use of the inventive beta-lactamases in a patient undergoing initial and/or adjunctive therapies.

In other aspects, the present invention provides the beta-lactamases and/or pharmaceutical compositions for use in treating an antibiotic-induced adverse effect in the GI tract and/or prevention or treatment of *C. difficile* infection (CDI) and/or a *C. difficile*-associated disease. In other aspects, there are provided uses of the beta-lactamases and/or pharmaceutical compositions for treating an antibiotic-induced adverse effect in the GI tract and/or preventing or treating a *C. difficile* infection (CDI) and/or a *C. difficile*-associated disease. Further, some aspects provide for the use of the disclosed beta-lactamases in the manufacture of a medicament for use in treating an antibiotic-induced adverse effect in the GI tract and/or prevention or treatment of *C. difficile* infection (CDI) and/or a *C. difficile*-associated disease.

In various embodiments, the present invention provides for compositions and methods that mitigate or prevent the overgrowth of various coliforms in a patient’s gut (including coliforms that are virulent and/or antibiotic resistant). In various aspects, the methods and compositions described herein prevent or diminish secondary infections with resistant organisms and may, in some embodiments, diminish beta-lactam resistance development. Further, the methods and compositions described herein may allow for use of beta-lactam antibiotics which are currently avoided due to resistance concerns and/or reduce the need for co-administration or co-formulation with one or more beta-lactamase inhibitors (e.g. AUGMENTIN is a mixture of amoxicillin and clavulanic acid).

In some embodiments, the terms “patient” and “subject” are used interchangeably. In some embodiments, the subject and/or animal is a mammal, e.g., a human, mouse, rat, guinea pig, dog, cat, horse, cow, pig, rabbit, sheep, or non-human primate, such as a monkey, chimpanzee, or baboon. In other embodiments, the subject and/or animal is a non-mammal,

such, for example, a zebrafish. In some embodiments, the subject and/or animal may comprise fluorescently-tagged cells (with e.g. GFP). In some embodiments, the subject and/or animal is a transgenic animal comprising a fluorescent cell.

In some embodiments, the subject and/or animal is a human. In some embodiments, the human is a pediatric human. In other embodiments, the human is an adult human. In other embodiments, the human is a geriatric human. In other embodiments, the human may be referred to as a patient.

In certain embodiments, the human has an age in a range of from about 6 to about 18 months old, from about 18 to about 36 months old, from about 1 to about 5 years old, from about 5 to about 10 years old, from about 10 to about 15 years old, from about 15 to about 20 years old, from about 20 to about 25 years old, from about 25 to about 30 years old, from about 30 to about 35 years old, from about 35 to about 40 years old, from about 40 to about 45 years old, from about 45 to about 50 years old, from about 50 to about 55 years old, from about 55 to about 60 years old, from about 60 to about 65 years old, from about 65 to about 70 years old, from about 70 to about 75 years old, from about 75 to about 80 years old, from about 80 to about 85 years old, from about 85 to about 90 years old, from about 90 to about 95 years old or from about 95 to about 100 years old.

In other embodiments, the subject is a non-human animal, and therefore the invention pertains to veterinary use. In a specific embodiment, the non-human animal is a household pet. In another specific embodiment, the non-human animal is a livestock animal.

The invention provides kits that can simplify the administration of any agent described herein. An exemplary kit of the invention comprises any composition described herein in unit dosage form. In one embodiment, the unit dosage form is a container, such as a pre-filled syringe, which can be sterile, containing any agent described herein and a pharmaceutically acceptable carrier, diluent, excipient, or vehicle. The kit can further comprise a label or printed instructions instructing the use of any agent described herein. The kit may also include a lid speculum, topical anesthetic, and a cleaning agent for the administration location. The kit can also further comprise one or more additional agent described herein. In one embodiment, the kit comprises a container containing an effective amount of a composition of the invention and an effective amount of another composition, such those described herein.

This invention is further illustrated by the following non-limiting examples.

EXAMPLES

The following abbreviations are used throughout:

Mutations relative to P1A (based on the Ambler classification)	Name
Wild type	RS310 (or P1A)
D276N	IS118 (or P3A)
I72S	IS222
T160F	IS203
R244T	IS217
R244T D276K	IS215
Q135M	IS197
G156R A238T	IS235
F33Y D276N	IS158
F33Y S240P D276N	IS230 (or IS181)
F33Y A238T D276N	IS232 (or IS180)
I72S Q135M T160F (Block 1 mutants)	IS227
A232G A237S A238G S240D (Block 2 mutants)	IS191

-continued

Mutations relative to P1A (based on the Ambler classification)	Name
A232G A237S A238G S240D R244T	IS229
A232G A237S A238G S240D D276R	IS219
A232G A237S A238G S240D D276K	IS221
A232G A237S A238G S240D Q135M	IS224
A238T	IS233
T243I S266N D276N	IS234 (or IS176)
A232G A237S A238G S240D D276N	IS288 (or P4A)

Example 1

Materials and Methods

Different P1A mutants were made either with site-directed mutagenesis using SOE or QUIKCHANGE Multi Site-Directed Mutagenesis Kit (Stratagene, Calif., USA) or with random mutagenesis using GeneMorph® II Random Mutagenesis Kit (Stratagene, Calif., USA). Mutated DNA was transformed into bacteria, either *E. coli* or *B. subtilis*, which were screened on cefotaxime (site-directed mutants) or ceftriaxone (random mutants). The presence of β -lactamase in colonies was further confirmed with a nitrocefin test and the positive colonies were selected for further analysis. The DNA sequences of the mutant β -lactamases were confirmed by sequencing.

β -lactamase activity of the mutants was further characterized from *E. coli* cell lysates and *B. subtilis* growth supernatant with ampicillin, cefotaxime, ceftriaxone, cefuroxime, ceftazidime, meropenem and imipenem. The activities were determined using only one substrate concentration of each antibiotic tested. Selection of the antibiotic concentration was done with the antibiotic concentration as high as was practical, $[S] \gg K_m$. Based on these preliminary experiments, more accurate kinetic parameters were determined for a couple of mutants from both the site-directed and the random mutagenesis research lines. All the mutant enzymes selected for more detailed study were expressed in the *B. subtilis* host strain RS303. The cells carrying the mutated proteins were grown in minimal medium over-night after which the cells were harvested and the supernatant was concentrated in 1:10 with Amicon Ultra-4 centrifugal devices, MWCO 10000 (Amicon). Protein concentration in each culture supernatant was estimated from 12% SDS-PAGE (Criterion XT Bis-Tris, Bio-Rad, Calif., USA) using the P1A reference material A18K31 as a standard. Kinetic parameters were measured for cefotaxime, ceftriaxone and cefuroxime for each selected mutant strain. The site-directed mutants were designed to hydrolyze cefotaxime and random mutants were screened with ceftriaxone, which made these antibiotics natural choices for determining kinetic parameters.

P1A Modifications

A series of mutations in P1A were made, including:

TABLE 1

Mutations on P1A from structure-based design and construction of cefotaxime degrading P1A derivatives,	
Block 1	I72S, Q135M, T160F
Block 2	A232, A237S, A238G, S240D
	R244T
	D276R/K
	and different combinations of these

Parallel to the site-directed mutagenesis, random mutagenesis was done on P1A and P3A with ceftriaxone as a screening antibiotic.

There are also two modification mutants of P1A that were made for the quality control studies: P1A-N293D (IS205, “the deamidation mutant”) and P1A-ΔMNGK (IS206, “the deletion mutant”). Both of these strains grew equally well as P1A-strain RS310 and the protein production was about the same as well. These mutations did not affect the crystallizability of the protein either.

Sequences of selected CTX-M-enzymes were derived from Swiss-Prot protein sequence data bank and the 3D structures were derived from the Protein Data Bank (PDB). Accession codes for individual sequences and structures are represented below. In some instances, this data was used to predict structure functional activity.

Illustrative Swiss-Prot or TrEMBL, GenBank and Protein Data Bank (PDB) accession codes for enzymes used in the study.

	Swiss-Prot/ TrEMBL	GenBank	PDB
CTX-M-9	Q9L5C8_ECOLI (TrEMBL)	AF174129	1YLJ
CTX-M-14	Q9L5C7_ECOLI (TrEMBL)	AF252622	1YLT
CTX-M-15	Q9EXV5_ECOLI (TrEMBL)	AY044436	—
CTX-M-16	Q939K2_KLEPN (TrEMBL)	AY029068	1YLW
CTX-M-25	Q8KSA6_ECOLI (TrEMBL)	AF518567	—
CTX-M-44	Q47066 (Swiss-Prot)	D37839	1IYO

A Linux workstation and Fedora Core 5 operating system was used as the study platform. The sequence alignments and analysis as well as the modelling and 3D structure analysis were performed with Bodil molecular modelling environment.

Additionally, several other P1A mutants were designed and made. One of these mutants was IS288, which has following mutations: A232G, A237S, A238G, A240S and D276N. The four mutations A232G, A237S, A238G and A240S are a so-called “Block2”-mutation. The fifth mutant, D276N, is the “P3A”-mutation, which alone gives the protein’s ability to degrade ceftriaxone. The P1A variant IS288 was a result of two rounds of mutagenesis on the P1A gene and preliminary activity assays of secreted P1A variants. Several P1A variants made by site-directed mutagenesis were cloned into the *B. subtilis* production host and produced in shaker flask cultivations. The growth supernatants of these mutant strains were concentrated and used to measure each of the strains’ ability to degrade selected cephalosporins. IS288 was produced in larger scale and purified using ammonium sulphate precipitation, hydrophobic interaction chromatography and ultra filtration. The study was set up to characterize the selected P1A variant IS288 in more detail.

In addition to purifying with HIC, methods were developed to further desalt and characterize the amount and purity of protein in the eluted protein peak.

Hydrophobic interaction chromatography, i.e. HIC, is a method, where the separation of bio-molecules is based on interplay between the hydrophobicity of the medium, the nature and composition of the sample, the prevalence and distribution of surface-exposed hydrophobic amino acid residues, and the type and concentration of salt in the binding buffer. The adsorption of protein into the gel matrix is dependent on the concentration of anti-chaotropic salts such as $(\text{NH}_4)_2\text{SO}_4$ and Na_2SO_4 ; the higher the salt concentration the stronger the adsorption capacity of the HIC medium. The HIC matrix can have different types of hydrophobic substituents,

like butyl, octyl, or phenyl groups. Butyl sepharose was selected as the HIC matrix as P1A binds tightly to octyl sepharose that it is not properly eluted.

IS288 Purification Procedure:

The first day, a small amount of a frozen IS288 *Bacillus subtilis* cell culture was streaked on a LB-agar plate containing 10 µg/ml kanamycin. The following morning single cultures were taken from this plate and inoculated into 10×3 ml of minimal growth medium for shaker flask cultivation. In the afternoon of the same day, a volume resulting in a 0.01 starting OD₆₀₀ in the 100 ml overnight (o/n) culture was taken from the 3 ml pre-culture. The 100 ml cultures were incubated in a shaker at +37° C. over night. From a couple of 3 ml pre-cultures OD₆₀₀=1 glycerol stocks (10% glycerol) were made. The glycerol stocks were stored at -70° C.

On the third day after ~16 hours of cultivation, the final OD₆₀₀ of the shaker flask cultivations were measured and cells were centrifuged (Sorvall RC6) with SLA-3000 rotor, 7000 rpm, 10 minutes at +4° C.). Culture supernatants were collected and combined and then filtrated with a 0.2 µm filter. The volume of the filtrate was measured to ~745 ml. 326.86 g of solid $(\text{NH}_4)_2\text{SO}_4$ resulting in 70% $(\text{NH}_4)_2\text{SO}_4$ saturation was weighed and added as ~33 g aliquots into the supernatant while continuously mixing. Once all salt had dissolved, the solution was transferred to a +4° C. refrigerator.

The following morning, the ammonium sulphate solution was centrifuged (Sorvall RC6) SLA-3000 rotor, 90 minutes, 10500 rpm, +4° C., acceleration 9 and deceleration 4). After centrifugation, the supernatants were combined and filtered through a 0.2 µm sterile filter. 72.5 ml of the filtrate was taken for hydrophobic interaction chromatography (HIC). The rest of the filtrate was divided into 17×50 ml aliquots and stored first at +4° C. and later transferred to -20° C. Moreover, two 1 ml samples of the filtrate were taken and stored at first at +4° C. and later transferred to -20° C.

The $(\text{NH}_4)_2\text{SO}_4$ concentration of the ammonium sulphate filtrate was estimated to be ~2.7 M and it was diluted so that the concentration of $(\text{NH}_4)_2\text{SO}_4$ decreased to ~2 M with 190 mM Na-H-Phosphate buffer, pH 6.8. The filtrate was subjected to HIC in a 5 ml HiTrap Butyl FF column equilibrated with 50 mM Na-H-Phosphate, pH 6.9, 2 M $(\text{NH}_4)_2\text{SO}_4$. A total of 50 ml of sample was pumped through the column and the flow-through was collected as 5 ml fractions. Unbound sample was washed out with 50 mM Na-H-Phosphate, pH 6.9, 2 M $(\text{NH}_4)_2\text{SO}_4$. Bound IS288 was eluted with decreasing $(\text{NH}_4)_2\text{SO}_4$ concentration using step gradient and 50 mM Na-H-Phosphate, pH 6.8. In the first run, elution fractions were 1 ml. Fractions containing the eluted protein were combined in three different sets; C3-C8, C9-D1 and D2-D8. Flow-through fractions and the rest of the elution fractions were stored at +4° C. Samples were taken from flow-through fractions B1 and B8 as well as the pooled protein fractions and stored at +4° C.

The HIC step was repeated two times for different sample volumes, namely 65 ml and 125 ml of equilibrated ammonium sulphate filtrate. These later HIC runs were somewhat modified from the first run. The wash volume for unbound sample was 3 and 4 CV, the volume of flow-through fractions was increased to 10 ml and the volume of the elution fractions was increased to 1.5 ml. No other modifications were made to the running conditions.

Pooled protein fractions as well as flow-through fractions of B1 and B8 of the first run were ultra filtrated and concentrated using Amicon Ultra-4 concentrators with 10000 NMWL. The samples were centrifuged six times and after each centrifugation, except the last one, the rest of the sample or buffer was added into the retentate so as to gain 4 ml

volume. After the sixth concentration step, retentates were collected, measured and stored at +4° C. Samples from succeeding HIC runs, as well as one more flow through fraction from the first run were ultra filtrated and concentrated with Amicon Ultra-15 with 10000 NMWL and only four centrifugation rounds were needed. These samples were also stored at +4° C. Ammonium sulphate filtrates were ultra filtrated and concentrated, and flow through fractions were once more concentrated with MultiScreen filter plates with Ultracell-10 membrane, 10000 NMWL. Protein concentration of the flow through samples was measured before this last concentration step whereas no protein concentration was measured for ammonium sulphate filtrates.

The protein concentration of the samples was measured with BCA Protein Assay-kit. Based on the elution peak obtained from Äkta, eluted protein fractions were diluted ~1:20 and ~1:40 while the flow-through fractions were used as such (1:1 dilution) or diluted in half (1:2 dilution).

Finally, samples from all three purifications were run on a Criterion XT 12% Bis-Tris SDS-PAGE using MOPS running buffer to optimally separate proteins of 6-66 kDa in size. The maximum amount of sample was added for ammonium sulphate filtrates and all flow through fractions except for B1₁₂₁₁₀₈ for which the protein content was determined to be 0.21 µg/µl in the 33× concentrated sample. The amount of protein peak fractions pipetted was calculated so that 0.5 µg of protein would be loaded on the gel.

A purified protein solution of IS288 was used to determine the activity of IS288 per mass unit. The stability of IS288 in storage conditions was examined to find out if the protein is degraded or by some other way modified during storage. The stability of the protein at the target site matrix was also determined. The results from the stability assay in target site matrix were combined with the results from the activity assay to estimate the time it takes for IS288 to eliminate antibiotic residues in the small intestine.

In addition to getting more detailed information on P1A variant IS288, the characterization scheme created during the study was used in the characterization of other subsequent P1A variants.

Method to Study Production Levels of P1A Variant IS288

During the *Bacillus subtilis* shaker flask cultivation, samples were taken at various time points to evaluate the production level of IS288. The OD₆₀₀ value was measured for each of the time points as well to see how the cells were growing. The amount of produced protein was estimated on SDS-PAGE.

Method to Study Enzyme Efficacy and Antibiotic Spectrum at Different pH Levels

The antibiotics used in this study were a selection of, e.g., intravenously administered cephalosporins. The concentrations of selected antibiotics were, without wishing to be bound by theory, set to as close to the in literature reported concentrations in bile or duodenum as possible. This experiment was also performed at varying pHs.

Method to Study Enzyme Stability in Human Ileal Chyme

Different amounts of purified enzyme were incubated in human ileal chyme at the appropriate temperature for a set period of time. Samples were taken at different time points. The degradation pattern of IS288 was analysed with SDS-PAGE and the activity of IS288 was analysed using nitrocefin as a substrate.

Compounds and Bacterial Strain: Ampicillin, Ceftriaxone, Cefazidime, Meropenem, Cefepime, Cefazolin, Amp/Sulbactam, Cefoperazone, Cefotaxime, and Cefuroxime were purchased from commercial sources and stored as frozen stocks of 15 mg/mL and 10 mg/mL, respectively. P1A was

supplied by Synthetic Biologics as a stock concentration of 32 mg/mL and stored at -80° C. *Escherichia coli* ATCC 25922 was obtained from American Type Culture Collection (ATCC, Manassas, Va.).

Determination of the Effective Inhibitory Concentration in the Absence and Presence of P1A (RS310), P3A (IS118), and P4A (IS288): The effective inhibitory concentrations (EC₅₀, 90) of ampicillin, ceftriaxone, ceftazidime, meropenem, cefepime, cefazolin, amp/sulbactam, cefoperazone, cefotaxime, and cefuroxime were determined by microbroth dilution analysis according to the CLSI guidelines (M07-A9: Methods for Dilution Antimicrobial Susceptibility Tests for Bacteria That Grow Aerobically; Approved Standard—Ninth Edition). *Escherichia coli* strain 25922 was cultured from glycerol stocks to logarithmic growth in cation adjusted Mueller-Hinton broth at 37° C. in an atmospheric environment and diluted in cation adjusted Mueller-Hinton broth to achieve a starting in well concentration of 5×10⁵ CFU/mL. Serial 2-fold dilution of antimicrobial compounds were prepared in cation adjusted Mueller-Hinton broth to achieve final in well concentration of 128 µg/mL to 0.13 µg/mL for ampicillin and 4 µg/mL 4 ng/mL in the absence or presence of serial 10-fold dilutions of P1A, P3A, or P4A β-lactamase standard, 5 pg/mL to 50 ng/mL for ampicillin and 50 pg/mL to 500 ng/mL for ceftriaxone, ceftazidime, meropenem, cefepime, cefazolin, amp/sulbactam, cefoperazone, cefotaxime, and cefuroxime. Antibiotics were either added to the P1A and incubated for 30 min at 37° C. (pre-incubation) prior to addition of bacteria, or added to the plates following the addition of P1A, P3A, P4A, and bacteria (simultaneous). All concentrations and combinations were evaluated in triplicate in a 96-well plate format that contained media only, compound plus media only (colorimetric control), and untreated bacterial controls. Following overnight incubation, the OD₆₂₅ of the plates was measured in a Spectramax 384 plate reader and the data were imported into a customized Excel spreadsheet for the determination of the effective concentration that inhibited bacterial growth by 50%, 90% and 99% (EC₅₀, 90, 99) by linear regression analysis.

Example 2

Results of Mutagenesis

Mutation design was based on, inter alia, structural data (e.g. crystal structure data, homolog models, etc.) of the following: P1A crystal structure (Knox and Moews, *J. Mol Biol.*, 220, 435-455 (1991)), CTX-M-44 (1BZA (Ibuka et al. *Journal of Molecular Biology* Volume 285, Issue 5 2079-2087 (1999)), 1IYS (Ibuka et al. *Biochemistry*, 2003, 42 (36): 10634-43), 1IYO, 1IYP and 1IYQ (Shimamura et al. 2002 *J. Biol. Chem.* 277:46601-08), *Proteus vulgaris* K1 (1HZO, Nugaka et al. *J Mol Biol.* 2002 Mar. 15;317(1):109-17) and *Proteus penneri* HugA (Liassine et al. *Antimicrob Agents Chemother.* 2002 January;46(1):216-9. 2002), and reviewed in Bonnet, *Antimicrob. Agents Chemother* 48(1): 1-14 (2004) (for CTM-X), the contents of all of these documents are hereby incorporated by reference in their entirety). In some embodiments, the present mutations are informed by analysis of structural data (e.g. crystal structure data, homolog models, etc.) of any one of the following β-lactamases: P1A, P2A, P3A, CTX-M-3, CTX-M-4, CTX-M-5, CTX-M-9, CTX-M-10, CTX-M-14, CTX-M-15, CTX-M-16, CTX-M-18, CTX-M-19, CTX-M-25, CTX-M-26, CTX-M-27, CTX-M-32, CTX-M-44, CTX-M-45, and CTX-M-54. Such information

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is available to one skilled in the art at known databases, for example, Swiss-Prot Protein Sequence Data Bank, NCBI, and PDB.

Results from site-directed and random mutagenesis are summarized in Tables 2 and 3, respectively. Enzymatic activity for selected mutants is presented in Table 4 and kinetic parameters are represented in Table 5. Kinetic parameters for previous enzymes P1A, P3A and P2A are also included in that table.

Mutation Block 1. I72S, Q135M and T160F

The first block of mutations including I72S, Q135M and T160F were introduced in mutagenic primers in one reaction using The QUIKCHANGE Multi Site-Directed mutagenesis kit (for short later called Multi kit), Stratagene. Plasmid carrying the penP gene was used as template in the mutagenesis reaction. Mutant single-stranded plasmids were transformed into *E. coli* XL10Gold and selected with kanamycin/chloramphenicol. Competent XL10 Gold cells are provided with the Multi kit. Colonies (10-30) were picked and tested with nitrocefin to verify β -lactamase activity. Part of the colony was used as template in PCR and the resulting fragment, if clean, was sent to DNA sequencing. The rest of the colony was cultivated in Luria supplemented with appropriate antibiotic, overnight, at 37° C., and a part of the culture was stored in glycerol ($\leq 10\%$, final), another part is used for plasmid isolation. After the desired mutations were confirmed with DNA sequencing, the mutant clones were characterized in more detail. Periplasmic lysates were prepared and analyzed in SDS PAGE and the kinetics of cefotaxime hydrolysis were measured spectrophotometrically.

Mutation Block 2. A232G, A237S, A238G and S240D

The target amino acids of the B3 β -strand of P1A are so closely situated to each other that the Multi-Site protocol was not suitable, at least in one round. Instead, all four amino acids were exchanged using the SOE technique. The region encompassing the exchanges contained 24 nucleotides (8 amino acid codons) and was included in the overlap extension part of two PCR primers. The mutated penP gene was first PCR'ed in two parts, using primers A+B and C+D (FIG. 7). The two PCR products were then combined and amplified using primers A and D, complementary to the 5' and 3' parts of the penP gene. Cloning sites were included in primers A and D. Once the construct was ready, further mutations were added using the Multi kit.

Further Mutations, R244T and D276R(K)

These mutations were added using the Multi-Site kit technique and the previously made mutant plasmids as templates. A list of the constructed mutations is shown in Table 1. The number of mutation combinations was increased easily and consequently, the characterization was done with more robust methods than measuring the cefotaxime degradation kinetics of individual clones. Primary characterization was the determination of the cefotaxime MIC.

TABLE 1

Constructed mutations and other possible combinations		
Clone	Technique	Amino acid replacements
Block 1	Multi kit	I72S, Q135M, T160F
	Multi kit	I72S, T160F
	Multi kit	I72S, Q135M
	Multi kit	I72S
	Multi kit	Q135M
	Multi kit	T160F
Block 2	SOE	A232G, A237S, A238G, S240D
	SOE + Multi kit	A232G, A237S, A238G, S240D + R244T
	SOE + Multi kit	A232G, A237S, A238G, S240D + D276R(K)

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TABLE 1-continued

Constructed mutations and other possible combinations		
Clone	Technique	Amino acid replacements
	SOE + Multi kit	A232G, A237S, A238G, S240D + R244T + D276R(K)
	SOE + Multi kit	A232G, A237S, A238G, S240D + I72S, Q135M, T160F
	SOE + Multi kit	A232G, A237S, A238G, S240D + I72S, Q135M, T160F + R244T
	SOE + Multi kit	A232G, A237S, A238G, S240D + I72S, Q135M, T160F + D276R(K)
	SOE + Multi kit	A232G, A237S, A238G, S240D + I72S, Q135M, T160F + R244T + D276R(K)
	Multi kit	R244T
	Multi kit	D276R(K)
	Multi kit	R244T + D276R(K)
	Multi kit	I72S, Q135M, T160F + R244T
	Multi kit	I72S, Q135M, T160F + D276R(K)
	Multi kit	I72S, Q135M, T160F + R244T + D276R(K)

TABLE 2

P1A mutants made using site-directed mutagenesis. <i>B. subtilis</i> and <i>E. coli</i> strain numbers, mutations and plasmid names are marked.			
<i>B. subtilis</i> strain	Amino acid replacements	Plasmid name in <i>B. subtilis</i>	<i>E. coli</i> strain
IS191	Block 2 = A232G, A237S, A238G, S240D	pCTX-42	IS189
IS197	Q135M	pIS197	IS193
IS215	R244T + D276K	pIS215	IS207
IS217	R244T	pIS217	IS108
IS219	Block 2 + D276T	pIS219	IS209
IS221	Block 2 + D276K	pIS221	IS210
IS222	I72S	pIS222	IS212
IS224	Block 2 + Q135M	pIS224	IS214
IS227	Block 1 = I72S, Q135M, T160F	pRSH227	SOE
IS229	Block 2 + R244T	pRSH229	SOE

TABLE 3

P1A mutants generated with random mutagenesis. <i>B. subtilis</i> and <i>E. coli</i> strain numbers, mutations and plasmid names are marked.			
<i>B. subtilis</i> strain	Amino acid replacements	Plasmid name in <i>B. subtilis</i>	<i>E. coli</i> strain
IS158	F33Y, D276N	pRSH158	
IS230	F33Y, S240P, D276N	pRSH230	IS181
IS232	F33Y, A238T, D276N	pRSH232	IS180
IS234	R55R, A123A, T2431, S266N, D276N	pRSH234	IS176

TABLE 4

Preliminary enzymatic activities of P1A mutants measured from <i>B. subtilis</i> supernatants. Values are represented as multiples of P1A. AMP = ampicillin, CTX = cefotaxime, CRO = ceftriaxone, CXM = cefuroxime and CAZ = ceftazidime. P1A values used for the calculations are means of two different experiments.						
<i>B. subtilis</i> strain	Amino acid replacements	AMP	CTX	CRO	CXM	CAZ
RS310	P1A strain	1.00	1.00	1.00	1.00	1.00
IS191	A232G A237S A238G S240D	0.16	2.17	2.91	3.22	1.28
IS197	Q135M	0.15	1.32	1.09	1.83	0.55
IS203	T160F	0.01	0.01	0.00	0.00	0.13
IS215	R244T D276K	0.44	0.50	0.93	0.81	0.26
IS217	R244T	0.22	0.00	0.00	0.05	0.11
IS219	A232G A237S A238G S240D D276R	0.05	7.22	15.99	7.67	0.27
IS221	A232G A237S A238G S240D D276K	0.06	9.39	15.95	9.87	0.38
IS222	I72S	0.71	0.00	0.15	0.18	0.00
IS224	A232G A237S A238G S240D Q135M	0.02	4.74	5.18	6.37	1.24
IS230	F33Y S240P D276N	0.02	8.07	9.26	4.76	8.98
IS232	F33Y A238T D276N	0.08	12.34	9.14	5.71	26.73
IS233	A238T	0.07	4.17	5.25	4.76	12.70
IS234	T243I S266N D276N	0.57	6.04	11.03	6.86	11.23
IS235	G156R A238T	0.03	1.66	2.09	1.34	5.42

TABLE 5

Kinetic parameters measured for the P1A mutant strains. Kinetic parameters for IS230 degrading cefuroxime are not reliable as the original curve did not obey Michaelis-Menten kinetics. Moreover, it should be noted that kinetic parameters for IS219 and IS221 are near one another except when cefotaxime is degraded; when the V_{max} for IS219 is increased, it is decreased for IS221. On the other hand, the affinity of IS219 towards cefotaxime has not increased with the same amount as IS221's. Low = not enough activity for measurements; ND or empty cell = not done.												
	Ceftriaxone				Cefotaxime				Cefuroxime			
	V_{max} (nmol/s)	K_m (mM)	k_{cat} (1/s)	k_{cat}/K_m ($M^{-1} * s^{-1}$)	V_{max} (nmol/s)	K_m (mM)	k_{cat} (1/s)	k_{cat}/K_m ($M^{-1} * s^{-1}$)	V_{max} (nmol/s)	K_m (mM)	k_{cat} (1/s)	k_{cat}/K_m ($M^{-1} * s^{-1}$)
P1A (A18K31)	2.81	179	25	$1.41 * 10^5$	1.18	232	16	$6.8 * 10^4$	1.25	107	23	$2.11 * 10^5$
IS219	0.37	47	171	$3.62 * 10^6$	1.51	164	60	$3.68 * 10^5$	0.94	140	126	$8.98 * 10^5$
IS221	0.30	30	103	$3.46 * 10^6$	0.43	66	43	$6.47 * 10^5$	0.90	108	89	$8.22 * 10^5$
IS227	Low	Low	Low	Low	Low	Low	Low	Low				
IS229	Low	Low	Low	Low	3.10	574	136	$2.37 * 10^5$				
P3A ⁽¹⁾	0.25	55	54	$9.89 * 10^5$	0.24	230	17	$7.3 * 10^4$	ND	ND	ND	ND
IS158	1.27	33	45	$1.36 * 10^6$	0.57	144	20	$1.41 * 10^5$	ND	ND	ND	ND
IS232 (IS180)	0.47	4	27	$6.25 * 10^6$	0.31	14	23	$1.69 * 10^6$	low	low	low	low
IS234 (IS176)	1.09	37	53	$1.45 * 10^6$	0.61	33	30	$8.88 * 10^5$	0.77	17	38	$2.19 * 10^6$
IS230 (IS181)	0.66	18	32	$1.80 * 10^6$	0.47	21	23	$1.09 * 10^6$	0.50	7	24	$3.27 * 10^6$
P2A ⁽¹⁾		68	95	$1.40 * 10^6$	0.79	66	479	$7.28 * 10^6$	0.37	27	221	$7.99 * 10^6$

TABLE 6

V_{max} and K_m values of selected mutants represented as relative values of P1A activities. If $K_m < 1.0$, the mutant's affinity towards its ligand increased. If $K_m > 1.0$, the affinity decreased. The opposite is true for k_{cat} and V_{max} . Low = low activity; — = not done.												
	Ceftriaxone				Cefotaxime				Cefuroxime			
	V_{max}	K_m	k_{cat}	k_{cat}/K_m	V_{max}	K_m	k_{cat}	k_{cat}/K_m	V_{max}	K_m	k_{cat}	k_{cat}/K_m
P1A (A18K31)	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
IS219	0.13	0.26	6.76	25.64	1.29	0.71	3.81	5.39	0.75	1.31	5.56	4.25
IS221	0.11	0.17	4.09	24.46	0.37	0.29	2.70	9.48	0.71	1.01	3.93	3.89
IS227	low	low	low	low	low	low	low	low	—	—	—	—
IS229	low	low	low	low	2.64	2.47	8.58	3.47	—	—	—	—
P3A	0.09	0.31	2.15	7.00	0.20	0.99	1.07	1.07	—	—	—	—
IS158	0.45	0.19	1.80	9.63	0.48	0.62	1.28	2.06	—	—	—	—
IS232 (IS180)	0.17	0.02	1.08	44.25	0.26	0.06	1.44	24.70	—	—	—	—

TABLE 6-continued

V_{max} and K_m values of selected mutants represented as relative values of P1A activities. If $K_m < 1.0$, the mutant's affinity towards its ligand increased. If $K_m > 1.0$, the affinity decreased. The opposite is true for k_{cat} and V_{max} . Low = low activity; — = not done.												
	Ceftriaxone				Cefotaxime				Cefuroxime			
	V_{max}	K_m	k_{cat}	k_{cat}/K_m	V_{max}	K_m	k_{cat}	k_{cat}/K_m	V_{max}	K_m	k_{cat}	k_{cat}/K_m
IS234 (IS176)	0.39	0.20	2.10	10.26	0.52	0.14	1.87	13.00	0.61	0.16	1.66	10.38
IS230 (IS181)	0.24	0.10	1.27	12.70	0.40	0.09	1.45	15.99	0.40	0.07	1.08	15.48
P2A	—	0.38	3.76	9.87	0.67	0.28	30.18	106.55	0.29	0.25	9.78	37.81

Enzymatic Activity and Kinetic Parameters

From the kinetic parameters measured for P1A against cefotaxime, ceftriaxone and cefuroxime, without wishing to be bound by theory, it was deduced that P1A has a low intrinsic cephalosporinase activity. In general, the mutants' affinity towards ligands has increased (i.e. decreased K_m). V_{max} increased and the affinity decreased (K_m increased) in the IS229 (Block 2+R244T) mutant (Table 5). For IS229, the k_{cat} increased the most.

The random mutagenesis screen with ceftriaxone preferred mutants with low K_m (high affinity) towards the antibiotic.

Decrease in K_m by subsequent mutation rounds was seen in the mutation series P3A→IS158→IS232; IS232 had extremely high affinity towards ceftriaxone. The V_{max} for ceftriaxone did not decrease sequentially being the highest for IS158 but k_{cat} , nevertheless, decreased through the mutation rounds. The first two mutants of this round did not have cefotaximase activity but IS232 with additional A238T had cefotaximase activity.

Site-directed Block2 (IS191) mutations alone did not markedly increase P1A-derivative's activity towards cephalosporins but the activity towards ampicillin was reduced 20 fold (Table 4). The cephalosporinase activity of Block2+D276R/K (IS219/IS221) was considerably better than that of IS191's. The kinetic parameters of the IS219 and IS221 site-directed mutants revealed that the site-directed mutagenesis preferably increased the enzyme's affinity towards the target antibiotics.

Stability Assay/Glucose Stress Test

To test the stability of mutants in *B. subtilis* growth conditions, a shaker flask cultivation with RS310, IS219 and IS232 was performed. The cultivation was made in two different glucose concentrations of 5 g/l and 10 g/l.

P1A (RS310) and the two mutants IS219 and IS232 were cultivated in shaker flasks (100 ml) in minimal medium with 5 or 10 g/l glucose. The condition of the cells was monitored with microscope, the glucose consumption measured from 5 g/l growth and samples from the culture supernatants (concentrated and non-concentrated) were run on 12% Bis-Tris SDS-PAGE (Criterion, BioRad) (FIGS. 1 and 2). The ODs of the cell cultures were also monitored.

In 10 g/l glucose cultures, P1A and IS232 were produced in comparable amounts but the production of IS219 was lower (lanes 2-4, FIG. 1). During concentration, the IS219 yield was further reduced (lanes 6, 9 and 16, FIG. 1). A faint band ~29 kDa can be seen on lane 16 in FIG. 1. P1A and IS232, again, could be concentrated (lanes 5, 7, 8, 10, 15 and 16 FIG. 1).

In the presence of 5 g/l glucose, P1A strain RS310 started growing slower than the other two strains. However, by the end of the cultivation the amount of P1A had superseded the amount of the other proteins (FIG. 2). The mutant strain IS219 started to produce roughly at the same velocity as the

strain IS232 but by 10 hours of cultivation its production had started to lag behind IS232 and its amount was decreased over time. All glucose was consumed by 9.5 hours of cultivation both in IS219 and IS232 cultivations and from that point on it is assumed that the amount of proteases started to increase. Because RS310 grew slower, it still had some glucose left at 11 hours of cultivation but also RS310 had consumed all glucose by the following morning. Because no samples were taken from RS310 during the night, the exact time-point at which RS310 had used all its glucose and started to produce proteases cannot be stated. By 27.5 hours of cultivation the amount of P1A had increased but the amount of both IS219 and IS232 had decreased from the previous day. From the study, without wishing to be bound by theory, it can be deduced that IS232 tolerates proteases better than IS219, which does not tolerate them at all.

Based both on 10 g/l glucose and 5 g/l glucose cultivations, without wishing to be bound by theory, it can be said that modifications to P1A reduce the protein's protease tolerance. Glucose deprivation lead to the appearance of degradation products in the case of IS219 and IS232 and the amount of degradation products was increased after all glucose was consumed. IS219 protein was more sensitive to degradation as its amount decreased more than that of IS232. Without wishing to be bound by theory, this may indicate either lower secretability or greater sensitivity to *Bacillus* proteases, or both. In conclusion, production of IS219 was not possible using the same growth conditions as for RS310 and IS232. The glucose stress test also demonstrated that this test can be used to estimate the protease tolerance of the generated mutants.

With the glucose stress test it was demonstrated that Block2+D276R=IS219 (and most likely Block2+D276K=IS221) mutation decreases the enzyme's protease tolerance. Most likely, without wishing to be bound by theory, the instability is due to D276R mutation, which creates a protease cleavage site on the surface of the protein.

A D276N mutation on IS191, which would result in a Block2+D276N-mutant was made. Without wishing to be bound by theory, an aspartic acid to asparagine mutation is expected to increase the ceftriaxone (and maybe cefotaxime) activity of IS191 but at the same time keep the enzyme more resistant to proteases than IS219 (and IS221) is. R244T mutation could be combined with Block2 and D276N to see whether we can increase the V_{max} (and k_{cat}) with this mutation. R244T mutation could also be combined with IS219 and IS221.

An activity assay (Table 4) R244T (IS217) mutation abolished nearly all enzymatic activity of P1A. Some of the activity was gained with a double mutant R244T+D276K (IS215),

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but not to the level of the wild type enzyme. Thus, R244T mutation alone cannot provide P1A with cephalosporinase activity.

Based on kinetic data (Table 5) Block1 mutation (I72S/Q135M/T160F, IS227) did not improve P1A's ability to hydrolyze cephalosporins. In other activity assays neither did Q135M or T160F alone increase P1A's ability to hydrolyze cephalosporins, instead, both single mutations abolished nearly all P1A's enzymatic activity, even towards ampicillin. There was, however, a Block2+Q135M-mutant (IS224), which had an activity towards cefotaxime and ceftriaxone between Block2 (IS191) and Block2+D276R/K (IS219/IS221) mutants. Without wishing to be bound by theory, Block1 may "loosen" the Ω -loop where the catalytic Glu166 is located. T160F mutation is quite a large modification in the core of a protein's structure. It might disrupt the structure totally or at least incapacitate the protein from functioning properly. To avoid this possibility, two compensatory mutations were added to accompany T160F mutation in P1A, namely I72S and Q135M and this combination of mutations was called "Block1".

Example 3

Results from Production and Purification of P1A Mutant IS288

The results of the purification are reported as elution peak figures (FIGS. 3, 4, and 5) and peak tables, as protein concentration and as SDS-PAGE picture(s) (FIG. 6). Chromatograms from Hydrophobic Interaction Chromatography

Hydrophobic interaction chromatography (HIC) was run on three separate occasions with differing parameters. In the first run (16.10.08, FIG. 3) the sample volume was ~50 ml, volume of flow through fractions 5 ml, wash volume for the unbound sample 2 CV, and the volume of eluted fractions 1.0 ml. In the following HIC run (4.11.08, FIG. 4) the sample volume was increased to 65 ml, flow-through fraction volume to 10 ml, and wash volume for unbound sample to 3 CV and fraction size to 1.5 ml. In the last run (12.11.08, FIG. 5) only the sample volume and wash volume for unbound sample differed from the preceding run being 125 ml and 4 CV, respectively.

All the graphs of HIC runs (FIGS. 3, 4, and 5) represent the same overall form. The flow through fraction gave constant absorbance ~700 mAU at 280 nm throughout the sample application while during washing of unbound sample, the absorbance at 280 nm dropped to zero rising again when the bound protein was eluted. The conductivity of the sample was

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slightly lower in samples applied in FIG. 3 and FIG. 4 than in running buffer A containing 2 M $(\text{NH}_4)_2\text{SO}_4$ while in the run on 12.11.2008 it stayed the same as in the A-buffer. In the two first runs the $(\text{NH}_4)_2\text{SO}_4$ concentration of the ammonium sulphate filtrate was estimated as 2.7 M and in the last run 2.6 M. It seems that the 2.6 M estimation was closer to the actual $(\text{NH}_4)_2\text{SO}_4$ concentration of the ammonium sulphate filtrate as the conductivity of the out coming solution did not drop during sample application (FIG. 5).

The elution that started right after the elution buffer (buffer B) was applied on the column. This indicated that the protein in the sample was barely bound to the column as it eluted so early in the gradient. Thus, it was vital that the conductivity of the sample never got less than ~240 mS/cm.

The bound protein eluted as a typical elution peak with steep ascending slope and a gentle descending slope giving slightly asymmetrical peak. The size of the peak was largest in the first run, ~18 ml (18×1 ml fractions) (FIG. 3). This peak was collected in three sets: first of 6 ml (fractions C3-C8), second of 5 ml (fractions C9-D1) and third of 7 ml (fractions D2-D8) as there was a "shoulder" in the elution peak between fractions C8 and C9 and it was considered that the last 7 fractions could be too diluted for concentration. In the two subsequent HIC runs (4.11.2008 and 12.11.2008, FIGS. 4 and 5, respectively) the elution peak was 13.5 ml (9×1.5 ml fractions, 4.11.2008) and 12 ml (8×1.5 ml fractions, 12.11.2008) and the whole peak was pooled to one fraction.

In Table 7, the statistical values for A_{280} graphs of the three HIC runs are given. The area of the peak was used to estimate the protein content of that sample and it was used in estimating the dilution factor for different samples in the BCA assay. The calculations required the theoretical extinction coefficient of the protein, which was calculated using ProtParam-Tool at ExPASy web service. For IS288, this coefficient was 25440 $\text{M}^{-1}\text{cm}^{-1}$ and gave A_{280} 0.869 for 0.1% (=1 g/l) solution.

Using the proportion above, the amounts of protein in eluted protein samples and the known volumes of these samples, the protein amount eluted from the column was calculated. The formula used was: $((\text{mAU} \cdot \text{ml}) \cdot 10^{-3} / V_{\text{fraction}}) / 0.869 \cdot V_{\text{fraction}} \rightarrow (\text{mAU} \cdot \text{ml}) \cdot 10^{-3} / 0.869$, which resulted in: 3.36 mg protein 161008, 2.95 mg 041108, and 3.60 mg 121108.

Table 7: Peak Table from Three Separate HIC Runs

The date of the run, peak name, retention time i.e. the volume at which the peak reaches its top, area of the peak, and the maximum height of the peak are given. The area of the peak was used to estimate the amount of eluted protein in the elution fraction.

TABLE 7

Peak table from three separate HIC runs					
The date of the run, peak name, retention time i.e. the volume at which the peak reaches its top, area of the peak, and the maximum height of the peak are given. The area of the peak was used to estimate the amount of eluted protein in the elution fraction.					
Date	Peak name	Retention (ml)	Area (mAU * ml)	Height (mAU)	Estimated protein amount (mg)
16 Oct. 2008	Flow through fraction	84.48	44814.9666	816.803	3.36
	Wash unbound sample	86.54	4640.2506	659.512	
	Eluted sample	93.08	2923.2825	814.968	
4 Nov. 2008	Flow through fraction	27.94	42450.2080	673.843	2.95
	Eluted sample	87.01	2562.0783	953.277	
12 Nov. 2008	Flow-through fraction	23.40	78004.2505	658.094	3.60
	Eluted protein	152.03	3126.5816	1214.232	

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Ultra Filtration and Concentration of Flow-Through and Eluted Protein Fractions

From each of the purifications one, two or three of the flow-through fractions were selected for concentration as well as all fractions containing the eluted protein peak. The selected samples are represented in Table 8. Also represented in the table are the starting volume of that sample as well as final volume and concentration factor.

Table 8: Samples Selected for Ultra Filtration and Concentration With Amicon Ultra-4, 10000 NMWL

The original volume of the sample and the final volume of the sample after concentration as well as the concentration factor are presented. The concentration factor was calculated by dividing the original volume by the final volume of the sample after concentration.

TABLE 8

Samples selected for ultra filtration and concentration with Amicon Ultra-4, 10000 NMWL				
The original volume of the sample and the final volume of the sample after concentration as well as the concentration factor are presented. The concentration factor was calculated by dividing the original volume by the final volume of the sample after concentration.				
Date of purification	Selected fractions	Original volume (ml)	Final volume (μl)	Conc. factor
16 Oct. 2008	B1 (FT)	5	750	6.7 X
	B7 (FT)	4.5	1000	4.5 X
	B8 (FT)	5	750	6.7 X
	C3-C8 (Protein Peak)	6	900	6.7 X
	C8-D1 (Protein Peak)	5	900	5.6 X
4 Nov. 2008	D2-D8 (Protein Peak)	7	900	7.8 X
	A2 (FT)	9.5	750	12.7 X
	A7 (FT)	9.5	750	12.7 X
	B2-B10 (Protein Peak)	13	700	18.6 X
12 Nov. 2008	B1 (FT)	9.5	900	10.6 X
	B8-C3 (Protein Peak)	11	1200	9.2 X

Flow-through fractions selected for the ultra filtration were from the beginning (B1₁₆₁₀₀₈ and A2₀₄₁₁₀₈) and from the end of sample application (B7₁₆₁₀₀₈, B8₁₆₁₀₀₈, A7₀₄₁₁₀₈ and B1₁₂₁₁₀₈). These samples were analysed in order to see if the capacity of the column was exceeded with that particular sample amount.

Moreover, ammonium sulphate filtrates for each of the purifications were ultra filtrated and concentrated and the once concentrated flow through fractions were still further concentrated with MultiScreen plates with Ultracell-10 membrane, 10000 NMWL. The results of that concentration are presented in Table 9.

Table 9: Samples Selected for Ultra Filtration and Concentration With MultiScreen Filter Plates With Ultracell-10 Membrane, 10000 NMWL

Both the original volume and the volume of the sample after concentration are indicated. Concentration factor was calculated by dividing the original sample volume with the final sample volume.

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TABLE 9

Samples selected for ultra filtration and concentration with MultiScreen Filter Plates with Ultracell-10 membrane, 10000 NMWL				
Both the original volume and the volume of the sample after concentration are indicated. Concentration factor was calculated by dividing the original sample volume with the final sample volume.				
Date of purification	Selected fractions	Original volume (ml)	Final volume (μl)	Conc. factor
16 Oct. 2008	AS-sup 161008	500	420	1.2 X
	B1 (FT)	400	125	3.2 X
	B7 (FT)	1200	300	4.0 X
	B8 (FT)	380	115	3.3 X
4 Nov. 2008	AS-sup 041108	500	440	1.1 X
	A2 (FT)	360	115	3.1 X
	A7 (FT)	700	265	2.1 X
12 Nov. 2008	AS-sup 121108	500	455	1.1 X
	B1 (FT)	430	165	2.6 X

The flow through fractions of the HIC runs were, thus, concentrated by a factor of 16 or more. The final concentration factors of the flow through fractions are listed in Table 10. Table 10: Final Concentration Factors of Flow Through Samples

The final concentration factors were calculated by multiplying the concentration factor of the first concentration with the concentration factor of the second concentration.

TABLE 10

Final concentration factors of flow through samples		
The final concentration factors were calculated by multiplying the concentration factor of the first concentration with the concentration factor of the second concentration.		
Date of purification	Selected fractions	Conc. factor
16 Oct. 2008	B1 (FT)	20.0 X
	B7 (FT)	16.7 X
	B8 (FT)	21.2 X
	A2 (FT)	39.1 X
4 Nov. 2008	A7 (FT)	35.8 X
12 Nov. 2008	B1 (FT)	33.0 X

BCA Assay Results

Results from the BCA assay are presented in Table 11. The protein concentration of the flow through fractions was measured before the last concentration step with MultiScreen plates with Ultracell-10 membrane.

Table 11: Protein Concentrations of HIC Flow-Through and Peak Fractions

The protein concentration was determined from two replicates of two different dilutions of a given sample. The final volume of the sample was measured after concentration.

TABLE 11

Protein concentrations of HIC flow-through and peak fractions			
The protein concentration was determined from two replicates of two different dilutions of a given sample. The final volume of the sample was measured after concentration.			
Sample name	Protein concentration in the sample (mg/ml)	Final volume of the sample (ml)	Amount of protein in the sample (mg)
B1 ₁₆₁₀₀₈	0.02	0.75	0.015
B8 ₁₆₁₀₀₈	0.01	0.75	0.008

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TABLE 11-continued

Protein concentrations of HIC flow-through and peak fractions The protein concentration was determined from two replicates of two different dilutions of a given sample. The final volume of the sample was measured after concentration.			
Sample name	Protein concentration in the sample (mg/ml)	Final volume of the sample(ml)	Amount of protein in the sample (mg)
C3-C8 ₁₆₁₀₀₈	9.59	0.9	8.63
C9-D1 ₁₆₁₀₀₈	0.10	0.9	0.09
D2-D8 ₁₆₁₀₀₈	0.72	0.9	0.65
Total protein amount in fractions C3-D8			9.37
A2 ₀₄₁₁₀₈	0.00	0.75	0.00
B2-B10 ₀₄₁₁₀₈	9.45	0.7	6.62
B1 ₁₂₁₁₀₈	0.08	0.9	0.07
B8-C3 ₁₂₁₁₀₈	7.56	1.2	9.07

Chromatograms from HIC runs gave rather high, ~700 mAU, A₂₈₀ values throughout sample application for the out coming solution. The eluted protein peaks contained sufficient amounts of protein. They were near the presumed binding capacity of the column, 10 mg. The area of the peak from the Äkta gave three times lower estimates for protein amount than the BCA assay.

SDS-PAGE

The maximum volume (17.5 µl) of sample was loaded on the SDS-PAGE (FIG. 6) for ammonium sulphate filtrates and flow through fractions, except for B1₁₂₁₁₀₈ for which 6.5 µl was loaded. The volume of the samples from protein peaks was calculated so that there would be ~0.5 µg of protein per lane and assumed that all protein in the sample would be P1A derivative IS288. The SDS-PAGE is represented in FIG. 6.

Ammonium sulphate filtrate samples all contain a double band near 30 kDa (MW of IS288 is 29288 Da, and of P1A 29273 Da). The same double band accompanied by a slightly smaller extra band can be clearly seen in flow through fraction B1₁₂₁₁₀₈ and somewhat more faintly in flow through fraction A7₀₄₁₁₀₈. On the other hand, both of these extra bands have been removed from eluted protein peaks. Some minor impurities can be seen on the last pooled fractions of purification D2-D8 161008, when ~10 times the intended amount of protein was loaded. It can be concluded that the purified protein was the major protein form and that by molecular weight analysis it was the P1A variant IS288.

Example 4

Test of a Microbiological Screening Assay for Beta-Lactamase Activity

The 50% effective inhibitory concentration (EC₅₀) of ampicillin and ceftriaxone against *Escherichia coli* was determined in the absence and presence of P1A β-lactamase. For the purposes of these studies, the stated effective concentration was the initial concentration of ampicillin or ceftriaxone added at the beginning of the experiment, and not a concentration measured during the course of the experiment or at the end of the incubation period. The results of the analysis are presented in Tables 12-14.

Initial evaluation utilized P1A at concentrations of 0.5 ng/mL to 5 µg/mL and ampicillin or ceftriaxone at concentrations of 0.13 µg/mL to 128 µg/mL. Addition of the lowest

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concentration of P1A (0.5 ng/mL) shifted the EC₅₀ of ampicillin from 16 µg/mL to greater than the highest test concentration of 128 µg/mL when added simultaneously and to 64 µg/mL when P1A was pre-incubated with ampicillin prior to addition of the bacteria (Table 12). Using the same concentrations of P1A with 0.13 µg/mL to 128 µg/mL of ceftriaxone, it was observed that ceftriaxone was completely inhibitory at the lowest concentration (EC₉₉ < 0.13 µg/mL) and the EC₅₀ of ceftriaxone was shifted to 0.25 µg/mL in the presence of 5 ng/mL P1A when added simultaneously and to 128 µg/mL when 50 ng/mL P1A was preincubated with the antibiotic. Based on this analysis, the concentrations of P1A and ceftriaxone were adjusted to bracket the inhibitory concentrations of both.

Using concentration of P1A from 5 pg/mL to 50 ng/mL with 0.13 µg/mL to 128 µg/mL of ampicillin, the EC₅₀ of ampicillin was shifted from 16 µg/mL in the absence of P1A to 128 µg/mL in the presence of 50 pg/mL when pre-incubated or added simultaneously. Lower concentrations of P1A (5 pg/mL) had no effect on the ampicillin EC₅₀. The EC₅₀ of ceftriaxone was shifted from 0.03 µg/mL in the absence of P1A to greater than 4 µg/mL in the presence of 50 ng/mL or greater P1A when preincubated or added simultaneously (Table 13). Lower concentrations of P1A (50 pg/mL to 5 ng/mL) had no effect on the ceftriaxone EC₅₀. Results with the simultaneous addition of P1A with ampicillin or ceftriaxone were confirmed through repeat analysis (Table 14).

TABLE 12

Effective inhibitory concentration (EC ₅₀) of ampicillin and ceftriaxone in the absence and presence of P1A determined by micro broth dilution analysis.				
P1A	Ampicillin EC ₅₀ (µg/mL)		Ceftriaxone EC ₅₀ (µg/mL)	
Con- centration	Pre incubation	Simultaneous	Pre incubation	Simultaneous
0	16	16	<0.13	<0.13
0.5 ng/mL	64	>128	<0.13	<0.13
5 ng/mL	128	>128	<0.13	0.25
50 ng/mL	>128	>128	>128	>128
500 ng/mL	>128	>128	>128	>128
5 µg/mL	>128	>128	>128	>128

Antibiotics were either added to the P1A and incubated for 30 min at 37° C. (Pre-incubation) prior to addition of bacteria, or added to the plates following the addition of P1A and bacteria (Simultaneous). All concentrations and combinations were evaluated in triplicate in a 96-well plate format. Following overnight incubation, the OD₆₂₅ of the plates was measured in a Spectramax 384 plate reader and the data were imported into a customized Excel spreadsheet for the determination of the 50% effective inhibitory concentration at (EC₅₀) by linear regression analysis. For the purposes of these studies, the stated effective concentration is the initial concentration of ampicillin or ceftriaxone added at the beginning of the experiment, and not a concentration measured during the course of the experiment or at the end of the incubation period.

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TABLE 13

Effective inhibitory concentration (EC ₅₀) of ampicillin and ceftriaxone in the absence and presence of P1A determined by micro broth dilution analysis.				
PIA	Ampicillin EC ₅₀ (μg/mL)		Ceftriaxone EC ₅₀ (μg/mL)	
Con- centration	Pre incubation	Simultaneous	Pre incubation	Simultaneous
0	16	16	0.02	0.02
5 pg/mL	16	16	NA	NA
50 pg/mL	128	128	0.02	0.02
500 pg/mL	>128	>128	0.03	0.03
5 ng/mL	>128	>128	0.03	0.03
50 ng/mL	>128	>128	>4	>4
500 ng/mL	NA	NA	>4	>4

Antibiotics were either added to the P1A and incubated for 30 min at 37° C. (Pre-incubation) prior to addition of bacteria, or added to the plates following the addition of P1A and bacteria (Simultaneous). All concentrations and combinations were evaluated in triplicate in a 96-well plate format. Following overnight incubation, the OD₆₂₅ of the plates was measured in a Spectramax 384 plate reader and the data were imported into a customized Excel spreadsheet for the determination of the 50% effective inhibitory concentration (EC₅₀).

TABLE 14

Effective inhibitory concentration (EC ₅₀) of ampicillin and ceftriaxone in the absence and presence of P1A determined by microbroth dilution analysis.		
PIA Concentration	Ampicillin EC ₅₀ (μg/mL) Simultaneous	Ceftriaxone EC ₅₀ (μg/mL) Simultaneous
0	16	0.03
5 pg/mL	16	NA
50 pg/mL	128	0.03
500 pg/mL	>128	0.03
5 ng/mL	>128	0.03
50 ng/mL	>128	>4
500 ng/mL	NA	>4

Antibiotics were added to the P1A bacteria (Simultaneous). All concentrations and combinations were evaluated in triplicate in a 96-well plate format. Following overnight incubation, the OD₆₂₅ of the plates was measured in a Spectramax 384 plate reader and the data were imported into a customized Excel spreadsheet for the determination of the 50% effective inhibitory concentration at (EC₅₀) by linear regression analysis.

These results demonstrated that P1A effectively inhibited the activity of ampicillin and ceftriaxone in an in vitro microbiological assay and indicated that P1A is 1,000-fold more active against ampicillin than ceftriaxone in the microbiological assay.

Next, the 90% effective inhibitory concentration (EC₉₀) of ampicillin, ceftriaxone, ceftazidime, meropenem, cefepime, cefazolin, amp/sulbactam, cefoperazone, cefotaxime, and cefuroxime against *Escherichia coli* was determined in the absence and presence of P1A, P3A, or P4A β-lactamase. For the purposes of these studies, the stated effective concentration was the initial concentration of ampicillin, ceftriaxone, ceftazidime, meropenem, cefepime, cefazolin, amp/sulbactam, cefoperazone, cefotaxime, and cefuroxime added at the beginning of the experiment, and not a concentration measured during the course of the experiment or at the end of the incubation period. The results of the analysis are presented in Tables 15-17. Each box in the below tables (Tables 15-17) has

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two numbers separated by a slash-the two numbers are duplicate assays, all of which showed tight agreement.

Specifically, the assay was performed by diluting (2-fold steps) the chosen antibiotic across the rows of a 96 well dish. The MIC was determined as the point at which the diluted antibiotic transitions from killing to not killing the bacteria (in this case, an *E. coli* strain that is sensitive to all antibiotics). P1A, P3A, and P4A were diluted (10 fold steps) down the columns (the β-lactamase concentrations increased down the columns). As the β-lactamase concentration increased, it reached a level at which it degraded the antibiotic sufficiently to alter the apparent MIC reading (the MIC didn't actually change, but it appeared to change because the amount of antibiotic remaining in the well was less than the amount loaded into the well). In most cases, there were two break points, the first in which the MIC appeared to increase a bit, and then at ten-fold concentration higher where the MIC exceeded the maximum amount of antibiotic tested. The latter point was used as the cut-off point. It is important to note that with this *E. coli*, the MIC was different for each antibiotic. However, for this assay, the absolute MIC is not the most important parameter. The most relevant readout was the β-lactamase concentration at which the apparent MIC increases to the maximum amount of antibiotic tested.

When the cut-off point occurred at low β-lactamase concentration, this implied that the β-lactamase was potent at degrading that antibiotic. When it took a lot of β-lactamase to move the MIC, then the β-lactamase was relatively weak for that antibiotic. This assay was designed, without wishing to be bound by theory to provide predictions of in vivo efficacy for each β-lactamase for a given antibiotic. These data can be combined with the in vitro kinetic data.

TABLE 15

Antibiotic activity of ampicillin and ceftriaxone in the presence of bacterial culture supernatants from <i>Bacillus</i> strains expressing soluble β-lactamase proteins P1A (RS310), P3A (IS118), or P4A (IS288).				
	Ampicillin EC ₅₀ (μg/mL)		Ceftriaxone EC ₅₀ (μg/mL)	
	Rep 1	Rep 2	Rep 1	Rep 2
RS310 Concentration				
0	2	4	0.008	0.02
5 pg/mL	4	16	NA	NA
50 pg/mL	32	128	0.008	0.008
500 pg/mL	>128	>128	0.008	0.008
5 ng/mL	>128	>128	0.06	0.008
50 ng/mL	>128	>128	>4	>4
500 ng/mL	NA	NA	>4	>4
IS118 Concentration				
0	2	4	0.008	0.02
5 pg/mL	4	8	NA	NA
50 pg/mL	16	64	0.008	0.008
500 pg/mL	>128	>128	0.5	0.06
5 ng/mL	>128	>128	NA	>4
50 ng/mL	>128	>128	>4	>4
500 ng/mL	NA	NA	>4	>4
IS288 Concentration				
0	2	4	0.02	0.02
5 pg/mL	2	16	NA	0.03
50 pg/mL	4	32	0.5	0.13
500 pg/mL	64	128	>4	4
5 ng/mL	>128	>128	>4	>4
50 ng/mL	>128	>128	>4	>4
500 ng/mL	NA	NA	>4	NA

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E. coli 29522 was incubated overnight in the presence of serial dilutions of antibiotic and the indicated amounts of β -lactamase protein from cell culture supernatants in a 96 well plate. Bacterial density (OD_{625}) of the cultures was measured and the concentration at which the density was reduced by 90% relative to an untreated bacterial control (EC_{90}) was determined. The assay was performed on two separate days and the data reported as Rep1 and Rep2. NA=not assayed.

TABLE 16

Antibiotic activity of ceftazidime, meropenem, cefepime, and ceftriazone in the presence of bacterial culture supernatants from <i>bacillus</i> strains expressing soluble β -lactamase proteins P1A (RS310), P3A (IS118), or P4A (IS288).				
	Ceftazidime EC_{50} (μ g/mL) (high test 4 μ g/mL)	Meropenem EC_{50} (μ g/mL) (high test 1 μ g/mL)	Cefepime EC_{50} (μ g/mL) (high test 2 μ g/mL)	Ceftriazone EC_{50} (μ g/mL) (high test 1 μ g/mL)
PS310 Concentration				
0	0.13/0.5	0.06/0.06	0.06/0.03	0.06/0.06
50 pg/mL	0.25/0.25	0.06/0.03	0.03/0.03	0.06/0.06
500 pg/mL	0.13/0.25	0.06/0.06	0.06/0.06	0.06/0.13
5 ng/mL	0.13/0.25	0.06/0.06	0.13/0.25	0.25/1.0
50 ng/mL	0.5/0.5	0.03/0.06	>2.0/>2.0	>1.0/>1.0
500 ng/mL	>4.0/>4.0	0.03/0.03	>2.0/>2.0	>1.0/>1.0
IS118 Concentration				
0	0.13/0.13	0.06/0.03	0.06/0.03	0.06/0.06

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TABLE 16-continued

Antibiotic activity of ceftazidime, meropenem, cefepime, and ceftriazone in the presence of bacterial culture supernatants from <i>bacillus</i> strains expressing soluble β -lactamase proteins P1A (RS310), P3A (IS118), or P4A (IS288).				
	Ceftazidime EC_{50} (μ g/mL) (high test 4 μ g/mL)	Meropenem EC_{50} (μ g/mL) (high test 1 μ g/mL)	Cefepime EC_{50} (μ g/mL) (high test 2 μ g/mL)	Ceftriazone EC_{50} (μ g/mL) (high test 1 μ g/mL)
10				
50 pg/mL	0.13/0.25	0.06/0.03	0.03/0.03	0.06/0.13
500 pg/mL	0.25/0.25	0.06/0.03	0.06/0.03	1.0/0.25
5 ng/mL	0.13/0.25	0.06/0.06	0.13/0.13	>1.0/>1.0
50 ng/mL	1.0/2.0	0.03/0.06	>2.0/>2.0	>1.0/>1.0
500 ng/mL	>4.0/>4.0	0.06/0.06	>2.0/>2.0	>1.0/>1.0
15 IS288 Concentration				
0	0.13/0.5	0.06/0.06	0.03/0.03	0.06/0.06
50 pg/mL	0.25/0.25	0.06/0.03	0.03/0.03	0.13/>1.0
500 pg/mL	0.13/0.25	0.06/0.03	0.06/0.06	>1.0/>1.0
5 ng/mL	0.25/0.5	0.25/0.25	0.13/0.25	>1.0/>1.0
50 ng/mL	>4.0/>4.0	0.03/0.06	>2.0/>2.0	>1.0/>1.0
500 ng/mL	>4.0/>4.0	0.06/0.06	>2.0/>2.0	>1.0/>1.0

E. coli 29522 was incubated overnight in the presence of serial dilutions of antibiotic and the indicated amounts of β -lactamase protein from cell culture supernatants in a 96 well plate. Bacterial density (OD_{625}) of the cultures was measured and the concentration at which the density was reduced by 90% relative to an untreated bacterial control (EC_{90}) was determined. The assay was performed on two separate days and the data reported as two numeric values separated by a slash. NA=not assayed.

TABLE 17

Antibiotic activity of cefazolin, amp/sublactam, cefoperazone, cefotaxime, cefuroxime, and ceftriazone in the presence of bacterial culture supernatants from <i>bacillus</i> strains expressing soluble β -lactamase proteins P1A (RS310), P3A (IS118), or P4A (IS288).						
Project 307-04-04 Results Tables 27 March 2014						
	Cefazolin EC_{50} (μ g/mL) (high test 64 μ g/mL)	Amp. Sublactam EC_{50} (μ g/mL) (high test 128:64 μ g/mL)	Cefoperazone EC_{50} (μ g/mL) (high test 8 μ g/mL)	Cefotaxime EC_{50} (μ g/mL) (high test 2 μ g/mL)	Cefuroxime EC_{50} (μ g/mL) (high test 128 μ g/mL)	Ceftriazone EC_{50} (μ g/mL) (high test 1 μ g/mL)
RS310 Concentration						
0	2.0/2.0	4.0:2.0/4.0:2.0	0.25/0.13	0.004/0.008	4.0/4.0	0.03/0.03
50 pg/mL	2.0/4.0	4.0:2.0/8.0:4.0	0.5/0.25	0.008/0.004	4.0/2.0	0.03/0.06
500 pg/mL	16/16	4.0:2.0/4.0:2.0	2.0/4.0	0.004/0.004	4.0/4.0	0.13/0.13
5 ng/mL	>64/>64	8.0:4.0/16:8.0	>8.0/>8.0	0.008/0.008	32/32	0.13/0.13
50 ng/mL	>64/>64	32:16/32:16	>8.0/>8.0	1.0/0.25	>128/>128	>1.0/>1.0
500 ng/mL	>64/>64	128:64/128:64	>8.0/>8.0	>2.0/>2.0	>128/>128	>1.0/>1.0
IS118 Concentration						
0	2.0/2.0	4.0:2.0/4.0:2.0	0.25/0.25	0.004/0.004	2.0/4.0	0.03/0.03
50 pg/mL	4.0/4.0	4.0:2.0/8.0:4.0	0.5/0.5	0.004/0.004	8.0/4.0	0.13/0.13
500 pg/mL	64/64	4.0:2.0/4.0:2.0	8.0/>8.0	0.008/0.008	8.0/8.0	0.25/0.5
5 ng/mL	>64/>64	8.0:4.0/16:8.0	>8.0/>8.0	0.06/0.03	128/128	>1.0/>1.0
50 ng/mL	>64/>64	32:16/32:16	>8.0/>8.0	>2.0/>2.0	>128/>128	>1.0/>1.0
500 ng/mL	>64/>64	128:64/128:64	>8.0/>8.0	>2.0/>2.0	>128/>128	>1.0/>1.0
IS288 Concentration						
0	2.0/2.0	4.0:2.0/4.0:2.0	0.25/0.13	0.004/0.004	4.0/2.0	0.03/0.06
50 pg/mL	4.0/4.0	4.0:2.0/4.0:2.0	0.5/0.5	0.02/0.008	8.0/8.0	0.25/0.25
500 pg/mL	>64/>64	4.0:2.0/4.0:2.0	>8.0/>8.0	0.25/0.25	32/32	>1.0/>1.0
5 ng/mL	>64/>64	4.0:2.0/8.0:4.0	>8.0/>8.0	>2.0/>2.0	>128/>128	>1.0/>1.0
50 ng/mL	>64/>64	8.0:4.0/8.0:4.0	>8.0/>8.0	>2.0/>2.0	>128/>128	>1.0/>1.0
500 ng/mL	>64/>64	32:16/32:16	>8.0/>8.0	>2.0/>2.0	>128/>128	>1.0/>1.0

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E. coli 29522 was incubated overnight in the presence of serial dilutions of antibiotic and the indicated amounts of β -lactamase protein from cell culture supernatants in a 96 well plate. Bacterial density (OD_{625}) of the cultures was measured and the concentration at which the density was reduced by 90% relative to an untreated bacterial control (EC90) was determined. The assay was performed on two separate days and the data reported as two numeric values separated by a slash. NA=not assayed.

Without wishing to be bound by theory, these data suggest more activity in vitro than predicted from the corresponding kinetic data. For example, the ceftazidime kinetic data predicted a lack of therapeutic efficacy, but these data showed activity at 500 ng/ml with a 10-fold improvement for P4A. Next, it was observed that P1A and P3A were quite effective for ampicillin, while P4A lost some activity for ampicillin. The amp/sulbactam data verified that sulbactam effectively inhibited P1A, P3A, and P4A. Whereas the amp/sulbactam data verified that sulbactam is an effective β -lactamase inhibitor, this should not be interpreted to mean that these β -lactamases will not be efficacious in vivo. Specifically, if the pharmacokinetics for biliary excretion are different for the antibiotic and the inhibitor, then the antibiotic can be found in the small intestine in the absence of its inhibitor. For example, P1A has been shown to be efficacious in humans with piperacillin/tazobactam. The meropenem data confirmed what was expected, that the P1A, P3A, and P4A β -lactamases are penicillinases and cephalosporinases. Finally, P4A was observed to be remarkably effective at degrading ceftriaxone, better than P3A. P4A was as good for ceftriaxone as it was for ampicillin. P4 also had improved activity for several important cephalosporins with the exception of cefepime. In comparison to P1A, P4A exhibited about 10-fold or greater β -lactamase activity for ceftriaxone, cefotaxime, ceftazidime, ceftazolin, cefoperazone, and cefurixine.

Example 5

Analysis of the Antibiotic Degradation Properties of P1A, P3A, and P4A

P1A and P4A were screened for degradation of ampicillin, ceftriazone, cefotaxime, ceftazolin, cefuroxime, cefoperazone, cefepime, and ceftazidime in a microtiter plate activity assay that was designed to mimic the activity of the β -lactamases in the gut in the presence of high antibiotic concentrations. The assay was performed by mixing 10, 100 or 1000 μ g/ml of the indicated antibiotics with P1A or P4A, at concentrations of 10 or 100 ng/ml. Plates were either incubated for one hour at 37° C. after which *E. coli* (ATCC 25922) was added, or *E. coli* was added immediately after the addition of the β -lactamase enzymes. The plates were incubated overnight and the bacterial growth quantified by measuring the absorbance at 625 nm (OD_{625}) in a Spectramax 384 Plus plate reader. The analysis was performed twice for each β -lactamase and antibiotic. The β -lactamase activity of the culture supernatants was determined as positive or negative based on the appearance of bacterial growth in the individual wells. An OD_{625} of 1.0 or greater indicated maximal bacterial growth, therefore complete antibiotic degradation and high β -lactamase activity. An OD_{625} of less than 1.0 indicated low bacterial growth therefore incomplete antibiotic degradation, hence low β -lactamase activity.

Results from the analysis are shown in FIG. 8. There was no difference in the readings obtained for the plates that were preincubated with the β -lactamase and antibiotic compared to the plates that did not undergo preincubation.

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For all antibiotics, except ampicillin, P4A showed improved activity relative to P1A. P1A showed no activity for ceftriaxone, cefotaxime, and ceftazidime. P4A shows greater efficacy for cefotaxime, ceftazolin, cefoperazone, cefepime, and ceftazidime.

Accordingly, modification of five (P4A) amino acids in P1A boosted cephalosporinase activity 10-1000-fold. Oral administration of P4A can therefore protect the gut flora from commonly used cephalosporins and extend the clinical utility of this prophylactic strategy to the prevention of, for example, CDI. Specifically, the assays presented herein show that, at concentrations that should be readily achievable in the small intestine, P4A can degrade cephalosporins and other antibiotics and have an improved substrate profile relative to P1A.

DEFINITIONS

The following definitions are used in connection with the invention disclosed herein. Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood to one of skill in the art to which this invention belongs.

As used herein, "a," "an," or "the" can mean one or more than one.

Further, the term "about" when used in connection with a referenced numeric indication means the referenced numeric indication plus or minus up to 10% of that referenced numeric indication. For example, the language "about 50" covers the range of 45 to 55.

An "effective amount," when used in connection with medical uses is an amount that is effective for providing a measurable treatment, prevention, or reduction in the rate of pathogenesis of a disease of interest.

As referred to herein, all compositional percentages are by weight of the total composition, unless otherwise specified. As used herein, the word "include," and its variants, is intended to be non-limiting, such that recitation of items in a list is not to the exclusion of other like items that may also be useful in the compositions and methods of this technology. Similarly, the terms "can" and "may" and their variants are intended to be non-limiting, such that recitation that an embodiment can or may comprise certain elements or features does not exclude other embodiments of the present technology that do not contain those elements or features.

Although the open-ended term "comprising," as a synonym of terms such as including, containing, or having, is used herein to describe and claim the invention, the present invention, or embodiments thereof, may alternatively be described using alternative terms such as "consisting of" or "consisting essentially of."

As used herein, the words "preferred" and "preferably" refer to embodiments of the technology that afford certain benefits, under certain circumstances. However, other embodiments may also be preferred, under the same or other circumstances. Furthermore, the recitation of one or more preferred embodiments does not imply that other embodiments are not useful, and is not intended to exclude other embodiments from the scope of the technology.

The amount of compositions described herein needed for achieving a therapeutic effect may be determined empirically in accordance with conventional procedures for the particular purpose. Generally, for administering therapeutic agents (e.g. inventive β -lactamases and/or pharmaceutical compositions (and/or additional agents) for therapeutic purposes, the therapeutic agents are given at a pharmacologically effective dose. A "pharmacologically effective amount," "pharmacologically effective dose," "therapeutically effective amount," or

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“effective amount” refers to an amount sufficient to produce the desired physiological effect or amount capable of achieving the desired result, particularly for treating the disorder or disease. An effective amount as used herein would include an amount sufficient to, for example, delay the development of a symptom of the disorder or disease, alter the course of a symptom of the disorder or disease (e.g., slow the progression of a symptom of the disease), reduce or eliminate one or more symptoms or manifestations of the disorder or disease, and reverse a symptom of a disorder or disease. For example, administration of therapeutic agents to a patient suffering from a GI tract disorder (e.g. CDI) provides a therapeutic benefit not only when the underlying condition is eradicated or ameliorated, but also when the patient reports a decrease in the severity or duration of the symptoms associated with the disease. Therapeutic benefit also includes halting or slowing the progression of the underlying disease or disorder, regardless of whether improvement is realized.

Effective amounts, toxicity, and therapeutic efficacy can be determined by standard pharmaceutical procedures in cell cultures or experimental animals, e.g., for determining the LD50 (the dose lethal to about 50% of the population) and the ED50 (the dose therapeutically effective in about 50% of the population). The dosage can vary depending upon the dosage form employed and the route of administration utilized. The dose ratio between toxic and therapeutic effects is the therapeutic index and can be expressed as the ratio LD50/ED50. In some embodiments, compositions and methods that exhibit large therapeutic indices are preferred. A therapeutically effective dose can be estimated initially from in vitro assays, including, for example, cell culture assays. Also, a dose can be formulated in animal models to achieve a circulating plasma concentration range that includes the IC50 as determined in cell culture, or in an appropriate animal model. Levels of the described compositions in plasma can be measured, for example, by high performance liquid chromatography. The effects of any particular dosage can be monitored by a suitable bioassay. The dosage can be determined by a physician and adjusted, as necessary, to suit observed effects of the treatment.

In certain embodiments, the effect will result in a quantifiable change of at least about 10%, at least about 20%, at least

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about 30%, at least about 50%, at least about 70%, or at least about 90%. In some embodiments, the effect will result in a quantifiable change of about 10%, about 20%, about 30%, about 50%, about 70%, or even about 90% or more. In certain embodiments, the effect will result in a quantifiable change of two-fold, or three-fold, or four-fold, or five-fold, or ten-fold. Therapeutic benefit also includes halting or slowing the progression of the underlying disease or disorder or reduction in toxicity, regardless of whether improvement is realized.

EQUIVALENTS

While the invention has been described in connection with specific embodiments thereof, it will be understood that it is capable of further modifications and this application is intended to cover any variations, uses, or adaptations of the invention following, in general, the principles of the invention and including such departures from the present disclosure as come within known or customary practice within the art to which the invention pertains and as may be applied to the essential features herein set forth and as follows in the scope of the appended claims.

Those skilled in the art will recognize, or be able to ascertain, using no more than routine experimentation, numerous equivalents to the specific embodiments described specifically herein. Such equivalents are intended to be encompassed in the scope of the following claims.

INCORPORATION BY REFERENCE

All patents and publications referenced herein are hereby incorporated by reference in their entireties.

The publications discussed herein are provided solely for their disclosure prior to the filing date of the present application. Nothing herein is to be construed as an admission that the present invention is not entitled to antedate such publication by virtue of prior invention.

As used herein, all headings are simply for organization and are not intended to limit the disclosure in any manner. The content of any individual section may be equally applicable to all sections.

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 7

<210> SEQ ID NO 1

<211> LENGTH: 263

<212> TYPE: PRT

<213> ORGANISM: *Bacillus licheniformis*

<400> SEQUENCE: 1

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Leu Gly Ile Phe Ala Leu Asp Thr Gly Thr Asn Arg Thr Val Ala Tyr
20 25 30

Arg Pro Asp Glu Arg Phe Ala Phe Ala Ser Thr Ile Lys Ala Leu Thr
35 40 45

Val Gly Val Leu Leu Gln Gln Lys Ser Ile Glu Asp Leu Asn Gln Arg
50 55 60

Ile Thr Tyr Thr Arg Asp Asp Leu Val Asn Tyr Asn Pro Ile Thr Glu
65 70 75 80

Lys His Val Asp Thr Gly Met Thr Leu Lys Glu Leu Ala Asp Ala Ser

-continued

85					90					95					
Leu	Arg	Tyr	Ser	Asp	Asn	Ala	Ala	Gln	Asn	Leu	Ile	Leu	Lys	Gln	Ile
			100					105					110		
Gly	Gly	Pro	Glu	Ser	Leu	Lys	Lys	Glu	Leu	Arg	Lys	Ile	Gly	Asp	Glu
		115					120					125			
Val	Thr	Asn	Pro	Glu	Arg	Phe	Glu	Pro	Glu	Leu	Asn	Glu	Val	Asn	Pro
		130					135					140			
Gly	Glu	Thr	Gln	Asp	Thr	Ser	Thr	Ala	Arg	Ala	Leu	Val	Thr	Ser	Leu
		145					150					155			160
Arg	Ala	Phe	Ala	Leu	Glu	Asp	Lys	Leu	Pro	Ser	Glu	Lys	Arg	Glu	Leu
			165					170						175	
Leu	Ile	Asp	Trp	Met	Lys	Arg	Asn	Thr	Thr	Gly	Asp	Ala	Leu	Ile	Arg
		180						185					190		
Ala	Gly	Val	Pro	Asp	Gly	Trp	Glu	Val	Ala	Asp	Lys	Thr	Gly	Ala	Ala
		195					200					205			
Ser	Tyr	Gly	Thr	Arg	Asn	Asp	Ile	Ala	Ile	Ile	Trp	Pro	Pro	Lys	Gly
		210					215					220			
Asp	Pro	Val	Val	Leu	Ala	Val	Leu	Ser	Ser	Arg	Asp	Lys	Lys	Asp	Ala
		225					230					235			240
Lys	Tyr	Asp	Asp	Lys	Leu	Ile	Ala	Glu	Ala	Thr	Lys	Val	Val	Met	Lys
			245					250						255	
Ala	Leu	Asn	Met	Asn	Gly	Lys									
			260												

<210> SEQ ID NO 2

<211> LENGTH: 792

<212> TYPE: DNA

<213> ORGANISM: Bacillus licheniformis

<400> SEQUENCE: 2

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gcttcgacga ttaagccttt aactgtaggc gtgcttttgc aacagaaatc aatagaagat      180
ctgaaccaga gaataacata tacacgtgat gatcttgtaa actacaaccc gattacggaa      240
aagcacgttg atacgggaat gacgctcaaa gagcttgccg atgcttcgct tcgatatagt      300
gacaatgcgg cacagaatct cattcttaaa caaattggcg gacctgaaag tttgaaaaag      360
gaactgagga agattggtga tgaggttaca aatcccgaac gattcgaacc agagttaaat      420
gaagtgaatc cgggtgaaac tcaggatacc agtacagcaa gagcacttgt cacaagcctt      480
cgagcctttg ctcttgaaaga taaacttcca agtgaaaaac gcgagctttt aatcgattgg      540
atgaaacgaa ataccactgg agacgcctta atccgtgccg gtgtgccgga cggttgggaa      600
gtggctgata aaactggagc ggcatcatat ggaacccgga atgacattgc catcatttgg      660
ccgccaaaag gagatcctgt cgttcttgca gtattatcca gcagggataa aaaggacgcc      720
aagtatgatg ataaacttat tgcagaggca acaaaggtgg taatgaaagc cttaaacatg      780
aacggcaaat aa                                           792

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<210> SEQ ID NO 3

<211> LENGTH: 299

<212> TYPE: PRT

<213> ORGANISM: Bacillus licheniformis

<400> SEQUENCE: 3

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attggtgatg aggttacaaa tcccgaacga ttccaaccag agttaaatga agtgaatccg 540
ggtgaaactc aggataccag tacagcaaga gcacttgta caagccttcg agcctttgct 600
cttgaagata aacttccaag tgaaaaacgc gagcttttaa tcgattggat gaaacgaaat 660
accactggag acgccttaat ccgtgccggt gtgccggacg gttgggaagt ggctgataaa 720
actggagcgg catcatatgg aaccgcgaat gacattgcca tcatttggcc gccaaaagga 780
gatectgtcg ttcttgcaat attatccagc agggataaaa aggacgcaa gtatgatgat 840
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<210> SEQ ID NO 5
<211> LENGTH: 262
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

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<400> SEQUENCE: 5

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Glu Met Lys Asp Asp Phe Ala Lys Leu Glu Glu Gln Phe Asp Ala Lys
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Leu Gly Ile Phe Ala Leu Asp Thr Gly Thr Asn Arg Thr Val Ala Tyr
20     25     30
Arg Pro Asp Glu Arg Phe Ala Phe Ala Ser Thr Ile Lys Ala Leu Thr
35     40     45
Val Gly Val Leu Leu Gln Gln Lys Ser Ile Glu Asp Leu Asn Gln Arg
50     55     60
Ile Thr Thr Arg Asp Asp Leu Val Asn Tyr Asn Pro Ile Thr Glu Lys
65     70     75     80
His Val Asp Thr Gly Met Thr Leu Lys Glu Leu Ala Asp Ala Ser Leu
85     90     95
Arg Tyr Ser Asp Asn Ala Ala Gln Asn Leu Ile Leu Lys Gln Ile Gly
100    105    110
Gly Pro Glu Ser Leu Lys Lys Glu Leu Arg Lys Ile Gly Asp Glu Val
115    120    125
Thr Asn Pro Glu Arg Phe Glu Pro Glu Leu Asn Glu Val Asn Pro Gly
130    135    140
Glu Thr Gln Asp Thr Ser Thr Ala Arg Ala Leu Val Thr Ser Leu Arg
145    150    155    160
Ala Phe Ala Leu Glu Asp Lys Leu Pro Ser Glu Lys Arg Glu Leu Leu
165    170    175
Ile Asp Trp Met Lys Arg Asn Thr Thr Gly Asp Ala Leu Ile Arg Ala
180    185    190
Gly Val Pro Asp Gly Trp Glu Val Gly Asp Lys Thr Gly Ser Gly Asp
195    200    205
Tyr Gly Thr Arg Asn Asp Ile Ala Ile Ile Trp Pro Pro Lys Gly Asp
210    215    220
Pro Val Val Leu Ala Val Leu Ser Ser Arg Asp Lys Lys Asp Ala Lys
225    230    235    240
Tyr Asp Asn Lys Leu Ile Ala Glu Ala Thr Lys Val Val Met Lys Ala
245    250    255
Leu Asn Met Asn Gly Lys
260

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<210> SEQ ID NO 6

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<211> LENGTH: 299
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 6

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Met Ile Gln Lys Arg Lys Arg Thr Val Ser Phe Arg Leu Val Leu Met
1           5           10           15

Cys Thr Leu Leu Phe Val Ser Leu Pro Ile Thr Lys Thr Ser Ala Gln
20           25           30

Ala Ser Lys Thr Glu Met Lys Asp Asp Phe Ala Lys Leu Glu Glu Gln
35           40           45

Phe Asp Ala Lys Leu Gly Ile Phe Ala Leu Asp Thr Gly Thr Asn Arg
50           55           60

Thr Val Ala Tyr Arg Pro Asp Glu Arg Phe Ala Phe Ala Ser Thr Ile
65           70           75           80

Lys Ala Leu Thr Val Gly Val Leu Leu Gln Gln Lys Ser Ile Glu Asp
85           90           95

Leu Asn Gln Arg Ile Thr Tyr Thr Arg Asp Asp Leu Val Asn Tyr Asn
100          105          110

Pro Ile Thr Glu Lys His Val Asp Thr Gly Met Thr Leu Lys Glu Leu
115          120          125

Ala Asp Ala Ser Leu Arg Tyr Ser Asp Asn Ala Ala Gln Asn Leu Ile
130          135          140

Leu Lys Gln Ile Gly Gly Pro Glu Ser Leu Lys Lys Glu Leu Arg Lys
145          150          155          160

Ile Gly Asp Glu Val Thr Asn Pro Glu Arg Phe Glu Pro Glu Leu Asn
165          170          175

Glu Val Asn Pro Gly Glu Thr Gln Asp Thr Ser Thr Ala Arg Ala Leu
180          185          190

Val Thr Ser Leu Arg Ala Phe Ala Leu Glu Asp Lys Leu Pro Ser Glu
195          200          205

Lys Arg Glu Leu Leu Ile Asp Trp Met Lys Arg Asn Thr Thr Gly Asp
210          215          220

Ala Leu Ile Arg Ala Gly Val Pro Asp Gly Trp Glu Val Gly Asp Lys
225          230          235          240

Thr Gly Ser Gly Asp Tyr Gly Thr Arg Asn Asp Ile Ala Ile Ile Trp
245          250          255

Pro Pro Lys Gly Asp Pro Val Val Leu Ala Val Leu Ser Ser Arg Asp
260          265          270

Lys Lys Asp Ala Lys Tyr Asp Asn Lys Leu Ile Ala Glu Ala Thr Lys
275          280          285

Val Val Met Lys Ala Leu Asn Met Asn Gly Lys
290          295

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<210> SEQ ID NO 7
 <211> LENGTH: 900
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 7

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atgattcaaa aacgaaagcg gacagtttcg ttcagacttg tgcttatgtg cacgctgtta      60

tttgtcagtt tgccgattac aaaaacatca gcgcaagctt ccaagacgga gatgaaagat      120

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gattttgcaa aacttgagga acaatttgat gcaaaactcg ggatctttgc attggatata	180
ggtaaaaacc ggacggtagc gtatcggccg gatgagcggt ttgcttttgc ttgcacgatt	240
aaggctttaa ctgtaggcgt gcttttgcaa cagaaatcaa tagaagatct gaaccagaga	300
ataacatata cacgtgatga tcttgtaaac tacaaccgga ttacggaaaa gcacgttgat	360
acgggaatga cgctcaaaga gcttgccgat gcttcgcttc gatatagtga caatgcggca	420
cagaatctca ttcttaaca aattggcgga cctgaaagt tgaaaaagga actgaggaag	480
attggtgatg aggttacaaa tcccgaaaga ttcgaaaccag agttaaata agtgaatccg	540
ggtgaaactc aggataccag tacagcaaga gcacttgta caagccttcg agcctttgct	600
cttgaagata aacttccaag tgaaaaacgc gagcttttaa tcgattggat gaaacgaaat	660
accactggag acgccttaat ccgtgccggt gtgccggacg gttgggaagt gggtgataaa	720
actggaagcg gagattatgg aacccggaat gacattgcca tcatttggcc gccaaaagga	780
gatactgtcg ttcttgcaat attatccagc agggataaaa aggacgcca gtatgataat	840
aaacttattg cagaggcaac aaaggtggta atgaaagcct taaacatgaa cggaataaa	900

What is claimed is:

1. A beta-lactamase comprising an amino acid sequence having at least 90% sequence identity with SEQ ID NO: 1 and having an A238T mutation according to Ambler classification.

2. The beta-lactamase of claim 1, further comprising a G156R mutation according to Ambler classification.

3. The beta-lactamase of claim 1, further comprising a D276N mutation according to Ambler classification.

4. The beta-lactamase of claim 3, further comprising a F33Y mutation according to Ambler classification.

5. The beta-lactamase of any one of claims 1-4, wherein the beta-lactamase hydrolyzes one or more of penicillins and cephalosporins.

6. The beta-lactamase of claim 5, wherein the penicillin is ampicillin.

7. The beta-lactamase of claim 5, wherein the cephalosporin is selected from ceftriaxone, cefotaxime, cefazolin, cefoperazone, cefepime, cefuroxime, and ceftazidime.

8. The beta-lactamase of any one of claims 1-4, wherein the beta-lactamase has improved enzymatic activity against a cephalosporin as compared to SEQ ID NO: 1.

9. The beta-lactamase of claim 8, wherein the cephalosporin is selected from cefotaxime, cefepime, and ceftriaxone.

10. The beta-lactamase of any one of claims 1-4, wherein the beta-lactamase has improved catalytic efficiency against a cephalosporin as compared to SEQ ID NO: 1.

11. The beta-lactamase of claim 10, wherein the beta-lactamase has about a 10-fold to about a 1000-fold improved catalytic efficiency against a cephalosporin as compared to SEQ ID NO: 1.

12. The beta-lactamase of claim 11, wherein the cephalosporin is ceftriaxone.

13. The beta-lactamase of any one of claims 1-4, wherein the beta-lactamase is active in the GI tract.

14. The beta-lactamase of claim 13, wherein the beta-lactamase is stable in the small intestine, optionally selected from one or more of the duodenum, jejunum, and ileum.

15. A polynucleotide comprising a polynucleotide sequence encoding the beta-lactamase of any one of claims 1-4.

25 16. A host cell comprising the polynucleotide of claim 15.

17. A pharmaceutical composition, comprising the beta-lactamase of any one of claims 1-4 and a pharmaceutically acceptable carrier or excipient.

30 18. The pharmaceutical composition of claim 17, wherein the composition is formulated for oral administration, optionally selected from a tablet, a multi-particulate sprinkle, and a multi-particulate capsule.

19. A method for preventing an antibiotic-induced adverse effect in the GI tract, comprising administering an effective amount of a beta-lactamase to a patient in need thereof, wherein the beta-lactamase comprises an amino acid sequence having at least 90% sequence identity with SEQ ID NO: 1 and having an A238T mutation according to Ambler classification.

20. The method of claim 19, wherein the beta-lactamase further comprises a G156R mutation according to Ambler classification.

21. The method of claim 19, wherein the beta-lactamase further comprises a D276N mutation according to Ambler classification.

22. The method of claim 21, wherein the beta-lactamase further comprises a F33Y mutation according to Ambler classification.

23. The method of any one of claims 19-22, wherein the subject is being administered or will be administered an antibiotic.

24. A method for treating or preventing an antibiotic induced *C. difficile* infection (CDI) and/or a *C. difficile*-associated disease, comprising administering an effective amount of a beta-lactamase to a patient in need thereof, wherein the beta-lactamase comprises an amino acid sequence having at least 90% sequence identity with SEQ ID NO: 1 and having an A238T mutation according to Ambler classification.

25. The method of claim 24, wherein the beta-lactamase further comprises a G156R mutation according to Ambler classification.

26. The method of claim 24, wherein the beta-lactamase further comprises a D276N mutation according to Ambler classification.

27. The method of claim 26, wherein the beta-lactamase further comprises a F33Y mutation according to Ambler classification.

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28. The method of any one of claims **24-27**, wherein the subject is being administered or will be administered an antibiotic.

29. The method of any one of claims **24-27**, wherein the *C. difficile*-associated disease is antibiotic-associated diarrhea. 5

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